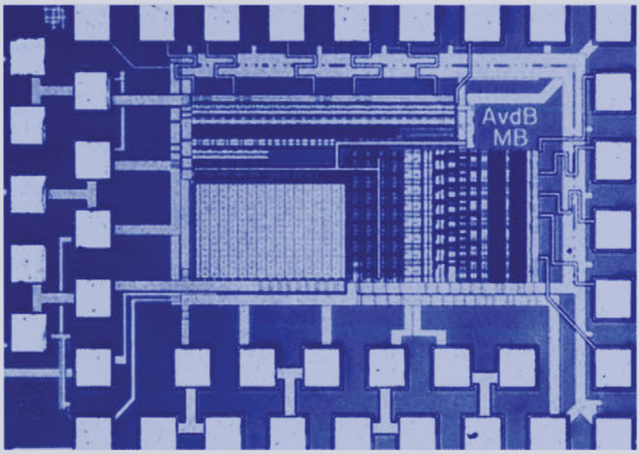
STATIC AND DYNAMIC PERFORMANCE LIMITATIONS FOR HIGH SPEED D/A CONVERTERS

Anne Van den Bosch, Michiel Steyaert
and Willy Sansen



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FOR HIGH SPEED D/A CONVERTERS

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Abstract

Although the digital world gains in importance due to the decreasing feature size of the transistors, the analog building blocks remain indispensable. The design of the interface between the two worlds has to comply with the strict requirements of both sides. The presented research focusses on the design of current steering D/A converters in the modern telecommunication systems of tomorrow.

An accurate description of both the static and the dynamic behaviour is of the utmost importance when designing a current steering D/A converter with a high performance. A new formula has been derived that accurately describes the INL-yield of a D/A converter as a function of the transistor mismatch behaviour in a given technology. The influence of systematic errors introduced by linear and quadratic gradients has to be minimised by the use of special switching schemes. Based on these elements, two 12 bit and one 14 bit current steering D/A converter have been implemented.

Apart from the three well known factors influencing the dynamic behaviour of the D/A converter, a fourth element has been introduced. The frequency dependency of the output impedance has a negative effect on high resolution, high speed current steering D/A converters. Taking this new element into account during the design resulted in the realisation of a 10 bit D/A converter with a Nyquist performance in the entire frequency band for a clock frequency up to 1 GS/s.

To determine the optimal segmentation level of a current steering D/A converter, a statistical analysis has been performed. It could be concluded that if the requirement dictated by the transistor mismatch of the unit current sources is fulfilled, both the INL and the DNL error are smaller than 1/2 LSB regardless of the number of bits implemented in a binary way. A low segmentation level implies a low decoder complexity and as such leads to a small power and area consumption. Based on these findings, a fully binary 10 bit D/A converter has been realised with a small power consumption and a good dynamic performance.

The second part of this work is focussed on transistor mismatch. An overview is given of the most important mismatch models and on the influence of both the immediate surroundings and the used transistor topology. Also the link between the static performance of a current steering D/A converter and the extraction of the transistor

mismatch parameters has been investigated. It is suggested that a current steering D/A converter could act as a test structure for matching characterisation.

Contents

List of Symbols and Abbreviations

Symbols

A_{VT}, A_{β}	MOS transistor mismatch parameters
β	Current factor of a MOS transistor
C_{gs}, C_{gd}, etc.	Device capacitances
dB	decibel
f	Frequency
gm	Transconductance
go	Transistor small-signal output conductance
L, W	Transistor gate-length and gate-width
V_T	Threshold voltage
q	Elementary Charge

Abbreviations

AC,ac	Alternating Current
ADC, A/D-converter	Analog-to-Digital Converter
ASIC	Application Specific Integrated Circuit
CMOS	Complementary Metal Oxide Semiconductor
CPU	Central Processing Unit
DAC, D/A-converter	Digital-to-Analog Converter
DC, dc	Direct Current
DNL	Differential non-linearity
DSP	Digital Signal Processing
ENOB	Effective Number of Bits
FSOC	Full scale output current
FOM	Figure-of-Merit
GBW	Gain-bandwidth product
GSM	Global System for Mobile communications

HF	High Frequency
IC	Integrated Circuit
INL	Integral non-linearity
LSB	Least Significant Bit
MSB	Most Significant Bit
nMOS	N-channel Metal Oxide Semiconductor
pMOS	P-channel Metal Oxide Semiconductor
RF	Radio Frequency
SFDR	Spurious Free Dynamic Range
SNDR	Signal to Noise and Distortion Ratio
S/N ratio, SNR	Signal-to-Noise Ratio
SR	Slew Rate
THD	Total Harmonic Distortion
VLSI	Very Large Scale Integration
xDSL	Any type of digital subscriber line

Chapter 1

Introduction

1.1 Introduction

During the last decade, the telecommunication market and especially the mobile telecommunication systems have known an unprecedented growth. New services are constantly being introduced since the evolution towards smaller technologies makes it possible to integrate millions of transistors on a single chip. The digital designers create new DSP (Digital Signal Processing) architectures that allow complex algorithms to be implemented at very high computational speeds. In parallel, analog designers have been putting an enormous effort in developing high speed, low distortion blocks that in combination with the digital building blocks will result in the high performance telecommunication systems of tomorrow. However, this implies that the design of the interface circuit between the analog and digital part of the system -the D/A converter and the A/D converter- is becoming more challenging in time. Besides a highly accurate circuit also a high operation speed achieved at a low power consumption are demanded. The combination of these constraints poses a real challenge for the designers of these circuits. Furthermore, additional problems -like the substrate noise coupling from the digital to the analog part on the chip and the scaling of the power supply voltage- add to the complexity of the design.

The presented work will focus on the design and implementation of high performance D/A converters and especially the current steering topology which offers at the moment the best result when realising the aforementioned constraints of speed, accuracy and power consumption in CMOS technologies. Since the static behaviour of a current steering architecture strongly depends on the mismatch behaviour of the transistors used to implement the current sources, this work will also address this issue in some more detail.

1.2 Outline of the Research Work

Fig.1.1 gives a schematic overview of this manuscript. First, chapter 2 will introduce the main set of specifications that are needed to fully describe the performance of a D/A converter. In the past, most publications only described the static behaviour of the presented D/A converter, while the telecommunication engineers of today also need information on the frequency domain behaviour of the circuit in order to accurately determine its impact on the whole system they are developing. Therefore, the second chapter of this work starts with a description of the basic functionality of an ideal D/A converter and the limitations this imposes on its attainable performance. Then, the different specifications that are commonly used (INL, DNL,...) to describe the static performance of a D/A converter, are discussed in more detail. The remainder of this chapter gives an overview of the dynamic specifications (SFDR,...).

Chapter 3 gives an overview of the different topologies existing today to implement a D/A converter. The basic functionality of this circuit is to generate for each digital input code a multiple of a certain reference quantity. Dependent on this quantity (a voltage, a charge or a current), three classes of D/A converters can be identified namely the resistor, the capacitor and the current steering architecture. In the first part of this chapter, both the resistor and the capacitor D/A converter will be discussed. It is the intention of the author to emphasise the existence of these architectures as useful alternatives for the current steering architecture. For a detailed study the reader is referred to [Razavi, Johns]. The remainder of this chapter describes the current steering topology. This topology will be analysed throughout this work.

The current steering topology is at the moment the preferred architecture for telecommunication applications requiring a high accuracy. In the fourth chapter, the emphasis will be put on the different factors that influence the static performance of this topology. This performance is mainly determined by the matching behaviour of the current source transistors. Since no two transistors behave exactly the same due to technological variations introduced during processing, it is important to know the impact of this phenomenon on the yield and the performance. This topic will be discussed in the first part of this chapter. Apart from the process variations which generate random errors, the current sources in the array are also influenced by systematic errors that are introduced by thermal, electrical and process gradients which will be discussed in detail in the second part of this chapter.

Chapter 5 will focus on the dynamic performance of the current steering D/A converter. The first part of this chapter will identify and discuss some design guidelines that will solve the generally known problems such as the imperfect synchronisation of the switch control signals, the digital signal feedthrough through the gate-drain capacitance of the switch transistors and the drain voltage variation of the current source transistor. The remainder of this chapter will focus on a fourth factor with a major impact on the frequency domain behaviour of the D/A converter, namely the dynamic output impedance. A formula will be derived for the spurious free dynamic range as a function of this impedance, providing the designer with the necessary information to design a high speed Nyquist D/A converter. For high resolution circuits, this implies the use of cascoded current cell structures as is discussed in the last part of this chapter.

The next chapter covers the design flow of a current steering topology. After discussing the approach to determine the segmentation level (number of binary and unary bits), the considerations that have to be made in choosing the thermometer decoder and the switch driver will be discussed. The remainder of this chapter describes the dimensioning of the transistors of the unit current cell (the switch, current source and cascode transistor).

In chapter 7, several implemented D/A converters will be discussed ranging from a high accuracy circuit (12 bits and 14 bits) to a high speed circuit (10 bits 1GSample/s) and to a combination of both (12 bits 500 MSamples/s). Also the design of a 10 bit low power D/A converter for telecommunication applications will be addressed. The last section of this chapter gives an overview of the most important static and dynamic specifications of the realised D/A converters and of the state-of-the-art published devices. In order to make a comparison possible, a figure of merit will be introduced that is based on the resolution, the dynamic behaviour and the power consumption.

It is easy to understand that the design of high performance analog circuits such as D/A and A/D converters, reference sources, ... requires the availability of reliable transistor mismatch models. Designers have to be able to rely on accurate simulation tools if they want to be successful in the realisation of a circuit with a performance that lies closely to the limits of the given technology. However, the accuracy of such simulation tools is determined by the underlying models. At this moment, the analog designer has to introduce large safety margins to guarantee the required performance of the circuit leading however to an unnecessary power consumption and an operation speed reduction [Kinge CICC96]. In literature, several models have been presented that describe the mismatch characteristics of the transistor by providing the designer with the standard deviation of the mismatch in a set of electrical parameters (like the threshold voltage V_T, the current factor β, the mobility degradation parameter θ, the bulk threshold parameter γ). In the first part of chapter 8, an overview of these models will be given starting with the basic models presented by Lakshmikumar and Pelgrom [Laksh JSSC86, Pelgr JSSC89]. However, going to deep submicron technologies, these models have to be adapted as to explain the transistor mismatch of short and narrow devices. The extraction of the transistor mismatch parameters for a 0.5 and a 0.4 μm standard CMOS technology and the transistor mismatch dependency on the used topology (wafer, quad or hexagonal structure) and its immediate surroundings (metal coverage, ...) will be discussed.

For the VLSI manufacturer, it is important to be able to provide his customers with the necessary quantitative data of the matching quality of his technology. This can be achieved by the use of dedicated test circuits that are especially designed for this purpose (low parasitics, high measurement accuracy, ...). Furthermore, it is necessary to follow the evolution of the matching performance of the technology over different runs in time. This requires a test circuit that gives a good indication of the matching technology for a low cost. However, these circuits are of no further use to the manufacturer or the designer and therefore an alternative structure has been sought for. A high performance current steering D/A converter is highly dependent on the matching quality of the technology. Furthermore, the evaluation of the D/A converter's performance poses no problem since there exist standardised test procedures. In the last part of chapter 8, the D/A converter's performance will be directly translated in the transistor mismatch characteristics of the used technology.

Finally, the last chapter of this work gives a summary of the main results that have been achieved in this work and some recommendations are formulated towards future research.

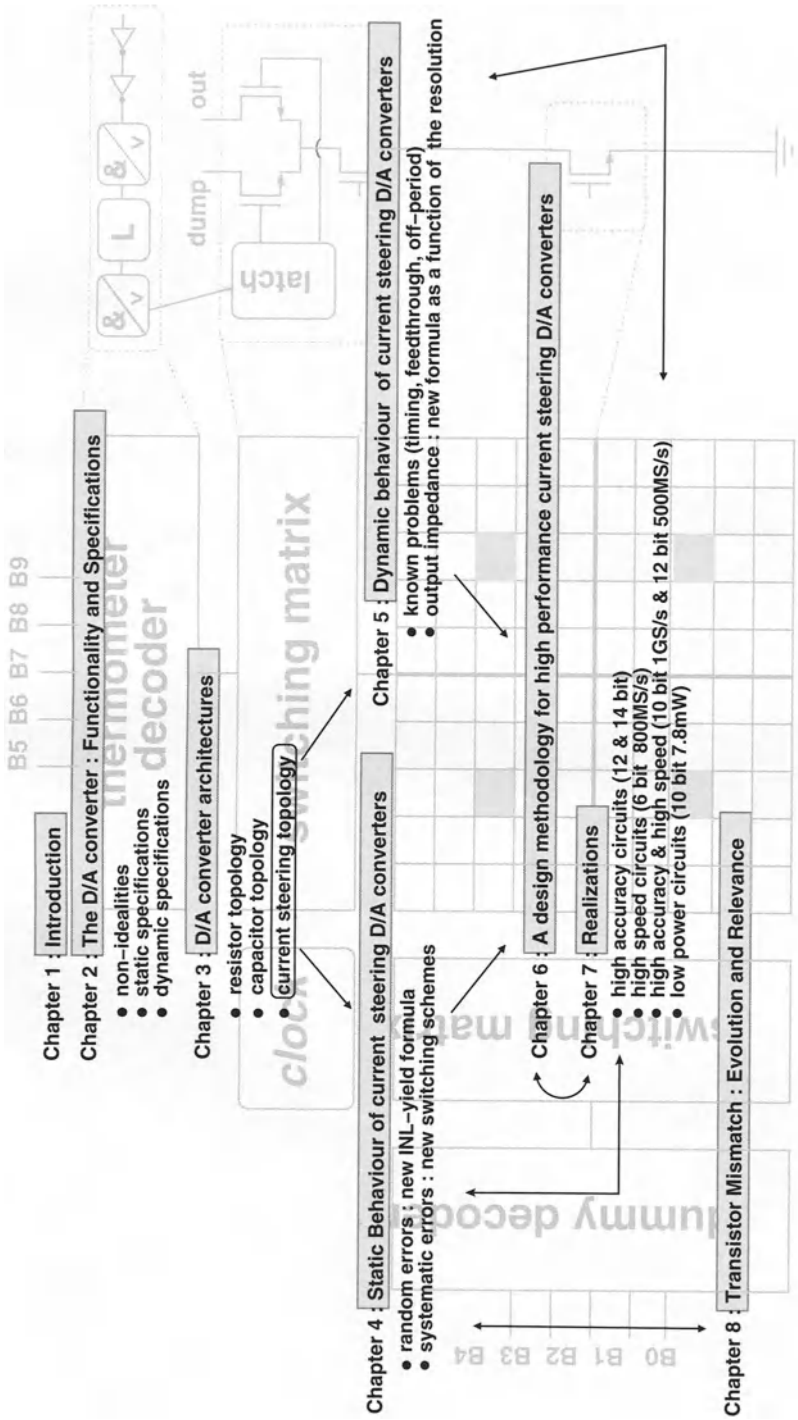


Figure 1.1: Outline of the presented work

Chapter 2

The D/A Converter: Functionality and Specifications

2.1 Introduction

Although the digital world is gaining in importance due to the decreasing feature size of transistors in the rapidly evolving semiconductor technology, the analog building blocks have proven to be indispensable. The advantages of digital processing (easier design, extensive programmability, ...) are counteracted by the fact that people perceive information in an analog form (f.i. speech). The design of the interface between the analog and the digital system (the A/D and the D/A converter) has to comply with the stringent specifications required by modern complex digital systems. Furthermore, additional problems -like the substrate noise coupling from the digital to the analog parts on the chip- add to the complexity of the design.

Where a few years ago, most papers only described the static behaviour of the D/A converter, the telecommunication engineers nowadays also need information on the frequency domain performance of these devices. The specifications describing both the static and the dynamic behaviour of a D/A converter will be discussed in this chapter.

2.2 The Basic D/A Converter Function

2.2.1 Analog and Digital Signals

Fig.2.1 gives a general schematic representation of the telecommunication systems that are used nowadays. All the signals in the our surrounding world have an analog

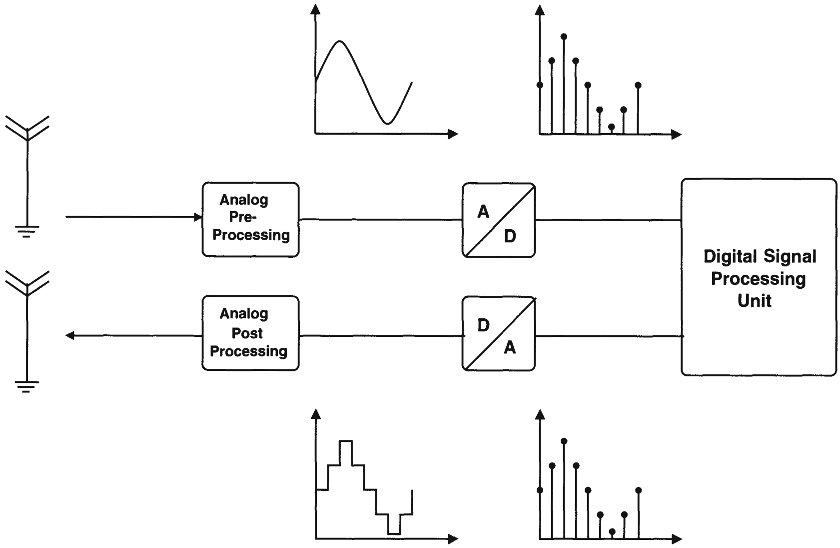


Figure 2.1: *The general schematic of a telecommunication system*

nature (like f.i. speech), which means that those signals have a continuous amplitude varying in time. However, in order to perform the required signal operations mostly digital algorithms are used since digital signal processing systems are relatively easy to design and are highly flexible due to their programmable software. Furthermore, they are capable of realising quite complex data manipulation algorithms and incorporate a high computing power on a small area. Digital signals can have only two values, namely, a high state ("1") and a low state ("0") and are only allowed to vary according to a specified clock. These signals are therefore discrete in time and amplitude. To transform an analog signal into a digital signal and vice versa, A/D and D/A converters are used.

However, since a N-bit digital signal can only have 2^N values, the amount of information at any given moment in time is limited and depends on the resolution (N) of the D/A converter. To illustrate this principle, fig.2.2 is given. Describing an analog signal at a given update rate with an one bit accuracy, only allows a coarse approximation as is indicated in fig.2.2.a. To improve the quality of the reconstructed analog signal, the resolution and/or the update rate of the D/A converter have to be increased (fig.2.2.b/c/d). It should be noted that a lower bound for the update rate is imposed by the Nyquist-Shannon theorem. This theorem states that the update rate of the D/A converter has to be larger than or equal to two times the highest frequency component of the signal in order to avoid aliasing and thus be able to fully reconstruct the signal. Although one can conclude from this discussion that a high accuracy D/A with a high update rate will give the best results, these devices will increase the cost

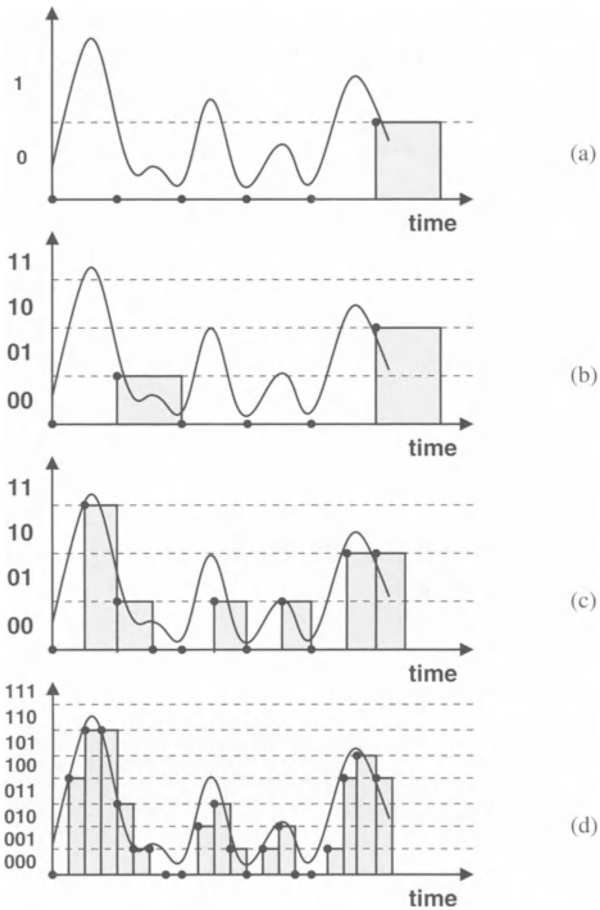


Figure 2.2: *Influence of the resolution and the update rate on the digital representation of an analog signal*

of the overall system. Therefore, the actual requirements of the D/A converter always depend on the application and the signals that need to be processed.

2.2.2 The D/A Converter as a Black Box

The input of the D/A converter is a N bit digital sequence that can be described by the following binary vector:

$$\bar{B} = (\, b_0\, b_1\, b_2 \cdots\, b_{N-1}) \qquad b_i \in \{0, 1\} \tag{2.1}$$

where b_0 is the least significant bit (LSB) and b_{N-1} is the most significant bit (MSB). The decimal number corresponding to this binary vector is equal to :

$$B = \sum_{i=0}^{N-1} 2^i \, b_i \tag{2.2}$$

The D/A converter's output signal can either be a charge, a voltage or a current and can be expressed as :

$$Y_{out}(B) = \sum_{i=0}^{N-1} 2^i \, b_i * Y_{ref} = B * Y_{ref} \tag{2.3}$$

where Y_{ref} denotes the unity (=LSB) charge, voltage or current depending on the D/A converter's architecture.

2.3 The Characteristics of an Ideal D/A Converter

2.3.1 Introduction

The function of an ideal D/A converter is the reconstruction of an analog waveform using sequences of digital words at the input. However, since these digital input words have a finite length and are updated at discrete time intervals, the output of the D/A converter has a shape similar to the one of a sample-and-hold circuit. The discrete nature of the D/A converter's waveform reconstruction process imposes quantifiable limitations on the accuracy with which even an ideal D/A converter can generate an analog waveform. These limitations will be discussed in more detail in this paragraph.

2.3.2 The Quantisation Error

By representing an analog signal with a finite number of bits, an error is introduced between the analog value and its digital representation. This error -which is called the quantisation error- inherently limits the maximum achievable dynamic range of the D/A converter. To better understand its impact on the dynamic behaviour of this device a short derivation is given [Razavi, VdPlassche, Stehr]. Fig.2.3 clearly shows that the quantisation process introduces an irreversible error since a signal with an amplitude $Y_j + \varepsilon$ will be quantised into the level Y_j. This procedure introduces a random error that can be considered to be equivalent to white noise in the frequency domain.

In this derivation ε denotes the quantisation error and Δ denotes the quantisation step (fig.2.3). Furthermore, it has been assumed that ε is :

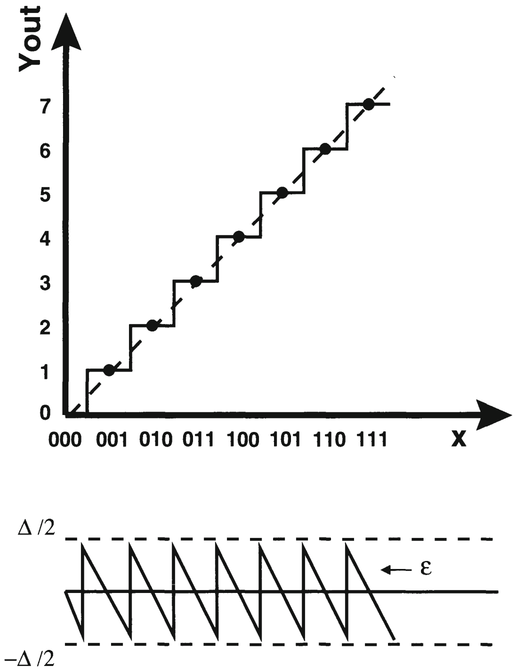


Figure 2.3: *The quantisation error of a D/A converter*

- a random variable that is uniformly distributed in the interval $[-\Delta/2, \Delta/2]$
- independent of the analog signal.

In this case, the quantisation noise power can be expressed as the mean square value of ε :

$$P_{noise} = E(\varepsilon^2) = \frac{1}{\Delta}\int_{-\Delta/2}^{\Delta/2} \varepsilon^2 d\varepsilon = \frac{\Delta^2}{12} \tag{2.4}$$

For a D/A converter with a high resolution ($N \geq 5$), the peak-to-peak value of a sinusoidal output signal can be approximated by :

$$V_{ptp} = 2^N * \Delta \tag{2.5}$$

The total signal power is then equal to :

$$P_{signal} = \frac{V_{ptp}^2}{8} = \frac{2^{2N}\Delta^2}{8} \tag{2.6}$$

The signal-to-noise ratio can be calculated using eq.(2.4) and eq.(2.6) :

$$SNR = \frac{P_{signal}}{P_{noise}} = \frac{3}{2} * 2^{2N} \tag{2.7}$$

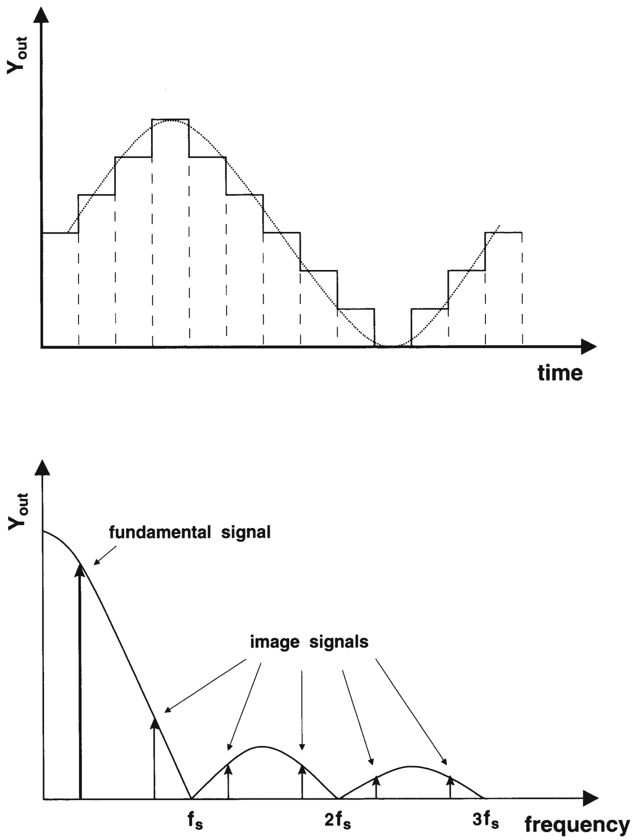


Figure 2.4: *The amplitude reduction error of an ideal D/A converter (f_s is the sampling frequency)*

or expressed in decibels :

$$SNR = 6.02N + 1.76dB \tag{2.8}$$

This equation is often used to compare the performance of a given D/A converter with that of an ideal D/A converter. From eq.(2.8) it can be concluded that the influence of the quantisation error on the signal-to-noise ratio decreases when the resolution of the D/A converter increases.

2.3.3 The Sample and Hold like Amplitude Distortion

During signal reconstruction, the D/A converter acts as a sample-and-hold circuit (fig.2.4). The analog output signal remains constant during the sampling time t_s. In the time domain the output is thus represented as a series of modulated rectangular pulses, while in the frequency domain the spectrum of the output is distorted by the sin(x)/x response (in which zeros appear at multiples of the sampling frequency). The frequency response of the output is given by :

$$|H(2\pi f_{in})| \sim \frac{sin(\pi f_{in}/f_s)}{\pi f_{in}/f_s} \tag{2.9}$$

When sampling at the Nyquist rate, the input frequency f_{in} equals half the sampling frequency f_s leading to an amplitude reduction of $2/\pi$ or 3.92 dB.

As a conclusion it can be stated that the sin(x)/x frequency response acts as a low pass filter that modifies the amplitude of the fundamental signal. In some applications, this amplitude distortion has to be corrected by the use of an inverse sin(x)/x filter or an equalizer. Furthermore, the reconstruction of the sampled signal typically requires the elimination of the image frequencies. For narrow-band signals ($f_{in} \ll f_s$) the sin(x)/x filter can already introduce a significant suppression of these unwanted signals. Otherwise, an explicit analog reconstruction filter is needed after the D/A converter.

2.4 The Performance Specifications of a D/A Converter

2.4.1 Introduction

In order to be able to compare different D/A converter architectures, a number of performance measures have been introduced. Each of these measures highlights a different aspect of the D/A converter. Their importance has to be determined according to the D/A converter's intended use. High accuracy D/A converters that are used for instrumentation purposes should have a low integral non-linearity, differential non-linearity and offset error while D/A converters used for waveform reconstruction applications (f.i. telecommunication) should have an excellent dynamic performance (low total harmonic distortion,...).

In literature different definitions for these specifications can be found. In this paragraph an overview of the most important static and dynamic measures are given based on [Hendriks, VdPlassche, Stehr, Razavi].

2.4.2 The Static Specifications

2.4.2.1 Introduction

The static specifications described in this paragraph give an idea of the D/A converter's distortion performance at low frequencies. It are these specifications that ultimately

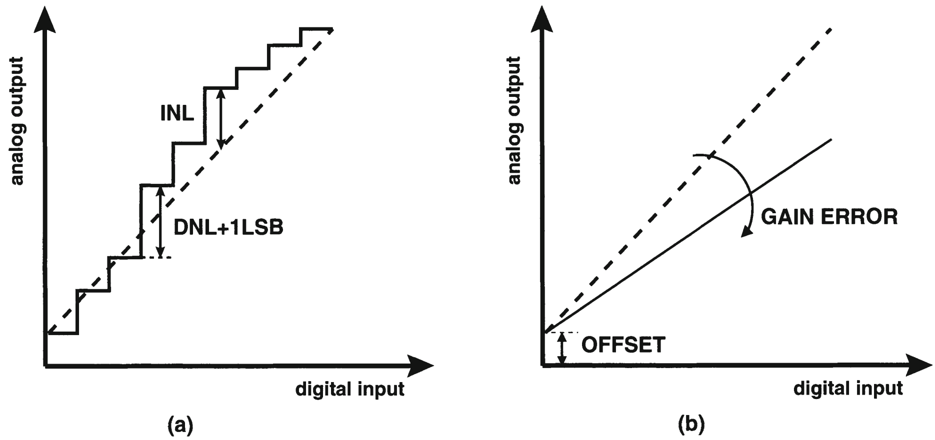


Figure 2.5: *The static specifications of a D/A converter*

impose the performance limit of the D/A converter. The most important static measures are the integral and differential non-linearity error.

2.4.2.2 The Offset Error and the Gain Error

The offset error is defined as the constant DC offset of the D/A converter transfer characteristic while the gain error is defined as the deviation of the slope of the measured D/A converter in comparison with its ideal characteristic (fig.2.5.b). Since these errors do not introduce any non-linearities, no effect is seen on the frequency domain characteristics.

2.4.2.3 The Differential Non-Linearity Error (DNL)

The differential non-linearity error is the worst case deviation between the actual and the ideal step size (= 1 LSB) between two adjacent codes. It can be described as :

$$DNL = max(Y_{out}(B) - Y_{out}(B-1) - 1LSB) \tag{2.10}$$

Fig.2.5.a gives a graphical representation of the DNL error.

2.4.2.4 The Integral Non-Linearity Error (INL)

The integral non-linearity error (INL) is defined as the maximum deviation of the measured D/A converter transfer characteristic from the ideal output curve (fig.2.5.a). This ideal curve is a straight line that is determined by the D/A converter's measured zero and full-scale output. This implies that nor the gain error nor the offset error has an impact on the INL value. This curve is described by :

$$Y_{out,id}(B) = Y_{out}(0) + \frac{Y_{out}(2^N - 1) - Y_{out}(0)}{2^N - 1} * B \tag{2.11}$$

Using eq.(2.11), the INL can be expressed by :

$$INL = max(Y_{out}(B) - Y_{out,id}(B)) \tag{2.12}$$

Using this definition for the ideal output curve implies a compensation of the offset and the gain errors since a zero output for $Y_{out}(0)$ excludes a potential offset error and the boundary condition for the maximal output eliminates the gain error. In practice the definition of the ideal output curve as the "best fit" through the D/A converter's measured code transitions is frequently used.

Both INL and DNL errors are measured in terms of LSB's and can have either positive or negative values. Although these measures give a good indication of the static behaviour of the D/A converter, in some cases they also provide the designer with some general information on the frequency domain behaviour of the converter. The shape of the INL characteristic gives an indication of the distortion component that will limit the D/A converter's dynamic performance. Fig.2.6.a shows the output spectrum for a D/A converter where the dynamic behaviour is determined by quantisation noise. The impact of the INL characteristic is illustrated in fig.2.6.b and fig.2.6.c. It is shown that a bow like INL characteristic gives rise to a spurious free dynamic range (defined in the section regarding dynamic specifications) that is determined by a second order harmonic (for a low frequency signal at a low update rate). This already indicates the maximum achievable limit for the SFDR since for high frequency signals and/or high update rates other factors like timing errors gain in importance and will further deteriorate the dynamic performance. It should however be clear that two D/A converters with the same INL error can have totally different distortion components since in both cases the INL characteristic can be completely different.

2.4.2.5 Monotonicity

A D/A converter is monotone when its output never decreases with an increasing digital input code. This implies that a minimum increase of zero is allowed when the input signal of the D/A converter increases with only one LSB. It can be proven for a binary D/A converter that it is always monotonic when the integral non-linearity error is less than or equal to 1/2 LSB [VdPlassche].

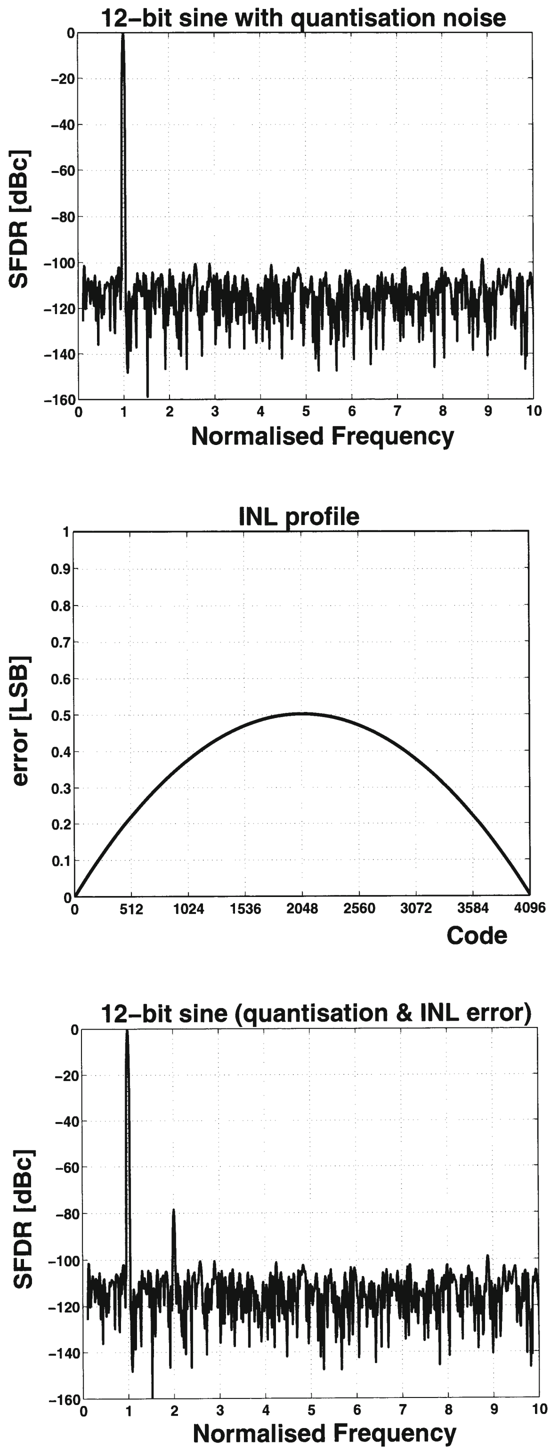


Figure 2.6: *The INL characteristic related to the dynamic performance of the D/A converter*

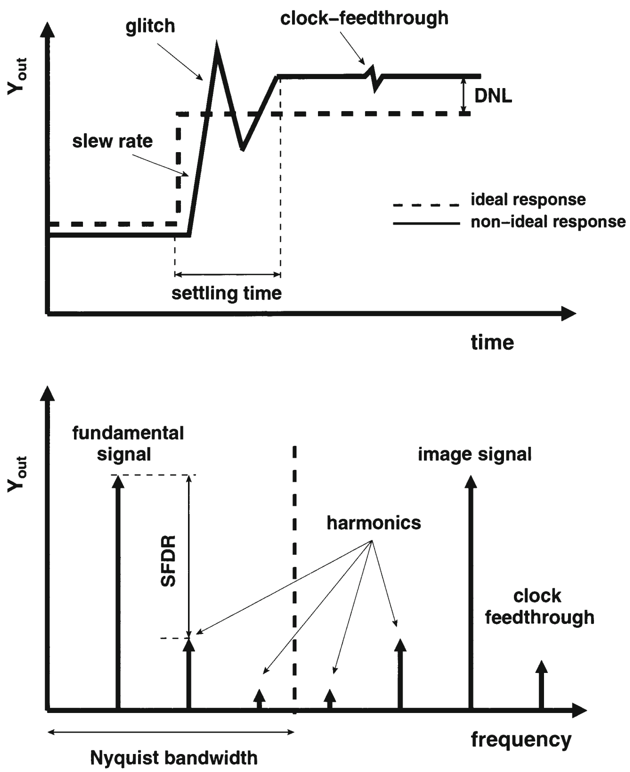


Figure 2.7: *The dynamic specifications of a D/A converter*

2.4.3 The Dynamic Specifications

2.4.3.1 Introduction

The influence of the dynamic non-linearities on the distortion performance of the D/A converter can be described by using measures in both the time and the frequency domain. Although the time domain specifications were frequently used in the past, the frequency domain specifications gain in importance.

The main purpose of D/A converters used in telecommunication systems is the reconstruction of waveforms from digital data. It is important that as a result of this conversion no new signal components are created that alter the information content of the original signal. In the remainder of this paragraph, the most important time and frequency domain specifications will be discussed in more detail.

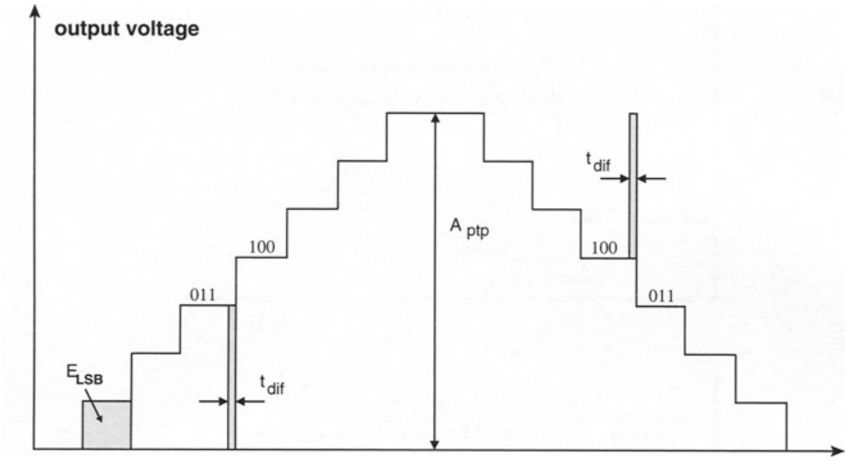


Figure 2.8: *The glitch energy error*

2.4.3.2 The Update Rate

The update rate is the rate at which the output is sampled. It therefore determines the maximal output signal frequency (which equals half of the update rate according to the Nyquist theorem).

2.4.3.3 The Settling Time

In general, the settling time of a D/A converter is defined as the time required for the output to experience a full scale transition and to settle within a specified error band around its final value.

However, also in the code-to-code transition response a settling time can be defined (fig.2.7). If this settling time is dependent on the applied code, it will affect the D/A converter's frequency domain behaviour.

2.4.3.4 The Glitch Energy

The glitch impulse gives an idea of the error generated by the earliest part of the transient response when the D/A converter switches between two consecutive output codes (fig.2.7). The glitch energy is defined as the area under this transient response. This error is mainly caused by timing errors within the D/A converter and results in a deterioration of the dynamic performance.

A qualitative description of the glitch energy for a binary D/A converter is given here under the following assumptions [VdPlassche]:

- the largest glitch impulse occurs at the MSB transition
- the glitch impulse has a square shape (worst case calculation)
- the timing error is represented by t_{dif}
- the peak-to-peak amplitude of the D/A converter is given by A_{ptp}

The glitch energy error can then be expressed as (fig.2.8):

$$E_{glitch} = t_{dif} * \frac{A_{ptp}}{2} \tag{2.13}$$

The energy of one LSB equals :

$$E_{LSB} = 2^{-N+1} * t_{sample} * \frac{A_{ptp}}{2} \tag{2.14}$$

To have a D/A converter with a good dynamic performance, the following ratio has to be minimised :

$$\frac{E_{glitch}}{E_{LSB}} = \frac{t_{dif}}{2^{-N+1} * t_{sample}} \tag{2.15}$$

The glitch energy can be reduced by placing a "deglitcher" circuit at the output of the D/A converter. However, since such highly accurate linear circuits are difficult to design, it is simpler to minimise the timing error t_{dif} by the use of well-placed synchronisation blocks.

2.4.3.5 The Slew Rate

The slew rate is defined as the maximal rate at which the output of the D/A converter can change with the varying input (fig.2.7).

$$SR = \frac{dV_{out}}{dt} = \frac{dV_{out}}{dq} * \frac{dq}{dt} = \frac{I}{C_{out}} \tag{2.16}$$

It should be noted that the slew rate can be different for the charging (SR+) or the discharging (SR-) of the output capacitance depending on the architecture of the D/A converter. If the output slewing has code dependent rise and fall times, it is also a contributor to distortion.

2.4.3.6 The Clock-Feedthrough

Due to parasitic capacitive coupling, the effect of the switching of the clock can be directly seen at the output of the D/A converter (fig.2.7). This clock feedthrough does not introduce any additional noise or distortion in the Nyquist baseband zone since it is not code dependent and manifests itself in the frequency domain as a component at the sampling frequency. This component can be easily removed by the use of a low pass filter at the output of the D/A converter.

Other feedthrough components that arise from coupling between the digital and the analog part of the D/A converter usually tend to increase the overall noise floor unless this feedthrough is directly related with the input codes.

2.4.3.7 The Signal to Noise Ratio (SNR)

The signal to noise ratio (SNR) is defined as the ratio of the power of the fundamental signal to the integrated noise power. Although this specification is rarely quoted on the D/A converter's data sheets, it is an important frequency domain specification. A lower limit of this specification is dictated by the quantisation noise. The quantisation contribution has already been calculated in paragraph 2.3.2.

$$SNR = 6.02N + 1.76dB \tag{2.17}$$

2.4.3.8 The Signal to Noise and Distortion Ratio (SNDR)

The signal to noise and distortion ratio (SNDR) is defined as the ratio of the power of the fundamental signal to the sum of the integrated noise power and the power in all distortion components.

2.4.3.9 The Spurious Free Dynamic Range (SFDR)

The spurious free dynamic range (SFDR) specification is defined as the ratio between the fundamental signal and the largest distortion component within a specified frequency band (fig.2.7). This distortion component does not necessarily has to be a harmonic of the fundamental signal nor does the frequency band has to be equal to the D/A converter's Nyquist baseband zone.

$$SFDR = \frac{P_{signal}}{P_{largest-dist}} \tag{2.18}$$

In some applications it is sufficient to have an idea on the SFDR specification within a certain frequency interval as any larger out-of-band spurs will be filtered out

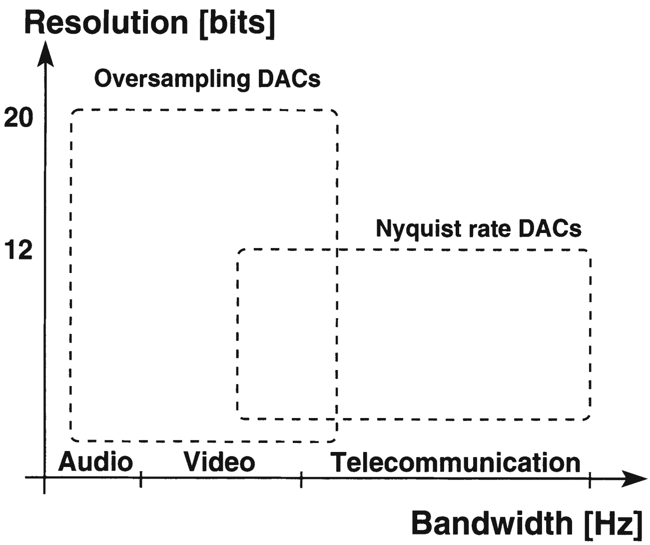


Figure 2.9: *The different D/A converter types as a function of speed and resolution [Wikner]*

in the next stage. In any case, it is imperative that the SFDR specification is always given together with the information concerning the measured frequency window. Furthermore, it should be noted that the SFDR is not a constant for a given D/A converter as the INL specification is. It depends on the operating conditions, the update rate, the digital sine wave and on the measurements (differential or single ended output). Therefore, in order to characterise the dynamic performance of a D/A converter, the SFDR should be given as a function of the update rate (for a number of fixed full-scale sinusoidal output signals) and as a function of the signal frequency (for different values of the update rate).

2.4.3.10 The Total Harmonic Distortion (THD)

The total harmonic distortion is defined as the ratio of sum of the power of the harmonic components to the power of the fundamental signal (eq.2.19). This specification is usually expressed in decibels and gives a more complete picture of the D/A converter's distortion performance than the SFDR.

$$THD = \frac{P_{H2+H3+H4+...}}{P_{signal}} \tag{2.19}$$

Since in most cases the worst distortion component is harmonically related and contains more than 80 % of the total harmonic energy, the total harmonic distortion is rarely plotted over frequency since its value is only 1-3 dB higher than the SFDR value. However, plotting the three most significant harmonic distortion components as a function of the frequency can be helpful in gaining more insight in the D/A converter's spectral performance.

2.5 The D/A converter specifications as a function of the application

Based on a literature study, a schematic overview of the required specifications in terms of resolution and bandwidth is given as a function of the application area (audio, video or telecommunication applications) in fig.2.9. This study has been done by [Wikner]. From this figure, it can be concluded that current steering Nyquist D/A converters are highly suited for high speed applications.

2.6 Conclusions

In this chapter the basic functionality and the characteristics of an ideal D/A converter have been described. To make a comparison between different D/A converters possible, a set of performance measures have been introduced that describe the static and the dynamic behaviour of these devices.

Chapter 3

CMOS D/A Converter Architectures

3.1 Introduction

In the previous chapter, the functionality of a D/A converter has been explained together with the specifications that describe its static and dynamic behaviour. This chapter discusses the different architectures for a D/A converter.

A D/A converter generates for each digital input code a multiple of a certain reference quantity. Dependent on this quantity (a voltage, a charge or a current), three classes of D/A converters can be identified namely the resistor, the capacitor and the current steering architecture. In the first part of this chapter, both the resistor and the capacitor D/A converter will be discussed. It is the intention of the author to emphasise the existence of these architectures as useful alternatives for the current steering architecture. For a detailed study the reader is referred to [Razavi, Johns]. The remainder of this chapter describes the current steering topology. Three possible implementations together with their advantages and disadvantages will be discussed in detail. During the remainder of this thesis, this architecture will be analysed with regard to its static and dynamic performance.

3.2 The Resistor D/A Converter

3.2.1 The resistor string D/A converter

In this type of D/A converter, a reference voltage is divided into $2^N - 1$ parts by selecting one tap of a segmented resistor string using a switching network (fig.3.1). Although this implementation provides a simple and inherently monotonic D/A conversion, it has some major drawbacks. For resolutions higher than eight bits, the

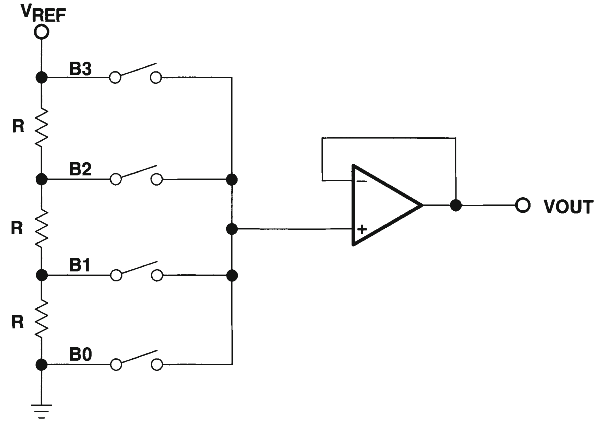


Figure 3.1: *The resistor string architecture*

occupied silicon area becomes fairly large. For a ten bit D/A converter, 1023 resistors and 1024 switches are required. Furthermore, the delay through the switching network poses a severe limitation on the update rate of the D/A converter.

The integral non linearity error is directly related to the matching precision of the used resistors. Due to uncertainties during processing, the values of the resistors in the resistor string will not be equal. The relative mismatch between two resistors is given by:

$$\frac{\Delta R}{R} = \frac{\Delta \rho}{\rho} + \frac{\Delta L}{L} - \frac{\Delta W}{W} - \frac{\Delta t}{t} + \frac{\Delta R_C}{R} \tag{3.1}$$

with R_C the contact resistance, L the length, W the width, t the thickness and ρ the resistivity of the resistor. The width, length and value of the resistor can be freely chosen by the designer as to minimise the mismatch. However, larger dimensions need more silicon area and create a higher capacitance to the substrate.

The influence of this mismatch on the integral non-linearity error of the resistor string D/A converter is given by [Razavi]:

$$INL = \frac{V_{ref}}{\sqrt{4N}} \frac{\Delta R}{R} \tag{3.2}$$

which is reached at the middle of the resistor string. Since this formula is a standard deviation, it has to be interpreted as follows. In 68 % of the cases, the non-linearity error will be smaller than or equal to the value calculated in eq.(3.2).

For high speed, high accuracy applications, this architecture is no longer the preferred solution.

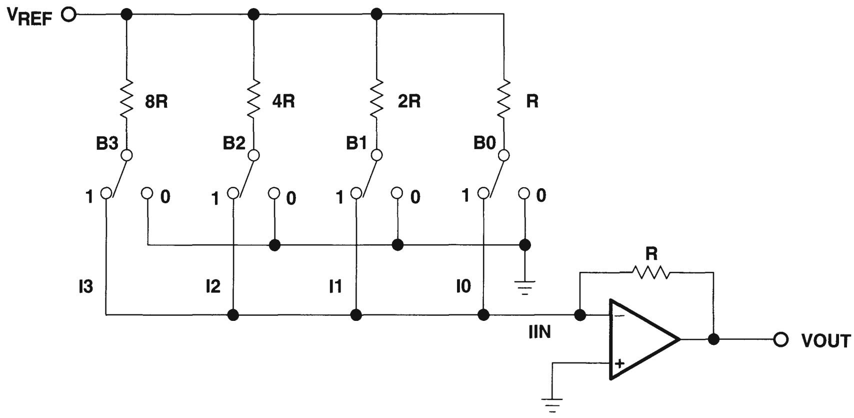


Figure 3.2: *The binary weighted resistor D/A converter*

3.2.2 The binary weighted resistor D/A converter

This type of converter is very similar to the resistor string structure. Each resistor in the string is given a value proportional to the binary value of the bit it represents (fig.3.2). The currents generated from each active bit are then summed to obtain the required output. The number of resistors and switches is now reduced to one per bit, but the range of the resistors is extremely wide for high resolution D/A converters. Furthermore, it is important to note that the feedback resistor R is implemented on chip. As a result, it experiences the same thermal drift as the resistor ladder and has no significant influence on the accuracy of the D/A converter. Besides the large resistor values for high resolution implementations, this architecture has no guaranteed monotonicity and is susceptible to glitches.

3.2.3 The R-2R based D/A converters

The large resistor values of the binary architecture can be reduced by using series resistors. This results in the very frequently used R-2R D/A converter (fig.3.3). Although the number of resistors has doubled in comparison to the binary structure, only a single size resistor is necessary since the 2R is realised by a series combination of two resistors with a value of R. The major advantages of this structure are its smaller area and higher accuracy. However, the resistors often exhibit a non linear behaviour. Furthermore, a time delay between the processing of the different bits adds to the generation of glitches and distortion components.

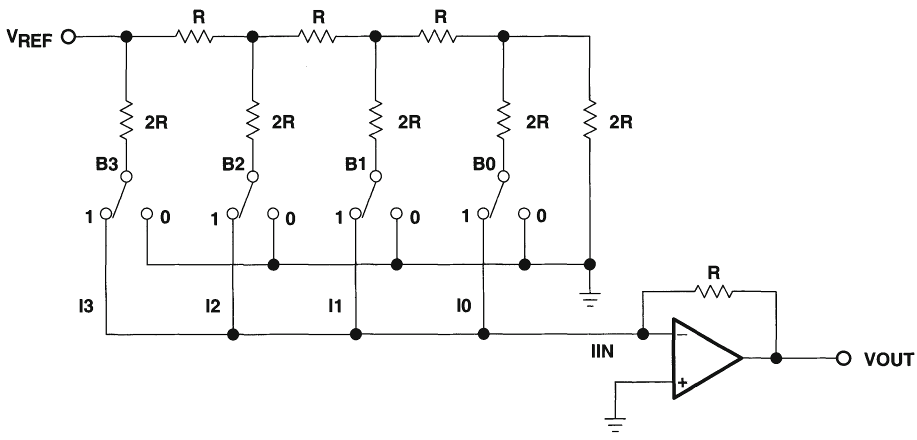


Figure 3.3: *The R-2R D/A converter*

3.3 The Capacitor D/A Converter

A simple architecture for the capacitor D/A converter is given in fig.3.4. The bottom plates of the capacitors switch from the ground to the reference voltage according to the digital input. If all the capacitors have the same value, a thermometer decoded input is necessary. Similar to the resistor string D/A converter, a binary structure can be implemented by adjusting the capacitor values. An example of a 3 bit charge redistribution D/A converter is given in fig.3.5.

One of the causes that influences the non linearity error of a capacitor D/A converter is the random mismatch caused by processing inaccuracies. This mismatch can be expressed as :

$$\frac{\Delta C}{C} = \frac{\Delta W}{W} + \frac{\Delta L}{L} - \frac{\Delta t_{ox}}{t_{ox}} \tag{3.3}$$

with W the width, L the length and t_{ox} the oxide thickness of the capacitors. The designer can minimise the mismatch by carefully choosing the dimensions of the capacitor. Besides the random errors, the non linearity error of the D/A converter is also determined by the voltage dependence of the capacitance modelled by

$$C \;=\; C_0 + C_0\,\alpha_1\,V + C_0\,\alpha_2\,V^2 + ... \tag{3.4}$$

with α_j the jth order voltage coefficient of the capacitor, and the non-linearity of the junction capacitance of the switches connected to the output expressed as

$$C_{junction} = \frac{C_0}{(1\,+\,V_j/\phi)^{m_j}} \tag{3.5}$$

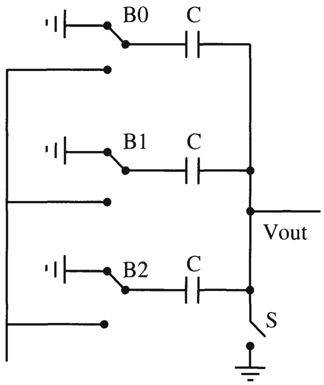


Figure 3.4: *The capacitor D/A converter*

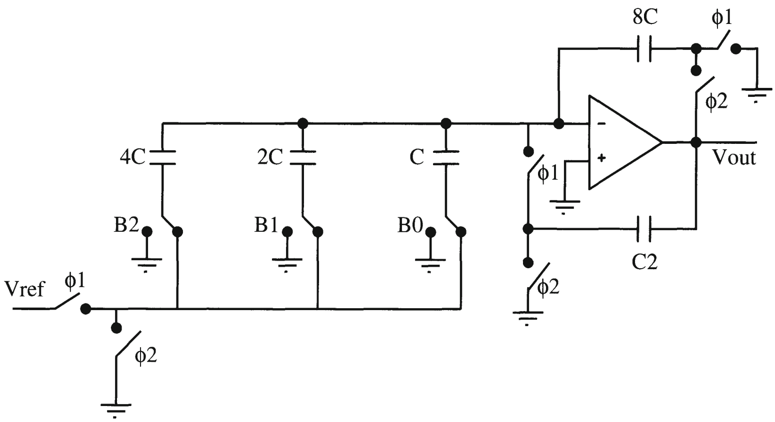


Figure 3.5: *A 3 bit charge redistribution D/A converter*

where C_0 is the zero bias capacitance, V_j is the voltage across the junction, ϕ is the built-in potential and m_j has a value that is typically between 0.3 and 0.5.

In recent years, this architecture has not been frequently used for telecommunication applications. However, [Fergu CICC00, Khano AACD01] presented two 10 bit D/A converters that possess a high linearity. Overcoming the problem of their non linear behaviour and the need for implementing a special driver for the D/A converter load, this architecture lends itself for low power applications.

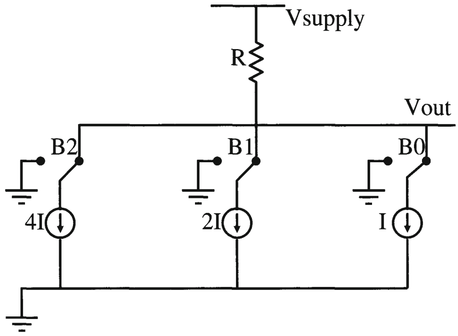


Figure 3.6: *The binary current steering D/A converter*

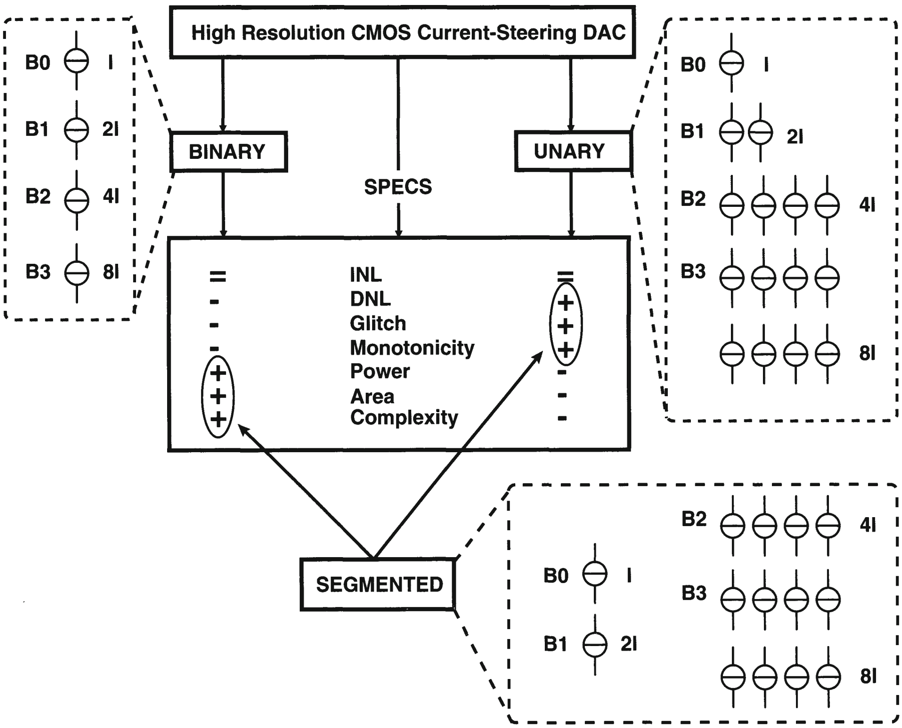


Figure 3.7: *The advantages/disadvantages of the unary, binary and segmented current steering D/A converter architecture*

3.4 The Current-Steering D/A Converter

3.4.1 Introduction

In the current steering architecture (fig.3.6), the reference quantity is given by a current. Also here different implementations are possible as will be discussed in the following paragraphs. This architecture has the advantage of combining a small silicon area with a high update rate. The resolution of the D/A converter is determined by the matching behaviour of the current sources. This issue is analysed in the next chapter.

3.4.2 The Binary Implementation

In the binary implementation, every switch switches a current to the output that is twice as large as that of the next less significant bit. The digital input code directly controls these switches. The advantages of this architecture are its simplicity (since no decoding logic is necessary) and the small required silicon area. However, this structure does not always exhibit a monotone behaviour and suffers from a large DNL and dynamic error. At the half scale transition, 2^{N-1} unit sources are switched on/off and $2^{N-1}-1$ other independent sources are switched off/on. Assuming a normal distribution for the unit current sources with a standard deviation $\sigma(I)$, this step has a $\sigma(\Delta I)$ determined by:

$$\begin{aligned}\sigma^2(\Delta I) &= (2^N-1)\sigma^2(I) \\ \Rightarrow \sigma(\Delta I) &= \sqrt{2^N-1}\,\frac{\sigma(I)}{I}\,LSB\end{aligned} \qquad (3.6)$$

This sigma, $\sigma(\Delta I)$, is a good approximation for the DNL error.

3.4.3 The Unary Implementation

In the unary decoded architecture every unit current source is addressed separately. The digital input code is converted to a thermometer code that controls the switches (table 3.1). The advantages of this architecture are its good DNL error and the small dynamic switching errors. Furthermore, the D/A converter has a guaranteed monotone behaviour since only one additional current source has to be switched to the output for one extra LSB. The major disadvantages of the unary decoded architecture are the complexity, the area and the power consumption of the thermometer decoder. Performing similar calculations as in eq(3.6) for the unary architecture leads to the following result:

$$\sigma(\Delta I) = \frac{\sigma(I)}{I}\,LSB \qquad (3.7)$$

Decimal	Binary			Thermometer Code						
	b_1	b_2	b_3	d_1	d_2	d_3	d_4	d_5	d_6	d_7
0	0	0	0	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0	0	0	1
2	0	1	0	0	0	0	0	0	1	1
3	0	1	1	0	0	0	0	1	1	1
4	1	0	0	0	0	0	1	1	1	1
5	1	0	1	0	0	1	1	1	1	1
6	1	1	0	0	1	1	1	1	1	1
7	1	1	1	1	1	1	1	1	1	1

Table 3.1: *The decimal, binary and thermometer code*

This formula mathematically represents the idea behind the unary decoding. The error between two consecutive codes is just the deviation on the additional unity current source. The DNL error was defined as the maximum deviation at a single LSB transition. For an N bit converter, this means that the DNL error is determined by the maximum when taking $2^N - 1$ samples from a normal distribution with the sigma defined in eq.(3.7).

3.4.4 The Segmented Implementation

To get the best of both worlds, most current-steering D/A converters are implemented using a segmented architecture. In this case, the D/A converter is divided into two sub-D/A converters : the B LSBs (least significant bits) are implemented using a binary architecture while the (N-B) MSBs (most significant bits) are implemented in a unary way. In this architecture, a balance between good static and dynamic specifications versus a reasonable decoder power, area and complexity can be found. This is illustrated in fig.3.7. Since the segmented architecture is a mixture of the previous two architectures, the result for the most critical transition is of the same form.

$$\sigma(\Delta I) = \sqrt{2^{B+1} - 1}\,\frac{\sigma(I)}{I}\,LSB \tag{3.8}$$

Note that the formula in eq.(3.8) for the segmented architecture is a general formula that is valid for the binary (B=N-1) and the unary implementation (B=0).

3.5 Conclusions

In this chapter, an overview has been given of the different D/A converter architectures. In the remainder of this work, the current steering topology has been studied.

The advantages and disadvantages of the unary, binary and segmented architecture have been analysed as to optimise the D/A converter's performance.

Chapter 4

Static Behaviour of Current Steering D/A converters

4.1 Introduction

As has been discussed in the previous chapter, the current steering topology is currently the preferred architecture for telecommunication applications requiring a high update rate and/or a high accuracy. Designing such a D/A converter requires a thorough understanding of both the static and the dynamic behaviour of this device.

In this chapter, the emphasis will be put on the static performance of the current steering D/A converter. This performance is determined by the matching behaviour of the current source transistors. Since no two transistors behave exactly the same due to technological variations introduced during processing, it is important to know the impact of this phenomenon on the yield and the performance of the circuit. This topic will be discussed in the first part of this chapter. Apart from the process variations which generate random errors, the current sources in the array are also influenced by systematic errors that are introduced by thermal, electrical and process gradients. The second part of this chapter discusses the errors generated by these gradients and will go into more detail on how to solve this problem.

4.2 Modelling of the random errors

4.2.1 Introduction

Due to the mismatch of the current source transistors, the INL performance of different D/A converters made in the same process technology will vary randomly. It is therefore important to be able to predict this performance within certain boundaries. For this purpose, the concept of the D/A converter's INL_yield will be introduced. This yield figure is defined as the percentage of functional D/A converters with an INL performance smaller than the specification of half a LSB (least significant bit).

In this section, several analytical expressions for the INL_yield found in open literature will be discussed and evaluated. Also a new model will be derived that is easy to use and gives the designer an accurate indication on the required matching behaviour of the current sources as a function of the resolution of the D/A converter. All the models - including the new one- start from the assumption that the unit current source errors have a Gaussian nature.

4.2.2 Lakshmikumar Approach

A first suggestion to analytically determine the yield of a D/A converter as a function of the matching parameters of the current source transistors was made in [Laksh JSSC86]:

$$INL\_yield = \prod_{i=2}^{2^N-1} erf(\frac{Q_i}{\sqrt{2}}) \tag{4.1}$$

with

- $\frac{1}{Q_i} = 2^{N+1}\sqrt{\frac{\overline{z_i}(1-\overline{z_i})}{2^N-1}}(\frac{\sigma(I)}{I})$
- N = the number of bits,
- $\overline{z_i}$ = the mean normalised output at code i,
- $\frac{\sigma(I)}{I}$ = the unit current relative standard deviation.

However, this formula is based on the assumption that there exists no correlation between the outputs of a current-steering D/A converter. The INL_yield can then be obtained by multiplying the probabilities that each output has an error smaller than half an LSB. To demonstrate that this assumption is not correct the following example is given. The outputs corresponding with the digital input word 01100 and 01101 are strongly correlated since they are both implemented using the same current sources.

Using eq.(4.1) to estimate the yield will impose too severe constraints in terms of matching accuracy on the current sources leading to an oversizing of these transistors. In [Laksh JSSC88], an adjustment of this model was presented. Here the MSB (most significant bit) transition is considered to be the most critical one since in a binary implementation this transition has the largest probability of generating an output error.

Furthermore, the D/A converter's outputs before and after the MSB transition are not correlated. The yield can then be described as :

$$INL\_yield = \prod_{i=2^{N-1}-1}^{2^{N-1}} erf(\frac{Q_i}{\sqrt{2}}) \tag{4.2}$$

with

- $\frac{1}{Q_i} = 2^{N+1}\sqrt{\frac{\overline{z_i}(1-\overline{z_i})}{2^N-1}}(\frac{\sigma(I)}{I})$
- N = the number of bits,
- $\overline{z_i}$ = the mean normalised output at code i,
- $\frac{\sigma(I)}{I}$ = the unit current relative standard deviation.

Fig.4.1 shows a comparison between the Monte Carlo simulations and the formulas derived in this section.The yield estimation given in eq.(4.2) is too optimistic since the influence of only two outputs is taken into account while the errors generated by the other outputs are being ignored.

Comparing the two approaches, one can conclude that designing a chip using the first approach can lead to a large but nearly fault-free D/A converter while using the second approach results in a compact but low yield circuit. The correct yield estimation is situated somewhere in between these two results.

4.2.3 Monte Carlo Approach

To obtain an accurate estimation of the INL_yield, the Monte Carlo simulation has been considered to be the best alternative [Conro JSSC88]. A lot of recent publications refer to this method to do their accuracy analysis. The following procedure can be used. Each of the $(2^N - 1)$ current sources has a random value that has been derived from a Gaussian distribution with a mean value I_{LSB} and a standard deviation $\sigma(I)$. For every digital code the output current of the D/A converter is calculated and compared to the ideal value. If the difference is larger than half an LSB - even for only one digital code - the D/A converter is regarded as not functional and is rejected. For every $\sigma(I)$ this procedure is repeated a large number of times (>100) to obtain reliable results. The INL_yield is then given by the ratio of the number of functional D/A converters ($INL < 1/2LSB$) to the total number of try- outs. In this way the relation between the unit current standard deviation and the INL_yield is determined. However, to obtain the results depicted in fig.4.1, a large amount of CPU time is necessary. Running a Monte Carlo simulation for a high resolution D/A converter takes several hours which is a major drawback.

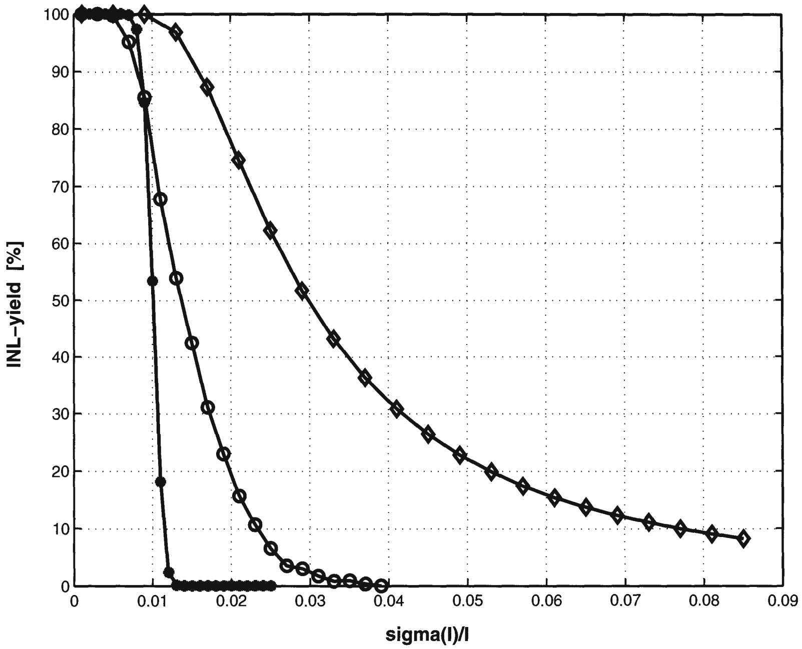


Figure 4.1: *Monte Carlo simulations (○) of the INL_yield compared to the results of the formulas in eq.4.1 (●) and eq.4.2 (◇) for a 10 bit D/A converter*

4.2.4 A new INL_yield Formula

4.2.4.1 Introduction

The models presented in the previous section either lack accuracy but are fast or are accurate but very time consuming. The model that will be derived in this section is both accurate and fast. The INL_yield calculated with this formula shows a good agreement with the Monte Carlo simulations and can be thousands of times faster for high resolutions (table 4.1).

4.2.4.2 Theory

The idea will be elaborated for an arbitrary number of bits N. For the simplicity of notation the following symbols are defined:

Definition 1: X(j) is the sum of j non-correlated unity current sources
Definition 2: Y(j) is the difference between X(j) and the sum of the ideal current sources

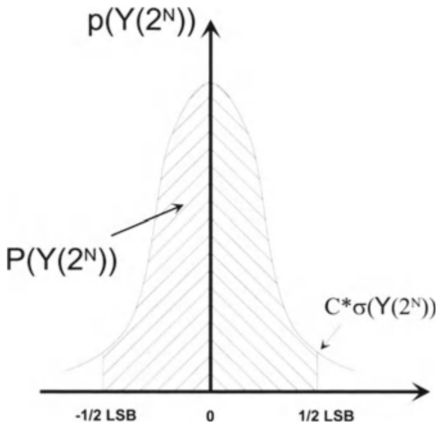


Figure 4.2: *The shaded area determines the probability* $P(Y(2^N) < 0.5LSB)$

Every current source has a normal distribution with a mean value I_{mean} and a standard deviation $\sigma(I)$.This implies that both X(j) and Y(j) also have a normal distribution with the following properties eq.(4.3):

$$\begin{aligned} X_{mean}(j) &= j * I_{mean} & Y_{mean}(j) &= 0 \\ \sigma(X(j)) &= \sqrt{(j)} * \sigma(I) & \sigma(Y(j)) &= \sigma(X(j)) \end{aligned} \tag{4.3}$$

The exact possibility that an INL error will occur is based on the fact that each current source generates an error and is given by the following sum of probabilities :

$$P(INL\_error) = \tag{4.4}$$

$$\begin{aligned} &P(|Y(1)| \geq 0.5\&|Y(2)| < 0.5\&\cdots\&|Y(2^N-1)| < 0.5) + \\ &P(|Y(2)| \geq 0.5\&\cdots\&|Y(2^N-1)| < 0.5) + \cdots + \\ &P(|Y(2^N)| \geq 0.5) \end{aligned}$$

However, this equation is not transparant and therefore not suitable for practical use. The basic idea behind the new theory is based on the fact that if at any point $|Y(j)|$ reaches half an LSB, there exists a 50% chance that the error increases and 50% chance that it decreases again since a normal distribution with mean value zero is used. Extending this line of thought, one can say that if an INL error occurs - when passing through all the possible codes generated by the $(2^N - 1)$ digital input words - there is a 50% chance that this error still exists for the (fictive) code 2^N. This idea can be expressed as:

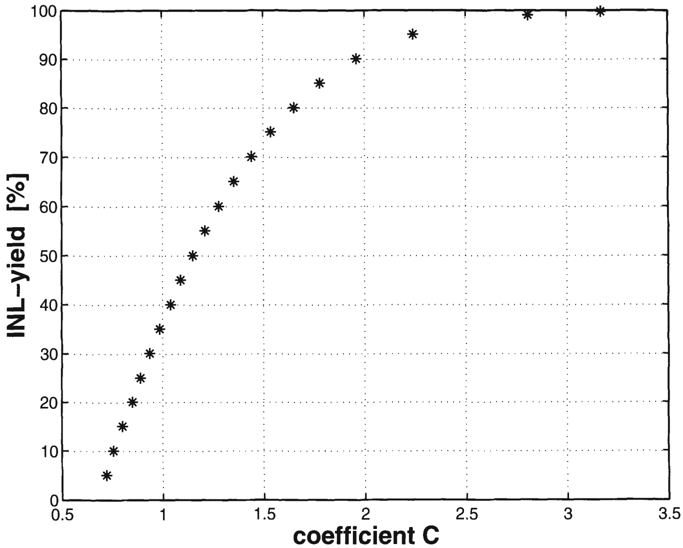


Figure 4.3: *INL_yield as a function of the coefficient C*

$$P(|Y(2^N)| \geq 0.5) = \frac{P(\exists j \in [1..(2^N - 1)] : |Y(j)| \geq 0.5])}{2} \qquad (4.5)$$

At this point the INL_yield of the D/A converter can be integrated in the calculation since there exists an obvious relation between the yield and the possibility of an INL error to occur:

$$INL\_yield = 1 - P(\exists j \in [1..(2^N - 1)] : |Y(j)| \geq 0.5]) \qquad (4.6)$$

Combining eq.(4.5) and eq.(4.6):

$$P(|Y(2^N)| \geq 0.5) = \frac{1 - INL\_yield}{2} \qquad (4.7)$$

From eq.(4.7), the possibility that no INL error occurs at code 2^N can be easily derived and equals:

$$P(|Y(2^N)| < 0.5) = 0.5 + \frac{INL\_yield}{2} \qquad (4.8)$$

At this point, the INL_yield is no longer described as a sum of probabilities (eq.(4.4)) but as the possibility that a sample from a normal distribution is smaller than half an LSB. This requirement can be written as:

$$\int_{-1/2LSB}^{1/2LSB} p(Y(2^N))dY(2^N) = \int_{-C*\sigma}^{C*\sigma} p(Y(2^N))dY(2^N) = 0.5 + \frac{INL\_yield}{2} \quad (4.9)$$

with $p(Y(2^N))$ the probability density function (fig.4.2). Eq.(4.9) directly gives the following result:

$$C * \sigma(Y(2^N)) \leq \frac{1}{2} LSB \quad (4.10)$$

with:

$C = inv\_norm_{(-x,x)}(0.5 + \frac{INL\_yield}{2})$

The $inv\_norm_{(-x,x)}$ is the inverse function of the normal cumulative function integrated from -x to x. In appendix 1, a table is given determining the value of C as a function of either the INL_yield or as a function of the value of $(0.5 + \frac{INL\_yield}{2})$. It should be further noted that if the normal cumulative function integrates from minus infinity to x the expression for C slightly alters and is given by:

$C = inv\_norm_{(-\infty,x)}(0.75 + \frac{INL\_yield}{4})$

Since $Y(2^N)$ has a normal distribution with a standard deviation $\sqrt{2^N}\sigma(I)$ (eq.(4.3)), the relation existing between the INL_yield of the current steering D/A converter and the matching properties of the current source transistors can be deduced from eq.(4.10).

$$\frac{\sigma(I)}{I} = \frac{1}{2\sqrt{2^N}C} \quad (4.11)$$

Fig.4.3 shows the INL_yield of the D/A converter as a function of the coefficient C. In fig.4.4 the INL_yield of a 10-bit D/A converter calculated using the new formula in eq.(4.11) and simulated using the Monte Carlo approach are depicted. From this figure, it can be concluded that the formula is in good agreement with the Monte Carlo simulations.

To gain more insight in eq.(4.11), the unit current standard deviation is plotted in logarithmic scale versus the resolution of the D/A converter (fig.4.5). As can be seen from this figure, the relationship is characterised by straight lines since

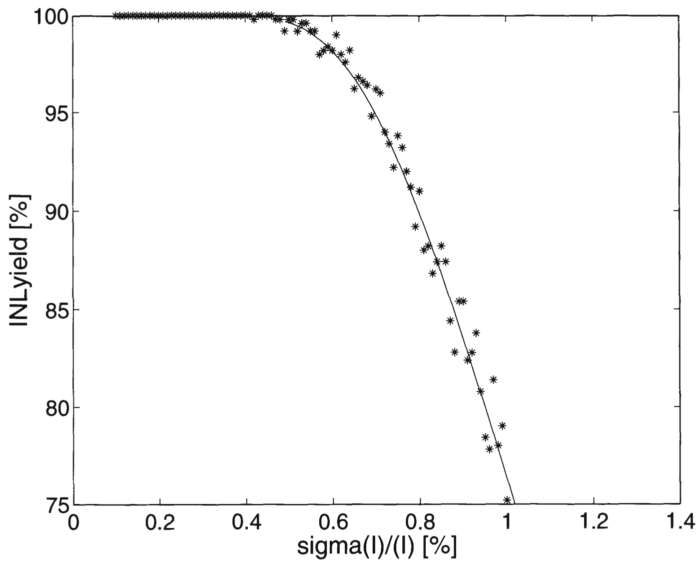


Figure 4.4: *Comparison between the Monte-Carlo simulations and the new formula for a 10-bit D/A converter*

$$\log(\frac{\sigma(I)}{I}) = A * N + B \tag{4.12}$$

with

$$\begin{aligned} A &= -\frac{log2}{2} \\ B &= -log2 - logC \end{aligned}$$

One can easily conclude from fig.4.5 that for the design of a high accuracy current steering CMOS D/A converter the matching parameters play a significant role. A small deviation of the required σ(I)/I can lead to severe yield degradation.

The time to create a figure like fig.4.5 using Monte Carlo simulations in MATLAB is given in table 4.1, where the results for an INL_yield from 100% to 10% for a current steering D/A converter with different resolutions can be found. For all the simulations twenty values for the relative unit current standard deviation were taken. This can be understood as follows. In a first coarse approximation, a simulation using 10 values for σ(I)/I -that span a wide range- is run. From the obtained result, the interval for the σ(I)/I that obtain a high INL_yield can be specified. In this interval another 10 points are simulated. Creating fig.4.5 using the new formula takes only a few minutes. The time to write the short MATLAB program is so to speak the most time consuming. It is also worth noting that the time necessary to calculate the INL_ yield is independent

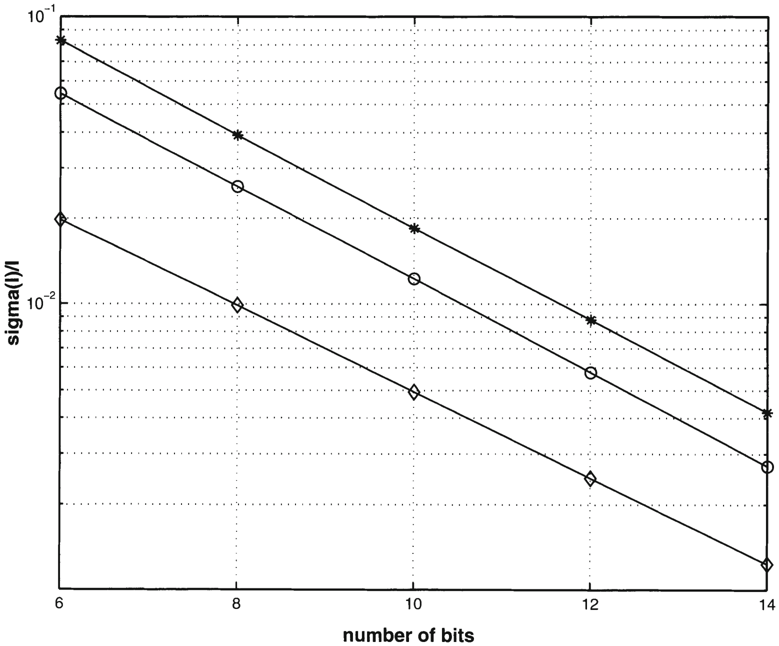


Figure 4.5: *The unit current relative standard deviation as a function of the resolution of the D/A converter for a yield of* 99.7% *(◇), of* 50% *(○) and of* 10% *(*)*

resolution	Monte Carlo	formula
8 bit	0.13 hours	seconds
10 bit	0.69 hours	seconds
12 bit	5.31 hours	seconds
14 bit	108.57 hours	seconds

Table 4.1: *Comparison between the CPU time consumption of the Monte Carlo simulations and the new formula*

on the resolution of the D/A converter while the time consumption of the Monte Carlo simulations "explodes" with an increasing D/A converter's accuracy.

4.2.5 Conclusion

In this section, the impact of random errors caused by the matching behaviour of the current source transistors on the yield of the D/A converter has been discussed. A new formula has been derived that provides a clear insight on this issue in a fast and accurate way. In chapter 6, that discusses the design flow for a current steering D/A converter, the usefulness of this formula will become clear during the dimensioning of the current source transistors. It will allow to determine the area of the unit current source transistor in such a way as to optimise the static performance of the D/A converter and to minimise its silicon area consumption.

4.3 Modelling of the systematic errors

4.3.1 Possible causes

Apart from the random errors, the static performance of the D/A converter is determined by the following systematic errors :

- Although the transistor mismatch effect of the current sources has already been taken into account during the sizing of these transistors, it can still have a negative influence on the static performance of the D/A converter due to the "edge effect" [Wong ICMTS95] which states that the mismatch behaviour of a transistor is dependent upon its surroundings. To avoid this error, the current source array has to be expanded by inserting dummy rows and columns as to provide identical surroundings for all the active current source transistors.

- The voltage drop along the ground line will slightly change the output current of the different current source transistors placed on the same row. This leads to an integral non linearity error that is given by [Miki JSSC86] :

  $$E_{INL-VD} = \frac{g_m R}{f \sqrt{3}} \tag{4.13}$$

  where g_m is the derivative of the full scale current to the current source bias voltage, R is the total resistance of the ground line and f is a factor depending on the used switching scheme. If all the current sources are switched sequentially from the left to the right, the value for f equals 9. This error can be reduced by either using sufficiently wide power supply lines (reducing the resistance R) or by using a special switching scheme (increasing the factor f).

- If the resolution of the D/A converter increases by a single bit, the number of current sources in the current source array doubles. The area occupied by a single unity current source also doubles because of the random matching constraint. This leads to a four-times area increase for the current source array for each additional bit. For D/A converters with a resolution of 10 bit and higher, the dimensions of the current source array become so large that process- and temperature gradients have to be considered. The non-linearity errors introduced by these gradients can be (partially) compensated by the introduction of a special switching scheme. How this is implemented is discussed in more detail in the next section.

4.3.2 Switching Schemes

4.3.2.1 Introduction

If the error contributions of the current sources are totally random and uncorrelated, the INL_yield of the D/A converter dictates the minimal requirement for the matching precision of the current sources as is indicated in the previous paragraph. The random error can then be kept within the specified boundaries ($INL < 0.5LSB$) by adjusting the active area. This implies that in order to guarantee a good static performance, the systematic errors introduced by linear and/or symmetrical gradients have to be compensated in order to keep the random errors dominant. This is done by using optimised switching schemes for the current sources. Several switching schemes have been presented in literature [Miki JSSC86, Nakam JSSC91, Marqu ISSCC98, VdPla JSSC99, VdBos ISSCC01]. These switching schemes will be discussed in more detail in the next paragraphs. First, a short introduction on the basic principles of gradients will be given.

4.3.2.2 The Gradient Error Distribution

It is generally assumed that the error distribution introduced by the gradients can be modelled by a superposition of both a linear and a quadratic component [Cong TCASII].

The linear gradient error can be expressed as :

$$\varepsilon_l(x, y) = a_l * cos\vartheta * x + a_l * sin\vartheta * y \tag{4.14}$$

with $\vartheta \in [0, 360°]$ is the angle of the gradient, a_l represents the slope of the gradient and (x,y) determines the position of the current source transistor. This type of gradient can be caused by f.i. a variation of the oxide thickness over the wafer or by the voltage drop along the ground line of the current source transistors [Miki JSSC86].

The quadratic gradient error can be expressed as :

$$\varepsilon_s(x, y) = a_s * (x^2 + y^2) - b_0$$

with b_0, a_s being technological parameters and (x,y) the position of the current sources.

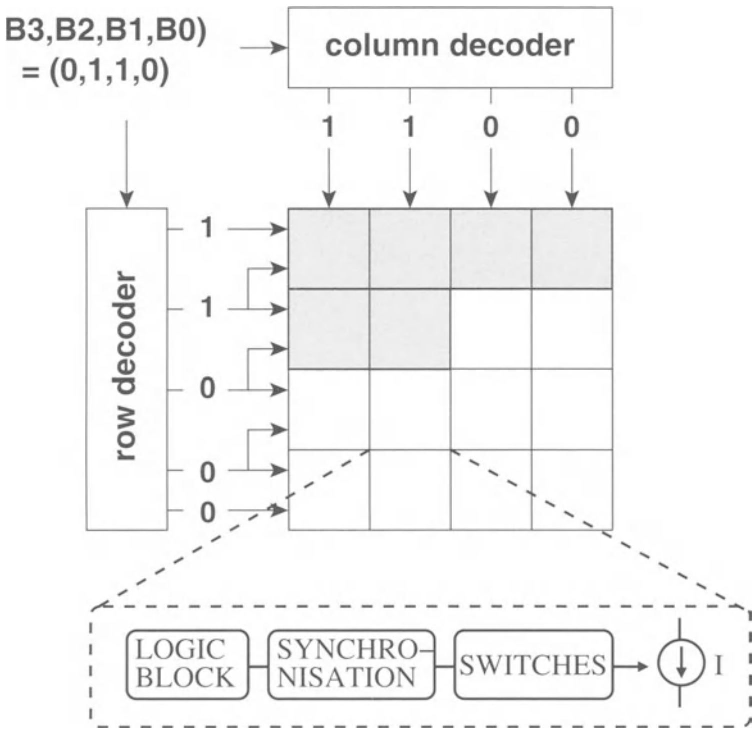


Figure 4.6: *The row and column decoding principle; every cell of the matrix contains some decoding logic, a synchronisation block, a switch and the current source*

This expression has been derived under the assumption that the symmetrical gradient is identical for both the x and the y dimension and that the current source array is located in the center of the die [Cong TCASII]. Examples of symmetrical gradients are f.i. the gradients introduced by temperature and by die stress [Basto TSM97].

Since the actual error distribution is a superposition of both the linear and the symmetrical gradient error, it can be expressed as :

$$\varepsilon(x, y) = \varepsilon_l(x, y) + \varepsilon_s(x, y) \tag{4.15}$$

4.3.2.3 The sequential, conventional and hierarchical symmetrical switching schemes

The switching schemes that will be discussed in the next two subsections are determined by the implementation of the thermometer decoder that transforms the digital input word in a decimal value. Fig.4.6 shows the schematic of a row and a column

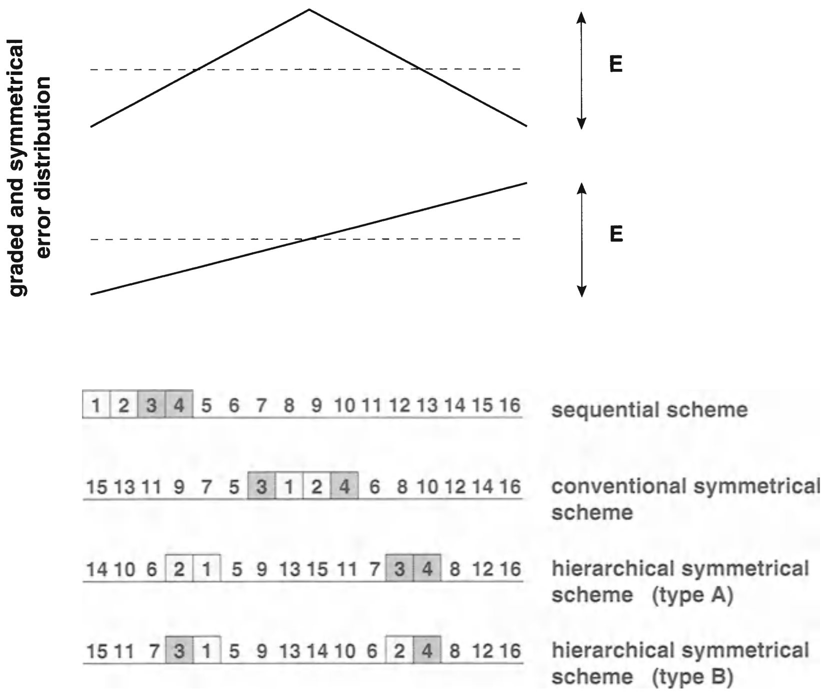


Figure 4.7: *The sequential, the conventional symmetrical and the hierarchical symmetrical switching scheme*

decoder [Miki JSSC86]. The working principle is based on comparing the generated row signals of two adjacent rows as follows :

- If both signals are high, the entire row of current sources is turned on.
- If both signals have different values, only the current sources that have a high column signal are turned on.

The above described principle can be implemented by using a simple digital logic block, leading to a high speed decoder circuit with a low power consumption. Three different switching schemes have been presented in [Miki JSSC86, Nakam JSSC91] using this decoder logic : the sequential, the conventional symmetrical and the hierarchical symmetrical scheme. A schematic representation of these switching sequences is given in fig.4.7.

In the sequential switching scheme, the current sources in a given row are turned on sequentially from the left to the right . The effect of this scheme on the integral nonlinearity error is given in fig.4.8 and fig.4.9. As can be clearly seen from these figures,

Switching Sequence	Graded Error	Symmetrical Error
sequential	$\frac{E}{8} * \frac{S^2}{S-1}$	$\frac{E}{16} * \frac{S^2}{S-2}$
conventional symmetrical	$\frac{E}{2}$	$\frac{E}{8} * \frac{S^2}{S-2}$
hierarchical symmetrical(A)	$\frac{E}{2} * (1 + \frac{1}{S-1})$	$\frac{E}{2}$
hierarchical symmetrical(B)	$\frac{E}{2}$	E

Table 4.2: *The INL errors of the three types of switching schemes [Nakam JSSC91] S denotes the number of current sources in a row of the current source array and E denotes the peak-to-peak error in both the graded and the symmetrical error distribution*

the sequential switching sequence causes large linearity errors due to the accumulation of both graded and symmetrical errors.

In the conventional symmetrical switching scheme, the current sources are turned on symmetrically around the center of the row. By using this scheme, the graded errors are cancelled at every two increments of the digital input (fig.4.8) but the errors generated by a symmetrical gradient will accumulate as is indicated in fig.4.9.

In the hierarchical symmetrical switching scheme, the current sources are turned on around the first and the third quarter of the current source row. Two different schemes are possible, depending on which error is cancelled first. Using switching scheme A, the symmetrical error generated by current source 1 is cancelled by current source 2 while the graded error caused by the current source pair (1,2) is cancelled by the current source pair (3,4). In switching scheme B, the errors caused by the linear gradient are cancelled out at current source level while the symmetrical errors are cancelled at the current source pair level. Comparing the linearity error of both switching schemes leads to the following result. In type A, the integral non-linearity error is approximately the same for both the graded and the symmetrical error distribution while in type B, the integral non-linearity error caused by the symmetrical error distribution is twice as large as the one caused by the graded error distribution (fig.4.8 and fig.4.9).

Table 4.2 gives an overview of the non-linearity errors of the three types of switching sequences where S denotes the number of current sources in a row of the current source array and E denotes the peak-to-peak error in both the graded and the symmetrical error distribution.

As a conclusion, it can be stated that if a 1-dimensional row and column decoder is used, it is best to implement the hierarchical symmetrical switching scheme type A since this scheme avoids the accumulation of linear and symmetrical errors resulting in a small integral non-linearity error.

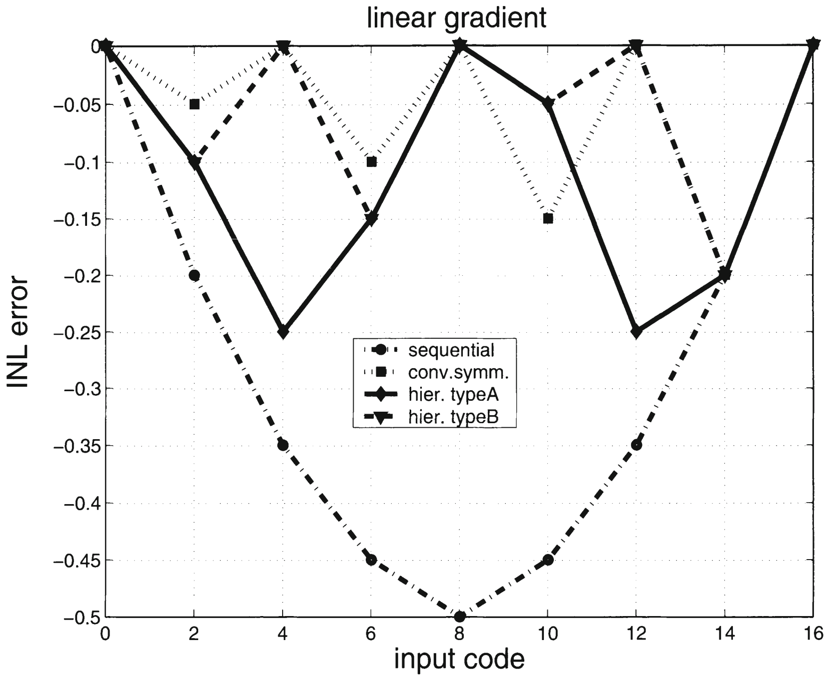


Figure 4.8: *The INL error caused by a linear gradient for the sequential, the conventional symmetrical and the hierarchical symmetrical (type A and type B) switching schemes (E=0.4)*

4.3.2.4 Switching Schemes for the 2-D Row and Column Decoding Principle

In the previous paragraph, the errors generated by the systematic gradients are only minimised in one dimension, namely the x-dimension. It is possible to adjust the decoder as to implement a hierarchical scheme in both the x and the y-direction. Although this implementation improves the overall linearity of the D/A converter, it is still not optimal.

In [Marqu ISSCC98] a switching scheme has been presented that preserves the simple row and column decoder but further reduces the INL error. The presented D/A converter has a 6-2-4 segmented architecture where the 6 most significant bits are implemented using a row and column decoder. Instead of using only one current source, the current is generated by four current sources that are placed symmetrically around the center of the array. Each current source is controlled by a separate decoder. This implies that instead of using only one decoder, four decoders have been placed on the chip. The D/A converter is actually built up out of four sub D/A converter blocks. By implementing the conventional symmetrical switching scheme in both the x and y direction, the linear errors have been completely cancelled out due to the spatial

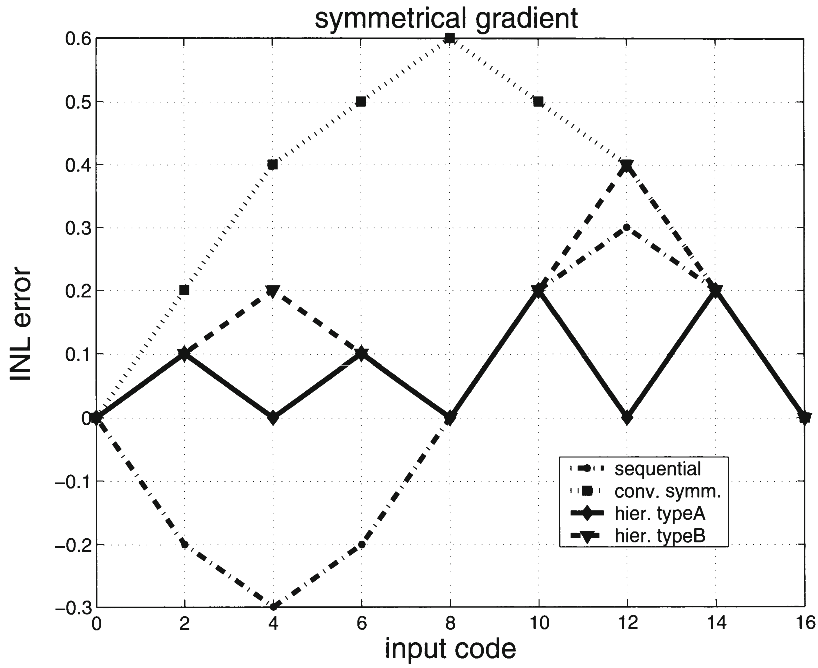


Figure 4.9: *The INL error caused by a symmetrical gradient for the sequential, the conventional symmetrical and the hierarchical symmetrical (type A and type B) switching schemes (E=0.4)*

symmetry while the symmetrical errors have been reduced significantly.

4.3.2.5 Decoder Independent 2-D Centroid Switching Schemes

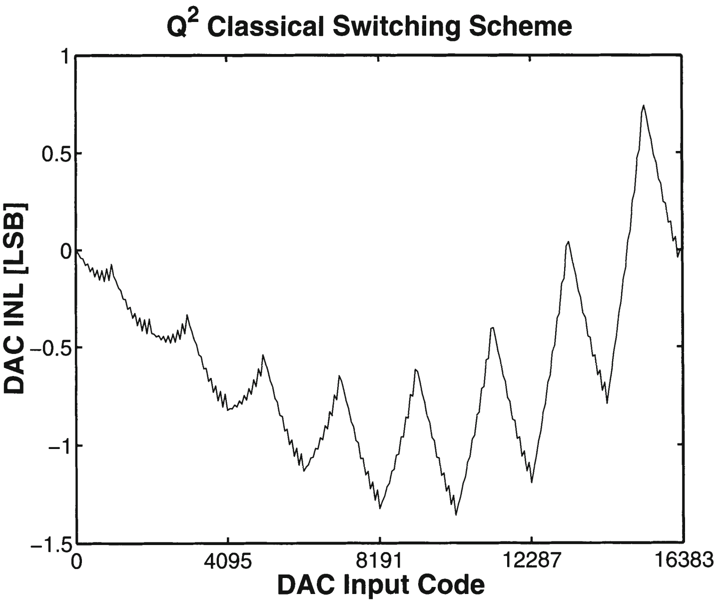
In most D/A converters implemented using a row and column decoder, the switches and the current source transistors are part of the same matrix. However, to minimise the coupling between the 'digital' switches and the 'analog' current sources and at the same time increase the flexibility of the switching schemes, two separate matrices for the current source and the switch transistors are used. In this way, the decoder does no longer determine the complexity of the switching scheme. This complexity is now dictated by the occupied silicon area for the interconnections (which can be done on top of the current source transistors), the used technology (especially the number of metal layers) and the creativity of the layout engineer.

Most current steering D/A converters have a segmented architecture implying that the unary current source in most cases equals a number of times the unit current source (= LSB current). It is therefore possible to divide this unary current source into a number of "sub"-current sources. For the 12 bit implementation [VdBos CICC98], 16 sub- current sources have been used that were switched on simultaneously around the center of each quadrant (double centroid switching scheme). In the 12 bit D/A converter described in [VdBos ISSCC01], the current source array was divided in 16 blocks and the sub-current sources were placed symmetrically around the center of each block leading to an INL error smaller than 0.3 LSB. This switching scheme is better known as the triple centroid switching scheme. To indicate the possibilities of defining alternative switching schemes, the example of a 14 bit D/A converter is given in fig.4.10. This figure shows a comparison between the switching scheme presented by [Miki JSSC86] and the Q^2 Random Walk switching scheme presented by [VdPla JSSC99] for a 14 bit D/A converter. To obtain such a high resolution without the use of any tuning or trimming, the systematic errors have to be made as small as possible. In this case, information regarding the gradients had been extracted from an earlier test chip. A Q^2 random walk switching scheme has been implemented. The resulting INL error for this switching scheme is about ten times smaller than for the classical switching scheme presented by Miki, which resulted in the first CMOS D/A converter with an intrinsic accuracy of 14 bit. More details on the here discussed switching schemes can be found in chapter 7.

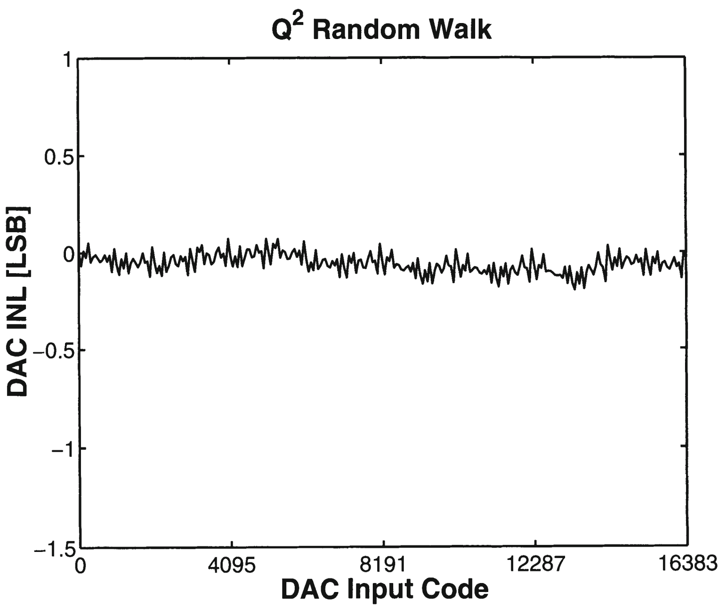
4.3.2.6 The Analytical Optimisation of a Switching Scheme

Up until now, deriving the optimal switching sequence has been done in a heuristic manner [Miki JSSC86, Nakam JSSC91, Marqu ISSCC98]. However, trying to solve this optimisation problem analytically can have several advantages. The problem of determining the switching scheme on sight is converted in solving a set of analytical equations that describe the gradients. A first advantage of this approach is obvious. This method allows to find a solution for every type of gradient by simply modifying and/or adding an extra equation. Another advantage is that solving this optimisation problem, a (near) optimal switching sequence will be found that is in most cases better than the heuristically derived one. For the case of the Q^2 random walk switching scheme of the 14 bit D/A converter [VdPla JSSC99] which has been determined using an optimisation algorithm which was heuristically constrained, it has been shown in [Cong TCASII] that this switching scheme is an optimal solution for the minimisation of the quadratic gradient errors but not for the minimisation of the linear graded errors. In the remainder of this paragraph, the mathematical ideas behind the optimisation problem will be highlighted.

In order to find the optimal switching sequence, a lower bound for the INL error of the D/A converter has to be determined. This is done by taking the following elements into account :



(a) Simulation of the INL for the Q^2 classical switching scheme



(b) Simulation of the INL for the Q^2 Random Walk switching scheme

Figure 4.10: A comparison between the switching scheme presented by [Miki JSSC86] and the Q^2 Random Walk switching scheme [VdPla JSSC99]

- The INL error of a D/A converter is by definition equal to :

$$INL = max(-INL_m, INL_M) \tag{4.16}$$

where INL_m is the minimal value and INL_M is the maximal value of the INL characteristic of the D/A converter.

- If the percentual errors of the different current sources are given [$\frac{\Delta I}{I_{ideal}} * 100 = F_i, i = 1 \cdots 2^N - 1$], one can state that the difference between the upper and the lower limit for the INL characteristic has to be larger than the maximum current source error.

$$F_{max} \leq INL_M - INL_m \tag{4.17}$$

- The INL error of the D/A converter will be minimal if INL_m and INL_M are located symmetrically around zero.

$$INL = INL_m = INL_M \tag{4.18}$$

Combining these three elements results in an absolute lower bound on the INL error :

$$INL_{LB} = \frac{F_{max}}{2} \tag{4.19}$$

Once the lower bound of the INL error is given, the (near) optimal switching sequence can be determined by using an INL bounded algorithm. Since it is not the intention of the author to go deeper into the computing algorithms for solving the optimisation problem, the reader is referred to [Cong TCASII] for more information.

It should be noted that some gradient information has to be available to tackle the switching scheme optimisation problem. This information can be extracted from a test chip but this method significantly increases the design time of the D/A converter circuit [VdPla JSSC99]. When this is not allowed, the heuristically derived schemes are at the moment still the best alternative. If extra silicon area and power consumption are not a problem, calibration circuits can be implemented that counteract the influence of the gradients (and even of the random mismatch errors) [Bugej JSSCC00].

4.4 Conclusion

A new INL_yield formula has been derived that in an accurate way describes the influence of the mismatch behaviour of the current sources on the yield of the D/A converter. In the second part of this chapter, the impact of the systematic errors on the static performance of a current steering D/A converter has been discussed. Implementing special switching schemes leads to a minimisation of these errors in such a way that the random errors become dominant. This is important since the derived yield formula will remain valid and can be used for determining the dimensions of the current source transistor.

Chapter 5

Dynamic Behaviour of Current Steering D/A Converters

5.1 Introduction

Where in the previous chapter, the static behaviour of the current steering D/A converter has been discussed, this chapter will focus on the dynamic performance of these architectures.

Recent papers [Marqu ISSCC98, Bavel CICC98, VdBos CICC98] reveal the problem that high speed Nyquist D/A converters are difficult to design. Analysis of the frequency domain performance of these architectures reveals that second and third order harmonic distortions limit the spurious free output signal bandwidth. This is caused by a combination of different factors that each have to be solved. The first part of this chapter will identify these performance degrading factors and some design guidelines will be proposed that solve the problems such as imperfect synchronisation of the switch control signals, the digital signal feedthrough through the gate-drain capacitance of the switch transistors and the drain voltage variation of the current source transistor.

The importance of the output impedance of the D/A converter will be discussed in the remainder of this chapter. A formula will be derived for the spurious free dynamic range as a function of the output impedance that allows the designer to calculate the minimal impedance that is required to obtain a certain frequency performance. For high resolution D/A converters, this implies that the use of cascoded structures becomes mandatory. These architectures will be discussed in detail in the final section of this chapter.

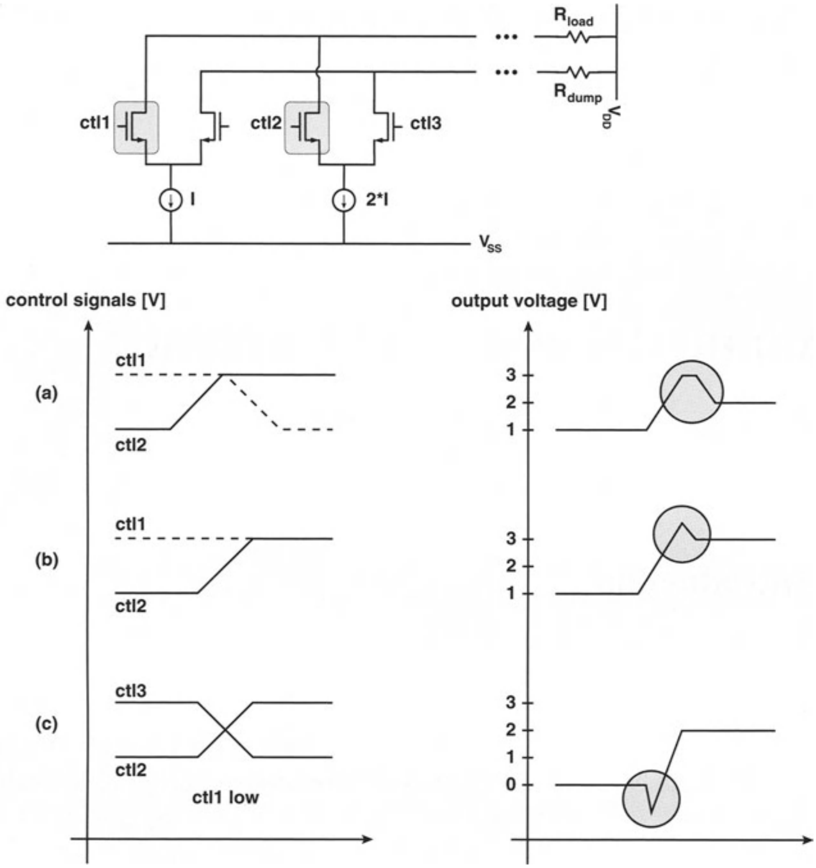


Figure 5.1: *(a) imperfect synchronisation of the switch control signals, (b) capacitive feedthrough, (c) drain voltage variation of the current source*

5.2 Major contributors

It has been generally accepted that the dynamic performance of a current steering D/A converter is mainly determined by either timing errors, by capacitive feedthrough from the digital control signals to the output node or by the voltage stability of the drain node of the current source transistors (fig.5.1). Different solutions for these problems can be found in open literature at the design level. However, in this paragraph a fourth element will be introduced that significantly degrades the dynamic performance of the D/A converter, namely the frequency dependency of its output impedance. These four factors will now be discussed in more detail based on the circuit schematics of a basic current cell block (fig.5.2.a).

5.2.1 The Imperfect Synchronisation of the Control Signals of the Switches

If the control signals of both switches (M_{swa} and M_{swb}) are not exactly matched in time, a glitch error will be directly visible at the output of the D/A converter (fig.5.1.a). This problem can be solved by placing a synchronisation block immediately in front of the switch transistors. In this way, any delay introduced by the digital decoding logic is cancelled out and the timing error is minimised. However, one should keep in mind that at the layout level, the implementation of this circuit has no use unless identical connections between the synchronising circuit and the switching transistors are drawn.

5.2.2 The Digital Signal Feedthrough via the C_{GD} of the Switch Transistors

The gate-drain capacitances of the switch transistors M_{swa} and M_{swb} form a feedthrough path that allows the digital control signals to have a direct impact on the output of the D/A converter (fig.5.1.b). The glitch energy error that is generated in this way can be significantly lowered by the use of a reduced voltage swing at the input of the switches or it can be minimised by placing a cascode transistor on top of the switch transistors [Marqu ISSCC98] . Since the introduction of these cascode transistors (that also have to be switched on/off) does not solve the problem entirely and leads to a higher area consumption and a distortion of the fully symmetrical operating principle of the basic current cell, recent D/A converter designs opt for the first solution since in some designs the implementation of a reduced voltage swing can be done by the same synchronisation circuit used to solve the problem described in the previous section (5.2.1).

5.2.3 The Voltage Variation at the Drain of the Current Source Transistors

If the crossing point of the switch control signals is situated at exactly the $\frac{V_{DD}+V_{SS}}{2}$ value, the following problem will occur. A time interval exists in which both switch transistors are simultaneously in the off-state. Since the current source transistor M_{cs} is still delivering current, the capacitance C_0 at its drain node will discharge. At the moment one of the switches starts conducting, an extra amount of current will flow through these transistors as to restore the DC voltage at that node. This will result in a glitch error at the output of the D/A converter leading to a deterioration of the dynamic performance (fig.5.1.c). This problem can be solved by the use of a special switch driver circuit [Kohno CICC95]. However, also for this building block

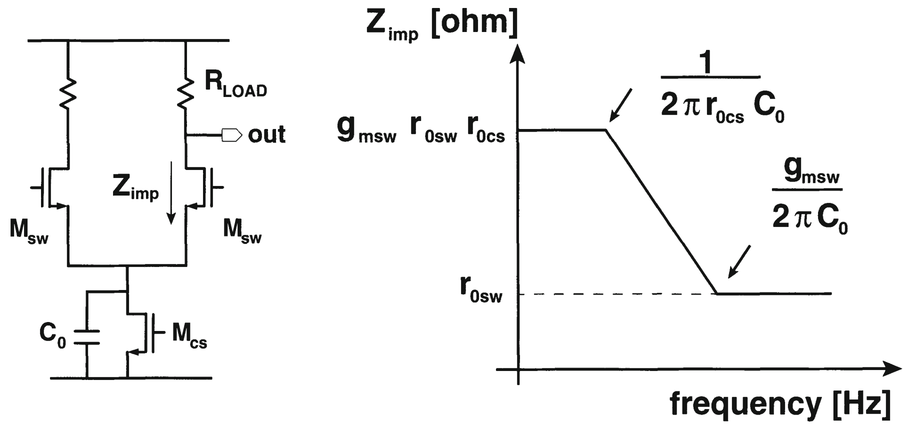


Figure 5.2: *(a) The basic current cell block (capacitance C_0 is the parasitic capacitance), (b) The impedance seen in the drain of the switch transistor versus frequency*

a trend exists towards an integration with the synchronisation circuit [Bavel CICC98, VdBos CICC98].

5.2.4 The Output Impedance

Since the first three problems causing a degradation of the dynamic performance of a current steering D/A converter can be solved by the use of one well designed and carefully layouted synchronised switch driver, the update rate of these devices is mainly determined by the design of the digital decoder. Therefore mixed-signal designers are pushing their designs to higher resolutions. Higher resolutions however do not only complicate the design of the decoder, it also introduces a fourth important factor that influences the dynamic performance of the D/A converter, namely the output impedance.

As is generally known, the output resistance R_{imp} (fig.5.2) of each current cell has to be made large so that its influence on the INL (integral non-linearity) specification of the D/A converter is negligible. The relation between this output resistance and the achievable INL specification is given by [Razavi]:

$$INL = \frac{I_{unit} R_{load}^2 N^2}{4 R_{imp}} \tag{5.1}$$

with R_{load} the load resistor, I_{unit} the LSB current and N the total number of unit current sources. In most cases, the cascode configuration of the switch and the current source transistors achieves the INL specification. However, this is only true over a limited frequency bandwidth as can be concluded from the following calculation. Fig.5.2.a shows the schematic of the unit current cell of a current steering D/A converter where the parasitic capacitance C_0 is indicated. The impedance Z_{imp} (the impedance seen from the output node into the drain of the switch transistor M_{swb}) can be calculated (fig.5.2.b) and equals :

$$Z_{imp} = r_{0sw}(1 + g_{msw}(r_{0cs}//C_0)) = r_{0sw}(1 + g_{msw}r_{0cs})\left(\frac{1 + j\omega C_0/g_{msw}}{1 + j\omega C_0 r_{0cs}}\right) \quad (5.2)$$

This formula indicates that the impedance has a pole and a zero at the following frequencies :

$$pole = \frac{1}{2\pi C_0 r_{0cs}} \quad and \quad zero = \frac{g_{msw}}{2\pi C_0} \quad (5.3)$$

The possibility to shift this pole and zero to a higher frequency is determined by the flexibility in adjusting the following four parameters : the output resistance of the current source transistor r_{0cs} and the switch transistor r_{0sw}, the transconductance of the switch g_{msw} and the capacitance C_0. According to eq.(5.3) the pole can be shifted towards a higher frequency by minimising the output resistance r_{0cs} of the current source transistor. However, the value of this resistance can not be freely adjusted since the gate-length L of this transistor is dictated by matching considerations [Pelgr JSSC89] and the current I_{DS} is determined by the full scale output signal. Since the current through the switches equals the current through the current source transistors and the gate-length L of the switch transistor is chosen to be minimal for speed reasons, nothing can be gained by the output resistance r_{0sw} of these transistors. Also the transconductance g_{msw} is largely determined since the gate overdrive voltage of the switches is the result of an optimisation process between the area occupied by the current sources and the useful utilisation of the limited power supply range of the D/A converter [VdBos ISDDMI98].

As can be concluded from the previous paragraph, the transistor parameters can be hardly adjusted to improve the negative impact generated by the pole of the output impedance. A whole different story applies to the capacitance C_0. Considering that for one extra bit of accuracy, not only the active area increases by a factor of four but also the area occupied by the decoding logic and the interconnections scales up, it is easy to understand that the total area needed for high resolution D/A converters (> 10 bit) becomes in the order of magnitude of several square millimeters. To reduce the dimensions of the current source matrix (which is beneficial for distance matching) and to minimise the coupling between the digital and the sensitive analog part of the chip, all decoding logic (including the switching transistors) and their interconnections are placed outside the current source array. Although this reduces the area occupied by the current source matrix considerably, it is not enough to ignore the influence of for example the parabolic gradients. To minimise the systematic errors introduced by these gradients special switching schemes have been devised as described in the previous chapter. These schemes introduce in almost all cases extra routing wires and thus an extra interconnect capacitance C_0 which is independent of the current source and switch transistor dimensions and is totally determined by the layout. The value for this interconnect capacitance can be several orders of magnitude larger than the intrinsic value of C_0.

5.3 SFDR-Bandwidth limitations

At this point, the frequency dependency of the output impedance has been discussed qualitatively but the question remains if this impedance has a significant effect on the dynamic performance of the D/A converter. In this paragraph, the value for the required minimal output impedance for a unit current cell will be calculated as a function of the resolution of the D/A converter. It will then become clear that for high resolutions and designs with a large interconnect capacitance C_0 the non-linearity introduced by the output impedance severely limits the output signal bandwidth.

For a $sin(\omega t)$ output signal, the number of switches T that conduct current at a time t equals :

$$T(t) = S\left(\frac{1 + sin(\omega t)}{2}\right) \tag{5.4}$$

with S the total number of current sources.

The total output impedance of the D/A converter is determined by the load impedance Z_L in parallel with T(t) parallel impedances Z_{imp}:

$$Y_{out}(t) = Y_L + Y_{imp}S\left(\frac{1+sin(\omega t)}{2}\right)$$

$$\Rightarrow V_{out}(t) = \frac{SI(1 + sin(\omega t))}{2Y_L + Y_{imp}S(1 + sin(\omega t))} \tag{5.5}$$

with I the current through one switch transistor and $Y_{imp} = 1/Z_{imp}$ and $Y_L = 1/Z_L$. A Taylor series expansion of the output voltage allows to determine the influence of Y_{imp} on the dynamic performance of the D/A converter.

On the condition that $SY_{imp} \ll Y_L$, the ratio Q of the fundamental signal to the second order harmonic is given by equation (5.6).

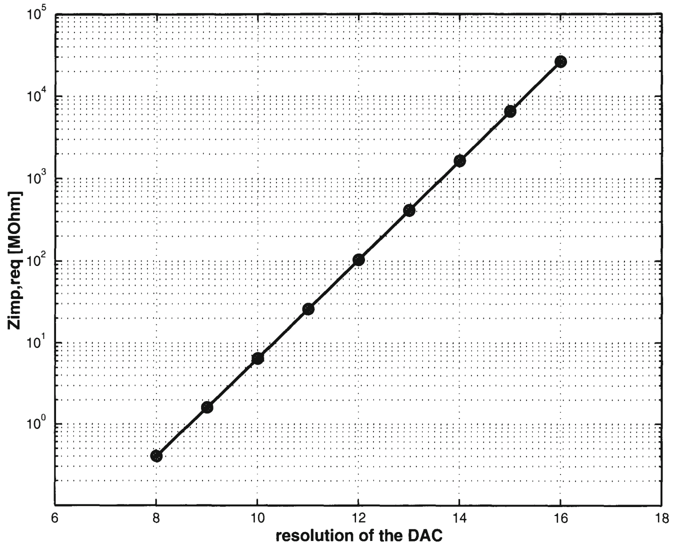


Figure 5.3: *The required output impedance as a function of the resolution of the D/A converter and for a load impedance of 25 ohm*

$$Q = \frac{4Y_L + 2SY_{imp}}{SY_{imp}} \tag{5.6}$$

Similar calculations can be performed to determine the influence of higher order harmonics on the SFDR of the D/A converter.

For the completeness of the presented work, the detailed calculations are given in Appendix 2. This Appendix will also show that the third order distortion component caused by the dynamic impedance effect is very small.

From formula (5.6), the value for the required Z_{imp} for a given resolution can be easily determined and equals :

$$Z_{imp,req} = \frac{SZ_L}{4}(Q-2) \approx \frac{SZ_L Q}{4} \tag{5.7}$$

Eq.(5.7) is plotted in fig.5.3 for a D/A converter with a resolution between 8 and 16 bits and for a load resistor of 25 ohm. For a resolution of 10 bits the Z_{imp} has to have a value of about 6.4 $M\Omega$ which is still relatively easy to implement. However, for a 12 bit current steering D/A converter with a load resistor of 25Ω, the value for the required Z_{imp} has to be at least 100 $M\Omega$ in the Nyquist frequency range. This is no longer a straightforward design specification since for high speed, high accuracy circuits the effect of the interconnect capacitance on the output impedance can no longer be neglected. From eq.(5.3), it can be derived that the total capacitance at the drain of the current source has to be smaller than the value given in eq.(5.8) otherwise the non-linearity introduced by the output impedance of the D/A converter poses a hard constraint on its dynamic performance.

$$C_0 \leq \frac{r_{0sw} g_{msw}}{2\pi f_N Z_{imp,req}} \tag{5.8}$$

with f_N the Nyquist frequency.

Eq.(5.7) can be rewritten in function of the SFDR.

$$SFDR \approx HD2 = 20\,log(Q) = 20\,log(\frac{4Z_{imp}}{SZ_L}) \tag{5.9}$$

The number of current sources S approximately equals 2^N (with N the number of bits) for high resolution D/A converters. Eq.(5.9) can then be rewritten as :

$$\begin{aligned} SFDR &= 20\,log(\frac{Z_{imp}}{Z_L}) + 20\,log4 - 20\,log(2^N) \\ SFDR &= 20\,log(\frac{Z_{imp}}{Z_L}) - 6.02\,(N-2) \end{aligned} \tag{5.10}$$

From this equation, it can be concluded that doubling the load resistance of the D/A converter can lead to a SFDR decrease of 6dB. Increasing the resolution of the D/A converter for a constant $\frac{Z_{imp}}{Z_L}$ ratio, also deteriorates the dynamic performance.

5.4 SFDR-Bandwidth Optimised Implementations

Since the values of the pole and zero in the current cell output impedance are practically solely determined by the interconnect capacitance C_0, an optimised layout can lead to a considerable improvement. However, this will only result in a few extra MHz SFDR for high resolution D/A converters. Therefore a solution has to be found at the design level that consists of changing the frequency dependency of the output impedance Z_{imp} by placing for example an extra cascode transistor either on top of the switching transistors or on top of the current source transistor. This will not only change the frequency behaviour of Z_{imp} but also increase its DC value.

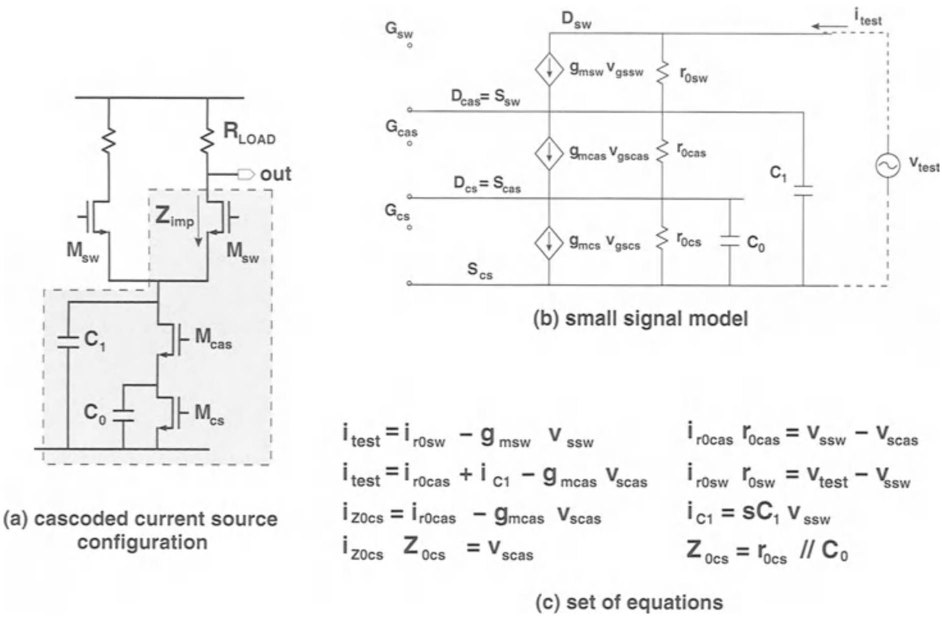


Figure 5.4: *The cascoded current source configuration (a) the circuit schematic (b) the small signal model and (c) the set of equations*

5.4.1 The Cascoded Current Source Transistor

A symmetrical solution consists of placing a cascode transistor on top of the current source transistor (fig.5.4.a). Only one additional transistor is necessary in comparison with the basic current cell. In order to be able to accurately describe the frequency behaviour of the impedance Z_{imp}, its algebraic expression has been calculated using the small signal model (fig.5.4.b). Solving the set of equations (fig.5.4.c) leads to the following result :

$$num(Z_{imp}) = g_{0cas}g_{0cs} + g_{0sw}g_{0cas} + g_{0sw}g_{mcas} + g_{0sw}g_{0cs} + g_{0cas}g_{msw} + g_{msw}g_{mcas} + g_{msw}g_{0cs} + s(g_{0cas}C_0 + g_{0cas}C_1 + g_{mcas}C_1 + g_{0cs}C_1 + g_{0sw}C_0 + g_{msw}C_0) + s^2C_0C_1 \tag{5.11}$$

$$denom(Z_{imp}) = g_{0sw}(g_{0cas}g_{0cs} + s(g_{0cas}C_0 + g_{0cas}C_1 + g_{mcas}C_1 + g_{0cs}C_1) + s^2C_0C_1) \tag{5.12}$$

Both the numerator and the denominator will now be separately discussed in some more detail.

5.4.1.1 The analysis of the zeroes of the impedance Z_{imp}

Since the transconductance g_m of a transistor is in general about 10 to 20 times larger than the output conductance of the same transistor and since for mismatch reasons the value of the conductance g_{0cs} is small (because of the large value for the gate-length L), the numerator of Z_{imp} can be simplified to :

$$T = num(Z_{imp}) = g_{msw}g_{mcas} + s(g_{mcas}C_1 + g_{msw}C_0) + s^2C_0C_1$$
$$= (sC_1 + g_{msw})(sC_0 + g_{mcas}) \qquad (5.13)$$

From eq.(5.13), it can be concluded that the impedance Z_{imp} has 2 zeroes, namely :

$$z_1 = \frac{g_{msw}}{2\pi C_1} \qquad and \qquad z_2 = \frac{g_{mcas}}{2\pi C_0} \qquad (5.14)$$

5.4.1.2 The analysis of the poles of the impedance Z_{imp}

For the denominator, two different cases have to be studied as will become clear from the analysis of the simplified expression:

$$D = denom(Z_{imp}) = g_{0sw}(g_{0cas}g_{0cs} + s(g_{0cas}C_0 + g_{mcas}C_1) + s^2C_0C_1) \qquad (5.15)$$

This equation can not be easily factored.

In the first case, it is assumed that the value of the product C_0g_{0cas} is much larger than the value of C_1g_{mcas}. The denominator can then be written as :

$$D = g_{0sw}(g_{0cas}g_{0cs} + s(g_{0cas}C_0 + g_{0cs}C_1) + s^2C_0C_1)$$
$$= g_{0sw}(sC_1 + g_{0cas})(sC_0 + g_{0cs}) \qquad (5.16)$$

It can be concluded that the impedance Z_{imp} has 2 poles located at the following frequencies :

$$p_{1a} = \frac{1}{2\pi C_1 r_{0cas}} \qquad and \qquad p_{2a} = \frac{1}{2\pi C_0 r_{0cs}} \qquad (5.17)$$

In the second case ($C_0g_{0cas} \ll C_1g_{mcas}$), the denominator can be factored in the following manner :

$$D = g_{0sw}(g_{0cas}g_{0cs} + s(\frac{g_{0cas}g_{0cs}C_0}{g_{mcas}} + g_{mcas}C_1) + s^2C_0C_1)$$

$$= \frac{g_{0sw} g_{0cas} g_{0cs}}{g_{mcas}} (sC_1 g_{mcas} r_{0cas} r_{0cs} + 1)(sC_0 + g_{mcas}) \qquad (5.18)$$

Notice that one of the poles of the impedance Z_{imp} will now coincide with one if its zeroes and as a consequence, a frequency behaviour similar to the one of the current source transistor without the cascode transistor is seen.

$$p_{1b} = \frac{1}{2\pi C_1 g_{mcas} r_{0cas} r_{0cs}} \quad and \quad p_{2b} = \frac{g_{mcas}}{2\pi C_0} \qquad (5.19)$$

The case where $C_0 g_{0cas} \simeq C_1 g_{gmcas}$ does not present itself in reality because for high resolution current steering D/A converters either C_0 or C_1 incorporates the interconnection capacitance making that term the dominant one.

5.4.1.3 A numerical example

A numerical example will now be given to illustrate the theory. In this example the impedance Z_{imp} will be calculated for a 12 bit current steering D/A converter with a full scale current of 20 mA that is designed in a $0.5\mu m$ technology. In a first step, the transistor parameters will determined.

The dimensions of the current source transistor are determined by the used technology mismatch parameters and the full scale current.

$$\begin{aligned}
A_\beta &= 1.9\,\%\,\mu m \\
A_{VT} &= 13\,mV\,\mu m \\
\frac{\mu C_{ox}}{2} &= 8.5 * 10^{-5}\,\frac{A}{V^2} \\
(V_{GS} - V_T)_{cs} &= 1\,V \\
I_{cs} &= 5\,\mu A
\end{aligned}$$

The values of the gate-length and gate-width can now be calculated and equal $W_{cs} = 2.3\mu m$ and $L_{cs} = 37\mu m$. The output conductance of the current source transistor equals $g_{0cs} = 8.5 * 10^{-8}\,S$

The current flowing through the cascode and the switch transistor is equal to the current flowing through the current source transistor and as a consequence also the values for the gate overdrive voltage, the dimensions, the transconductance and the output conductance of these transistors can be determined.

$$\begin{aligned}
(V_{GS} - V_T)_{cas} &= 0.24\,V \\
W_{cas} &= 0.8\,\mu m \\
L_{cas} &= 0.7\,\mu m \\
g_{mcas} &= 42\,\mu S \\
g_{0cas} &= 1.6\,\mu S
\end{aligned}$$

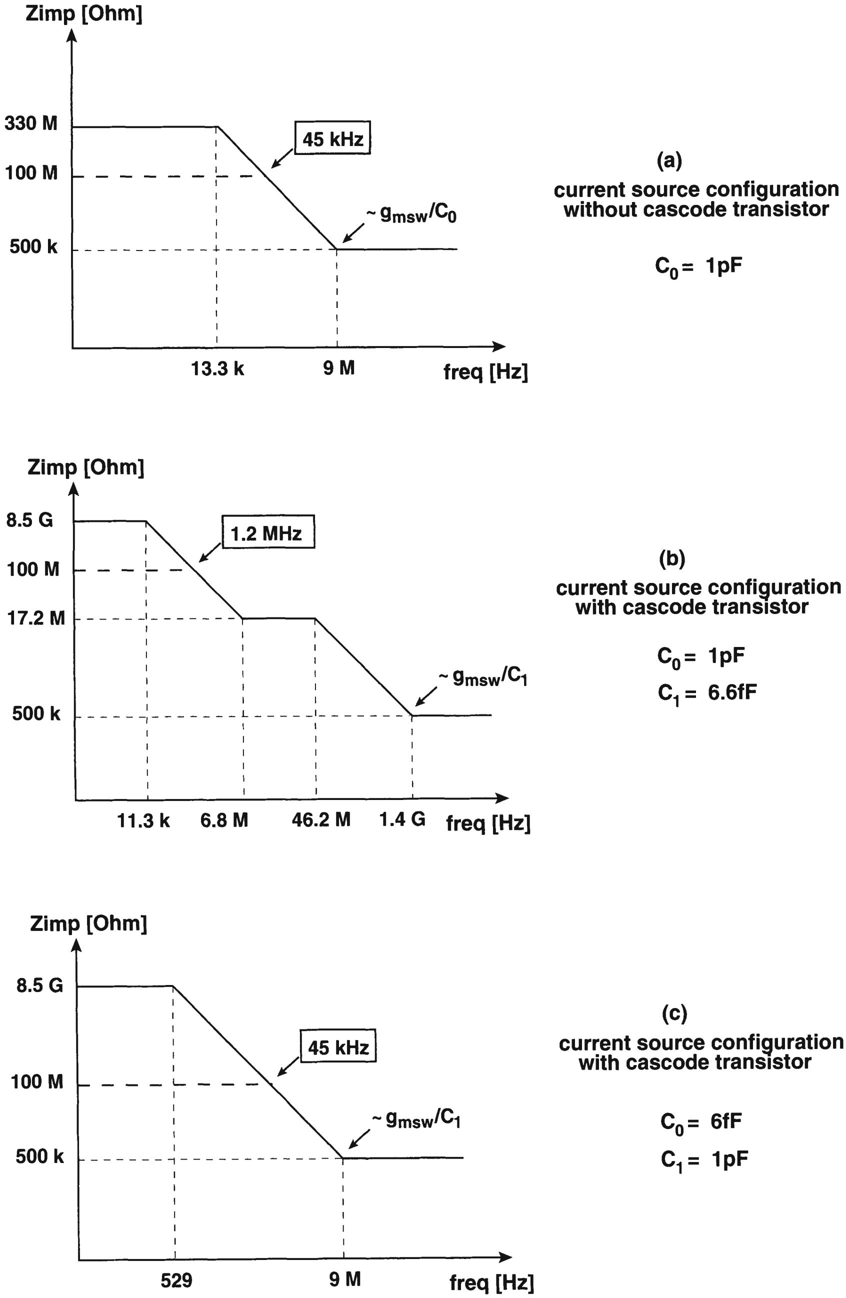


Figure 5.5: *Bode diagrams for the numerical example in section 5.4.1.3*

$$
\begin{aligned}
(V_{GS} - V_T)_{sw} &= 0.2\,V \\
W_{sw} &= 0.8\,\mu m \\
L_{sw} &= 0.5\,\mu m \\
g_{msw} &= 55\,\mu S \\
g_{0sw} &= 2\,\mu S
\end{aligned}
$$

The mimimum values (without layout parasitics) of the parasitic capacitances C_0 and C_1 are given by :

$$
\begin{aligned}
C_1 &= 2 * C_{S-sw} + C_{D-cas} \approx 6.6\,fF \\
C_0 &= C_{S-cas} + C_{D-cs} \approx 6\,fF
\end{aligned}
$$

Now, the Bode diagram will be calculated for the unit current cell without any additional cascode transistors. The value of the interconnect capacitance C_0 is chosen to be 1pF. The impedance Z_{imp} is then given by :

$$
Z_{imp} = 3.2 * 10^8 * (\frac{1 + 1.8 * 10^{-8} * s}{1 + 1.2 * 10^{-5} * s}) \tag{5.20}
$$

The values of the poles and zeroes are :

$$
\begin{aligned}
p &= 13.3\,kHz \\
z &= 9\,MHz
\end{aligned}
$$

The required value for the impedance Z_{imp} is $100M\Omega$ to obtain a good dynamic performance for the 12 bit D/A converter. This impedance is reached for a signal bandwidth frequency of about 45 kHz. The Bode diagram is given in fig.5.5.a.

The second case that will be discussed is the case where the interconnection between the switches and the current sources is done at the source node of the cascode transistors. The value of the capacitance C_0 is chosen to be 1pF. The impedance Z_{imp} can then be calculated and equals :

$$
Z_{imp} = \frac{2.5 * 10^{-9} + 5.9 * 10^{-17} * s + 6.6 * 10^{-27} * s^2}{2.7 * 10^{-19} + 3.8 * 10^{-24} * s + 1.3 * 10^{-32} * s^2} \tag{5.21}
$$

The exact values for the poles and the zeroes can be derived from this expression :

$$
\begin{aligned}
p_{1a} &= 46.2\,MHz \\
p_{2a} &= 11.3\,kHz \\
z_1 &= 6.8\,MHz \\
z_2 &= 1.4\,GHz
\end{aligned}
$$

The required impedance of $100M\Omega$ is reached for a signal bandwidth of 1.2 MHz. This is a more than 25 times improvement compared to the case where the current source transistor is implemented without the cascode transistor. The Bode diagram is given in fig.5.5.b.

The third case describes the situation where the interconnection between the current source array and the switches is made at the drain node of the cascode transistor. The value of the capacitance C_1 is chosen to be 1pF. The impedance Z_{imp} is then given by :

$$Z_{imp} = \frac{2.5 * 10^{-9} + 4.4 * 10^{-17} * s + 6 * 10^{-27} * s^2}{2.7 * 10^{-19} + 8.7 * 10^{-23} * s + 1.2 * 10^{-32} * s^2} \tag{5.22}$$

The exact values of the poles and zeroes are :

$$\begin{aligned} p_{1b} &= 529\,Hz \\ p_{2b} &= 1.2\,GHz \\ z_1 &= 9\,MHz \\ z_2 &= 1.2\,GHz \end{aligned}$$

From these numbers it can be concluded that p_2 and z_2 coincide. The Bode diagram for this case is given in fig.5.5.c. The required impedance of $100M\Omega$ is reached for a signal bandwidth of only 45 kHz. This result is similar to the one of the current source transistor without the cascode transistor. The zero is located at the frequency given by $g_{msw}/2\pi C_1$, which is exactly the same frequency as the zero of the current source configuration without the cascode transistor $g_{msw}/2\pi C_0$ since both capacitances are in this case determined by the interconnections. As is generally known, the product of this frequency and the drain-source resistance of the switch transistor equals the product of the required impedance and its corresponding frequency for a 20 dB slope in the Bode diagram. As a consequence, both configurations have the same signal frequency bandwidth.

It can be concluded from this example that placing an extra cascode transistor on top of the current source is only useful if the interconnection between the switches and the current source array is done at the source node of the cascode transistor. Furthermore, it is important to note that this is only an example. During the design phase of the 12 bit D/A converter, the values for the poles and zeroes (= the values of the transistor parameters) together with the value of the interconnect capacitance C_0 have to be optimised to obtain a maximum frequency bandwidth.

5.4.1.4 Fault analysis of the presented theory

The solutions of the numerical example are exact since no terms in the numerator or the denominator of Z_{imp} have been eliminated during the calculation. If the values for the poles and the zeroes are calculated using the derived simplified analytic formula of eq.(5.14) and eq.(5.17), the following results are found :

$$p_{1a} = 38.6\,MHz \qquad and \qquad p_{2a} = 13.5\,kHz$$
$$z_1 = 6.7\,MHz \qquad and \qquad z_2 = 1.4\,GHz$$

There exists almost no difference between the values from the numerical example and the theoretical equation for the frequency of the zeroes but for the frequency of the poles this is not entirely true. This can be understood as follows.

The transfer function has the following form:

$$a + bs + cs^2 = 0$$

It is easy to derive that the sum of the two solutions of this equation equals b/c while the product equals a/c. If the exact formula of the denominator of Z_{imp} is compared to the simplified one, it becomes clear that for both equations the coefficients a and c remain the same and only coefficient b is different.

$$D_{real} = g_{0sw}(g_{0cas}g_{0cs} + s(g_{0cas}C_0 + g_{0cas}C_1 + g_{mcas}C_1 + g_{0cs}C_1) + s^2C_0C_1$$
$$D_{est} = g_{0sw}(g_{0cas}g_{0cs} + s(g_{0cas}C_0 + g_{0cs}C_1) + s^2C_0C_1)$$

This means that the product of the solutions of the exact equation and of the simplified one are equal. If we now assume that one pole is dominant over the other one, it can be stated that the pole with the highest frequency equals b/c. This directly implies that the dominant pole has a frequency equal to a/b. If the coefficient b of the simplified expression is x times smaller than the coefficient b of the exact expression, the calculated dominant pole will be a factor of x larger than the real dominant pole. In this case,

$$p_{real,dom} = p_{est,dom} * \frac{g_{0cas}C_0 + g_{0cs}C_1}{g_{0cas}C_0 + g_{0cas}C_1 + g_{mcas}C_1 + g_{0cs}C_1} \tag{5.23}$$
$$p_{real,dom} = p_{est,dom} * p_{corr1}$$

which for the numerical example leads to a correction factor p_{corr1} that equals 0.85. The real non-dominant pole has a value of $p_{real,nondom} = p_{est,nondom} * 1/p_{corr1} =$

$p_{est,nondom} * 1.18$ since the product of the two poles has to remain constant as was mentioned earlier.

Another important implication of this correction factor p_{corr1} is that the impedance levels of Z_{imp} shift! The middle level of Z_{imp} is in this case no longer equal to $g_{mcas} r_{0cas} r_{0sw}$ but to $g_{mcas} r_{0cas} r_{0sw} * p_{corr1}$. It is important to take this into account during the design of the D/A converter.

In the second case ($g_{0cas} C_0 \ll g_{mcas} C_1$), the exact and the simplified expressions for the denominator of Z_{imp} are given by

$$D_{real} = g_{0sw}(g_{0cas} g_{0cs} + s(g_{0cas} C_0 + g_{0cas} C_1 + g_{mcas} C_1 + g_{0cs} C_1) + s^2 C_0 C_1$$

$$D_{est} = g_{0sw}(g_{0cas} g_{0cs} + s(\frac{g_{0cas} g_{0cs} C_0}{g_{mcas}} + g_{mcas} C_1) + s^2 C_0 C_1)$$

The correction factor is determined in the same way as in the previous paragraph and equals

$$p_{corr2} = \frac{\frac{C_0 g_{0cas} g_{0cs}}{g_{mcas}} + g_{mcas} C_1}{g_{0cas} C_0 + g_{0cas} C_1 + g_{mcas} C_1 + g_{0cs} C_1} \qquad (5.24)$$

which has a value of about $p_{corr2} \approx 1(0.98)$ for the numerical example. In this case the estimated values of the pole and zero ($p_{1b} = 517Hz$ and $z_{2b} = 8.8MHz$) frequencies are practically the same as the numerical calculated ones.

5.4.1.5 Conclusion

To solve the problem of the output impedance, the cascoded current source is a good option. However, it is important to note that this implementation only provides a good solution if the cascode transistor is part of the switch array and not of the current source array. Analysing the Bode plots leads to the conclusion that the best results will be obtained by using a minimised interconnect capacitance C_0 and a small parasitic capacitance C_1.

5.4.2 The Cascoded Switch Transistor

The second solution that will be discussed is the use of a cascode transistor on top of the switch transistors (fig.5.6.a). This architecture is already frequently used because it reduces the glitch energy error caused by the digital feedthrough through the gate-drain capacitance of the switches [Marqu ISSCC98]. In fig.5.6.a/b, the schematic representation of the unit current cell is given together with the small signal model of

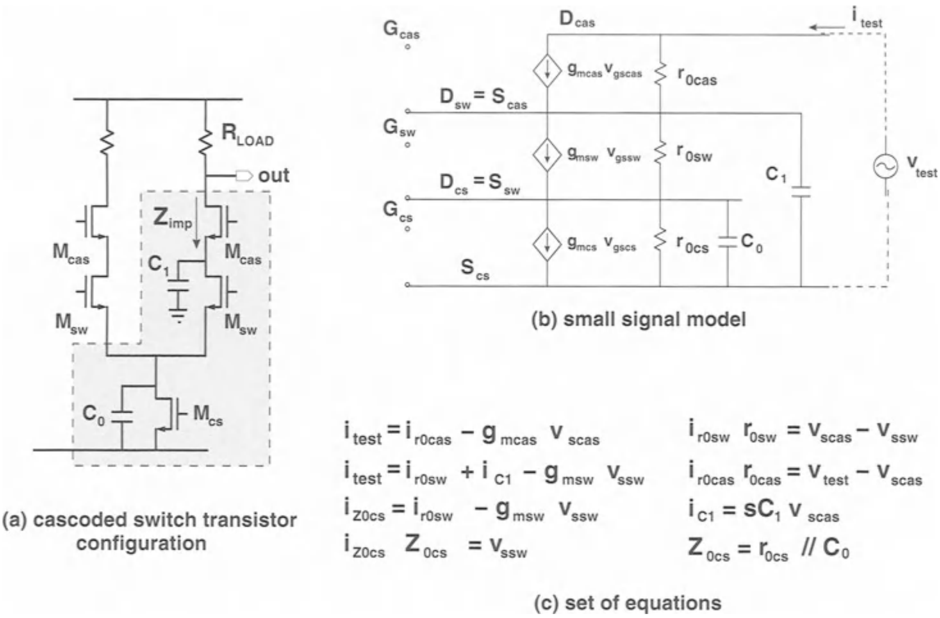


Figure 5.6: *The cascoded switch transistor configuration (a) the circuit schematic (b) the small signal model and (c) the set of equations*

the circuit. This model is similar to the one of the cascoded current source transistor (fig.5.4) that has been discussed in the previous section. The impedance Z_{imp} is given by:

$$\begin{aligned} num(Z_{imp}) = {} & g_{0sw}g_{0cs} + g_{0sw}g_{0cas} + g_{0cas}g_{msw} + g_{0cas}g_{0cs} + g_{0sw}g_{mcas} + \\ & g_{msw}g_{mcas} + g_{mcas}g_{0cs} + s(g_{0sw}C_0 + g_{0sw}C_1 + g_{msw}C_1 + g_{0cs}C_1 + \\ & g_{0cas}C_0 + g_{mcas}C_0) + s^2C_0C_1 \end{aligned} \tag{5.25}$$

$$\begin{aligned} denom(Z_{imp}) = {} & g_{0cas}(g_{0sw}g_{0cs} + s(g_{0sw}C_0 + g_{0sw}C_1 + g_{msw}C_1 + g_{0cs}C_1) \\ & + s^2C_0C_1) \end{aligned} \tag{5.26}$$

Also for this case, the numerator and the denominator are discussed separately.

5.4.2.1 The analysis of the zeroes of the impedance Z_{imp}

The numerator of the impedance Z_{imp} can be factored in the same way as for the cascoded current source implementation. This leads to the following result :

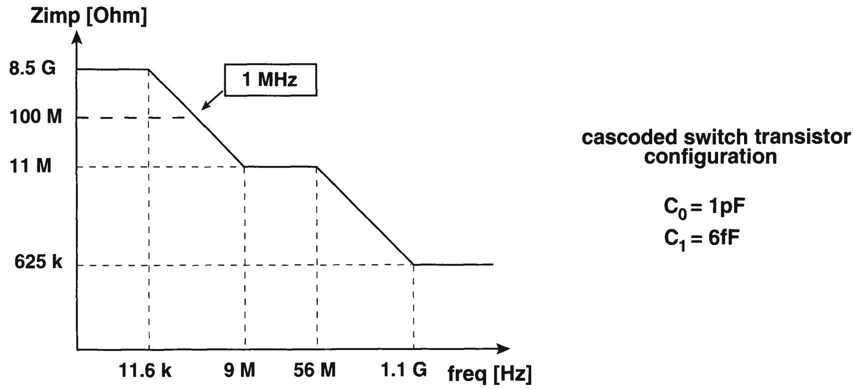


Figure 5.7: *Bode diagrams for the numerical example in section 5.4.2.3*

$$T = (sC_1 + g_{mcas})(sC_0 + g_{msw}) \tag{5.27}$$

The values of the zeroes is then given by :

$$z_1 = \frac{g_{mcas}}{2\pi C_1} \quad \textit{and} \quad z_2 = \frac{g_{msw}}{2\pi C_0} \tag{5.28}$$

5.4.2.2 The analysis of the poles of the impedance Z_{imp}

Since the switch transistors are never incorporated in the current source array because of decoupling reasons, there is actually only one case that has to be considered, namely the case where $g_{0sw}C_0 \gg g_{msw}C_1$ (the interconnection is made at the drain node of the current source and thus is part of the capacitance C_0). The denominator can be factored :

$$D = g_{0cas}(sC_1 + g_{0sw})(sC_0 + g_{0cs}) \tag{5.29}$$

and as a consequence the poles are given by :

$$p_1 = \frac{1}{2\pi C_1 r_{0sw}} \quad \textit{and} \quad p_2 = \frac{1}{2\pi C_0 r_{0cs}} \tag{5.30}$$

5.4.2.3 A numerical example

In order to make a comparison between the different implementations, the same values for the transistor parameters are chosen as in section 5.4.1.3. The value of the interconnect capacitance C_0 is chosen to be 1 pF. The impedance Z_{imp} is then given by :

$$Z_{imp} = \frac{2.5 * 10^{-9} + 4.6 * 10^{-17} * s + 6.6 * 10^{-27} * s^2}{2.7 * 10^{-20} + 3.8 * 10^{-24} * s + 10^{-32} * s^2} \tag{5.31}$$

The values of the poles and zeroes are :

$$\begin{aligned} p_1 &= 56\,MHz \\ p_2 &= 11.6\,kHz \\ z_1 &= 1.1\,GHz \\ z_2 &= 9\,MHz \end{aligned}$$

The Bode diagram is given in fig.5.7. From this figure it can be concluded that the non-linearity introduced by the output impedance limits the output signal bandwidth to a value of approximately 1 MHz which is a significant improvement compared to the topology without the cascode transistor.

5.4.2.4 Fault analysis of the presented theory

The values calculated using eq.(5.28) and eq.(5.30) are :

$$\begin{aligned} p_1 &= 48.2\,MHz \\ p_2 &= 13.5\,kHz \\ z_1 &= 1\,GHz \\ z_2 &= 8.8\,MHz \end{aligned}$$

The correction factor p_{corr3} is derived in a similar way as for the cascoded current source topology and equals :

$$p_{corr3} = \frac{g_{0cs}C_1 + g_{0sw}C_0}{g_{0cs}C_1 + g_{0sw}C_0 + g_{0sw}C_1 + g_{msw}C_1} = 0.86 \tag{5.32}$$

The value for $g_{0sw}C_0$ equals $2*10^{-18}$ and the value for $g_{msw}C_1$ is about a factor of five smaller and equals $3.6 * 10^{-19}$. If the difference between these two values becomes larger, the correction factor will approach one. On the other hand, if the difference decreases, the value of the correction factor becomes important and has to be taken into account!

5.4.2.5 Conclusion

Cascoding the switch transistors will be an effective solution for the problem of the dynamic output impedance. However it requires the implementation of two extra transistors for each current source which leads to a considerable increase of the occupied

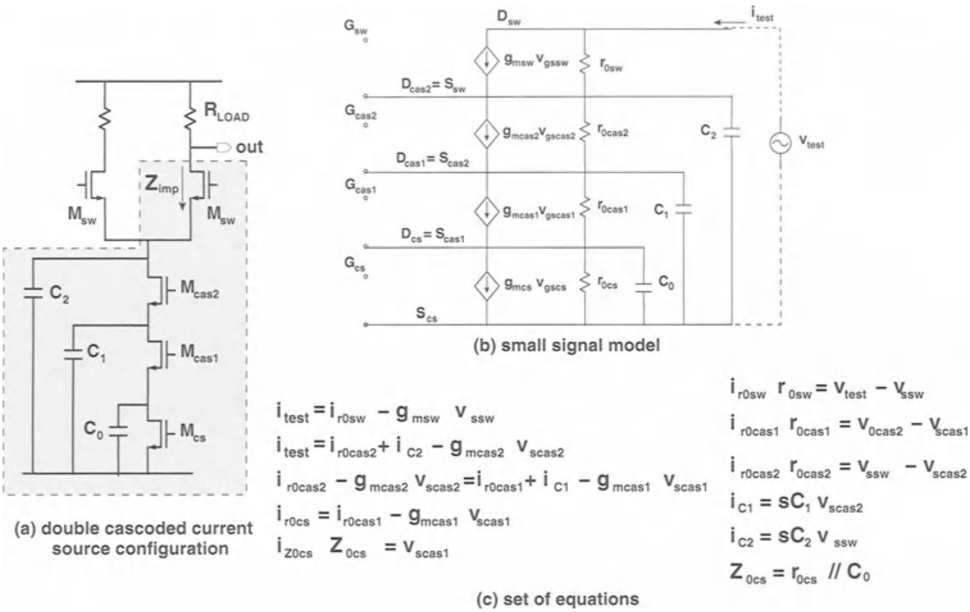


Figure 5.8: *The double cascoded current source configuration (a) the circuit schematic (b) the small signal model and (c) the set of equations*

silicon area. Furthermore, the on- and off-switching of the cascode transistors disturbs the fully symmetrical operation principle of the D/A core, leading to a deteriorated differential behaviour.

5.4.3 The Double Cascoded Current Source Transistor

In this section, the effect of placing a double cascoded current source configuration is investigated (fig.5.8). Throughout the calculations, it has been assumed that the interconnection between the current source and the switch/latch array is located at the drain of the current source transistor (large C_0). The output impedance has been calculated using the small signal equivalent depicted in fig.5.8.b. This leads to the following result:

$$
\begin{aligned}
num(Z_{imp}) = {} & g_{0cas1}g_{0cas2}g_{0cs} + (g_{0sw} + g_{msw})(g_{0cas1}g_{0cas2} + g_{0cas1}g_{mcas2} + \\
& g_{0cas1}g_{0cs} + g_{mcas1}g_{0cas2} + g_{mcas1}g_{mcas2} + g_{mcas1}g_{0cs} + g_{0cas2}g_{0cs}) + sC_1(g_{0cas1} \\
& g_{0cas2} + g_{0cas1}g_{mcas2} + g_{0cas1}g_{0cs} + (g_{0sw} + g_{msw})(g_{0cas2} + g_{mcas2} + g_{0cs})) + sC_0(\\
& g_{0cas1}g_{0cas2} + (g_{0sw} + g_{msw})(g_{0cas1} + g_{mcas1} + g_{0cas2})) + sC_2(g_{0cas1}g_{0cas2} + g_{0cas1} \\
& g_{mcas2} + g_{0cas1}g_{0cs} + g_{mcas1}g_{0cas2} + g_{mcas1}g_{mcas2} + g_{mcas1}g_{0cs} + g_{0cas2}g_{0cs}) + \\
& s^2C_0C_1(g_{0cas1} + g_{msw} + g_{0sw}) + s^2C_0C_2(g_{0cas1} + g_{mcas1} + g_{0cas2}) + s^2C_1C_2 \\
& (g_{0cas2} + g_{mcas2} + g_{0cs}) + s^3C_0C_1C_2
\end{aligned}
\tag{5.33}
$$

$$
\begin{aligned}
denom(Z_{imp}) = {} & g_{0sw}(g_{0cas1}g_{0cas2}g_{0cs} + sC_1(g_{0cas1}g_{0cas2} + g_{0cas1}g_{mcas2} + g_{0cas1} \\
& g_{0cs}) + sC_0(g_{0cas1}g_{0cas2}) + sC_2(g_{0cas1}g_{0cas2} + g_{0cas1}g_{mcas2} + g_{0cas1}g_{0cs} + g_{mcas1} \\
& g_{0cas2} + g_{mcas1}g_{mcas2} + g_{mcas1}g_{0cs} + g_{0cas2}g_{0cs}) + s^2C_0C_2(g_{0cas1} + g_{mcas1} \\
& + g_{0cas2}) + s^2C_0C_1(g_{0cas1}) + s^2C_1C_2(g_{0cas2} + g_{mcas2} + g_{0cs}) + s^3C_0C_1C_2)
\end{aligned}
\tag{5.34}
$$

5.4.3.1 The analysis of the zeroes of the impedance Z_{imp}

Eq.(5.33) can be drastically simplified by taking into account the fact that the transconductance of a transistor is about a factor of 20 larger than its conductance. The numerator of Z_{imp} can then be factorised in the following manner:

$$num(Z_{imp}) = (sC_0 + g_{mcas2})(sC_1 + g_{mcas1})(sC_2 + g_{msw}) \tag{5.35}$$

The zeroes are given by:

$$z_1 = \frac{g_{mcas2}}{2\pi C_0} \qquad z_2 = \frac{g_{msw}}{2\pi C_2} \qquad z_3 = \frac{g_{mcas1}}{2\pi C_1} \tag{5.36}$$

5.4.3.2 The analysis of the poles of the impedance Z_{imp}

Eq.(5.34) can be simplified in the same manner as the numerator of Z_{imp}. Taking into consideration the large value of the capacitance C_0, this results in:

$$
\begin{aligned}
denom(Z_{imp}) = {} & g_{0cas1}g_{0cas2}g_{0cs} + sC_0g_{0cas1}g_{0cas2} + sC_2g_{mcas1}g_{mcas2} \\
& + s^2C_0C_2g_{mcas1} + s^3C_0C_1C_2
\end{aligned}
\tag{5.37}
$$

Depending on the value of $C_0g_{0cas1}g_{0cas2}$ and $C_2g_{mcas1}g_{mcas2}$, two cases are possible.

In the first case, $C_2 g_{mcas1} g_{mcas2} \gg C_0 g_{0cas1} g_{0cas2}$. Eq.(5.37) further simplifies to:

$$denom(Z_{imp}) = g_{0cas1} g_{0cas2} g_{0cs} + sC_2 g_{mcas1} g_{mcas2} + s^2 C_0 C_2 g_{mcas1} + s^3 C_0 C_1 C_2 \tag{5.38}$$

For frequencies smaller than $g_{mcas1}/2\pi C_1$, the third order term in this equation can be neglected. This assumption is valid since C_1 has a small value (fF) and thus the pole g_{mcas1}/C_1 is located in the GigaHertz range.

$$denom(Z_{imp}) = g_{0cas1} g_{0cas2} g_{0cs} + sC_2 g_{mcas1} g_{mcas2} + s^2 C_0 C_2 g_{mcas1} \tag{5.39}$$

The solutions of this quadratic equation are given by:

$$p_{2,3} = \frac{g_{mcas2}}{4\pi C_0}(1 \pm \sqrt{1-x}) \qquad x = \frac{4C_0 g_{0cas1} g_{0cas2} g_{0cs}}{C_2 g_{mcas1} g_{mcas2}^2} \tag{5.40}$$

Since the value of x is much smaller than one, the Taylor series expansion of the square root can be used, which directly gives us the value of the two poles.

$$\begin{aligned} p_2 &= \frac{g_{mcas2}}{4\pi C_0}(1 + (1 - \frac{x}{2})) \simeq \frac{g_{mcas2}}{2\pi C_0} \\ p_3 &= \frac{g_{mcas2}}{4\pi C_0}(1 - (1 - \frac{x}{2})) \simeq \frac{g_{0cas1} g_{0cas2} g_{0cs}}{2\pi C_2 g_{mcas1} g_{mcas2}} \end{aligned} \tag{5.41}$$

Remember that the value of the first pole was given by:

$$p_1 = \frac{g_{mcas1}}{2\pi C_1} \tag{5.42}$$

In the second case, $C_2 g_{mcas1} g_{mcas2} \ll C_0 g_{0cas1} g_{0cas2}$. Eq.(5.37) further simplifies to:

$$denom(Z_{imp}) = g_{0cas1} g_{0cas2} g_{0cs} + sC_0 g_{0cas1} g_{0cas2} + s^2 C_0 C_2 g_{mcas1} + s^3 C_0 C_1 C_2 \tag{5.43}$$

The same assumption can be made as in the first case (relating to the third order term) resulting in the same high frequency pole.

$$denom(Z_{imp}) = g_{0cas1} g_{0cas2} g_{0cs} + sC_0 g_{0cas1} g_{0cas2} + s^2 C_0 C_2 g_{mcas1} \tag{5.44}$$

The solutions of this equation are given by:

$$p_{2,3} = \frac{g_{0cas1} g_{0cas2}}{4\pi C_2 g_{mcas1}}(1 \pm \sqrt{1-x}) \qquad x = \frac{4C_2 g_{mcas1} g_{0cs}}{C_0 g_{0cas1} g_{0cas2}} \tag{5.45}$$

Using the Taylor series expansion results in the following three poles

$$p_1 = \frac{g_{mcas1}}{2\pi C_1} \qquad p_2 = \frac{g_{0cs}}{2\pi C_0} \qquad p_3 = \frac{g_{0cas1} g_{0cas2}}{2\pi C_2 g_{mcas1}} \tag{5.46}$$

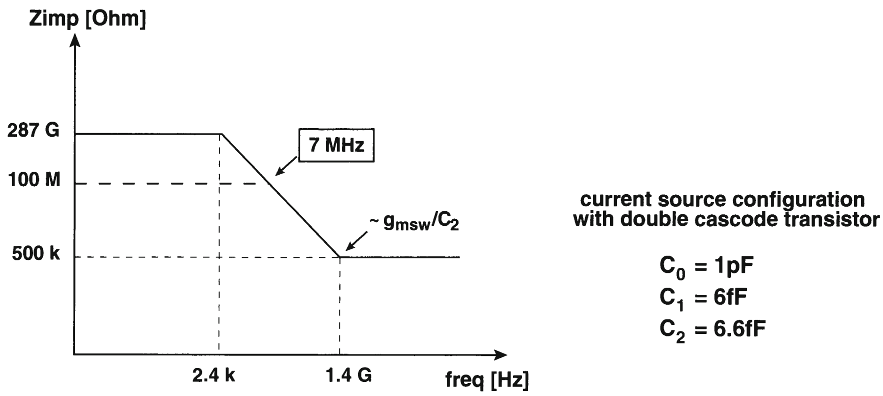


Figure 5.9: *Bode diagrams for the numerical example in section 5.4.3.3*

5.4.3.3 A numerical example

This example is a continuation of the example given in the previous sections (sections 5.4.1.3 and 5.4.2.3). For reasons of clarity, the values for the different parasitic capacitances are repeated.

$$C_0 = 6\,fF$$
$$C_1 = 6\,fF$$
$$C_2 = 6.6\,fF$$

The value for the transconductance and the output conductance of the second cascode transistor are chosen identical to the ones of the first cascode. The value of the interconnect capacitance is chosen to be 1 pF (C_0=1 pF). Substituting the numerical values in eq.(5.33) and eq.(5.34) gives the following result for the impedance Z_{imp}:

$$Z_{imp} = \frac{4.4 * 10^{-25} + 3.1 * 10^{-29} * s + 6.1 * 10^{-37} * s^2 + 3.3 * 10^{-47} * s^3}{1.1 * 10^{-13} + 2.6 * 10^{-21} * s + 4.5 * 10^{-31} * s^2 + 1.7 * 10^{-41} * s^3} \tag{5.47}$$

The exact values of the poles and zeroes are :

$$\begin{aligned} p_1 &= 2263\,Hz \\ p_2 &= 6.7\,MHz \\ p_3 &= 2.9\,GHz \\ z_1 &= 6.7\,MHz \\ z_2 &= 1.4\,GHz \\ z_3 &= 2.9\,GHz \end{aligned}$$

Poles 2 and 3 coincide with zeroes 1 and 3. A first order frequency behaviour is seen. The Bode diagram is given in fig.5.9. From this figure, it can be concluded that the non-linearity introduced by the output impedance limits the output signal bandwidth to a value of approximately 7 MHz which is a significant improvement compared to the topology with only one cascode transistor. The values of the poles and zeroes are equal to the ones calculated using the theoretical formulas.

If this result is compared to the Bode diagram presented in fig.5.5.b, it can be seen that the double cascoded configuration performs better than the cascoded one if the middle plateau has an impedance level that is lower than the required impedance. Otherwise, the two configurations would result in the same signal frequency bandwidth. In order to have 3 pole-zero pairs, the value of $C_0 g_{0cas1} g_{0cas2}$ has to be larger than the value of $C_2 g_{mcas1} g_{mcas2}$. This means that the interconnection capacitance has to be a factor of $g_{mcas1} g_{0cas1} g_{mcas2} g_{0cas2} (\approx 600)$ larger than the capacitance C_2 seen at the sources of the switch transistors. For a parasitic capacitance C_2 of a few femtoFarad, the capacitance C_0 has to have a value of at least a few picoFarad. This is almost never the case.

It should also be noted that for deep submicron technologies ($0.25\mu m$ and below), the implementation of a double cascode on top of the current source transistor is limited by the low power supply voltage. In this case alternative solutions have to be found to improve the negative impact of the frequency dependent output impedance of the D/A converter.

5.5 Conclusion

In this chapter, the SFDR-bandwidth limitations of high resolution D/A converters have been analysed. A main fundamental limitation is the dynamic output impedance of the circuit. The impact of this output impedance on the SFDR has been calculated. Based on this analysis the requirements for the value of the output impedance of each unit current branch has been derived. The frequency domain behaviour of the circuit's output impedance has been analysed for different current cell topologies and the effect on the SFDR has been illustrated with practical examples.

Chapter 6

A Design Methodology for High Performance CMOS Current Steering D/A Converters

6.1 Introduction

While in chapter 4 and 5, the emphasis has been put on modelling the static and the dynamic behaviour of the current steering D/A converter architecture, this chapter will cover the design flow for these circuits. In the first section of this chapter, the problem of determining the level of segmentation is addressed. In the remainder of the text, it has been assumed that a segmentation level of 0% matches a fully binary implementation and a segmentation level of 100 % represents a fully unary architecture. Both an area based as a mathematically based approach to determine the segmentation degree are discussed. In the following two sections, the choice of the thermometer decoder and the switch driver are addressed. The remaining sections of this chapter describe in detail the dimensioning of all the transistors in the unit current cell (the switch, current source and cascode transistors).

6.2 Determining the level of segmentation in a current steering D/A converter

6.2.1 The area approach

This approach is based on the silicon area consumption of the current steering D/A converter [Lin JSSC98]. The method targets a minimal chip area that still guarantees the required static resolution and has an optimal frequency domain performance. It consists of the following design flow.

- Determine the area necessary to achieve the INL specification.
  As has been already mentioned in chapter 3, the INL specification of a current steering D/A converter is independent on its architecture and is as such independent of the segmentation level.

- Determine the area necessary to achieve the DNL specification.
  Unlike the INL specification, the DNL error is dependent on the segmentation level of the D/A converter architecture. If the minimum analog area for a fully unary implementation equals A_{unit}, then the analog area of a fully binary architecture equals $2^N * A_{unit}$ based on the DNL performance only.

- Determine the area necessary to implement the thermometer decoder.
  It is evident that this area increases with the segmentation level of the D/A converter. The value of the digital area relative to the analog area is dependent on the circuit implementation and the used technology.

In fig.6.1, the impact of the different area constraints is depicted for a 10 bit implementation. The thick line indicates the area that is required to obtain a maximum DNL error of 0.5 LSB and a maximum INL error of 1 LSB. For an implementation with a minimum chip area, different segmentation levels (the flat part of the curve) are possible as shown in the figure. However, shifting the segmentation level to a higher value has a positive effect on the glitch and thus on the dynamic performance of the D/A converter. As a consequence, the optimal point is situated on the right hand side of the flat part of the curve. Although this method provides the designer with a tool to determine the optimal segmentation degree, it is based on an inaccurate starting point [Lin JSSC98]. In order to determine the area for implementing a certain INL and DNL specification, several Monte Carlo simulations have been performed. The expected value for both the INL and DNL error are then calculated by taking the root mean square (RMS) of the simulation results for every input code instead of first determining the INL/DNL error for every simulation and then extracting the expected values. As a consequence, the relative position of the minimum required analog area for the INL and DNL specification differ from the one presented in fig.6.1.

6.2.2 The mathematical approach

This approach is based on the fact that for a current steering D/A converter with differential switches (the current source is always turned on), a decoupling between the resolution and the update rate specification is possible. The resolution of the D/A

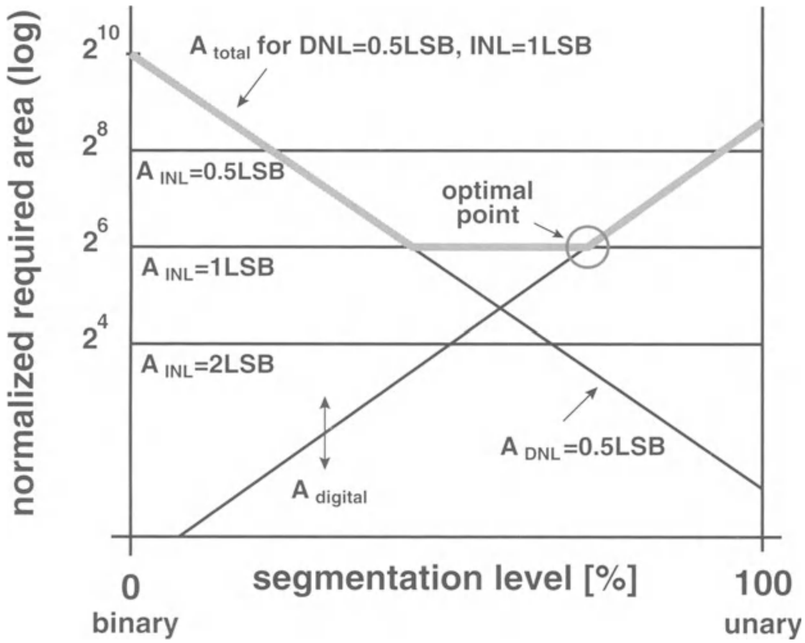


Figure 6.1: *Normalised required area versus the percentage of segmentation for a 10 bit D/A converter [Lin JSSC98]*

converter is determined by the mismatch behaviour of the current sources while the update rate is dictated by the switch transistors. In a first step, the dependency of the DNL error on the segmentation level is analysed. A first analysis that can be made to get insight in the expected inaccuracy at a certain transition is to determine which transition will be responsible for the maximum of the DNL profile. For the unary implementation, each transition has an equal probability of determining the DNL error. However, for the binary implementation, the probabilities for each of the transitions to determine the DNL error is different and has been simulated leading to the results depicted in fig.6.2 for a 10 bit implementation. This figure clearly indicates that in only 49% of the cases the MSB transition will determine the DNL error, while one intuitively would focus on this transition.

At this point, a mathematical derivation of the DNL error for the unary, the binary and the segmented architecture is given. The calculation is based on the fact that the more current sources switch at the same moment, the larger the variation on the transition ($\sigma_n = sqrt(n) * \sigma$) will be. For a fully unary implementation, there are $2^N - 1$ transitions where each transition is modelled as a normal distribution with $\sigma = \sigma_{unity}$. For a fully binary implementation, we only have N different transitions each modelled as a normal distribution with $\sigma_i = sqrt(2^i - 1) * \sigma_{unity}$ where i is the bit number. To determine the DNL error, a sample has to be taken from each distribution. For the unary implementation, this results in a set of $2^N - 1$ values while for the binary implementation this results in a set of N values. The DNL error is then determined by the maximum value of this set of samples. Based on basic statistical operations, theorems and functions, the DNL error versus the segmentation level can be calculated.

To derive a closed formula for the expected DNL error for the fully unary architecture [Borre PhD], we start from the assumption that each current source can be modelled by a normal gaussian distribution with a mean value equal to 1 LSB and a standard deviation σ. For a N-bit resolution D/A converter, the $2^N - 1$ different current sources are assumed to be uncorrelated. The definition of the DNL error can be translated in the following experiment.

Experiment: Take $2^N - 1$ samples from a normal distribution with standard deviation σ and mean 1 LSB. The largest sample in absolute value determines the DNL error.

The question that has to be solved is to find the number of σ where the expected value of the DNL error will be located. This can be solved by using the following statistical theorem :

Theorem: If P is the possibility that an event happens, the average number-of-experiments that has to be done to make the event happen is 1/P.

We define T as the possibility that a sample from the normal distribution is larger than $z.\sigma$ with z the number of sigma determining the DNL limit. Note that this z is the parameter to be determined in this analysis. Based on the theorem, we know that $1/T$ experiments are required to have one sample that is larger than $z.\sigma$. Actually the number of experiments is known and given by: $2^N - 1$.

$$\frac{1}{T} = 2^N - 1 \tag{6.1}$$

This equation can be rewritten as:

$$T = \frac{1}{2^N - 1} = 1 - \int_{LSB-z\sigma}^{LSB+z\sigma} p(\Delta I) d\Delta I \tag{6.2}$$

Based on statistical tables or on the inverse cumulative density function, the value of z can be determined. Multiplying this value of z with σ gives the expected value for the DNL error.

$$DNL_{exp} = z\sigma = inv\_norm_{(-x,x)}(1 - \frac{1}{2^N - 1}) \tag{6.3}$$

For a D/A converter with a 10 bit resolution, the expected DNL error equals approximately 3.3σ.

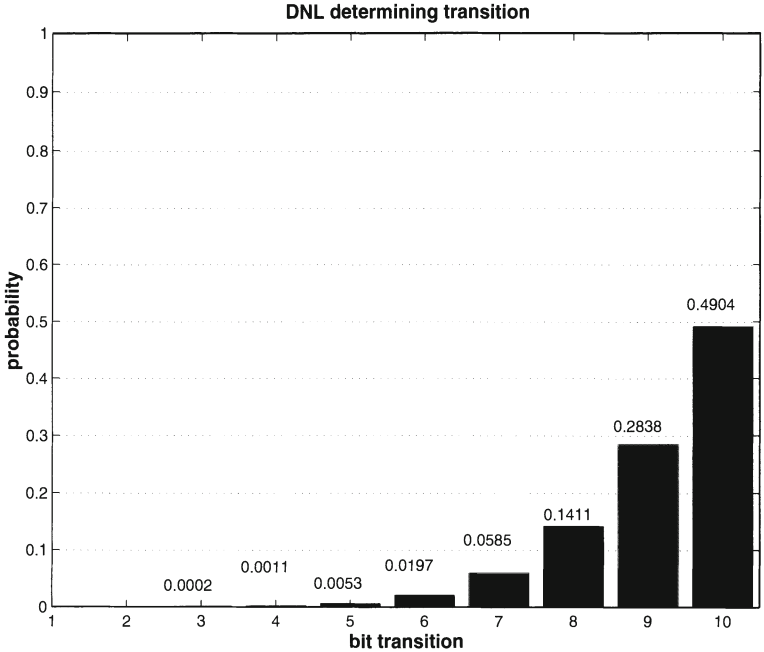


Figure 6.2: *Histogram of the probability that the DNL error is caused by transition i as a function of the transition for a 10 bit D/A converter*

For the fully binary architecture the expected value for the DNL error is given by:

$$DNL_{exp} \approx \sqrt{2^N - 1}\,\sigma \tag{6.4}$$

Fig.6.3 shows the expected DNL values as a function of the resolution for a binary and a unary architecture. The DNL values are expressed as a function of the unity current source accuracy σ_{unity}. This figure gives the results of both the Monte Carlo simulations (lines) as of the derived formulas in eq.(6.3) and eq.(6.4) (discrete points). As can be seen from this figure, a good agreement exists between the formulas and the simulations. Further analysis of this figure reveals that the expected DNL error for a unary implementation is only a few times the unity standard deviation, while for the binary implementation the expected DNL error strongly increases with the total number of bits. This is the reason why designers traditionally do not choose a binary implementation. However, based on the INL-yield considerations, one knows that the unity sigma decreases with an increasing number of bits since a higher resolution inherently requires a better accuracy. Therefore, fig.6.4 shows the expected DNL value expressed in LSB. From this figure, it can be concluded that the expected DNL performance for a binary architecture is relatively independent of the number of bits and is always better than the required 1/2 LSB accuracy. The expected DNL error of a unary implementation is even lower and improves with increasing resolution.

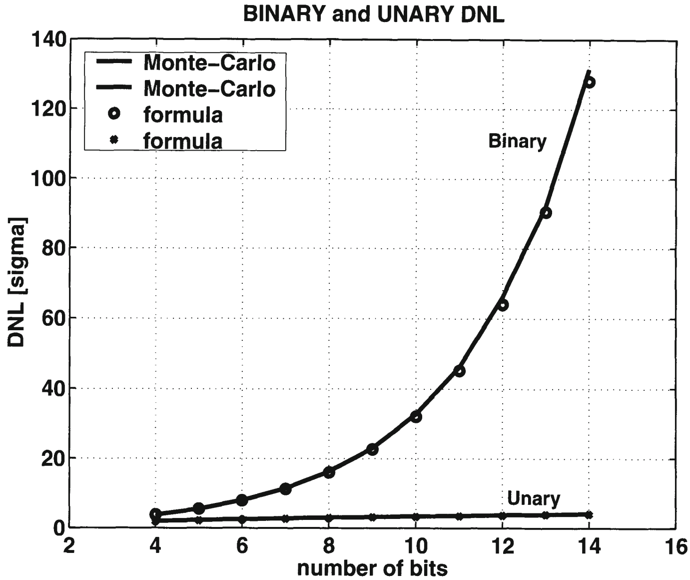


Figure 6.3: *The expected value of the DNL error expressed in sigma as a function of the resolution of the DAC for a fully binary and a fully unary architecture*

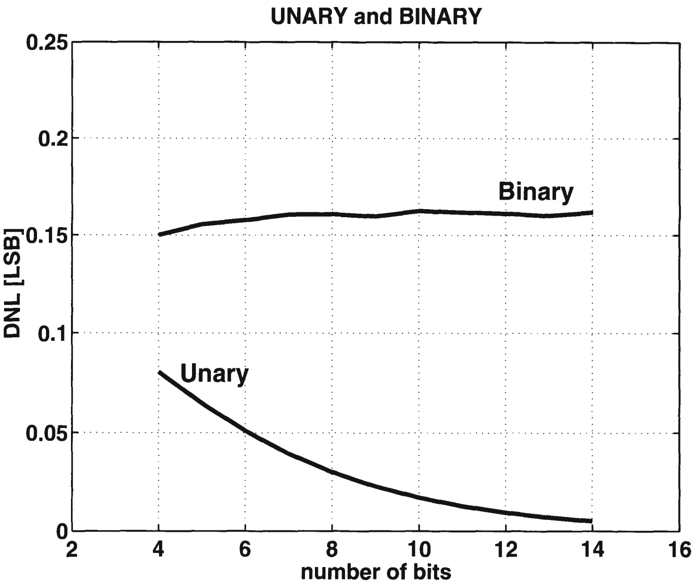


Figure 6.4: *The expected value of the DNL error expressed in LSB as a function of the resolution of the DAC for a fully binary and a fully unary architecture*

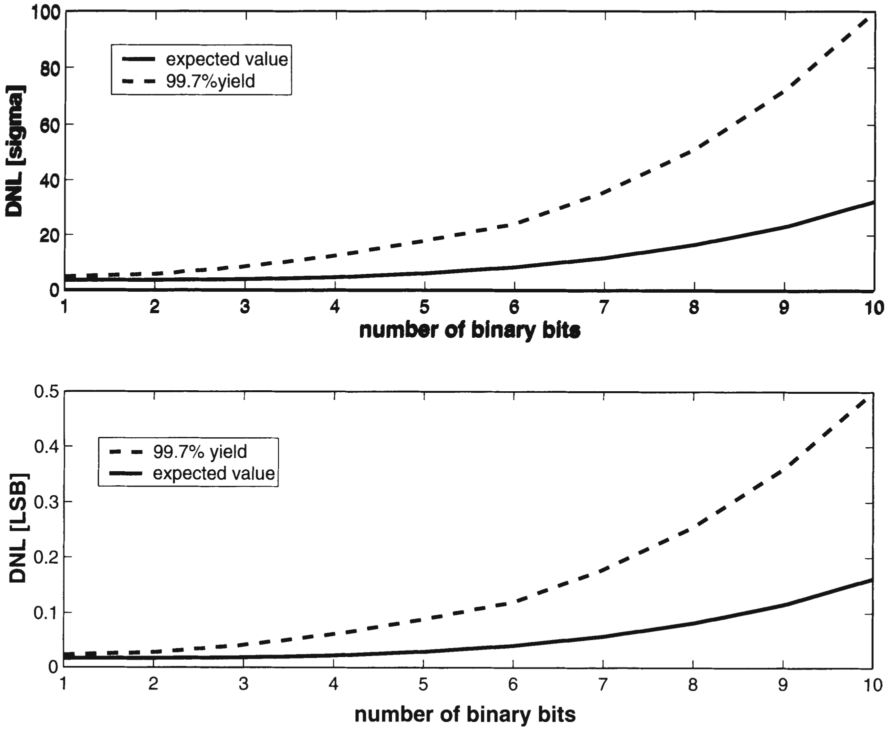


Figure 6.5: *Statistical analysis of the DNL yield*

Finally, also the yield for this DNL specification has been simulated. The results are presented in fig.6.5. Fig.6.5.a shows the expected value for the DNL error for a 10 bit D/A converter (expressed as a function of the unit current source standard deviation σ_{unity}) and the 99.7% yield limit as a function of the segmentation level (1=fully unary; 10=fully binary). It shows the strong DNL error increase when shifting towards more binary bits. Based on the INL yield versus accuracy relationship (section 4.2.4), fig.6.5.b has been calculated. It shows both the expected value of the DNL error and its 3σ limit, expressed in LSB, as a function of the segmentation. It indicates that the 1/2 LSB DNL specification is inherently achieved even for a fully binary implementation.

To complete this analysis, it can be concluded that if the unit current source of the D/A chip has been designed to achieve the INL yield specification, the DNL specification will automatically be achieved within the same yield requirement even in the extreme case of a fully binary topology.

6.2.3 Conclusion

From the mathematical point of view, the binary implementation seems to be the most advantageous one, especially since it also has a low power consumption. However, the dynamic performance of the D/A converter is heavily dependent on the segmentation level. The smallest glitches at the output and thus the best performance is achieved for the fully unary architecture. Therefore, a certain degree of segmentation is called for. For each bit that is added at the binary sub-D/A converter, the distortion due to the non ideal switching errors will increase by a factor of two. It can be concluded that in fact the level of segmentation is determined by the dynamic limitations. The optimal segmentation point has to be found through simulations where a trade-off has to be made between the area and power consumption and the SFDR specification.

6.3 Architectural choice of the thermometer decoder

In time, different architectures to implement the thermometer decoder have been presented. In this section, a short overview is given of the three most important ones.

- the row and column decoder
  The thermometer decoder consists of both a row decoder and a column decoder [Nakam JSSC91, Miki JSSC86]. The combination of the control signals of these two decoders determines if the current source is switched to the output or not (as has been explained in section 4.3.2.3). Although this decoder allows a simple implementation with a low power consumption, it inherently lacks the flexibility to realise optimal switching schemes.

- the VHDL decoder
  In order to implement any given switching scheme, the current sources are placed in a different array from the switches and their drivers [VdBos CICC98]. For high resolution D/A converters, the row and column decoder can then be integrated in one block using a VHDL implementation. Although the update rate of the circuit is determined by the performance of the available standard cell library, the major advantage is the high level of automation that can be obtained using this approach [VdPla JSSC99].

- the custom made decoder
  For high speed D/A converter architectures, a custom made thermometer decoder is used. This decoder allows an optimal exploitation of the symmetry by a detailed analysis of the required logic expressions. This results in a number of custom made "standard cells" that can be optimised towards a high update

rate. Furthermore, this approach gives the designer the opportunity to check the timing constraints on every point within the decoder leading to an improved dynamic behaviour of the D/A converter. An example of this type of decoder is given in chapter 7. The major drawback of this method is the increased design effort that is involved.

6.4 Design of the synchronised switch driver

To overcome the dynamic problems described in section 5.2, a synchronised driver has to be placed immediately in front of the switches. Depending on the update rate of the D/A converter different implementations can be used. Some of these drivers will be discussed in more detail in chapter 7, where several realisations with different resolutions and different update rates will be presented.

6.5 Dimensioning the unit current cell

In this section, the design constraints will be discussed that determine the dimensions of the current source, the switch and the cascode transistor.

6.5.1 The current source transistor

6.5.1.1 The area constraint

The dimensions of the current source transistor are dependent on the full scale output current of the D/A converter and the technology in which the chip will be implemented. According to the mismatch equations [Pelgr JSSC89, Laksh JSSC86], the gate area of the current source is determined by :

$$WL = \frac{1}{2\,(\sigma_I/I)^2}[A_\beta^2 + \frac{4\,A_{VT}^2}{(V_{GS}-V_T)^2}] \tag{6.5}$$

The values for the mismatch parameters A_β and A_{VT} are provided by the foundry. The relative unit current standard deviation is dependent on the resolution and the desired yield of the D/A converter (cfr. section 4.2.4) and can be easily calculated using eq.(4.11). As an example, the unit current source area is plotted versus the yield of the D/A converter with a 8 bit, 10 bit and 12 bit resolution (fig.6.6) for a gate overdrive voltage of 1V and for an implementation in a standard CMOS $0.35\mu m$

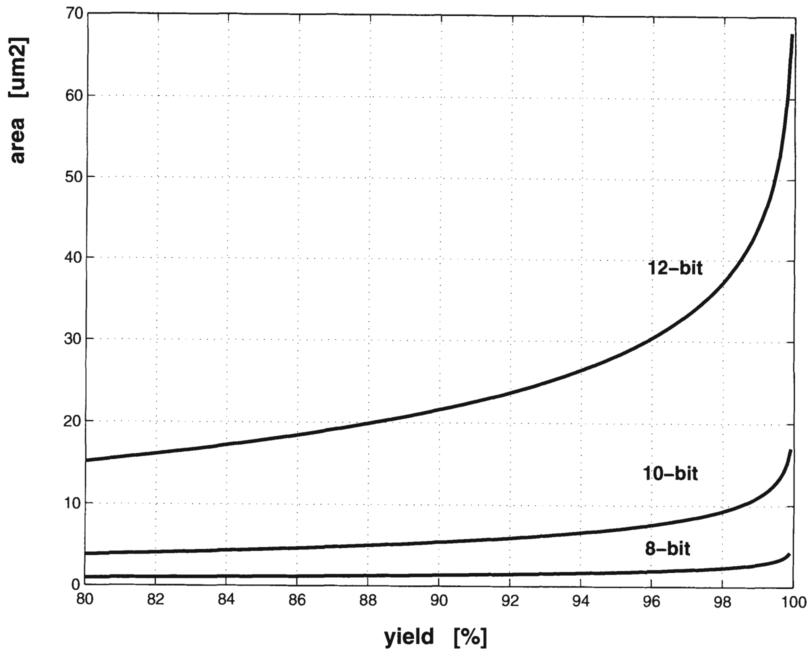


Figure 6.6: *The unit current source area as a function of the number of functional D/A converters for a 8 bit, a 10 bit and a 12 bit resolution*

technology. This figure clearly indicates that the unit current source area undergoes a large increase for yields above 95 %. Eq.(6.5) clearly indicates that the unit current source area is dependent on the used technology through the technological mismatch parameters A_β and A_{VT}. The A_{VT} parameter scales linearly down with technology [Laksh JSSC86] while the A_β parameter remains approximately the same [Laksh JSSC86, Bult ESSCIRC00]. For a certain yield and a given resolution of the D/A converter, it can then be stated that the unit current source area is proportional to the $\frac{A_{VT}^2}{(V_{GS}-V_T)^2}$ ratio. Since the supply voltage also scales down with technology and a certain output swing is required for the D/A converter, the gate overdrive voltage $V_{GS} - V_T$ will also scale down in approximately the same way as the mismatch parameter A_{VT} does. It can be concluded that the required area of the unit current source remains about the same. In fig.6.7, the unit current source area as a function of the used technology is given. From this figure, it can be clearly seen that going to a smaller technology with a scaled supply voltage will give no area advantages. When the supply voltage doesn't scale down, the improved matching behaviour of the transistors can have a beneficial effect. However, for future deep submicron technologies like the 0.12 μm, 0.10 μm and the 0.07 μm technology the supply voltage scales down and no decrease of the area of the current source array can be realised in comparison to 0.18 μm or 0.25 μm current-steering D/A converter designs.

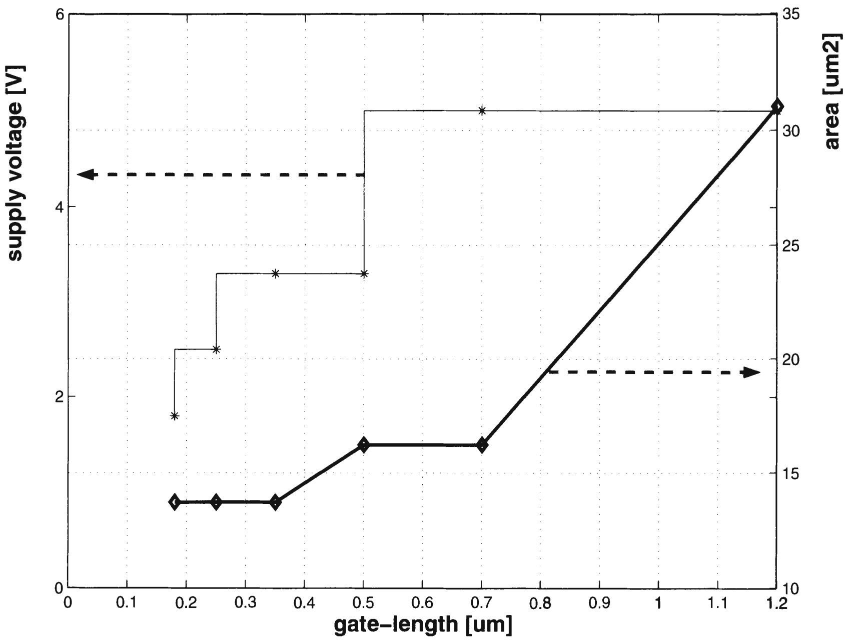


Figure 6.7: *The unit current source area as a function of the used technology (min. gate-length) for a 10-bit D/A converter*

6.5.1.2 The output voltage swing

A second constraint is given by the value of the full scale current I_{FS} of the D/A converter (or the wanted output voltage swing over a given resistor).

$$\frac{W}{L} = \frac{2\, I_{FS}}{\mu\, C_{ox}\, K\, P\, (2^N - 1)\, (V_{GS} - V_T)^2} \tag{6.6}$$

Combining eq.(6.5) and eq.(6.6) results in a value for the gate-length and the gate-width of the current source transistor.

The value for the gate overdrive voltage is determined as a trade-off between the area of the current source transistor (large $V_{GS} - V_T$) and the limit imposed by the fact that all the transistors (switch, cascode and current source) have to operate in the saturation region (low $V_{GS} - V_T$) for a given supply voltage.

6.5.2 The switch and cascode transistor

While the dimensions of the current source transistors are determined by mismatch considerations, the dimensions of the switch and cascode transistors are mainly determined by the update rate specification. For this reason, the gate-length of the switch transistor is chosen to be minimal. Since the current through the switch transistor equals the current through the current sources, the gate-width can be easily calculated. The main objective in dimensioning the switch transistors is in keeping both the gate-length and the gate-width as small as possible. In this way the parasitic capacitance at the output node is reduced which is beneficial for the settling time of the D/A converter. Furthermore, the gate-drain capacitance is minimised which in its turn reduces the digital feedthrough of the switch control signals to the output and as such has a positive impact on the dynamic behaviour of the D/A converter.

The dimensions of the cascode transistor are determined by the fact that the same current flows through both the current sources and the switches and by the dynamic performance constraint discussed in chapter 5. The gate-length of the cascode transistor is in most cases larger than the minimal value since the output resistance of the cascode transistor is directly proportional to this transistor parameter. In this way, a large output impedance (seen in the drain of the switch transistors) can be realised in order to overcome the dynamic limitations. The main objective in dimensioning the cascode transistor is finding a trade-off between the required output impedance and a small parasitic drain capacitance.

6.6 Conclusion

In this chapter, the design of the different sub-blocks of a current steering D/A converter has been discussed (the thermometer decoder, the switch driver and the unit current cell). Furthermore, a mathematical approach has been derived to determine the level of segmentation for a high resolution D/A converter. From this calculation, it can be concluded that this level is not determined by the static performance but is mainly determined by the dynamic performance of the driver.

Chapter 7

Realisations

7.1 Introduction

In this chapter, the design methodology discussed in chapter 6 has been put into use through the implementation of six CMOS current steering D/A converters. These realisations can be divided into four classes.

In the first section of this chapter, the emphasis is put on attaining a highly accurate D/A converter. The design and measurement results of a 12 bit and 14 bit D/A converter will be discussed. However, these devices have a poor frequency domain behaviour. In the second section of this chapter, a high speed circuit has been designed. For this purpose, a low resolution (6 bit) D/A converter has been chosen as to keep the complexity of the design (thermometer decoder, ...) within boundaries. In the third section, a combination of a high resolution and a high update rate has been realised in a 10 bit 1 GS/s and a 12 bit 500 MS/s current steering D/A converter. Measurements of both chips show a frequency domain behaviour that is better than all recently published D/A converters [Lin JSSC98, Vital AACD01, Bugej JSSC99, Khano AACD01]. The fourth section describes the design of a 10 bit low power D/A converter for telecommunication applications. This is the first chip (to the author's knowledge) that combines a good dynamic performance and a low power consumption.

In the fifth section, an overview is given of the most important static and dynamic specifications of the six presented realisations. In order to make a comparison possible, an objective figure of merit has been introduced in the last section of this chapter. The expression for this figure of merit is based on the resolution, the dynamic behaviour, the area and the power consumption of the D/A converters under comparison. This figure is then compared to other published D/A converter designs.

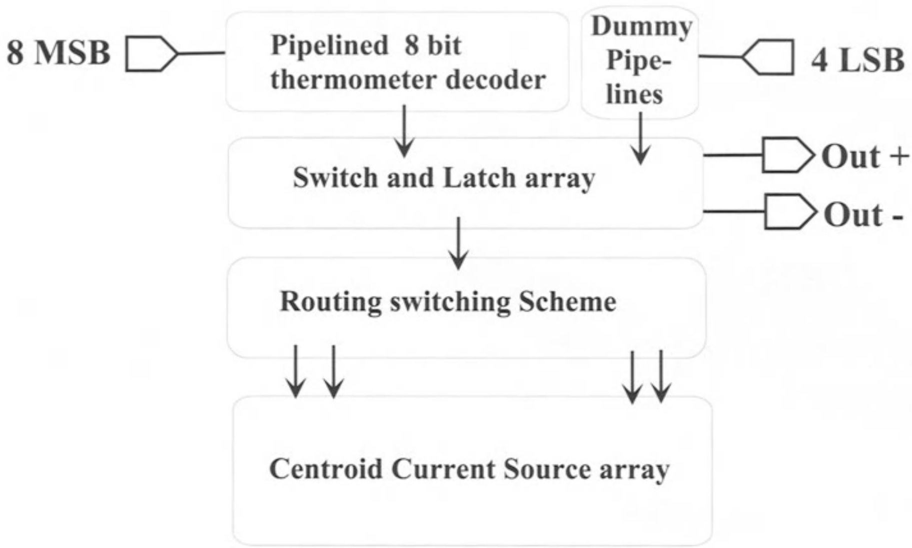


Figure 7.1: *The floorplan of the 12 bit D/A converter*

7.2 High Accuracy D/A Converters

7.2.1 First Design of a 12 bit D/A Converter

7.2.1.1 The Floorplan

The first D/A converter design that is discussed in detail is a 12 bit segmented current steering D/A converter. Fig.7.1 gives a schematic representation of the realised chip. As has become clear in chapter 6, the DNL error does not determine the number of binary implemented bits but the dynamic performance is the limiting factor. From simulations, the segmentation level was determined. As a result, the D/A converter has been divided into two sub-DACs, where the four LSBs were implemented in the binary and the 8 MSB's were implemented in the unary part. The output of the current sources drives a 50Ω double terminated load. Due to the complexity of the decoding logic, the conventional row and column decoder could no longer be used to generate the thermometer code and was replaced by a pipelined full custom decoder synthesised in a standard 0.5 μm CMOS logic. This D/A converter was designed to have a 0.5 V single ended output swing implying a 20 mA full scale drain current specification.

A_{VT}	$12\, mV\mu m$
A_{β}	$1.9\ \%\mu m$
$\sigma(I)/I$	0.25 %
$(V_{GS} - V_T)_{cs}$	1.4 V
I_{FS}	20 mA
segmentation	8-4
$(W/L)_{cs}$	$1.5\mu m/35.5\mu m$
$(W/L)_{sw}$	$0.8\mu m/1\mu m$
technology	$0.5\mu m$

Table 7.1: *The unit current cell specifications for the 12 bit DAC*

7.2.1.2 Design of the swatch cell

Fig. 7.2 shows the swatch cell consisting of the unit current cell and the latch (synchronised switch driver). To determine the dimensions of the transistors M_{CS},M_{swa} and M_{swb} of the unit current cell, the design procedure described in chapter 6 has been followed. For an INL related yield specification of 99.7 % within a 0.5 LSB INL error, the dimensions of the unit current source transistor have been calculated using the following parameters:

$$\begin{aligned} A_{\beta} &= 1.9\,\%\mu m \\ A_{VT} &= 12\, mV\mu m \\ (V_{GS} - V_T)_{CS} &= 1.4\, V \\ I_{FS} &= 20\, mA \end{aligned}$$

From these parameters, the following two constraints can be derived.

$$WL = 53\,\mu m^2 \ and\ \frac{W}{L} = 0.042$$

The resulting gate-width of the current source transistor equals 1.5 μm and the gate-length equals 35.5 μm.

The dimensions of the switch transistors have been chosen to be as small as possible while taking the current constraint into consideration.

$$(V_{GS} - V_T)_{sw} = 0.3\, V \ and\ \frac{W}{L} = 0.8$$

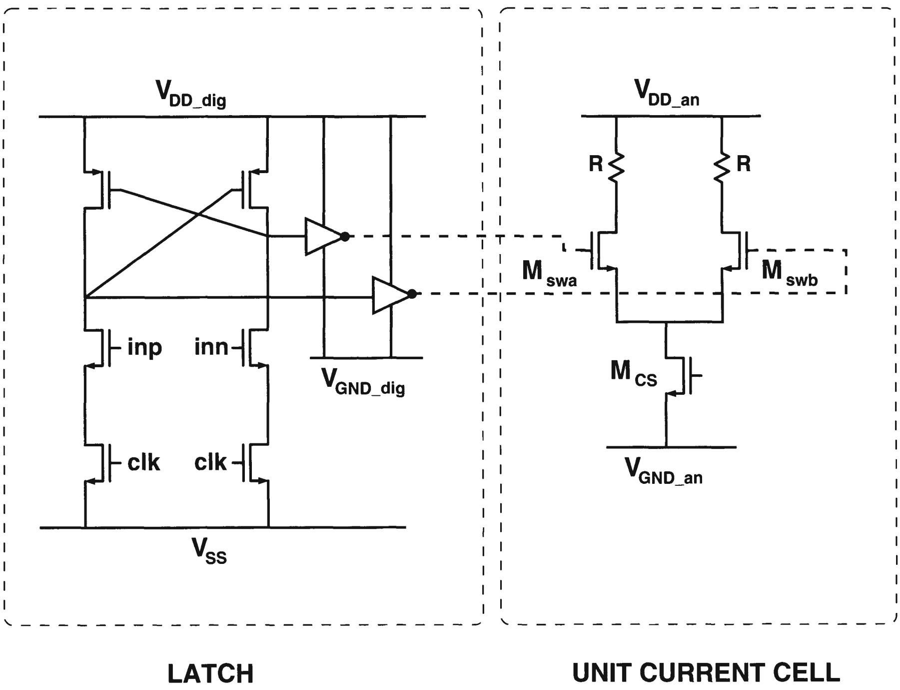


Figure 7.2: *The latch and the current cell*

For a minimal value of the gate-width (W=0.8 μm) of the switch transistors, the gate-length equals 1 μm. As a consequence, the switch transistor has a small parasitic drain capacitance which has a beneficial effect on the settling time behaviour of the D/A converter. An overview of the unit current cell dimensions is given in table 7.1.

The synchronisation of the input signals of the current switch transistors is guaranteed by placing a latch in front of the current switches, as indicated in fig.7.2. Apart from synchronising all signals, this latch also reduces the output current variation due to the drain voltage variation of the current sources. In case of conventional switch drivers, both switching transistors will be switched off simultaneously for a short period. This results in a discharge of the capacitance on the drain node of the current source transistor, which deteriorates the dynamic performance of the D/A converter. In literature, several driving schemes are documented to solve this problem, all of them requiring additional circuitry [Nakam JSSC91, Kohno CICC95, Wu JSSC95, Basti JSSC91]. In this circuit, the intrinsic delay between the complementary outputs is used to lower the crossing point. The value of this point can be further adjusted (if necessary) by changing the V_{SS} voltage of the latch as is indicated in fig.7.3. The crossing point for a V_{SS} voltage of 0 V, 0.2 V and 0.4 V is given on this figure. In this case, a higher V_{SS} leads to a lower crossing point due to the inverters that are placed between the latch and the switch transistors. Since these inverters switch be-

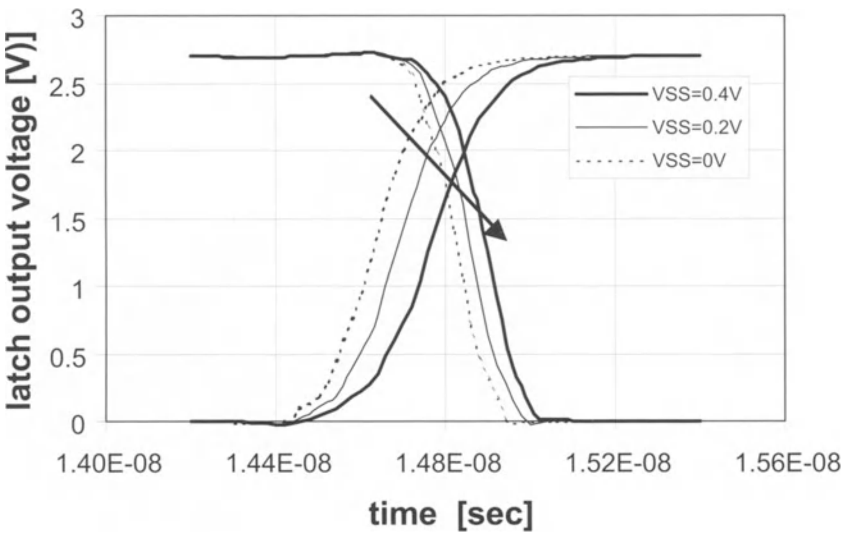


Figure 7.3: *The output voltage of the latch for different values (0 V, 0.2 V, 0.4 V) of the V_{SS} voltage*

tween $V_{GND-dig}$ and the power supply V_{DD-dig}, the control signals at the input of the switches have an amplitude of $V_{DD-dig} - V_{GND-dig}$. In fig.7.3, $V_{GND-dig}$ equals 0 V. To minimise the feedthrough through the C_{GD}, the value of $V_{GND-dig}$ can be adjusted to reduce the voltage swing. Furthermore, the inverters act as a buffer to minimise the clock feedthrough to the output of the D/A converter. Proper sizing of the latch has resulted in a high voltage crossing point at the input of the switches in order to prevent a simultaneous off-state of the switches.

7.2.1.3 Design of the thermometer decoder

In the classical row-column decoder, a complete row of cells has to be turned on before switching to the following row, which results in an accumulation of systematic and graded errors [Basto CICC96, Lin JSSC98, Nakam JSSC91, Miki JSSC86]. Since the number of output lines of the thermometer decoder increases with an increasing number of bits, the complexity of the decoding logic increases drastically resulting in a large input capacitance that has to be buffered. A VHDL implementation based on lookup tables was developed using a standard cell library. The unary thermometer decoder was synthesised from this VHDL code using Synopsys. The placement and routing was done with the Cell Ensemble 3 tool within the Cadence Framework [VdPla JSSC99]. To complete the thermometer decoder, a dummy decoder has been

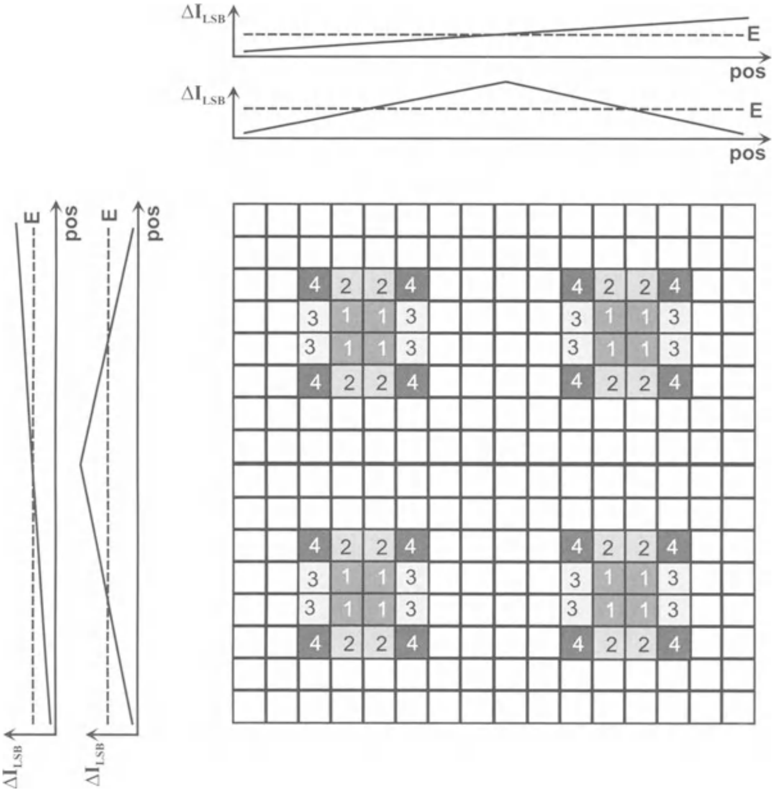


Figure 7.4: *The double centroid switching scheme*

made for the 4 LSBs as to prevent any latency problems that could otherwise occur between the unary and the binary part of the D/A converter. The maximum achievable update rate of the D/A converter is limited by the performance of the available standard cell library. Only 2 or 3 gates can be cascaded in a 5 nsec time frame which limits the update rate to 200 MS/s. A higher speed could be obtained if a faster digital cell library had been available.

7.2.1.4 The switching scheme

If the resolution of the D/A converter increases by a single bit, the number of current sources in the current source array doubles. The area occupied by a single unit current source also doubles because of the random matching constraint. This leads to a four-times area increase for the current source array for each additional bit. For D/A converters with a resolution of 10 bits and higher, the dimensions of the current source array become so large that process-, temperature- and electrical gradients have to be considered. The non-linearity errors introduced by these gradients can be (partially) compensated by the introduction of a special switching scheme as has been explained in section 4.3.

Since in this design, the four LSBs are implemented in a binary way, the value of the unary current source equals sixteen times the LSB current. This current source is divided into sixteen LSB current sources that are placed symmetrically around the center of each quadrant and are switched to the output at the same time as is indicated in fig.7.4. This switching scheme compensates for the errors introduced by linear and symmetrical gradients both in the horizontal or vertical direction. Furthermore, the accumulated error is minimised since the current sources in the four quadrants are simultaneously switched to the output.

To implement this switching scheme, the three available metal layers of the 0.5 μm technology have been used. The first metal connects the ground line of the current source transistors in the matrix. The second and the third metal have been used to implement the unary current source. Four vertical lines each connect four current sources in a selected column, while the horizontal line connects the four vertical lines connecting hereby the 16 unit current source transistors of a unary current source. As a consequence, the current source matrix has been completely covered with a fully symmetrical metal interconnection pattern that was identical for each current source transistor. In this way, a degradation of the mismatch behaviour caused by an asymmetrical metal coverage of these transistors has been avoided [Tuinh ICMTS97].

7.2.1.5 The layout

To obtain the 12 bit accuracy, several measures have been taken at layout level. The coupling between the analog and the digital part of the chip has been minimised by :

- using a separate array for the swatches and the current sources. This has the additional advantage of lowering the influence of the distance matching effect [Pelgr JSSC89],
- using different power supply lines for the analog and the digital block of the D/A converter,
- placing a guard ring around the sensitive analog current source array as to avoid any digital coupling through the substrate.

To reduce the voltage drop in the ground lines of the current source transistors, sufficiently wide supply lines have been used that are drawn on top of these transistors.

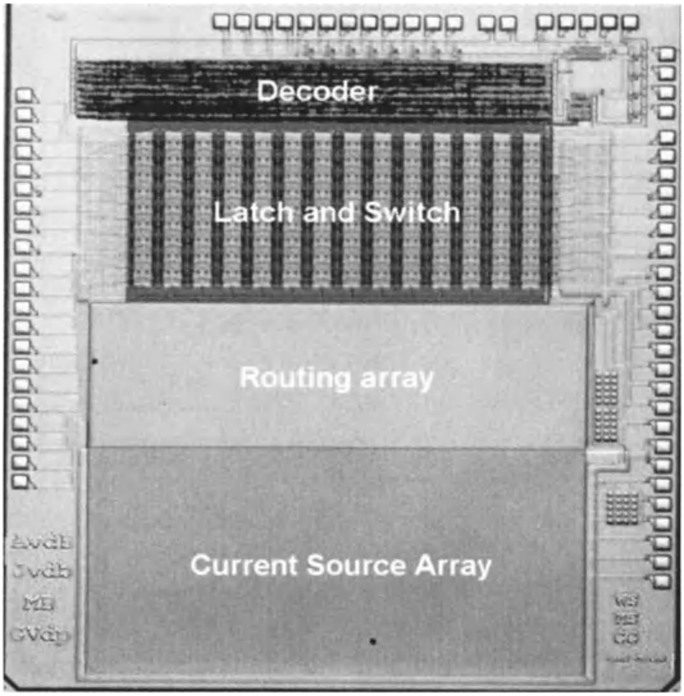


Figure 7.5: *The chip photograph of the 12 bit D/A converter*

To avoid any coupling from the clock and/or any other digital switching line to the output, a comb-like layout scheme has been used that is depicted in fig.7.6. From this figure, it can be seen that the swatch cells are placed together in groups of two that share the same clock line but have different lines to the output. In this way, a decoupling of the output from the clock has been realised.

The chip has been processed in a double poly standard 0.5 μm CMOS technology with 3 metal layers. The chip photograph of the 12 bit D/A converter is shown in fig.7.5. The decoding logic, the swatch array and the current source array can be clearly distinguished on this photograph. Between these two arrays, a considerable amount of area was used for routing the complex switching scheme. Since an accuracy of 12 bit was the main goal of this first design, a minimisation of the area was not targeted. However, the routing area can be completely eliminated by integrating the routing in the digital decoder. In this way an area reduction of 25 % can be achieved.

7.2.1.6 Measurement results

7.2.1.6.1 Static measurements The 12 bit D/A converter has been measured at a 2.7 V power supply for both the analog and the digital part of the chip. All measure-

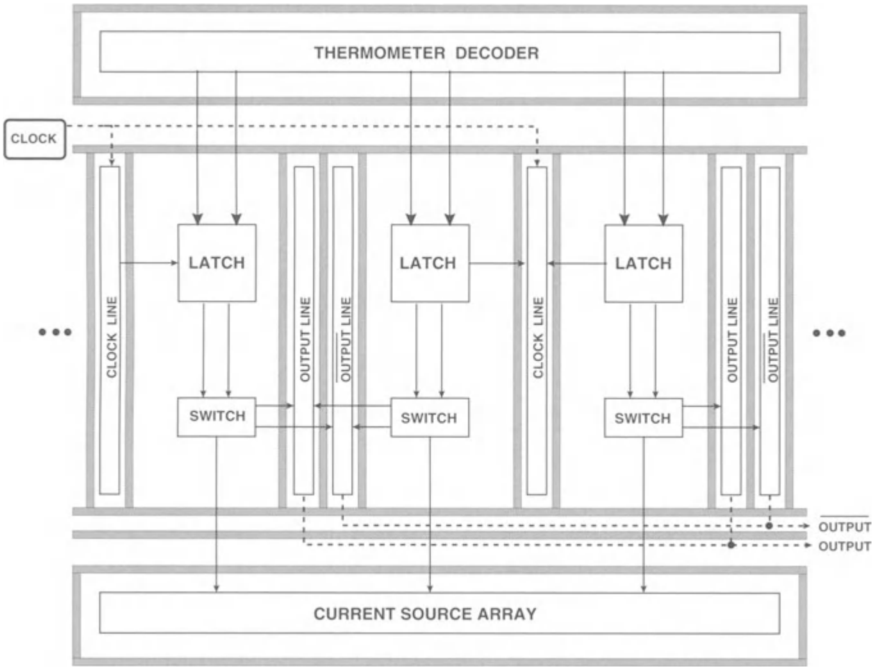
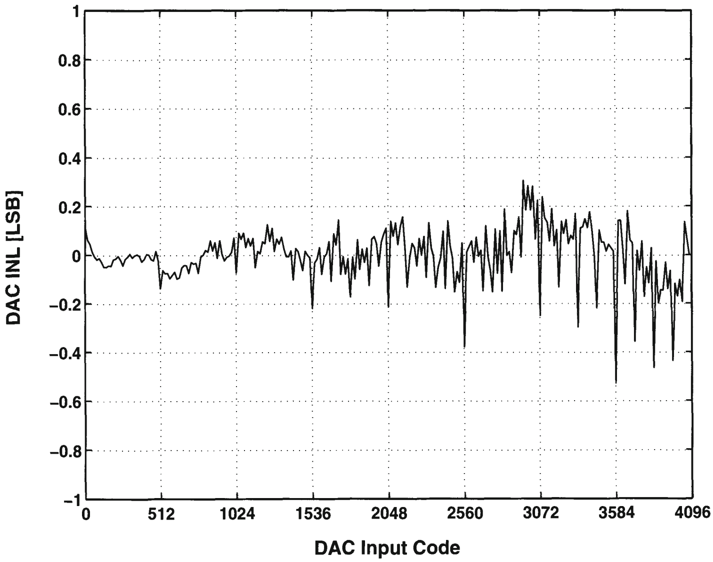


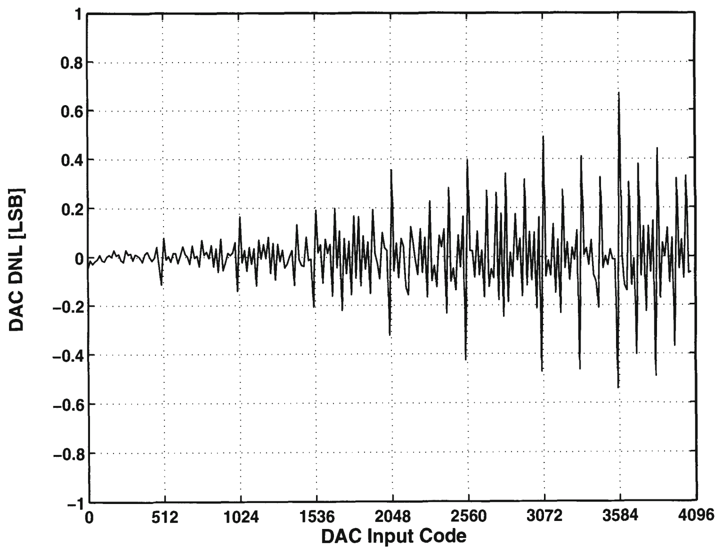
Figure 7.6: *A simplified scheme of the layout of the 12 bit D/A converter (the grey area represents the substrate straps)*

ments have been performed on a single-ended output for a full scale output current of 20 mA. To shield the D/A converter from any external noise coupling, the circuit has been mounted on a ceramic substrate that was encapsulated in a copper-beryllium case. Fig.7.7.a and fig.7.7.b show the measured integral non-linearity and differential non-linearity profile of the 12-bit D/A converter. The INL error is approximately 0.5 LSB, implying a monotone behaviour of the D/A converter. The DNL error equals 0.7 LSB.

7.2.1.6.2 Dynamic measurements In this paragraph, the measured dynamic behaviour of the D/A converter is discussed in detail. In fig.7.8.a the measured spurious free dynamic range (SFDR) of the D/A converter as a function of the input frequency is given for an update rate of 200 MHz. The SFDR remains above 70 dB up to an input frequency of 250 kHz. For larger frequencies the SFDR starts dropping and for a 2 MHz input signal its value equals 59 dB. On the same figure, the results of the model of the dynamic impedance are shown. As has been explained in chapter 5, the SFDR is limited due to the frequency dependency of the output impedance. The difference between the measurements and the model at low frequencies is mainly determined by the mismatch errors of the current sources. In fig.7.8.b, the frequency of the input sig-

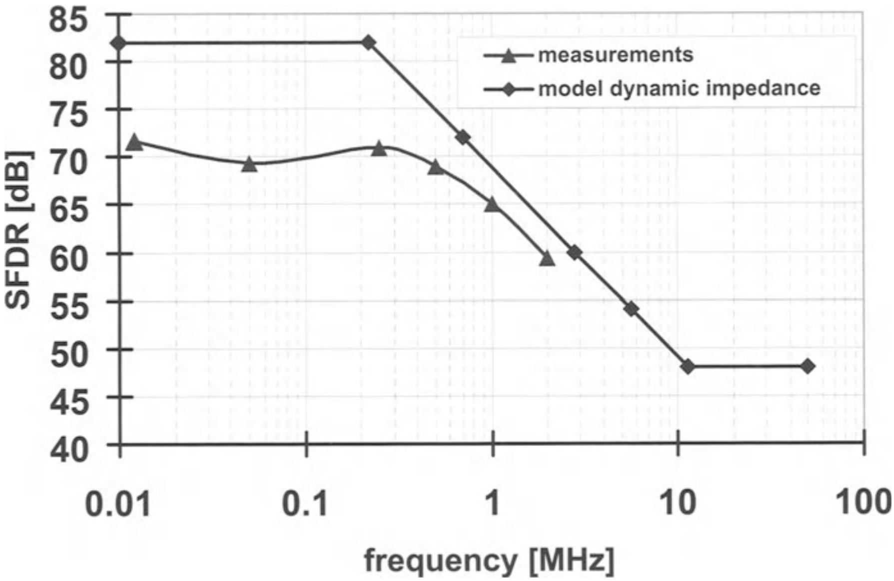


(a) The measured INL of the 12 bit D/A converter

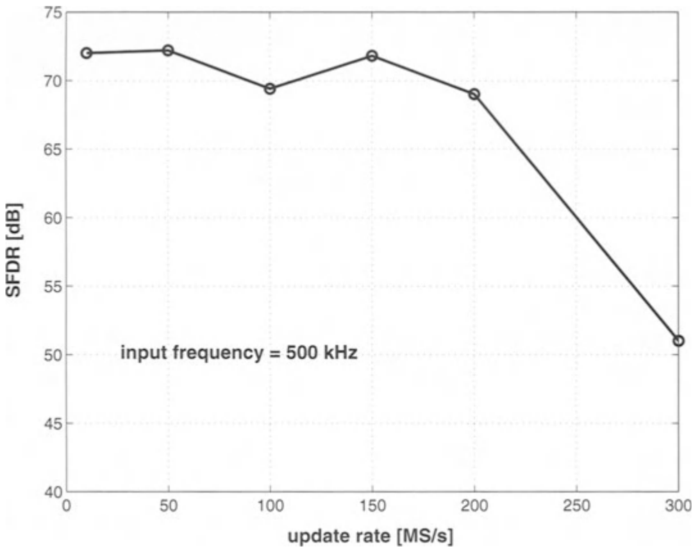


(b) The measured DNL of the 12 bit D/A converter

Figure 7.7: The static performance of the 12 bit D/A converter



(a) The SFDR versus the input frequency at an update rate of 200 MS/s



(b) The SFDR versus the update rate for a sinusoidal input signal frequency of 500 kHz

Figure 7.8: The dynamic performance of the 12 bit D/A converter

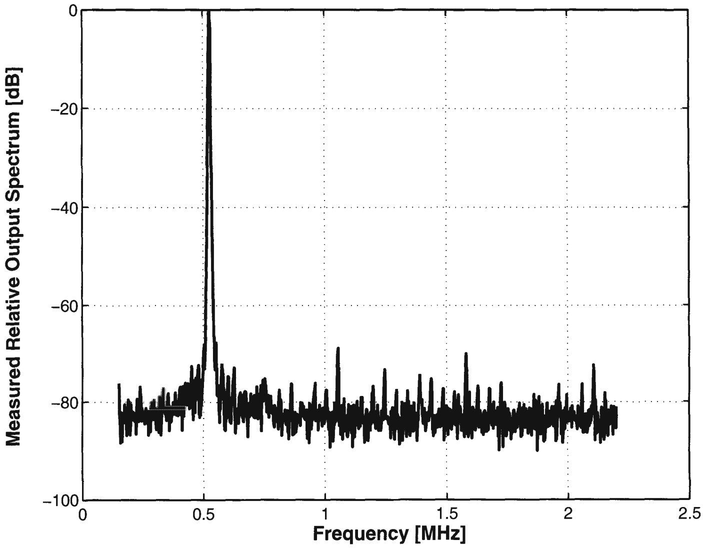


Figure 7.9: *The measured relative output spectrum @ approx.500kHz signal frequency and @ 200MHz update rate*

nal is kept constant at a value of 500 kHz while the update rate of the D/A converter is swept from 10 MHz to 300 MHz. The SFDR varies around the 70 dB level for update rates up to 200 MHz.

Fig.7.9 shows a typical output spectrum of the D/A converter for a 500 kHz full scale sinusoidal input signal clocked at a frequency of 200 MHz. The SFDR is dominated by the second order harmonic distortion component which lies 69 dB below the signal level. The worst case power consumption has been measured at a 200 MS/s update rate of the D/A converter and equals 140 mW.

7.2.1.7 Conclusion

The followed design procedure to determine the dimensions of the unit current cell has been verified by the static measurement results. The dynamic performance of this D/A converter is far from optimal what can be contributed to several causes. For example, the control signals of the switches don't have a reduced swing as to lower the capacitive feedthrough to the output of the D/A converter. Furthermore, no cascode transistors have been implemented to counteract the effect of the varying output impedance. However, the main goal of processing this chip was to investigate the static behaviour of high resolution current steering D/A converters. In the third section of this chapter, a second 12 bit D/A converter with a state-of-the-art dynamic performance will be discussed. The measurement results of this chip are summarised

resolution	12 bit
update rate	200 MS/s
INL error	0.5 LSB
DNL error	0.7 LSB
SFDR (1MHz@200MS/s)	65 dB
SFDR (500kHz@300MS/s)	50 dB
power consumption	140 mW
active area	14 mm^2
process	0.5 μm

Table 7.2: *Summary of the 12 bit DAC performance*

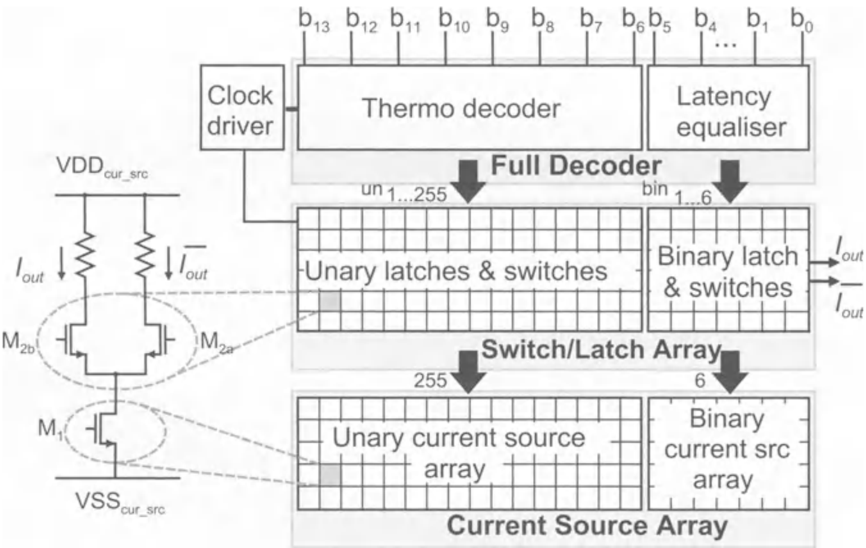


Figure 7.10: *The floorplan of the 14 bit D/A converter [VdPla JSSC99]*

in table 7.2.

A_{VT}	$12\, mV\mu m$
A_β	$1.9\,\%\mu m$
$\sigma(I)/I$	$0.125\,\%$
$(V_{GS}-V_T)_{cs}$	1 V
I_{FS}	20 mA
segmentation	8-6
$(W/L)_{cs}$	$1.1\mu m/104\mu m$
$(W/L)_{sw}$	$12.8\mu m/1\mu m$
technology	$0.5\mu m$

Table 7.3: *The unit current cell specifications for the 14 bit DAC*

7.2.2 A 14-bit Intrinsic Accuracy Q2 Random Walk CMOS D/A converter

7.2.2.1 The Floorplan

The block diagram of the 14 bit segmented D/A converter architecture is depicted in fig.7.10. The 8 MSB's (referred to as b6-b13 in fig.7.10) are decoded from binary to thermometer code in the thermometer decoder, which steers the unary weighted current source array. The 6 LSB's (referred to as b0-b5 in fig.7.10) are delayed by the latency equalizer block in order to have an equal delay with the MSB's. Similar to the 12 bit design all digital coding (thermometer decoder and latency equalizer block) has been grouped at the top of the chip. The latches and the current switches M2a and M2b are grouped in the switch/latch array in the middle of the chip. Finally, all current sources (binary weighted as well as unary weighted) can be found on the bottom of the chip in the current source array. A new switching scheme capable of obtaining a 14 bit intrinsic static linearity has been implemented and will be discussed in section 7.2.2.3 [VdPla JSSC99].

7.2.2.2 Design of the swatch cell and the thermometer decoder

Based on eq.(4.11), a relative unit current source standard deviation of 0.125 % is required to obtain a yield of 99.7 %. The 14 bit D/A converter has been implemented in a standard 0.5 μm CMOS process and has a full scale current of 20 mA. The dimensions of the current source transistor can be determined from the following equations :

$$WL = 116\,\mu m^2 \qquad \frac{W}{L} = \frac{1}{96} \tag{7.1}$$

From eq.(7.1) the gate-length and the gate-width of the current source transistor (W and L) have been calculated:

$$W = 1.1\,\mu m \qquad L = 104\,\mu m \tag{7.2}$$

An overview of the current cell dimensions is given in table 7.3.

The circuit of the latch driver used in this design was identical to the one of the first 12 bit design and as such an optimal gain in terms of IP reuse could be realised. The reader is referred to section 7.2.1.2. Since the current switches of the binary current sources are smaller, dummy switches have been added. So all latches, binary and unary, have the same load and delay.

Since the dummy LSB decoder has been expanded from four to six bits, the thermometer decoder for the unary block of the 14 bit D/A converter has been designed according to the same principles as was the case for the 12 bit design.

7.2.2.3 The switching scheme

Even more so than for the 12 bit D/A converter, the errors introduced by process, electrical and thermal gradients can degrade the static performance of the 14 bit D/A converter. It is therefore essential to keep these errors under control. First hand information on the gradients was available from a test chip that had already been processed. From measurements, the values for the different unary current sources could be estimated. This data has been used to extract the optimal switching scheme. In this way, the different error contributions have been randomised as to minimise any error accumulation in both the horizontal and the vertical direction.

The switching scheme has been given the name Random Walk Quad Quadrant (Q^2) switching scheme since four units placed in every quadrant of the current source array are needed to implement one unary current source and the current sources are switched on in a kind of random walk order over the array when the input code is increased. In fig.7.11, this scheme has been visualised. The regions indicated with the symbols A through P have been optimised as to compensate for the quadratic errors while the subregions 0 through 15 have been optimised towards the minimisation of the remaining linear errors. In practice this means that for the first unary current source subregion 0 in region A will be activated, for the second unary current source subregion 0 in region B, ..., for the 255th unary current source subregion 15 in region O. For more details, the reader is referred to [VdPla JSSC99].

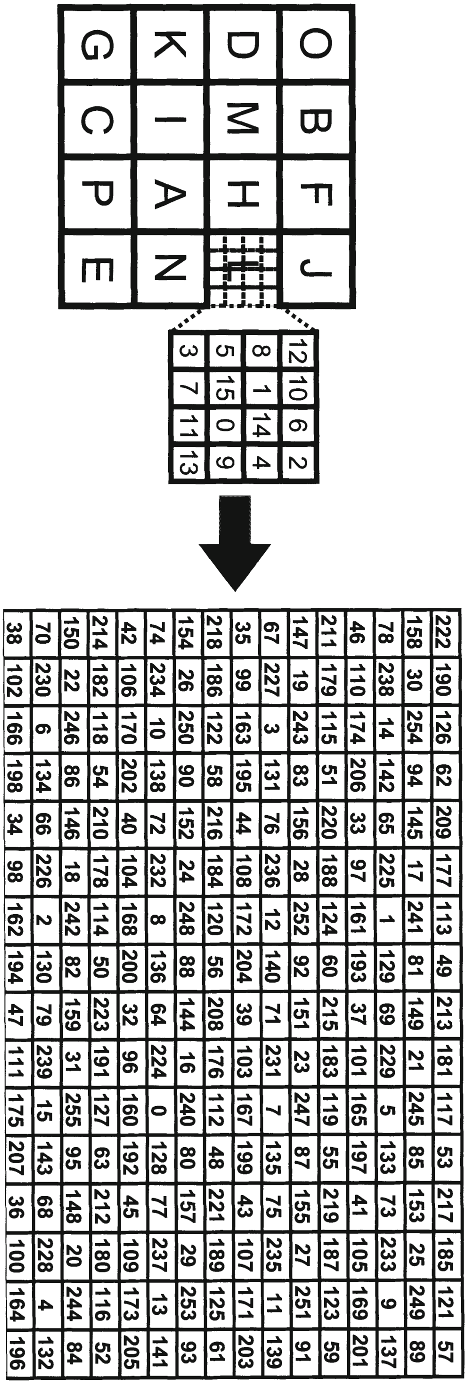


Figure 7.11: *Switching sequence of the Q^2 Random Walk switching scheme*

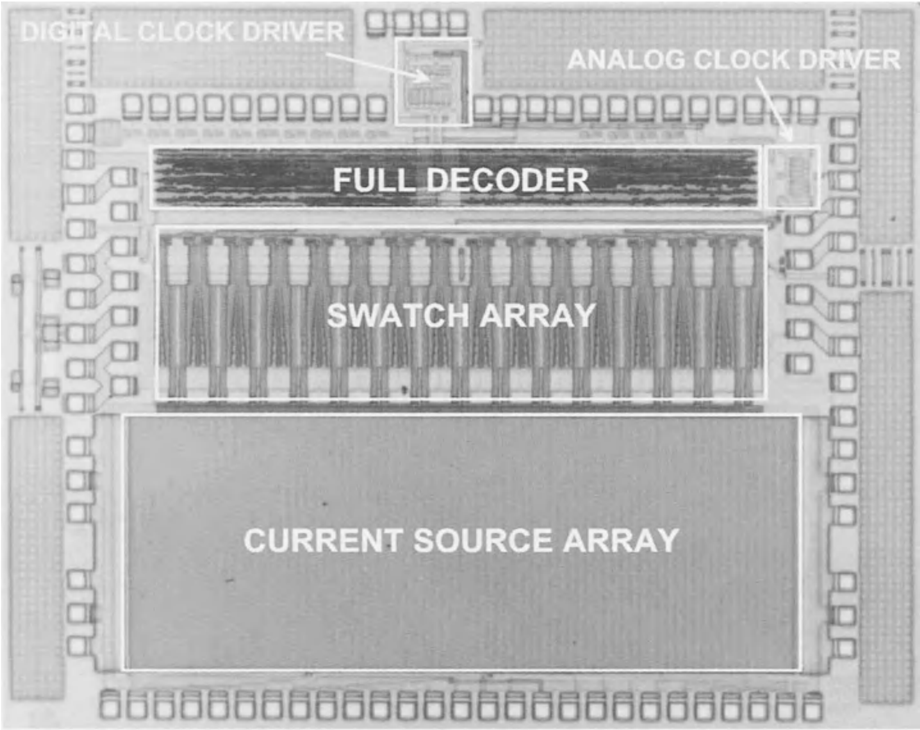


Figure 7.12: *The chip photograph of the 14 bit current steering CMOS D/A converter*

7.2.2.4 The Layout

To obtain a 14 bit intrinsic accuracy, the layout of the chip has been optimised. Some important issues are given:

- At layout level the D/A converter is subdivided into three functional blocks as to minimise any coupling that could otherwise occur between the digital and the analog part of the chip. The blocks are placed in the following way : the thermometer decoder at the top, the swatch array in the middle and the current source matrix at the bottom of the chip.

- Two separate power supplies have been used that are decoupled on chip by a 2.5 nF capacitance.

- Under the power supply lines and inside the swatch cells a decoupling capacitance of 750 pF is inserted. The swatch array (and the current source array) have been automatically generated by the Mondriaan tool [VdPla CICC98].

- The output signal has been shielded from the influence of other busses by using a similar comb structure as was discussed for the 12 bit design.
- The analog and the digital part of the chip have separate clock drivers. A binary tree has been used to distribute the analog clock over the chip as to minimise the clock skew between the different latches in the swatch array.
- To minimise the error introduced by edge effects, three dummy rows and four dummy columns have been added at each side of the current source matrix.
- A single current source transistor has a current that is equivalent to 4 LSB. Since its width to length ratio is extremely small, its layout has been folded three times to obtain an acceptable aspect ratio for the current source cell and current source array.

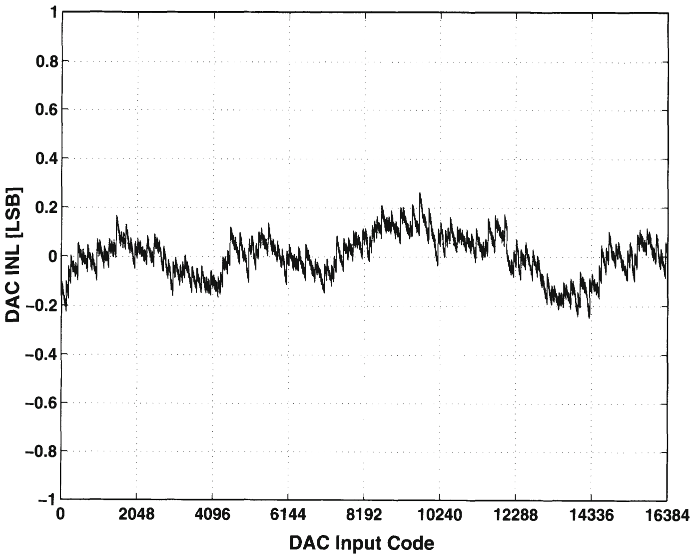
The chip has been processed in a single-poly, triple-metal 0.5 μm CMOS process. The chip photograph is shown in fig.7.12. The total chip area (bonding pads included) is only 13.1 mm^2.

7.2.2.5 Measurements

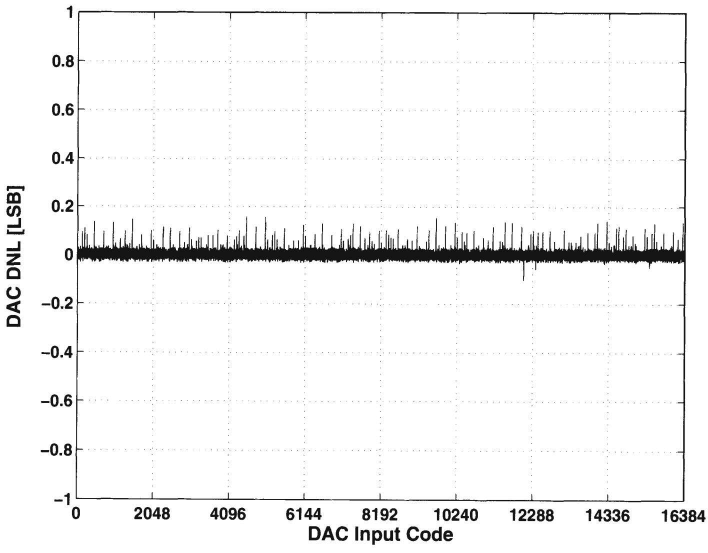
7.2.2.5.1 Static measurements The measured INL and DNL performance of the 14 bit D/A converter are shown in fig.7.13.a and fig.7.13.b. From this figure it can be concluded that the INL error is smaller than 0.3 LSB and the DNL error is smaller than 0.2 LSB, which proves the 14 bit intrinsic accuracy of the circuit and this without the use of any tuning or trimming.

7.2.2.5.2 Dynamic measurements The dynamic measurements were performed with an HP 3585 spectrum analyser having a frequency range of 40 MHz, and having a guaranteed dynamic range of 80 dB, with a typical value of 84 dB. All measurements have been performed with a single power supply of 2.7 V.

First, the SFDR of the D/A converter is given as a function of the input signal frequency for a fixed update rate of 150 MS/s. As can be seen from fig.7.14.a, a SFDR higher than 80 dB can be obtained for a sinusoidal input signal up to 800 kHz and higher than 70 dB for signals up to about 2 MHz. The SFDR as a function of the update rate is given in fig.7.14.b for a full scale sinusoidal input signal of 500 kHz. The SFDR remains well above 80 dB for update rates up to 150 MS/s. Fig.7.15.a and fig.7.15.b show the output spectra for the 14 bit D/A converter at an update rate of 150 MS/s for an input signal of 500 kHz and 5 MHz. In the first case, the SFDR equals 84 dB (no measurable spurs occurred) while in the second case the SFDR is dominated by both the second and the third harmonic and equals 61 dB.



(a) The measured INL performance of the 14 bit D/A converter



(b) The measured DNL performance of the 14 bit D/A converter

Figure 7.13: The static performance of the 14 bit D/A converter

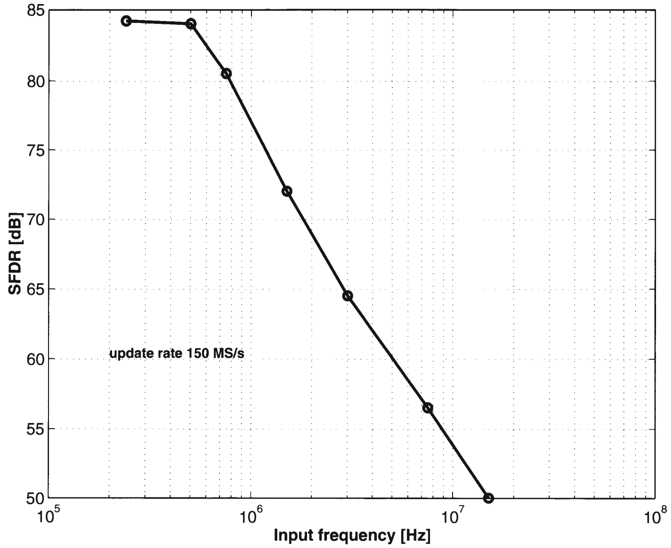
resolution	14 bit
update rate	150 MS/s
INL error	0.3 LSB
DNL error	0.2 LSB
SFDR (2MHz@150MS/s)	70 dB
SFDR (500kHz@200MS/s)	70 dB
power consumption	300 mW
active area	13.1 mm^2
process	0.5 μm

Table 7.4: *Summary of the 14 bit DAC performance*

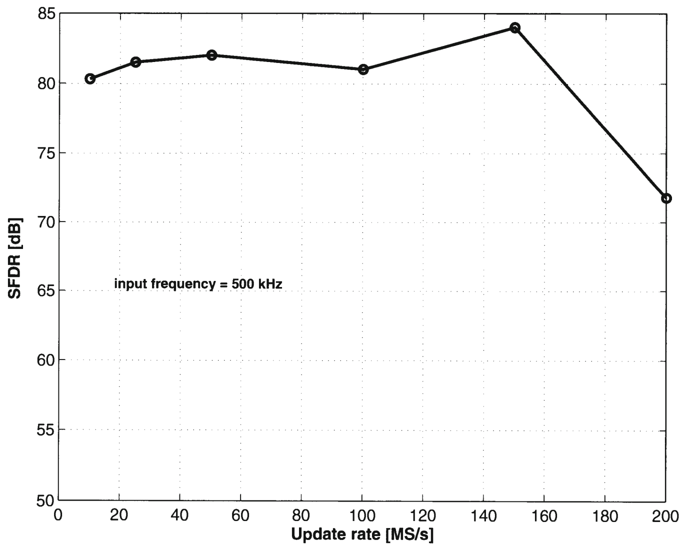
The measured power consumption is 70 mW at a 500 kHz output signal and 300 mW at a 15 MHz output signal.

7.2.2.6 Conclusion

As was the case for the 12 bit design, the main goal of realising this chip was to investigate the static behaviour of high resolution current steering D/A converters. This chip is unique in the sense that it was the first design (to the author's knowledge) that achieved the presented 14 bit specifications without the use of any tuning, trimming or calibration circuits in CMOS technology [Bugej JSSC99]. This chip proves the possibility of designing D/A converters with a resolution of 14 bit based on the mismatch behaviour of the current sources. However, it has to be noted that the layout plays a significant role in achieving such a state-of-the-art performance. The measurement results are summarised in table 7.4.

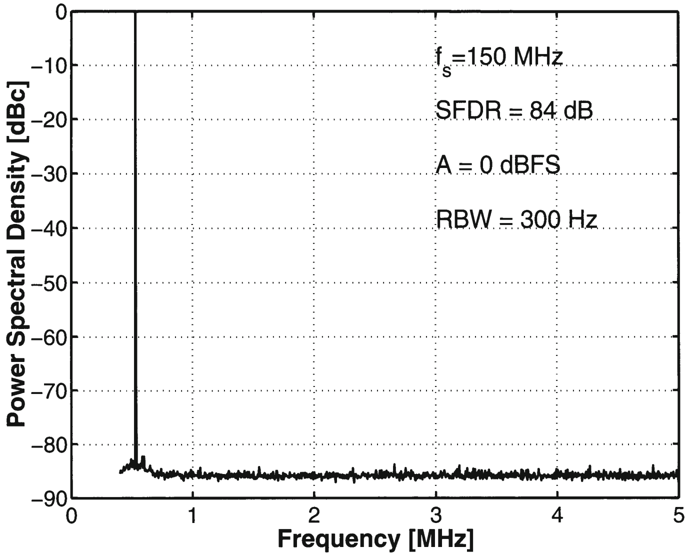


(a) The SFDR as a function of the input signal frequency @ 150MS/s update rate

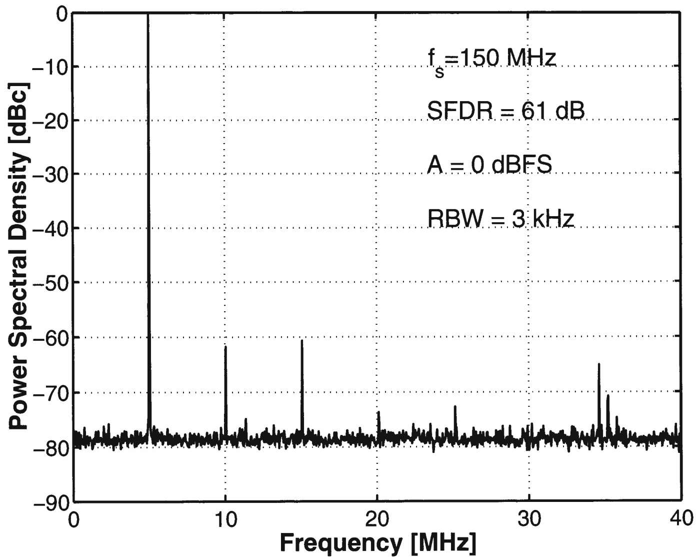


(b) The SFDR as a function of the update rate (input signal of 500 kHz at 0 dBFS)

Figure 7.14: The dynamic performance of the 14 bit D/A converter



(a) Output spectrum at a 150 MS/s update rate (500kHz signal, 0 dBFS)



(b) Output spectrum at a 150 MS/s update rate (5 MHz signal, 0 dBFS)

Figure 7.15: Some output spectra

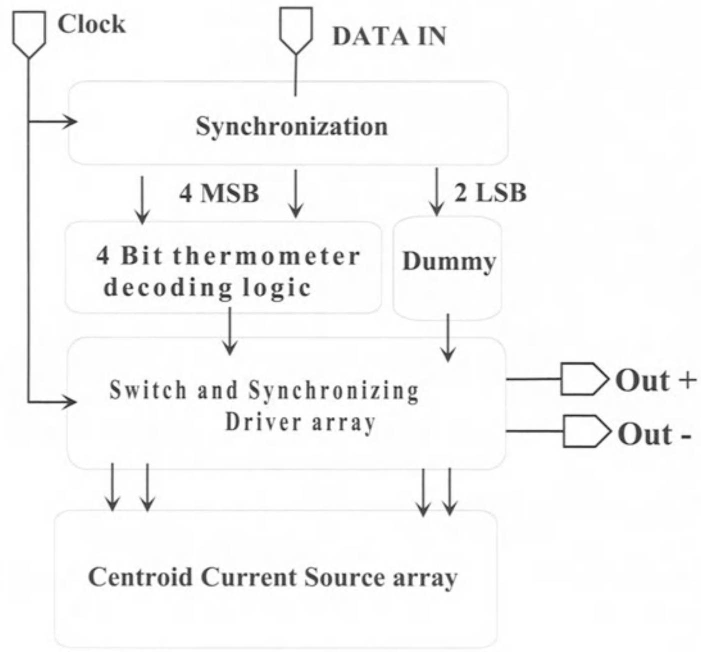


Figure 7.16: *Floorplan of the 6 bit D/A converter*

7.3 High Speed D/A Converters

7.3.1 A 800 MHz Ultra Low Glitch Energy 6-bit CMOS D/A Converter

7.3.1.1 The Floorplan

Fig.7.16 shows the floorplan of the realised 6 bit D/A converter. This chip has a segmented architecture, where the 2 LSB's have been implemented in a binary way while the 4 MSB's have been implemented in a unary way. Since the decoding logic of a 6 bit D/A converter is relatively simple, the thermometer code is generated by a row and column decoder which has the extra advantage of allowing us to target higher update rates. The synchronisation of the control signals of the selection logic and the use of a deglitch driver circuit placed immediately before the switches has led to a very low glitch energy specification.

A_{VT}	$12\,mV\mu m$
A_β	$1.9\,\%\mu m$
$\sigma(I)/I$	$2\,\%$
$(V_{GS}-V_T)_{cs}$	0.8 V
I_{FS}	20 mA
segmentation	4-2
$(W/L)_{cs}$	$6\mu m/0.7\mu m$
$(W/L)_{sw}$	$6\mu m/0.7\mu m$
technology	$0.5\mu m$

Table 7.5: *The unit current cell specifications for the 6 bit DAC*

7.3.1.2 Design of the swatch cell and the thermometer decoder

The relation between the gate-width W and the gate-length L of the current source transistor and the process technology, the full scale current of the D/A converter and the V_{GS} of the current source is given by:

$$L^2 = \frac{\alpha}{\gamma} + \frac{\delta}{\gamma}(V_{GS}-V_T)^2$$

$$W^2 = \frac{\alpha\gamma}{(V_{GS}-V_T)^4} + \frac{\delta\gamma}{(V_{GS}-V_T)^2} \tag{7.3}$$

$$\alpha = \frac{4A_{VT}^2}{2(\frac{\sigma(I)}{I})^2} \qquad \delta = \frac{A_\beta^2}{2(\frac{\sigma(I)}{I})^2} \qquad \gamma = \frac{I_{TOTAL}}{2^{N-1}KP}$$

where $\sigma(I)/I$ is the relative unit current source standard deviation, A_{VT} and A_β are mismatch related technology constants and N is the number of bits.

For this design, the gate overdrive voltage of the current source transistors has been chosen to be 0.8 V. The dimensions of the current source transistor have been calculated for an implementation in a standard 0.5 μm technology with a full scale current of 20 mA and an INL yield of 99.7 %. The gate-width equals 6 μm and the gate-length equals 0.7 μm. An overview of the unit current cell dimensions is given in table 7.5.

A very low glitch energy has been obtained by synchronising and adjusting the input signals of the switching transistors of the D/A converter. Fig.7.17 shows the circuit of the flipflop together with the driver. The master-slave flipflop has been designed to minimise the clock feedthrough and to maximise the achievable speed. The core of this flipflop is based on the combination of two dynamic single transistor clocked latches (DSTC-2-p/DSTC-1-n [Yuan VLSI96]). However that combination suffers from races under certain conditions (high clock and changing input signals). Therefore the clock transistor of the master latch has been replaced by the clocked pass-transistors M1a and M1b. Additional level restores have been placed to optimise the speed. After the master-slave flip flop a driver circuit has been placed that determines the crossing point of the switch input signals [Basto CICC96]. The digital feedthrough through the C_{GD} capacitance of the switches is minimised by reducing the voltage swing at the input of the switch transistors. This has been realised by taking an appropriate value for the VSS voltage of the driver.

7.3.1.3 The switching scheme

Since in this case the current source array is relatively small, no complicated switching scheme has been implemented. The current of a unary current source equals 4 times the LSB current. Four separate current sources have been placed around the center of the array. A representation of the switching scheme is given in fig.7.18. The boxes indicated with a 'B' represent the current sources used to implement the binary bits and the box indicated with a 'M' represents a transistor that is not used as a current source but that is configured as a MOS diode which acts as a biasing reference for the current source array.

7.3.1.4 Layout

The chip photograph of the 6 bit D/A converter is shown in fig.7.19. The different building blocks in the layout are the row and column decoder, the switching cells (consisting of the selection logic, the flipflop and the switches) and the current source array. The decoding logic of the D/A converter has been implemented using two identical row and column decoders, placed symmetrically around the current source array. This has been done to minimise the propagation delay of the control signals generated by these decoders. Several measures have been taken to avoid coupling from the digital part of the chip identical to the ones used in the previous designs.

7.3.1.5 Measurements

In this paragraph, the measurement set-up will be discussed in some more detail. To generate the digital input signals at update rates (clock rates) up to 1 GS/s, a HP 80000 datagenerator has been used. The static measurements have been performed using a HP 3457A multimeter. The dynamic measurements have been performed using a HP 3585B and/or Tektronix 2755P spectrum analyser and a Tektronix TDS 680B oscilloscope. The D/A converter itself is mounted on a ceramic substrate

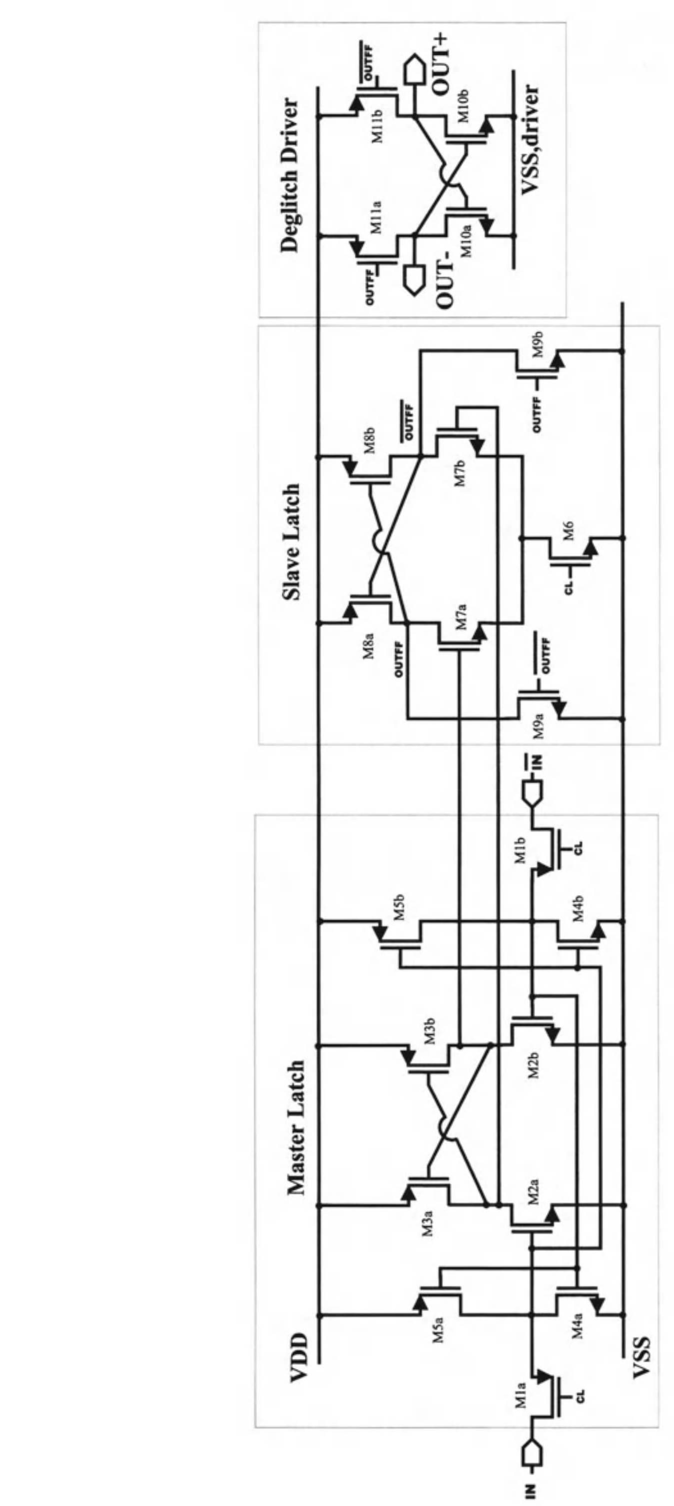


Figure 7.17: *The synchronising flipflop and the deglitch driver*

B	10	13	15	14	12	11	M
10	9	5	7	8	6	9	11
13	5	4	2	3	4	6	12
15	7	2	1	1	3	8	14
14	8	3	1	1	2	7	15
12	6	4	3	2	4	5	13
11	9	6	8	7	5	9	10
B	11	12	14	15	13	10	B

Figure 7.18: *The centroid switching scheme(B=binary, M=current reference)*

in order to shield the chip from external noise coupling. This substrate is encapsulated in a copper-beryllium case where all power supplies have been locally decoupled (fig. 7.20).

7.3.1.5.1 Static measurements The 6 bit D/A converter has been measured at a single 3 V power supply. All measurements have been performed on a single ended output with a load resistor of 25Ω. Fig.7.21.a and fig.7.21.b show the measured INL and DNL characteristic of the 6 bit D/A converter. From these figures, it can be concluded that the INL error is smaller than 0.15 LSB and the DNL error is smaller than 0.13 LSB which guarantees the 6 bit intrinsic accuracy of the presented D/A converter.

7.3.1.5.2 Dynamic measurements The full scale settling time of the D/A converter is depicted in fig.7.22. The D/A converter settles within 0.5 LSB in a time frame of 4.5 ns. The rise time (10 %-90 %) of the D/A converter equals 1.2 ns and the fall time (90 %-10 %) equals 2.5 ns. Fig.7.23 shows the worst case glitch energy which has been measured at the transition from input code 3 to code 4. The measured value for the glitch energy, including both the coupling and the code transition, equals only 0.2 pVs. This result has been obtained due to the synchronisation of all signals and a careful layout.

Fig.7.24.a shows the measured SFDR of the 6 bit D/A converter as a function of

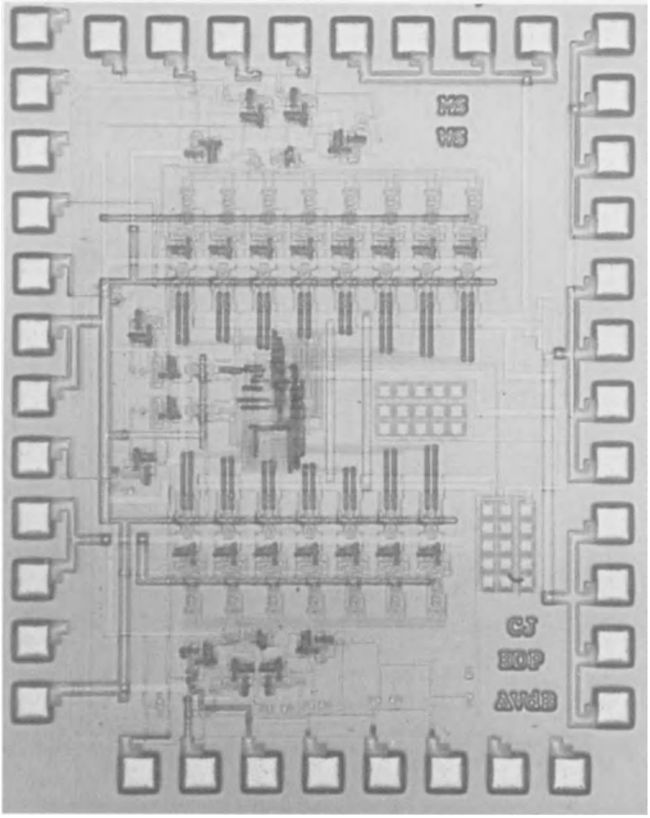


Figure 7.19: *The chip photograph of the 6 bit DAC*

the input frequency for a 800 MS/s update rate. The SFDR remains well above the 36 dB line for input frequencies up to 15 MHz. For frequencies higher than 15 MHz, the SFDR rapidly decreases. Fig.7.24.b shows the measured SFDR as a function of the update rate for a full scale sinusoidal input signal of 1 MHz. The SFDR is higher than 36 dB up to an update rate of 800 MS/s. For an update rate of 900 MS/s, a SFDR of about 35 dB has been measured. Fig.7.25 shows the output spectrum of the D/A converter for a full scale 12.5 MHz sinusoidal input clocked at a frequency of 800 MS/s. The overall signal to noise ratio is dominated by the second order harmonic distortion component which is 38 dB below the signal level. The maximum measured power consumption is only 86 mW.

7.3.1.6 Conclusion

The main focus for this D/A converter was to probe the design space for the limits in terms of update rate. For this reason a low resolution D/A converter has been chosen as

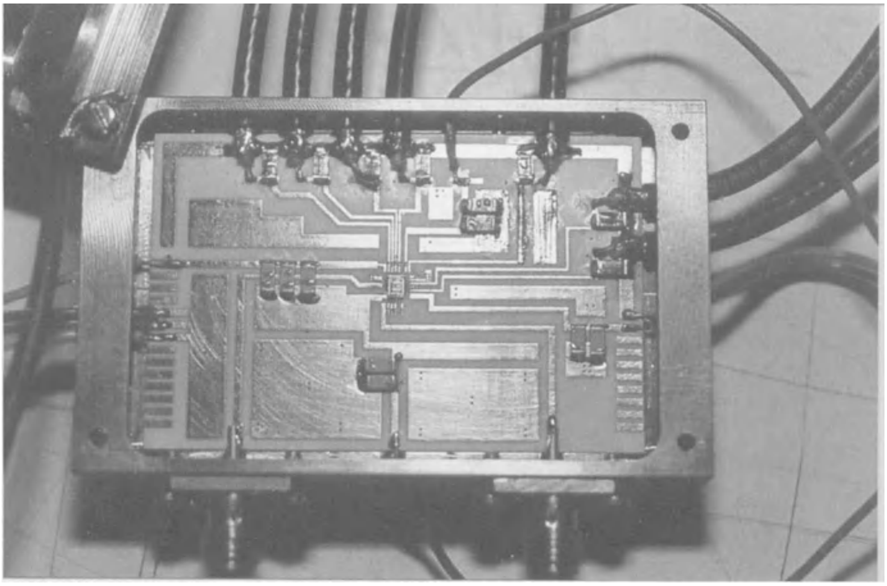
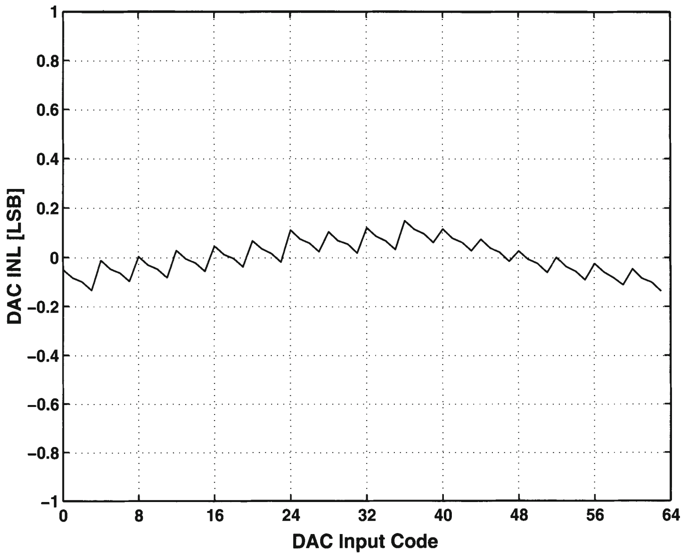


Figure 7.20: *The measurement set-up*

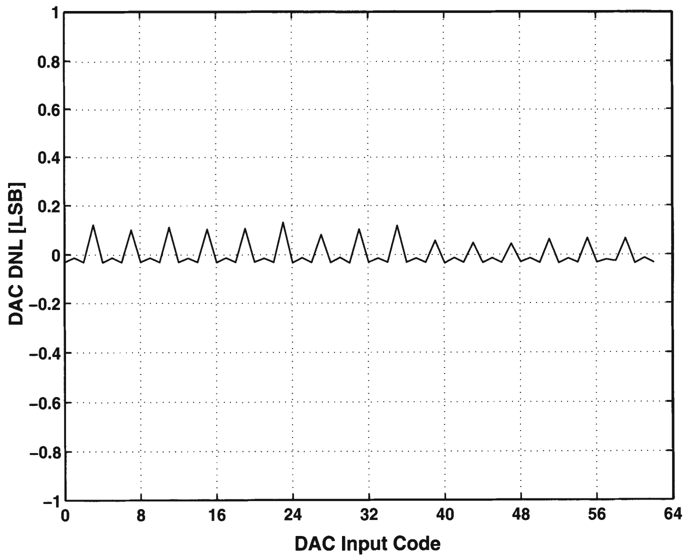
resolution	6 bit
update rate	800 MS/s
INL error	0.15 LSB
DNL error	0.13 LSB
SFDR (20MHz@800MS/s)	$> 30\,dB$
SFDR (1MHz@800MS/s)	$> 36\,dB$
power consumption	86 mW
active area	$1.4\ mm^2$
process	$0.5\ \mu m$

Table 7.6: *Summary of the 6 bit DAC performance*

to simplify the thermometer decoder design. This design demonstrates the possibility of achieving update rates in the order of hundreds of MHz. The next step will be to combine both a high resolution with a high update rate. The measurement results are summarised in table 7.6.



(a) The measured INL performance of the 6 bit D/A converter



(b) The measured DNL performance of the 6 bit D/A converter

Figure 7.21: The static performance of the 6 bit D/A converter

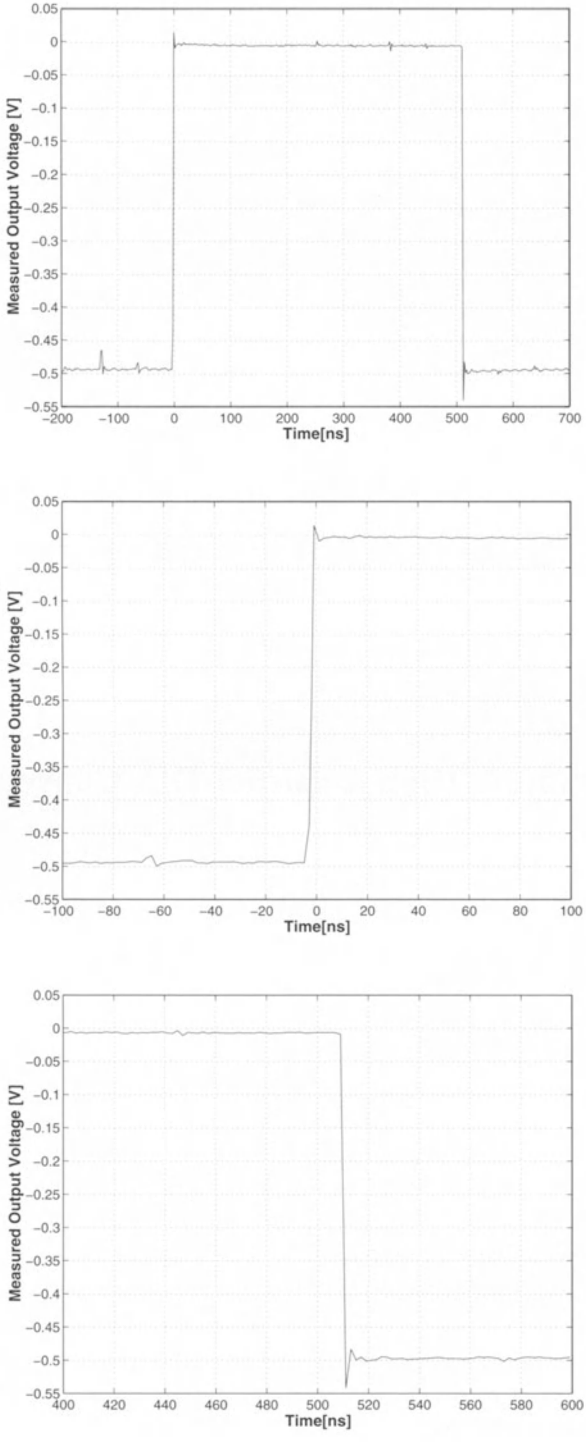


Figure 7.22: The settling behaviour of the 6 bit D/A converter

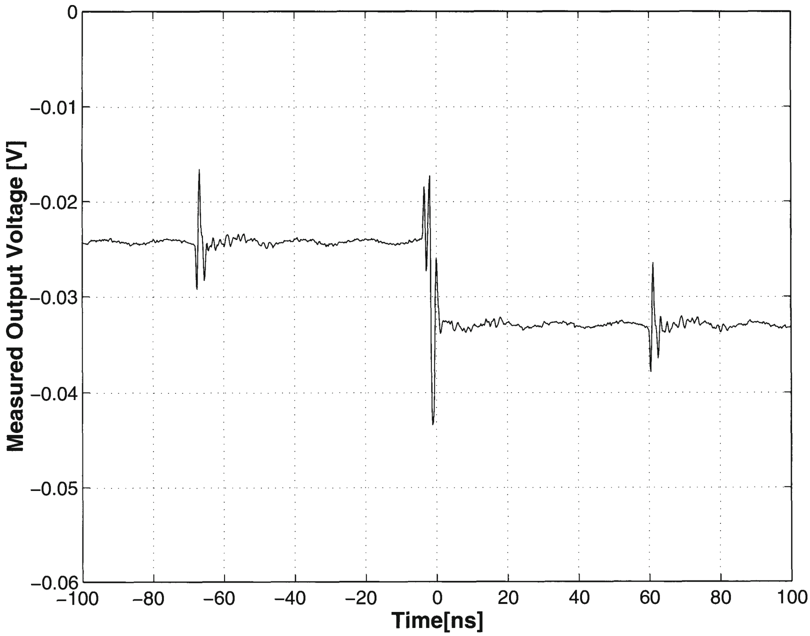


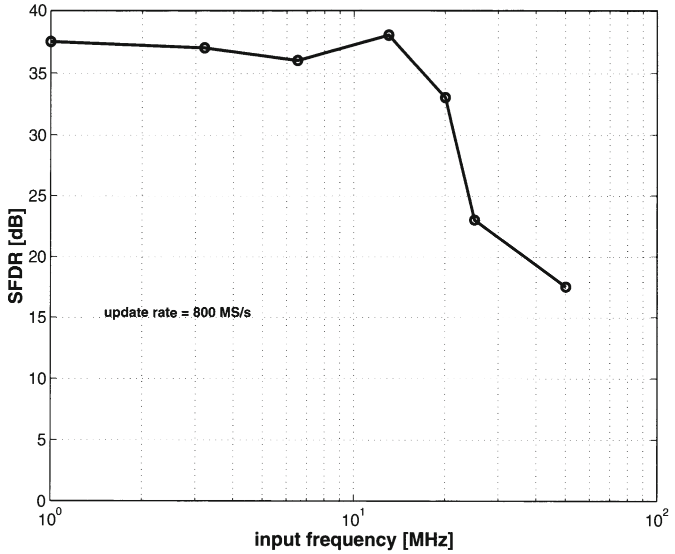
Figure 7.23: *The measured worst case glitch*

7.4 High Speed, High Accuracy D/A Converters

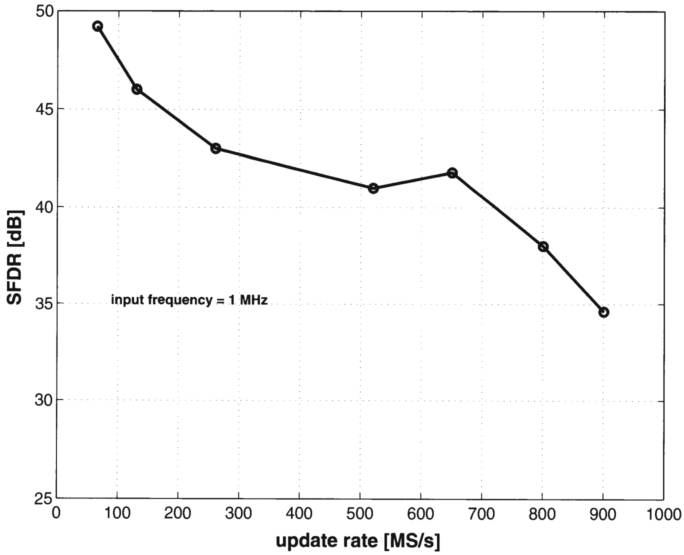
7.4.1 A 10-bit 1-GS/s Nyquist Current-Steering CMOS D/A Converter

7.4.1.1 The Floorplan

Fig.7.26 shows the floorplan of a high speed 10 bit D/A converter. The chip consists of a 5 bit binary and a 5 bit unary sub-DAC. For the five most significant bits, the input bit streams are converted to a 32 bit thermometer decoded output. For the 5 bit binary LSB processing, the outputs are logically the same as the inputs. Here, a dummy decoder has been added to minimise latency problems between the signals generated by the MSB decoder and the binary LSB bits. Due to the 1 GHz high speed specification, the decoder has been designed manually at transistor level taking the layout parasitic capacitances of both the active elements and the interconnections into consideration [VdBos JSSC01].



(a) The SFDR versus the input frequency for an update rate of 800 MS/s



(b) The SFDR versus the update rate for a full scale sinusoidal input frequency of 1 MHz

Figure 7.24: The dynamic performance of the 6 bit D/A converter

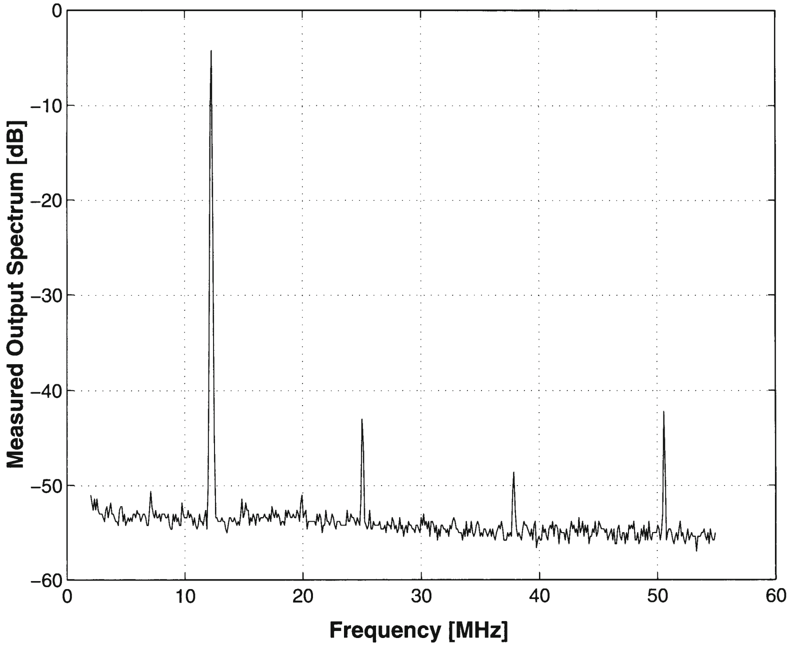


Figure 7.25: *The measured output spectrum for a full scale 12.5 MHz sinusoidal input signal @800 MS/s*

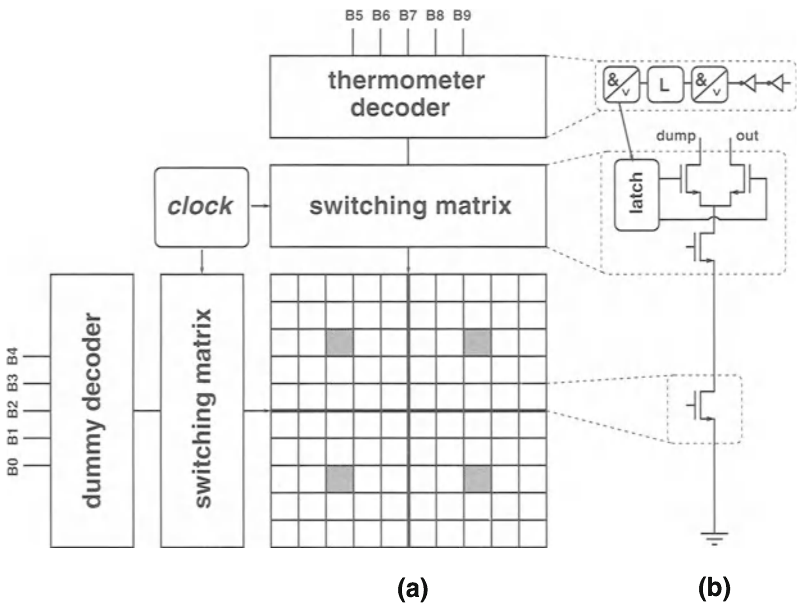
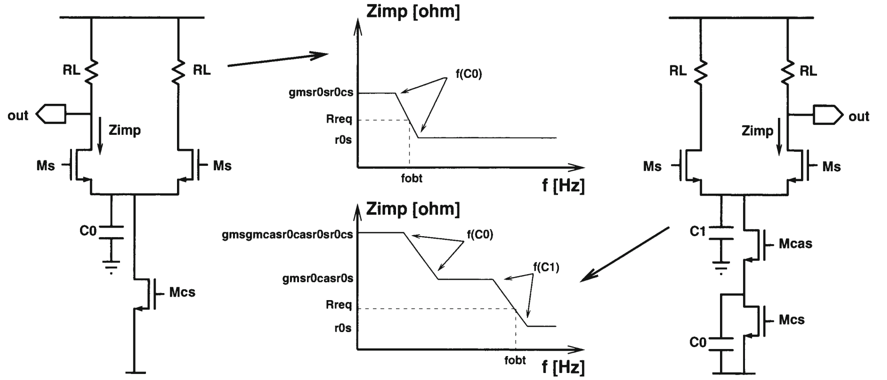
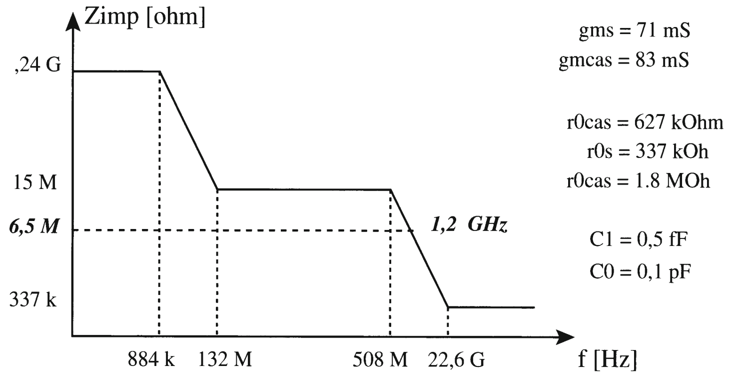


Figure 7.26: *The floorplan of the realised 10 bit segmented D/A converter*



(a) The frequency behaviour of the output impedance for a current cell with and without a cascode transistor



(b) The output impedance as a function of the frequency for the 10 bit D/A converter

Figure 7.27: The output impedance of the 10 bit D/A converter design

7.4.1.2 Design of the swatch cell

To achieve a 99.7% INL yield specification, the specification for the unity current source matching is 0.5%. Based on this value and the size versus matching relation [Pelgr JSSC89] for MOS transistors, the dimensions of the current source transistor have been calculated. For the presented design, the dimensions of the unit current source transistor are given by a 2 μm gate-width and an 8 μm gate-length. An overview of the dimensions of the unit current cell is given in table 7.7.

Beyond the bandwidth of the frequency dependent Z_{imp}, a severe linearity degradation is caused in the current to voltage conversion. Based on a detailed linearity

A_{VT}	$8.9\ mV\mu m$
A_β	$1.9\ \%\mu m$
$\sigma(I)/I$	$0.5\ \%$
$(V_{GS}-V_T)_{cs}$	1 V
I_{FS}	20 mA
segmentation	5-5
$(W/L)_{cs}$	$2\mu m/8\mu m$
$(W/L)_{cas}$	$1\mu m/0.7\mu m$
$(W/L)_{sw}$	$0.5\mu m/0.35\mu m$
technology	$0.35\mu m$

Table 7.7: *The unit current cell specifications for the 10 bit DAC*

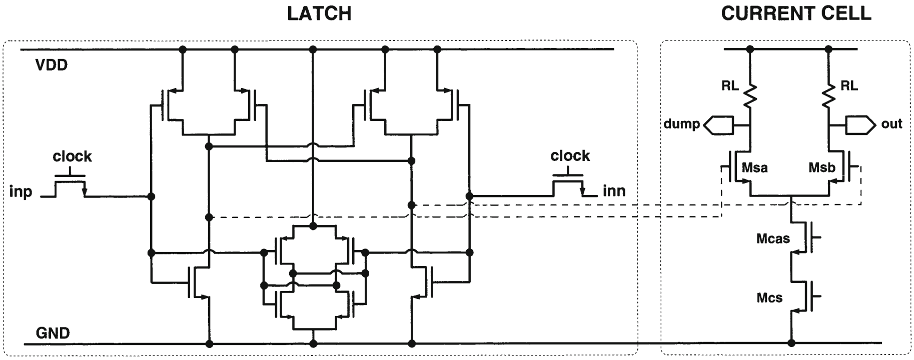


Figure 7.28: *The switch driver and the current cell*

analysis, the realised circuit has been optimised to achieve an improved linearity behaviour up to the Nyquist frequency of 500 MHz. A graphical representation of this impedance bandwidth effect is given in fig.7.27.a. It shows a non-cascoded and a cascoded current cell with a qualitative bode diagram of their output impedance behaviour. The diagrams also indicate the required impedance value to achieve the linearity specification. For a 10 bit D/A converter (SFDR of 60 dB) and a load resistor of 25Ω, this impedance level is 6.5 $M\Omega$. This high frequency output impedance design specification is one of the critical elements in achieving the improved high frequency linearity behaviour. The Bode diagram for this design is given in fig.7.27.b. Notice that the value of the capacitance C0 is a factor of ten smaller than the C0 of the 12 bit implementation discussed in the numerical examples of chapter 4. This can be explained by the fact that for a 12 bit D/A converter, the area of the current source matrix is 16 times larger compared to the 10 bit implementation. As a consequence,

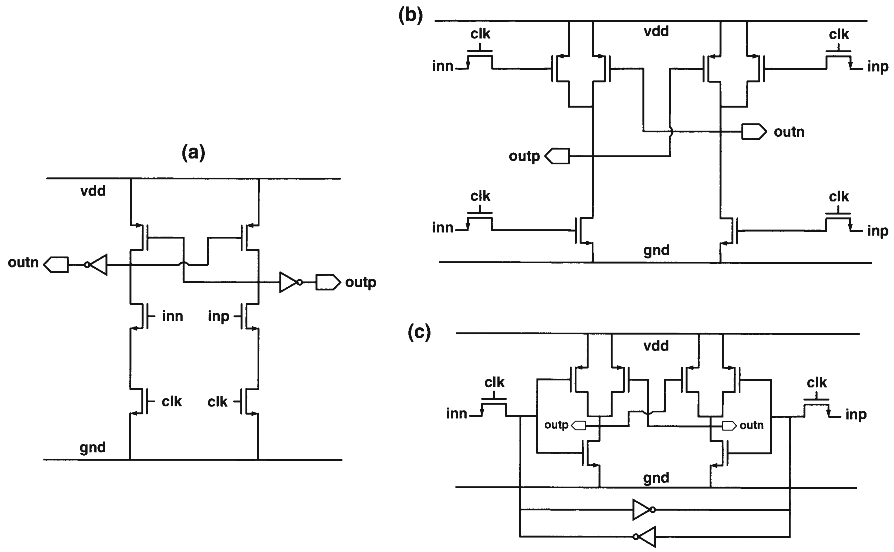


Figure 7.29: *Design of the rise/fall time based giga-latch*

the interconnection capacitance increases significantly.

Apart from the constraint dictated by the output impedance, also the other three effects discussed in section 5.2 (timing and feedthrough of the switch control signals and the drain voltage fluctuation of the current source transistor) have been taken into account in order to achieve a state-of-the-art performance. The design of the synchronised driver circuit will now be discussed in some more detail (fig.7.28).

In [Kohno CICC95, Marqu ISSCC98, VdBos CICC98] relatively simple but effective driver circuits have been used. In those circuits, the intrinsic delay between the two complementary outputs is used to lower the crossing point of the control signals of the switch transistors. In the nMOS implementations, an inverter is placed after the outputs of the driver to invert the crossing point of the differential driving voltages to a high value. However, for very high speeds, this driver can no longer be used. The combination of the intrinsic delay and the feedthrough of the steep control signals through the gate-drain capacitance of the switches, limits the operation frequency to a few 100 MHz. The problem of the too large required time period can be solved by using another type of driver that sets the crossing point of the control signals by using different rise and fall times for the driver's differential output [Chin JSSC94, Wu JSSC95].

To design a high speed rise/fall time based circuit, the delay based circuit in fig.7.29(a) has been modified. An extra pMOS input circuit (input transistor + clock) can be placed in parallel with each of the cross-coupled pMOS transistors situated at the top of the circuit, to obtain instantaneous charging of the output nodes with falling

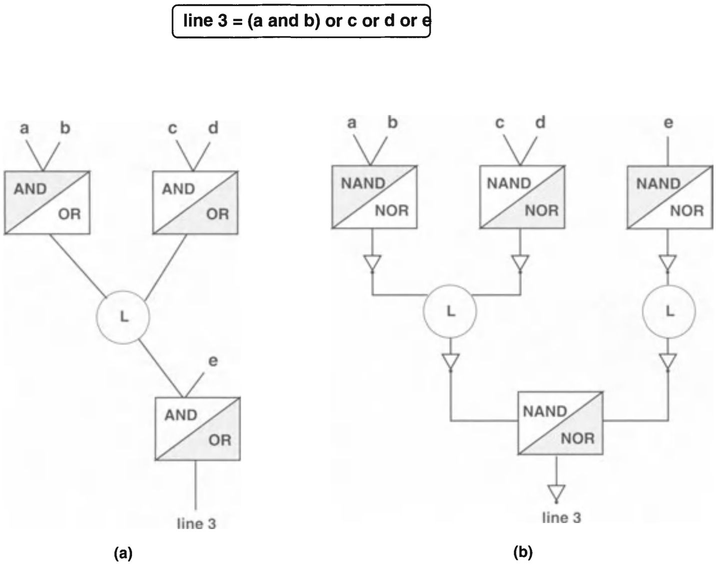


Figure 7.30: *Graphic representation of the logic equation for line three (L represents a logic function, in this case a simple OR function)*

inputs. In this way, the intrinsic delay is removed from the circuits operation as charging and discharging starts at the same moment. To deal with a low power supply and to keep a single phase clock, the cascaded nMOS and pMOS clock transistors have been replaced by nMOS pass-transistors in the input path. This intermediate circuit is shown in fig.7.29(b). The combination of the μ_n/μ_p scaled pMOS transistors and of the pMOS positive feedback loop, results in a rise time that is much faster than the fall time of the driver circuit. In this way, a high crossing point of the differential outputs is directly available at the output of the latch. Notice that also a lower crossing point can be realised by scaling the gate-width of the pMOS and/or nMOS transistors respectively down and up. The driver topology can thus be used for both nMOS and pMOS implementations of the D/A converter. The additional feedback by the small inverters (fig.7.29(c)) suppresses the clock feedthrough by the pass-transistors and stabilises the synchronised inputs. The circuit is fully functional for clock speeds exceeding 1 GHz. Since the analog part has been optimised towards a high speed, the same has to be done for the digital part of the 10 bit D/A converter.

7.4.1.3 Design of the thermometer decoder

Designing a thermometer decoder using standard cell libraries, has the disadvantage that the achievable operation speed is limited by the timing constraints of the used cells. To achieve an increased update rate, the decoder has been manually designed and has been layouted at transistor level.

Before going into detail on the derivation of the functional building blocks of the thermometer decoder, a general view on the architecture of this decoder is given by using as an example the logic equation of line three. This line has to be active if the digital input code of the D/A converter is larger than or equal to three.

$$line3 = (a\&b) \vee c \vee d \vee e \tag{7.4}$$

where a equals the LSB, b equals the LSB+1, c equals the LSB+2, ...

A possible graphic representation is given in fig.7.30.a. Since the digital decoder will be implemented using CMOS circuits, fig.7.30.b shows the implementation for an equal load using NOR and NAND blocks. The large number of inverters results in a decoder with a large area and power consumption. These inverters can and have been eliminated from the decoder based on the following theorem:

$$\begin{aligned} \overline{X \vee Y} &= \overline{X} \& \overline{Y} \\ \overline{X \& Y} &= \overline{X} \vee \overline{Y} \end{aligned} \tag{7.5}$$

Placing an inverter after a NAND block is identical to placing an inverter at the input of a NOR block. Since both the NAND and NOR function are present in this design, almost all inverters can be eliminated. In fig.7.31.a, this is shown in more detail. Shifting the inverter to the input of the last NAND/NOR block, results in two inverters placed immediately after each other and as a consequence these inverters can be dropped. The same holds for the inverters of the first NAND/NOR block. In this way only the input bits have to be inverted which results in only 5 inverters for the full thermometer decoder (fig.7.31.b). Furthermore, placing these inverters at the input has the additional advantage that these circuits can also act as a buffer that regenerates the input signal. Also note that bit e has been given the same load as bit a and bit b as to avoid any timing errors that could otherwise occur.

The remainder of this paragraph will discuss into more detail the way the logic building blocks were chosen and how the final architecture of the thermometer decoder looks like.

In a first step, the equations that the thermometer decoder has to realise are written down (table 7.8).

From these equations, one can clearly see that bit e can be separated from the rest of the equations by implementing the last logical level of the thermometer decoder as a block that has both a NAND and a NOR function. Since the target of this design is a simple and compact decoder, it has been investigated if the NAND/NOR block could

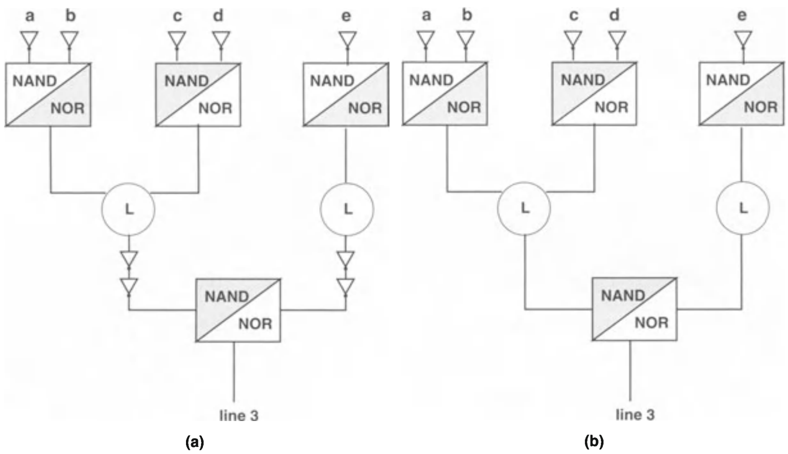


Figure 7.31: *Graphic representation of the logic equation for line three*

$1 = a \vee b \vee c \vee d \vee e$	$17 = a \vee b \vee c \vee d\&e$
$2 = b \vee c \vee d \vee e$	$18 = b \vee c \vee d\&e$
$3 = (a\&b) \vee c \vee d \vee e$	$19 = (a\&b) \vee c \vee d\&e$
$4 = c \vee d \vee e$	$20 = c \vee d\&e$
$5 = ((a \vee b)\&c) \vee d \vee e$	$21 = ((a \vee b)\&c) \vee d\&e$
$6 = (b\&c) \vee d \vee e$	$22 = (b\&c) \vee d\&e$
$7 = (a\&b\&c) \vee d \vee e$	$23 = (a\&b\&c) \vee d\&e$
$8 = d \vee e$	$24 = d\&e$
$9 = ((a \vee b \vee c)\&d) \vee e$	$25 = ((a \vee b \vee c)\&d)\&e$
$10 = ((b \vee c)\&d) \vee e$	$26 = ((b \vee c)\&d)\&e$
$11 = ((a\&b \vee c)\&d) \vee e$	$27 = ((a\&b \vee c)\&d)\&e$
$12 = (c\&d) \vee e$	$28 = (c\&d)\&e$
$13 = ((a \vee b)\&(c\&d)) \vee e$	$29 = ((a \vee b)\&(c\&d))\&e$
$14 = (b\&c\&d) \vee e$	$30 = (b\&c\&d)\&e$
$15 = (a\&b\&c\&d) \vee e$	$31 = (a\&b\&c\&d)\&e$
$16 = e$	

Table 7.8: *Logic equations for a 5 bit thermometer decoder*

$$1 = \overline{\overline{\overline{\overline{a}\&\overline{b}} \vee \overline{\overline{c}\&\overline{d}}}\&\overline{e}}$$

$$2 = \overline{\overline{\overline{\overline{b}\&\overline{c}} \vee \overline{\overline{c}\&\overline{d}}}\&\overline{e}}$$

$$3 = \overline{\overline{\overline{\overline{a} \vee \overline{b}} \vee \overline{\overline{c}\&\overline{d}}}\&\overline{e}}$$

$$4 = \overline{(\overline{c}\&\overline{d})\&\overline{e}}$$

$$5 = \overline{\overline{(\overline{\overline{a}\&\overline{b}}\&c) \vee d}\&\overline{e}}$$

$$6 = \overline{\overline{\overline{\overline{b} \vee \overline{c}} \vee d}\&\overline{e}}$$

$$7 = \overline{\overline{(\overline{\overline{a} \vee \overline{b}})\&(\overline{\overline{b} \vee \overline{c}}) \vee d}\&\overline{e}}$$

$$8 = \overline{\overline{d}\&\overline{e}}$$

$$9 = \overline{\overline{(\overline{\overline{a}\&\overline{b}}) \vee (\overline{\overline{b}\&\overline{c}})\&d}\&\overline{e}}$$

$$10 = \overline{\overline{\overline{\overline{b}\&\overline{c}}\&d}\&\overline{e}}$$

$$11 = \overline{\overline{(\overline{\overline{a} \vee \overline{b}})\&\overline{d} \vee \overline{\overline{c} \vee \overline{d}}}\&\overline{e}}$$

$$12 = \overline{(\overline{c} \vee \overline{d})\&\overline{e}}$$

$$13 = \overline{\overline{\overline{\overline{a}\&\overline{b}}\&\overline{\overline{c} \vee \overline{d}}}\&\overline{e}}$$

$$14 = \overline{\overline{\overline{\overline{b} \vee \overline{c}}\&d}\&\overline{e}}$$

$$15 = \overline{\overline{\overline{\overline{a} \vee \overline{b}}\&\overline{\overline{c} \vee \overline{d}}}\&\overline{e}}$$

$$16 = e$$

$$17 = \overline{\overline{\overline{\overline{a}\&\overline{b}} \vee \overline{\overline{c}\&\overline{d}}} \vee \overline{e}}$$

$$18 = \overline{\overline{\overline{\overline{b}\&\overline{c}} \vee \overline{\overline{c}\&\overline{d}}} \vee \overline{e}}$$

$$19 = \overline{\overline{\overline{\overline{a} \vee \overline{b}} \vee \overline{\overline{c}\&\overline{d}}} \vee \overline{e}}$$

$$20 = \overline{(\overline{c}\&\overline{d}) \vee \overline{e}}$$

$$21 = \overline{\overline{(\overline{\overline{a}\&\overline{b}}\&c) \vee d} \vee \overline{e}}$$

$$22 = \overline{\overline{\overline{\overline{b} \vee \overline{c}} \vee d} \vee \overline{e}}$$

$$23 = \overline{\overline{(\overline{\overline{a} \vee \overline{b}})\&(\overline{\overline{b} \vee \overline{c}}) \vee d} \vee \overline{e}}$$

$$24 = \overline{\overline{d} \vee \overline{e}}$$

$$25 = \overline{\overline{(\overline{\overline{a}\&\overline{b}}) \vee (\overline{\overline{b}\&\overline{c}})\&d} \vee \overline{e}}$$

$$26 = \overline{\overline{\overline{\overline{b}\&\overline{c}}\&d} \vee \overline{e}}$$

$$27 = \overline{\overline{(\overline{\overline{a} \vee \overline{b}})\&\overline{d} \vee \overline{\overline{c} \vee \overline{d}}} \vee \overline{e}}$$

$$28 = \overline{(\overline{c} \vee \overline{d}) \vee \overline{e}}$$

$$29 = \overline{\overline{\overline{\overline{a}\&\overline{b}}\&\overline{\overline{c}\&\overline{d}}} \vee \overline{e}}$$

$$30 = \overline{\overline{\overline{\overline{b} \vee \overline{c}}\&d} \vee \overline{e}}$$

$$31 = \overline{\overline{\overline{\overline{a} \vee \overline{b}}\&\overline{\overline{c} \vee \overline{d}}} \vee \overline{e}}$$

Table 7.9: *Re-written logic equations for a 5 bit thermometer decoder*

be re-used at another level of the logic. This is the case for the first level as will be discussed in the next paragraph. The equations have been re-written as to prove the concept and to determine the logic functions that are necessary to make the connection between the first and the last logic level in the decoder. The result is given in table 7.9.

Since at this point, it is already known that the last level is implemented using a block with both a NAND and NOR function, bit e is not taken into account so that only the first 15 equations in table 7.9 have to be considered. From these equations 8 blocks can be extracted that can fully realise the thermometer logic. In table 7.10, these blocks are given together with the number of times they occur.

From the equations in table 7.9, it becomes clear that the first logic level of the thermometer decoder can be built up using a combination of both the NAND and NOR function. However, the fan-out and fan-in of each block has to be optimised in order to achieve the hard giga-sample timing constraint. In this case, the load of each bit is determined by the load for bit c which requires three NAND/NOR blocks

logic block	occurrence
$\overline{\bar{a} \vee \bar{b}}$	4
$\overline{\bar{a} \& \bar{b}}$	4
$\overline{\bar{b} \vee \bar{c}}$	3
$\overline{\bar{b} \& \bar{c}}$	3
$\overline{\bar{c} \vee \bar{d}}$	4
$\overline{\bar{c} \& \bar{d}}$	4
c	1
d	8

Table 7.10: *First level logic*

(combinations (b,c),(c,c),(c,d)). Based on table 7.10, it can be concluded that each NAND/NOR block has to have a load of 4 NANDNOR blocks. Dummy blocks have been added for combinations where this is not the case, f.e. for $\overline{\bar{b} \vee \bar{c}}$. The intermediate level can be realised using a four input NANDNOR block. At this level also dummy cells have been used to create identical loads for all the logic combinations. In fig.7.32 a graphical representation of the decoder is given to illustrate its final architecture.

During the layout phase, an additional first order extraction and estimation of the parasitic load by the interconnects, has been performed at the critical points, which are the inter-logic level connections. This has resulted in a scaling of some sub-circuits and in additional drivers to deal with the increased capacitive load (at some points, the interconnect capacitance became as important as the parasitics of the active elements). It can be concluded, that the entire decoder is constructed based on only three functional levels. The layout of these blocks has been optimised towards area and symmetry in order to achieve a very high operation speed at a moderate power consumption.

For the binary part of the 10 bit converter, a dummy decoder has been inserted that is built up in the same manner as the thermometer decoder but that consists out of dummy cells that pass the value of the binary input bit directly to the latches. Using this method avoids any timing problems that could otherwise occur between the unary and the binary part of the chip.

7.4.1.4 The switching scheme

In this design, the current source of the unary array is divided into 4 current sources each delivering 8 times the LSB current. In every quadrant, a current source is placed based on a centroid scheme [Nakam JSSC91]. Since 31 current sources have to be

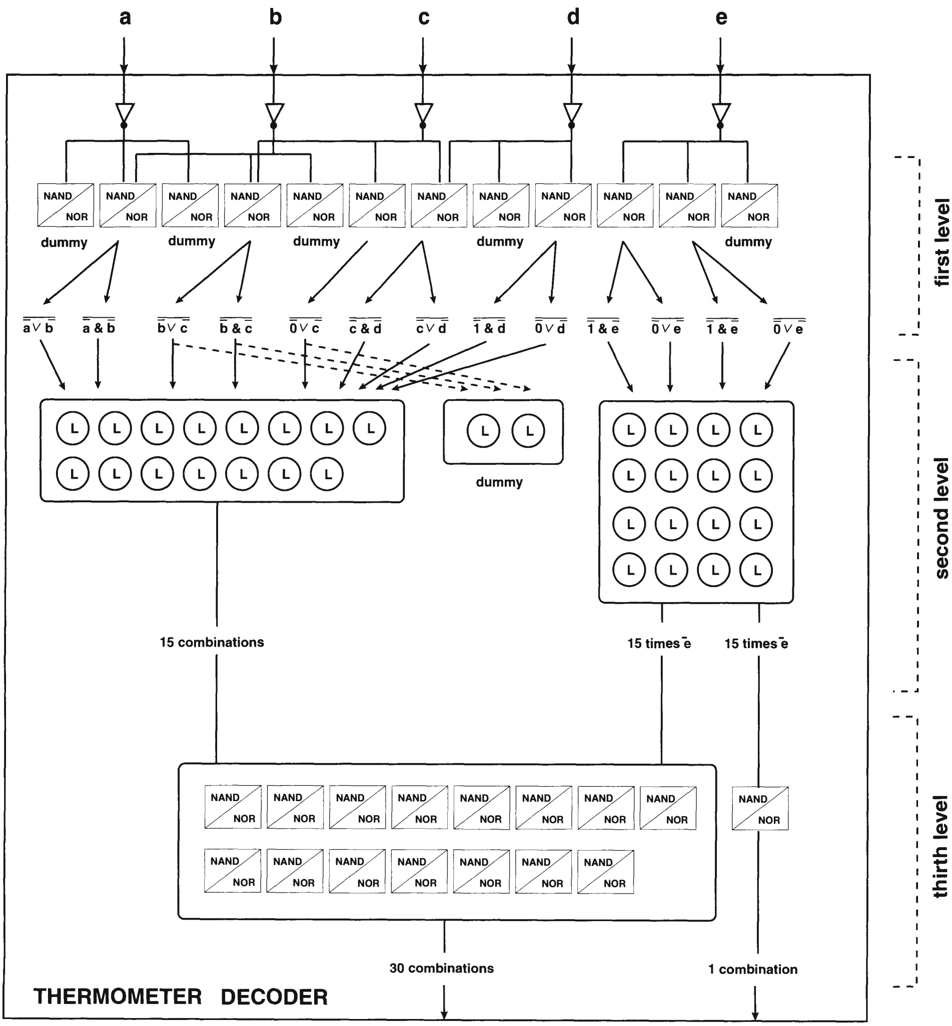


Figure 7.32: *Final architecture of the 5 bit thermometer decoder*

placed in each quadrant, a 6x6 array is used. The five remaining places per quadrant are occupied by the binary bits and the bias transistor Mo, which is also placed over the four quadrants to compensate for systematic errors. Four dummy rows and columns have been added as to avoid edge effects. A graphical representation of the double centroid structure of the current source array is given in fig.7.33. The shaded area represents the dummy cells. It is clear that the use of dummy current sources is at the cost of a considerable area increase. Simple mathematics show a 77% area increase when using a 16x16 array instead of an 12x12 implementation.

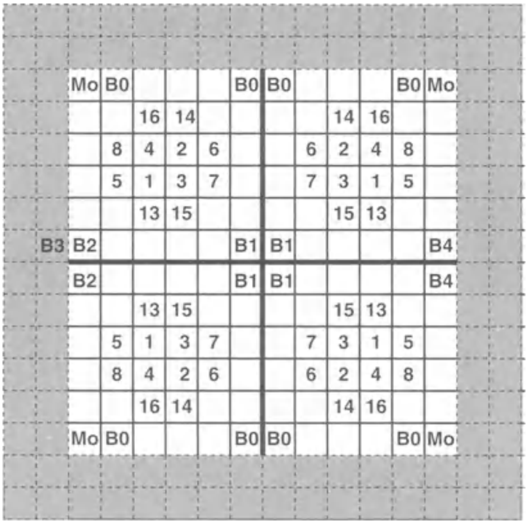


Figure 7.33: *The double centroid switching scheme*

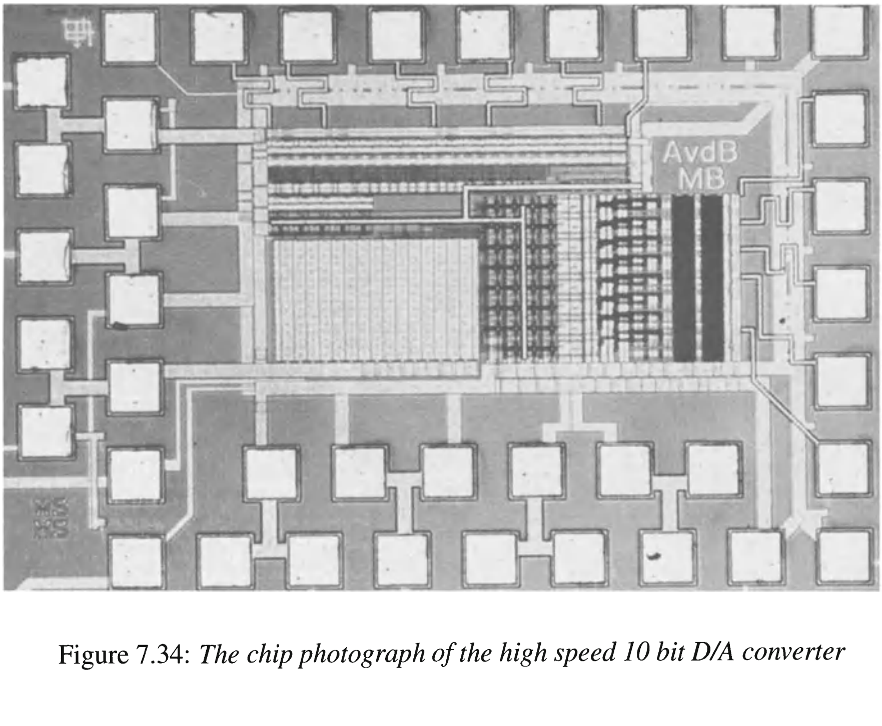


Figure 7.34: *The chip photograph of the high speed 10 bit D/A converter*

7.4.1.5 The layout

- The coupling between the digital and the analog part of the chip has been minimised. This is not only done by using different power supply lines but also by placing guard rings around the analog and the digital part of the chip and by using a separate array for the switches together with their drivers. Another advantage of these separate arrays is that the layout area of a unity cell in the current source array can be minimised. In this way the distances between the transistors are reduced resulting in improved matching properties.
- To reduce the voltage drop in the ground line of the current source transistors, wide supply lines are used. These are drawn on top of the transistors together with the interconnections needed to implement the switching scheme. In this way, a very compact current source array can be realised. Special care has been taken to realise a symmetrical interconnection array in order not to degrade the matching performance.
- To avoid any edge effects, the current source array has been expanded with a number of additional rows and columns.
- A multiple number of bonding pads is used at the output of the D/A converter as to lower the inductance of the wire bonding and as a result to minimise any ringing effects that could otherwise occur.
- Wherever possible, all interconnections have been made identical. In this way, no timing and/or load differences have been introduced.

Throughout the whole design, layout parasitics have been taken into account. They have been manually extracted at each critical node and iterated in the electrical simulations. Symmetry was not only introduced for the interconnections between any of the digital logic blocks, but in the whole layout (for e.g. the interconnection between the switch driver and the switch transistors). Dummy switch transistors are used to match the load of the latch for the binary bits. Another solution would be to scale the binary latches, proportional to the load. But as it is extremely difficult to scale the parasitics of the interconnections in the same way, the dummy load topology has been chosen. Much attention has been paid at the final layout of the chip, resulting in a very compact chip. The presented 10 bit D/A converter has an active area of only 0.35 mm^2. Fig.7.34 shows a chip photograph of the realised design.

7.4.1.6 Measurement results

7.4.1.6.1 Static measurements All measurements have been performed on a single ended 50Ω double terminated output. The analog voltage supply is 3 V, while the digital part of the chip operates at only 1.9 V. Unless otherwise stated, the full scale output current is 16mA in all measurements. Fig.7.35.a shows the measured INL profile versus the input code and fig.7.35.b shows the measured DNL profile of the 10 bit CMOS D/A converter. The INL error is smaller than 0.2 LSB while the DNL error equals 0.14 LSB which proves the intrinsic 10 bit accuracy. Both the INL and the DNL error specification have been measured for several values of the bias current of the D/A converter.

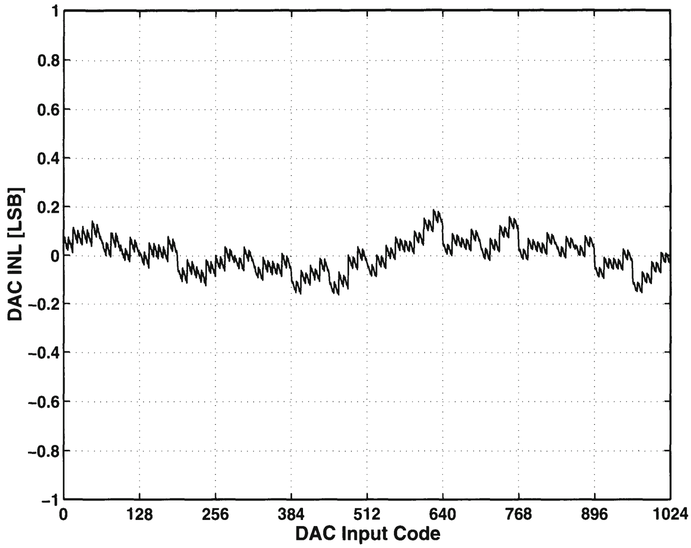
In fig.7.36, it is shown that the INL error stays smaller than 0.5 LSB for bias currents as low as 1 mA (=full scale current) and as large as 27 mA. It also shows that the performance improves with increasing current. This is evident since the relative current accuracy improves with increasing $V_{GS} - V_T$. For full scale currents larger than 4 mA, the INL error is even smaller than 0.25 LSB.

7.4.1.6.2 Dynamic measurements The single tone output spectrum has been measured for several values of the signal frequency and of the update rate up to 1 GS/s. For all the measurements, a SFDR better than 61 dB has been measured in the interval from DC to the Nyquist frequency. An overview of the measured dynamic results for a 100 MS/s and a 1 GS/s clock are given in fig.7.37. On the horizontal axis, the relative output frequency is plotted. This relative output frequency is defined as the ratio between the single tones fundamental frequency and the clock frequency. According to the Nyquist theorem the maximum x-axis value is 0.5. In this way, e.g. the 0.1 value on the x axis corresponds to respectively a 10 MHz (for a 100 MS/s clock) and a 100 MHz (for a 1 GS/s) output signal.

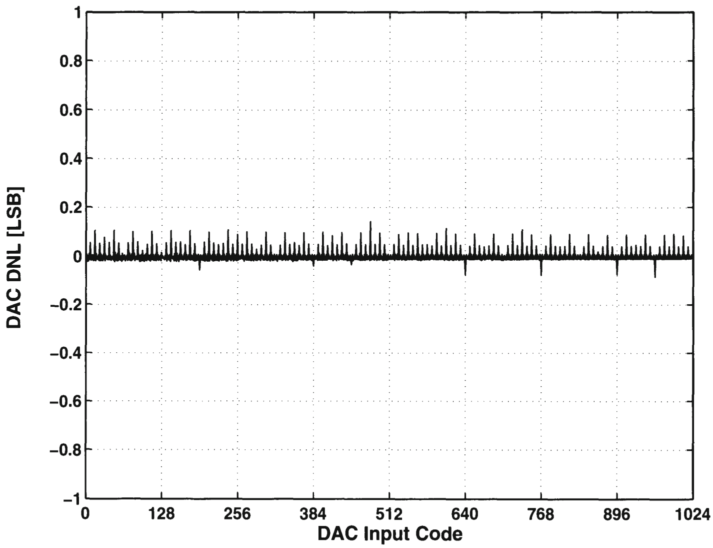
Fig.7.38.a and fig.7.38.b show some typical measured output spectra for the chip operating at a 1 GS/s clock rate for a 100MHz and for a near Nyquist full scale sinusoidal input signal frequency.

The SFDR has also been measured as a function of the bias current. Fig.7.39 shows the measurement results for a 1 GS/s clock speed and a 100 MHz output signal. For all full scale currents between 2 and 20 mA, the measured SFDR was better than 60 dB. For full scale currents larger than 16 mA , the measured SFDR is even better than 70 dB.

High speed D/A converters used in wideband transmitter applications require a wide dynamic range since they have to deal with multiple channels. In this case, a multi-tone SFDR plot can give extra information on the performance capability of the D/A converter. Most multi-tone testing consists of generating a series of equally spaced tones having equal amplitude at a specified update rate depending on the application. Due to the inherent D/A converters non-linearity, various spurs will appear in the spectral plot caused by intermodulation products. In fig.7.40, a dual tone SFDR measurement is shown. Two 8 mA sinusoidal signals around 105 MHz with a 5 MHz

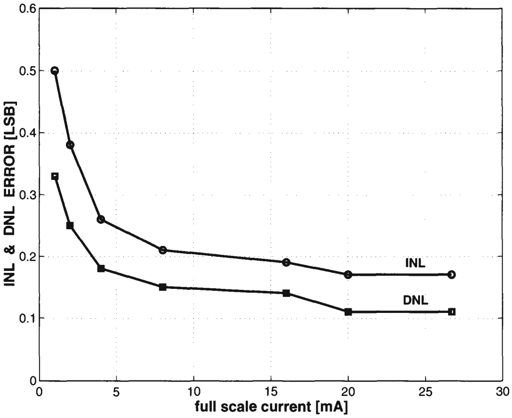


(a) The measured INL performance of the 10 bit D/A converter

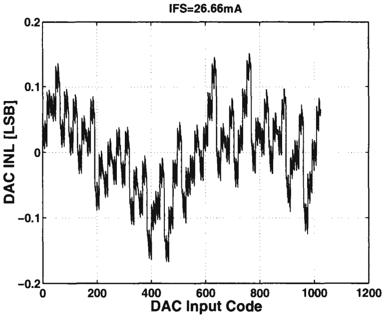


(b) The measured DNL performance of the 10 bit D/A converter

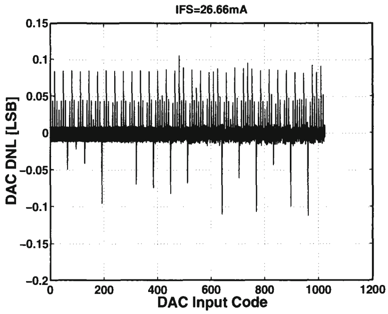
Figure 7.35: The static performance of the 10 bit D/A converter



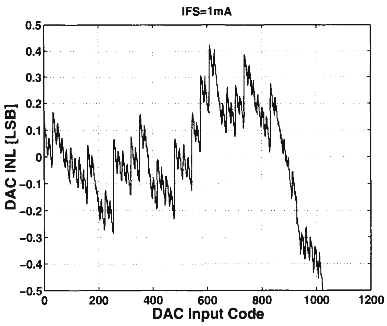
(a) The measured INL and DNL versus the full-scale current



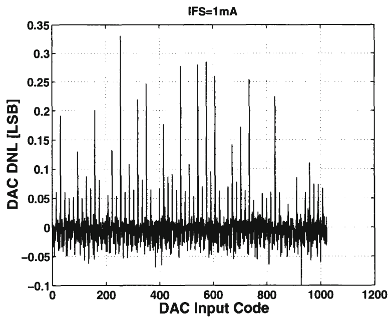
(b) The INL profile for $I_{FS} = 27mA$



(c) The DNL profile for $I_{FS} = 27mA$



(d) The INL profile for $I_{FS} = 1mA$



(e) The DNL profile for $I_{FS} = 1mA$

Figure 7.36: The INL and DNL error as a function of the full scale current for the 10 bit D/A converter

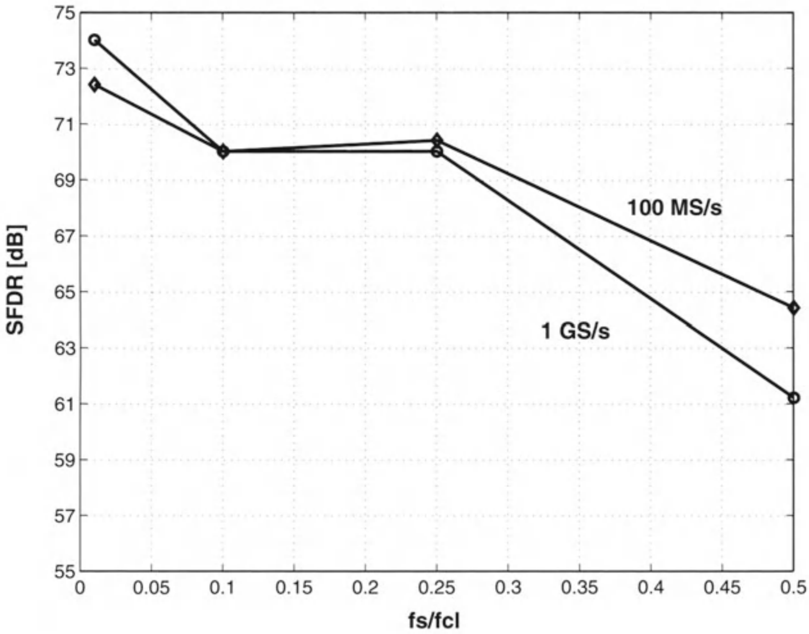


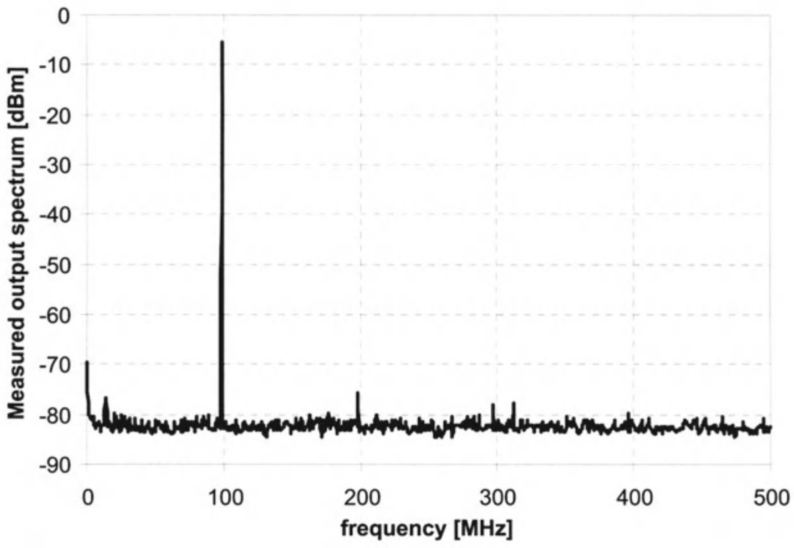
Figure 7.37: *The measured SFDR for a conversion rate of 100 MS/s and 1GS/s*

spacing have been applied to the D/A converter at an update rate of 1 GS/s. The measured SFDR equals 65 dB.

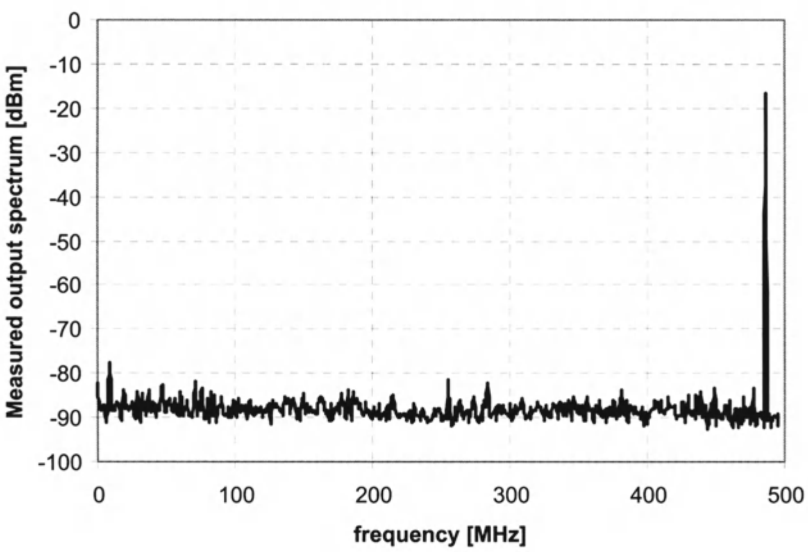
The power consumption has been measured for a 490 MHz input signal at a sample frequency of 1 GS/s. The digital part of the chip (the regenerative buffers at the input, the clock driver, the two decoders and the switch driver) consumes only 62 mW from a 1.9 V power supply, while the analog part consumes between 6 and 60 mW for a full scale output current between 2 and 20 mA. For a 16 mA full scale current, for which most of the measurements were done, the total power consumption equals 110 mW.

7.4.1.7 Conclusion

This D/A converter achieves a state-of-the-art static and dynamic performance. Measurements indicate that this chip has an intrinsic accuracy of 10 bit and a SFDR higher than 60 dB over the Nyquist frequency range up to a 1 GS/s update rate. Table 7.11 gives an overview of the realised performance. This is to the author's knowledge the first current steering D/A converter that is published in literature [VdBos CICC00] which achieves such a broadband dynamic performance.



(a) The output spectrum for a 100 MHz signal at a 1GS/s clock



(b) The output spectrum for a 490 MHz signal at a 1GS/s clock

Figure 7.38: Some measured output spectra of the 10 bit D/A converter

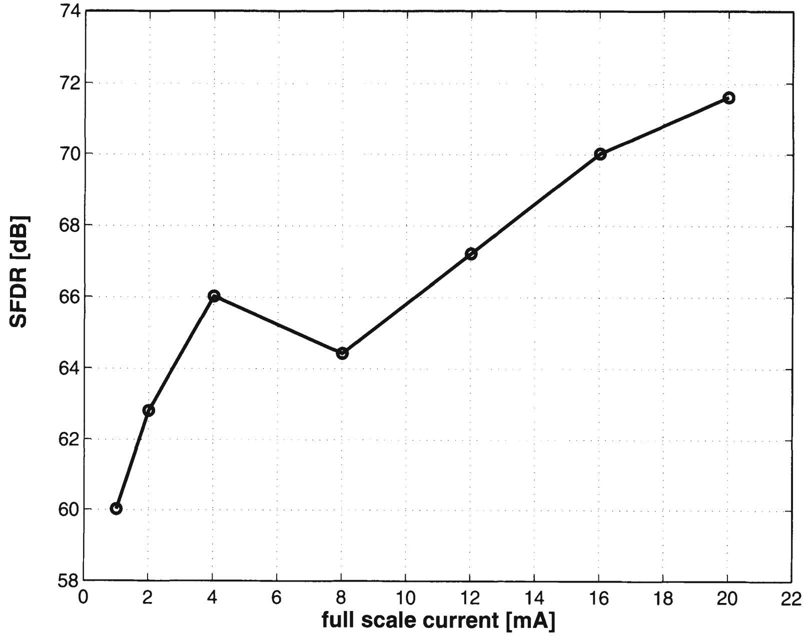


Figure 7.39: *The measured SFDR versus the full scale current for a signal of 100 MHz at a 1GS/s update rate*

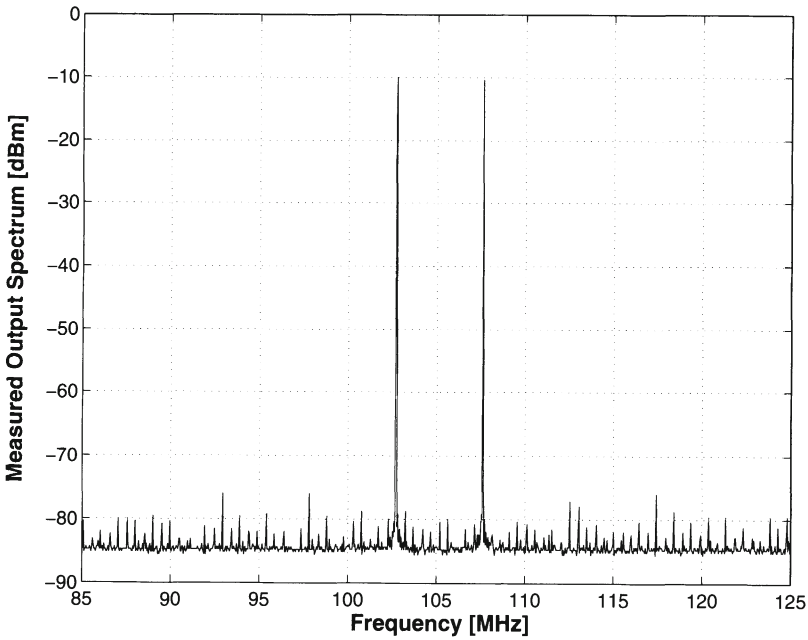


Figure 7.40: *A dual tone spectrum measured at a clock rate of 1GSample/s*

resolution	10 bit
update rate	1000 MS/s
INL error	0.2 LSB
DNL error	0.14 LSB
SFDR (50MHz@100MS/s)	64 dB
SFDR (500MHz@1GS/s)	61 dB
power consumption	110 mW
active area	0.35 mm^2
process	0.35 μm

Table 7.11: *Summary of the 10 bit DAC performance*

A_{VT}	8.9 $mV\mu m$
A_β	1.9 $\%\mu m$
$\sigma(I)/I$	0.25 %
$(V_{GS} - V_T)_{cs}$	1 V
I_{FS}	20 mA
segmentation	5-7
$(W/L)_{cs}$	$1.8\mu m/30\mu m$
$(W/L)_{sw}$	$0.5\mu m/0.7\mu m$
$(W/L)_{cas}$	$0.5\mu m/1.4\mu m$
technology	$0.35\mu m$

Table 7.12: *The unit current cell specifications for the 12 bit DAC*

7.4.2 A 12-bit 500-MS/s Current-Steering CMOS D/A Converter

7.4.2.1 The Floorplan

Fig.7.41 shows the floorplan of the 12 bit 500 MS/s CMOS D/A converter. The 5 MSBs are converted using the unary approach while the 7 LSBs are converted using the binary approach. Using this architecture, a trade-off between a good static and dynamic performance and a moderate power/complexity of the D/A converter has been achieved.

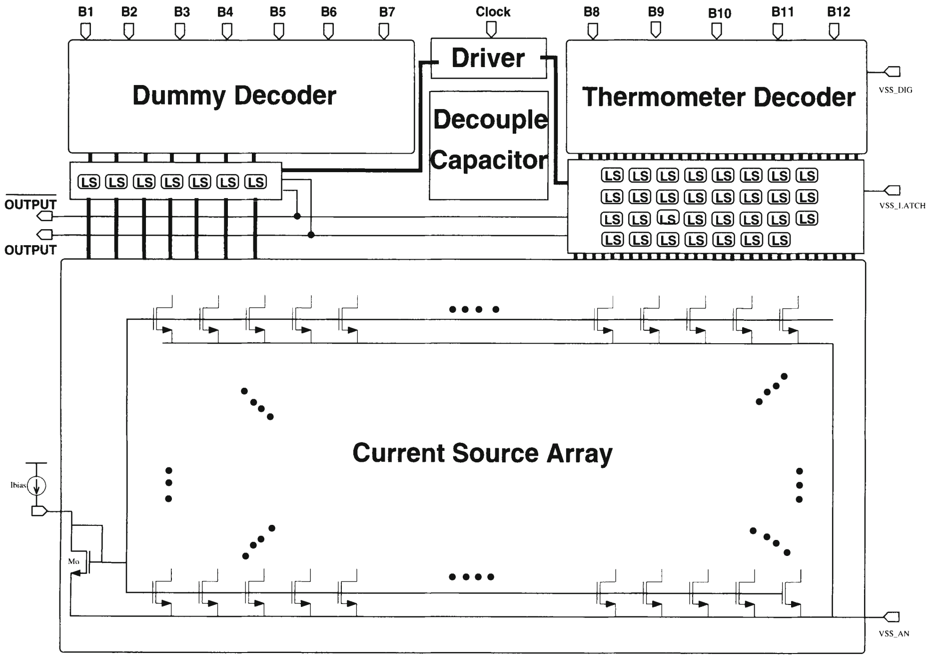


Figure 7.41: *The floorplan of the 12 bit D/A converter (LS = latch and switch)*

7.4.2.2 Design of the swatch cell and the thermometer decoder

Based on the combination of a 99.7% yield specification for the D/A converter and the transistor mismatch equations [Pelgr JSSC89], the dimensions of the unity current source have been determined (W=1.8 μm, L=30 μm). An overview of the dimensions of the unit current cell is given in table 7.12. The dynamic performance of the D/A converter has been obtained by the use of a well designed synchronised switch driver and by a careful design of the DAC's output impedance as to minimise any non-linearity caused by its frequency dependent value. To obtain a second order harmonic distortion that is better than 72 dB, the required output impedance of the D/A converter has to be larger than 100 $M\Omega$ over the frequency area of interest (here 250 MHz). To achieve such a hard requirement, the frequency dependency of the output impedance has been optimised by using a cascoded current source (cfr. chapter 5). Fig.7.42 shows the basic schematic of the current cell and the switch driver. To achieve the very stringent impedance requirement, the cascode transistors have twice the minimal gate-length, as a result of the design trade-off between the resistive output impedance (required large value) of the D/A converter and the parasitic capacitance at the internal node (required low value).

The same thermometer decoder architecture has been chosen as for the 10 bit design. The NAND/NOR circuits of the last logic level have been adjusted to drive

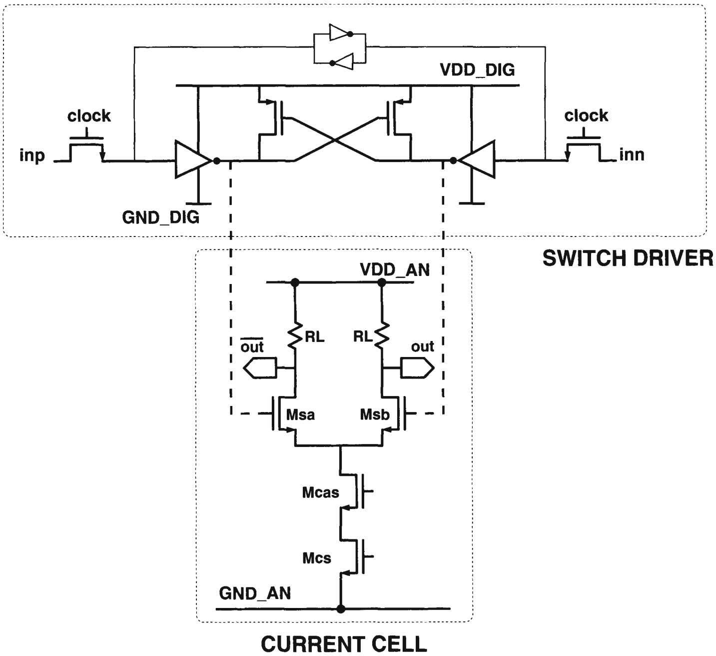


Figure 7.42: *The current cell and the switch driver circuit*

a higher load. Furthermore, symmetry has been fully exploited both at the logical design level as at the final layout stage. A dummy decoder has been inserted for the binary bits as to avoid timing problems.

7.4.2.3 The switching scheme

Apart from the random matching errors, the systematic errors caused by technological, electrical and temperature gradients over the die have been compensated by the implementation of a special triple centroid switching scheme. Since the first 7 LSBs are implemented in a binary way, the value of the unary current source equals 128 times the LSB current (I_{LSB}). This unary current source has been split up into 16 current sources with a value of $8{*}I_{LSB}$. The current source array has been divided into 16 squares and the current sources are placed symmetrically around the center of each square as is indicated in fig. 7.43. As a result, any two dimensional symmetrical and/or graded error is fully compensated. Four additional dummy rows and columns have been added to create identical surroundings for the current sources situated at the edge of the current source array.

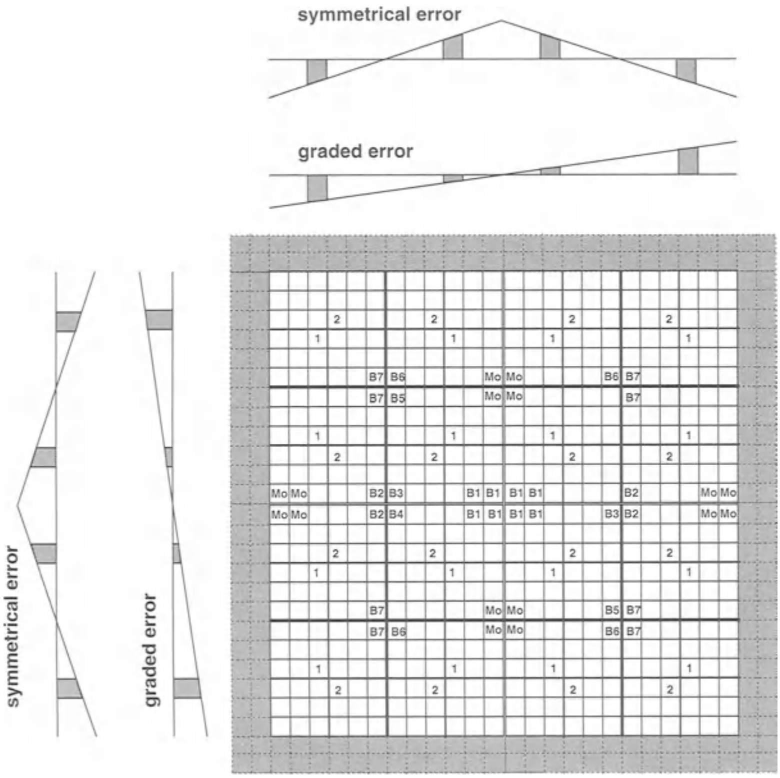


Figure 7.43: *The triple centroid switching scheme*

7.4.2.4 The layout

The chip has been realised in a single-poly five-metal layer standard 0.35 μm CMOS technology. The analog circuits are isolated from the noise generated by the digital part of the chip by the use of guard rings, separate power supplies and by placing the latches in a separate array from the current sources. Furthermore, all the interconnections between the different building blocks of the thermometer decoder have been made identical and have been put on top of the blocks. The same principle has been used for the analog part of the chip. The routing needed for realising the switching scheme has been done on top of the current source array. In this way, the additional silicon area used for interconnect purposes has been strongly reduced, resulting in a very compact chip. The total active area of the 12 bit D/A converter is only 1 mm^2. Fig.7.44 shows a chip photograph.

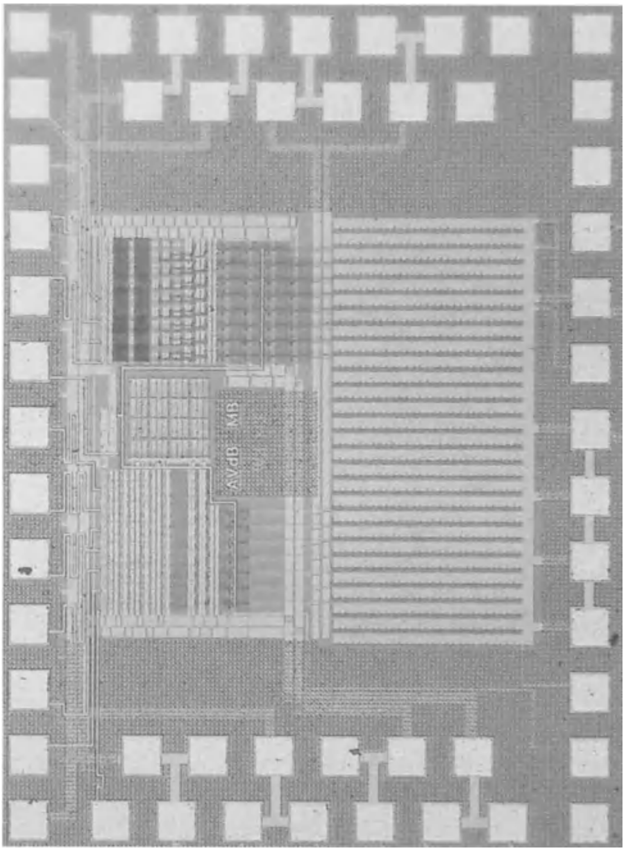
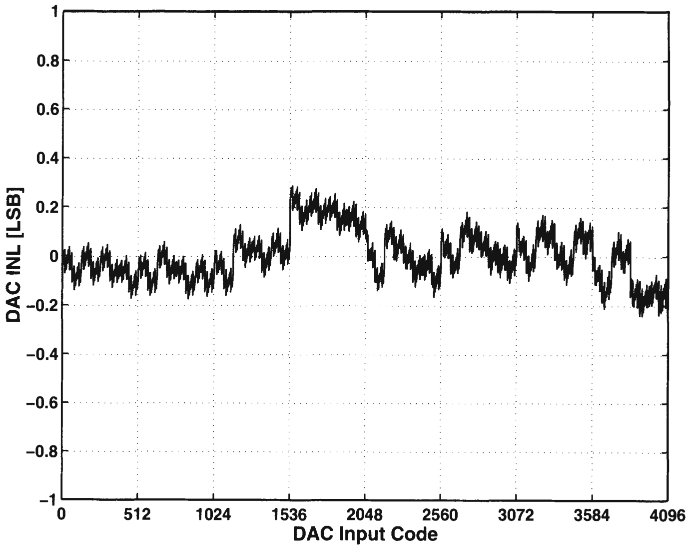


Figure 7.44: *The chip photograph of the 12 bit D/A converter*

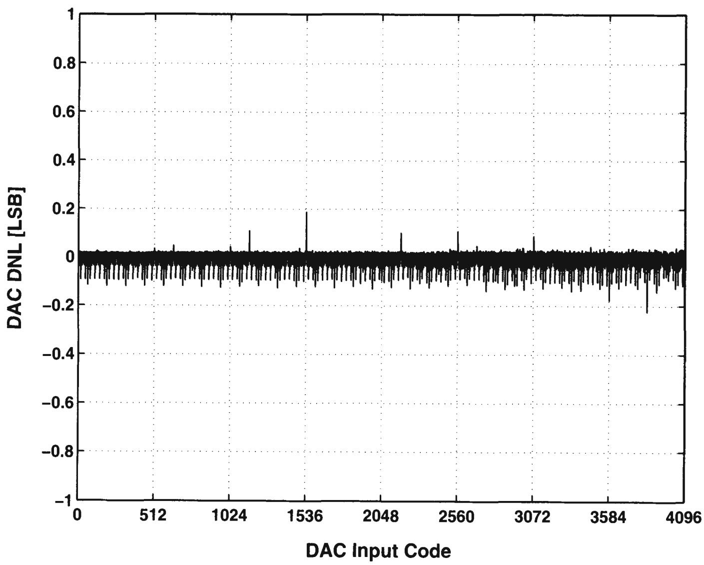
7.4.2.5 Measurement results

7.4.2.5.1 Static measurements All measurements are single ended and have been performed with a 3 V analog power supply and a 2.2 V digital power supply. The load resistor equals 25Ω. Fig.7.45.a and fig.7.45.b show the measured INL and DNL profile of the 12 bit D/A converter. The INL error is better than 0.3 LSB proving the 12-bit accuracy. The DNL error is better than 0.25 LSB.

7.4.2.5.2 Dynamic measurements The dynamic performance of the presented D/A converter is summarised in fig.7.46 and fig.7.47. In fig.7.46, the SFDR for different update rates as a function of the input signal frequency is given. At an update rate of 500 MS/s, the SFDR is still 62 dB for a sinusoidal input signal of 125 MHz. The dynamic performance of the presented D/A converter is better than all recently in open literature published 12 bit current-steering D/A converters [Basto CICC96,



(a) The measured INL performance of the 12 bit D/A converter



(b) The measured DNL performance of the 12 bit D/A converter

Figure 7.45: The static performance of the 12 bit D/A converter

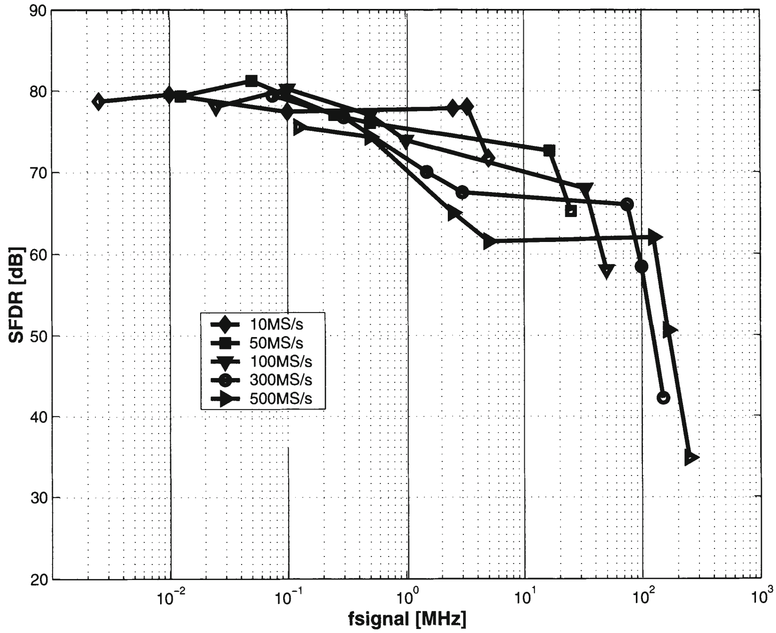
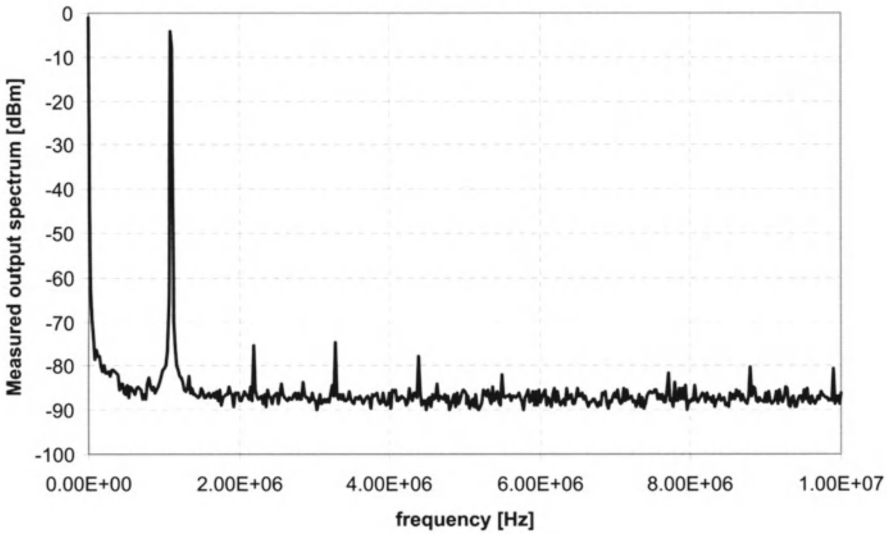


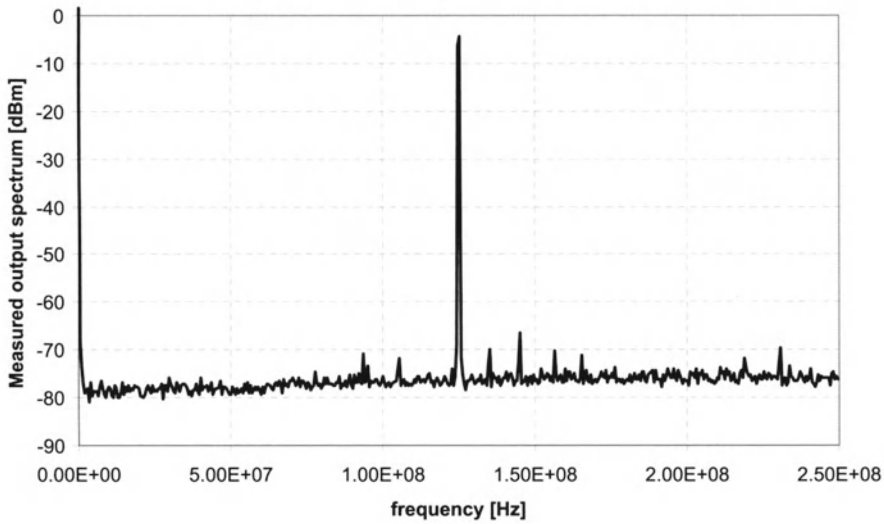
Figure 7.46: *The SFDR as a function of f_{signal} for different update rates*

VdBos CICC98] (to the author's knowledge). Fig.7.47.a and fig.7.47.b show the measured output spectrum for a 1 MHz and a 125 MHz sinusoidal output signal at an update rate of 500 MS/s. The SFDR for the 1 MHz signal equals 70.5 dB while the SFDR is still 62 dB for a 125 MHz output signal.

To give a more complete image of the dynamic performance of the presented D/A converter in comparison with existing 12 bit designs, fig.7.48 is given. Fig.7.48.a shows the SFDR in function of the update rate for a 1 MHz output signal. The measurement results of [Marqu ISSCC98] are also indicated on the same figure. It can be clearly seen that the SFDR for the 1 MHz output signal remains above 70 dB up to a 600 MS/s update rate for the presented D/A converter where previous designs reach this limit for update rates around 300 MS/s . Fig.7.48.b shows the SFDR in function of the output signal for an update rate of 300 MS/s in order to make a further comparison with [Marqu ISSCC98] possible. This figure clearly shows the superior dynamic performance of the presented D/A converter. In fig.7.49, the maximal reported SFDR is shown in function of the input signal frequency for different 12 bit designs. It can be concluded from this figure that the presented chip has a dynamic performance competitive with the best chips on the market [datasheet AD9753, datasheet AD9752, datasheet AD9765] for output signals up to 50 MHz. The power consumption has been measured for a near Nyquist full-scale sinusoidal output signal at a 500 MS/s update rate and equals 110 mW. The digital part of the chip consumes 63 mW.

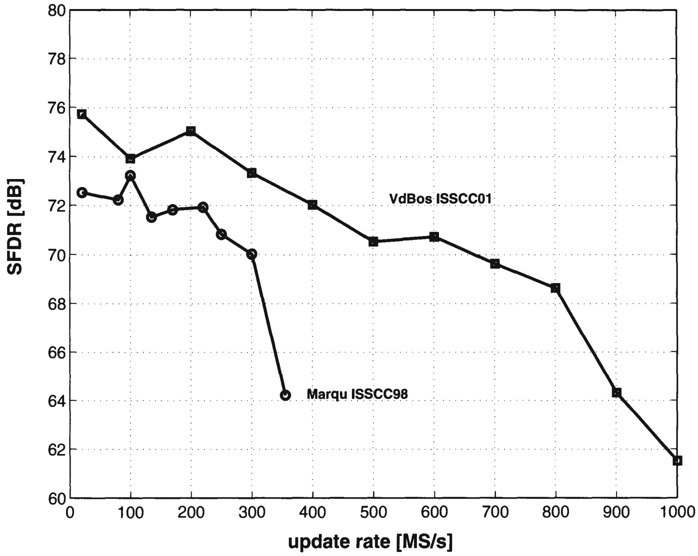


(a) The measured output spectra for a 1MHz output signal at a 500MS/s update rate

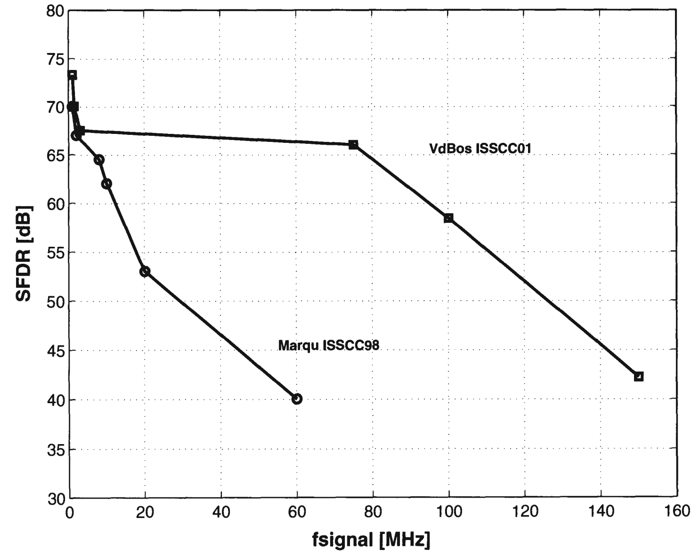


(b) The measured output spectra for a 125MHz output signal at a 500MS/s update rate

Figure 7.47: Some measurement spectra of the 12 bit D/A converter



(a) The SFDR (@1MHz output) versus the update rate



(b) The SFDR (@300MS/s clock) versus the output signal

Figure 7.48: The SFDR as a function of the update rate and the input signal frequency

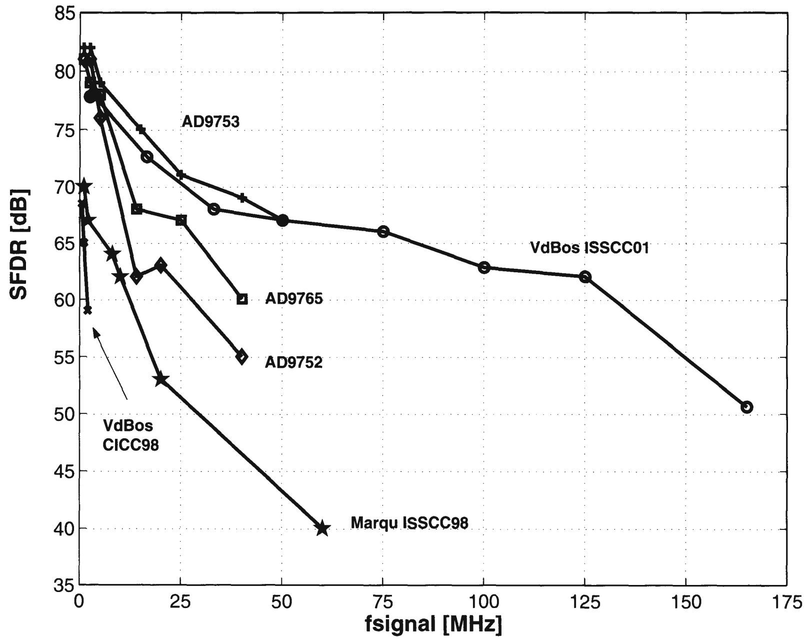


Figure 7.49: *The maximum reported SFDR as a function of the output signal for different 12 bit designs*

7.4.2.6 Conclusion

This D/A converter together with the 10 bit design of the previous section validates the fact that the dynamic behaviour of a current steering D/A converter is dependent on its output impedance. The constraint imposed by this impedance becomes more difficult to achieve for higher resolutions. However, this design proves that the combination of a high resolution and a good frequency domain performance at a high update rate is no longer an illusion. The measurement results are summarised in table 7.13.

7.5 Low Power High Speed D/A Converters

7.5.1 Introduction

Typical low power D/A converters, available in open literature and/or in major vendors' data sheets, are only functional for output signals up to at most a few 100 kHz. On the other hand, high speed converters have been reported that are able to generate linear output signals in the MHz range. However, these realisations have the main drawback that they consume a considerable amount of power. The presented D/A

resolution	12 bit
update rate	500 MS/s
INL error	0.3 LSB
DNL error	0.25 LSB
SFDR (125MHz@500MS/s)	62 dB
SFDR (1MHz@500MS/s)	70 dB
power consumption	110 mW
active area	1 mm^2
process	0.35 μm

Table 7.13: *Summary of the 12 bit DAC performance*

converter offers a unique combination of a linear high speed output signal and a low power consumption. This is achieved by using a fully binary implementation.

7.5.2 A Low Power 10 bit fully binary D/A Converter

7.5.2.1 Floorplan

A schematic overview of the data converter chip is shown in fig.7.50. It shows the cascoded current source matrix, the differential switches and the synchronising switch drivers. Also an on-chip clock driver was implemented. The 10 bit D/A converter has a fully binary topology. No thermometer decoder had to be implemented leading to a considerable area and power consumption decrease.

7.5.2.2 Design of the swatch cell

Since this D/A converter has been implemented in the same standard $0.35\mu m$ CMOS technology as the 1 GS/s 10 bit D/A converter, the dimensions of the current source transistor are the same ($W = 2\mu m; L = 8\mu m$). Also the same driver topology has been used. An overview of the dimensions of the unit current cell is repeated in table 7.14.

7.5.2.3 The switching scheme

In this design, all binary current sources are implemented as a number of parallel unity current sources with common drain interconnection. Thanks to the symmetrical

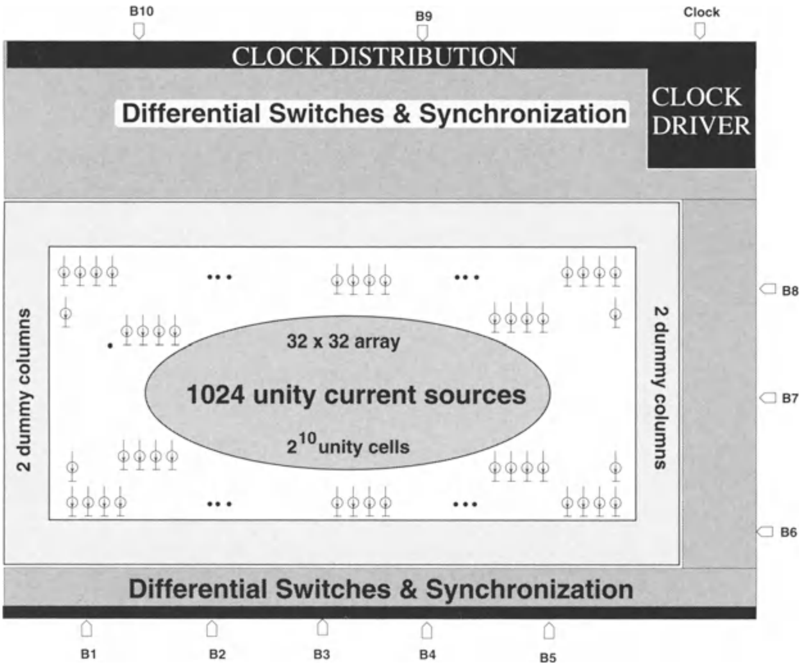


Figure 7.50: *Topology of the realized 10 bit binary DAC*

A_{VT}	$8.9\ mV\mu m$
A_β	$1.9\ \%\mu m$
$\sigma(I)/I$	0.5 %
$(V_{GS} - V_T)_{cs}$	1 V
I_{FS}	20 mA
segmentation	0-10
$(W/L)_{cs}$	$2\mu m/8\mu m$
$(W/L)_{cas}$	$1\mu m/0.7\mu m$
$(W/L)_{sw}$	$0.5\mu m/0.35\mu m$
technology	$0.35\mu m$

Table 7.14: *The unit current cell specifications for the 10 bit DAC*

current source pattern, the systematic effects have been minimised. Fig.7.51 shows the switching scheme for the implemented chip. In both the horizontal and the vertical direction, additional dummy current source transistors have been placed to provide identical surroundings for the current sources at the edge of the array (the shaded area). It can be noticed that the insertion of the dummy rows and columns implies a 50 % area increase of the current source array (42x36 versus 32x32).

7.5.2.4 The Layout

The chip microphotograph is shown in fig.7.52. The chip has been implemented in a standard 0.35 μm CMOS process. Due to the fully binary architecture of the D/A converter, the layout has proven to be less complex than the previous ones. However, also here the decoupling between the noise generated by the digital part of the chip and the analog part is essential. In this design symmetry has been exploited wherever possible. Dummy loads have been placed at the driver level for the LSB's. In order to minimise the timing inaccuracies internally in the synchronising drivers, the MSB drivers consist of a parallel sequence of less significant bit drivers.

7.5.2.5 Measurement Results

7.5.2.5.1 Static measurements All measurements have been performed on a single-ended output. A 2.7 V power supply has been used for the current cell and for the clock driver. The load resistor equals 25Ω. The synchronising circuits before the differential switches operate at a reduced 1.3 V in order to optimise the dynamic performance. For a 30 MS/s near Nyquist operation, the synchronising latches consume 66 μA and 342 μA is used for the clock driver, resulting in a total digital power consumption as low as 1 mW. The analog full scale output current (FSOC) is 2.5 mA, resulting in a maximal 7.8 mW total power consumption (FSOC*2.7 V+ 1 mW) at the 30 MS/s Nyquist operation. Five chips have been measured. For all the samples, both the INL and DNL error were better than 0.5 LSB (fig.7.53 and fig.7.54).

7.5.2.5.2 Dynamic measurements The dynamic performance of a current steering D/A converter is dependent on the amount of segmentation. The dynamic error due to the switching operation is proportional to the number of simultaneous switching current sources. Also the performance sensitivity to the timing accuracy of the switch control signals increases with lower segmentation. These elements indicate that a design trade-off exists between a high SFDR, a low area and a low power consumption. Due to a detailed analysis of the non linearity effects for current steering D/A converters and an appropriate switch driver design, good dynamic specifications have been achieved.

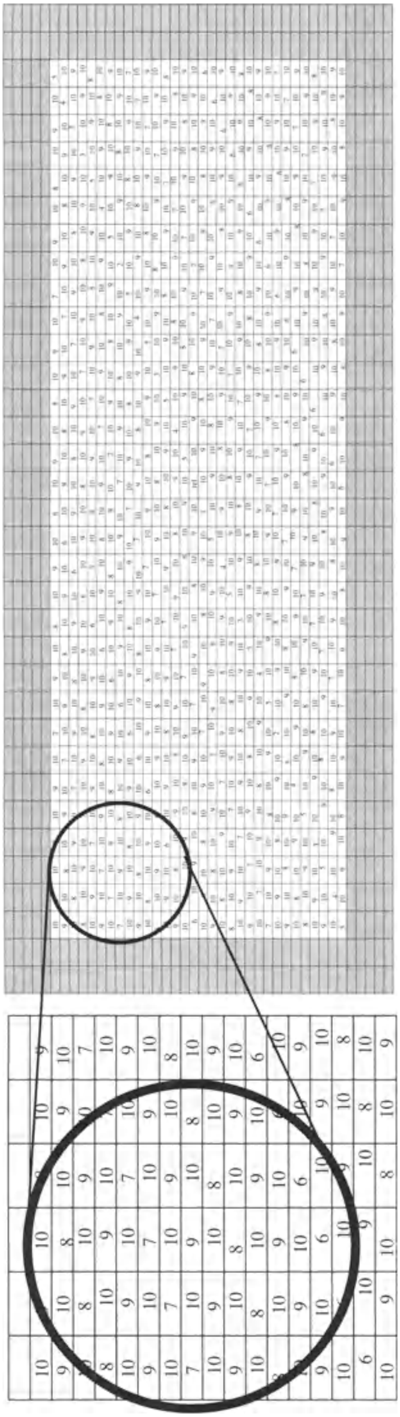


Figure 7.51: *The current source pattern*

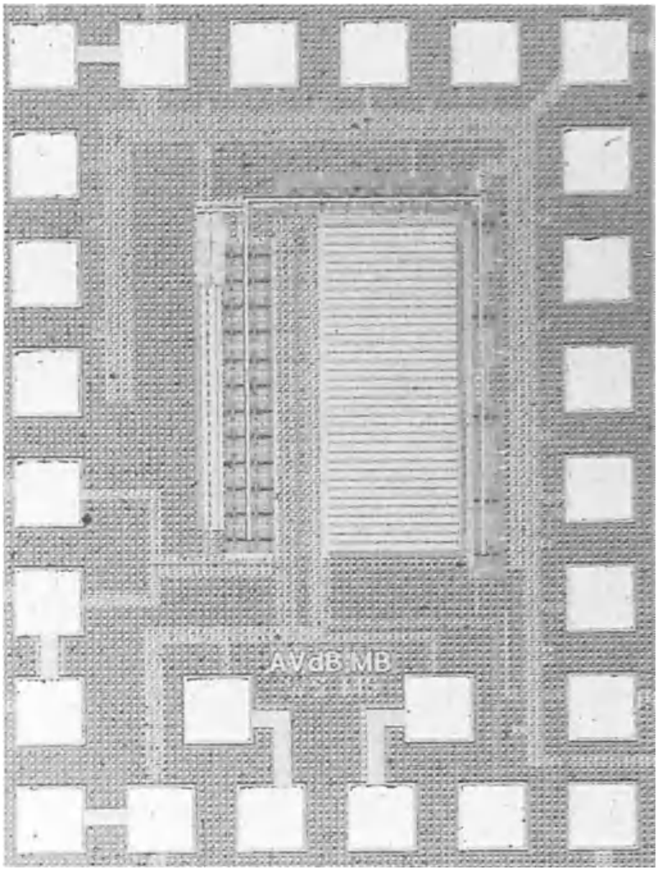


Figure 7.52: *The chip photograph of the 10 bit D/A converter*

The measured SFDR at a 30 MS/s and at a 50 MS/s update rate is shown in fig.7.55.a. It indicates that a better than 60 dB SFDR is achieved for all input signal frequencies up to a 15 MHz. To relax the system specifications on e.g. filters in the signal path, it can be beneficial to use a higher clock speed. Fig.7.55.b shows the measured SFDR for a 1 MHz signal for various values of the update rate. It shows that more than 60 dB is achieved up to an update rate of 800 MS/s. Fig.7.56 shows some typical measured output spectra for a Nyquist, a Nyquist/2 and a Nyquist/50 input signal frequency at a 30 MS/s update rate. They all achieve more than 60 dB SFDR over the whole Nyquist frequency band.

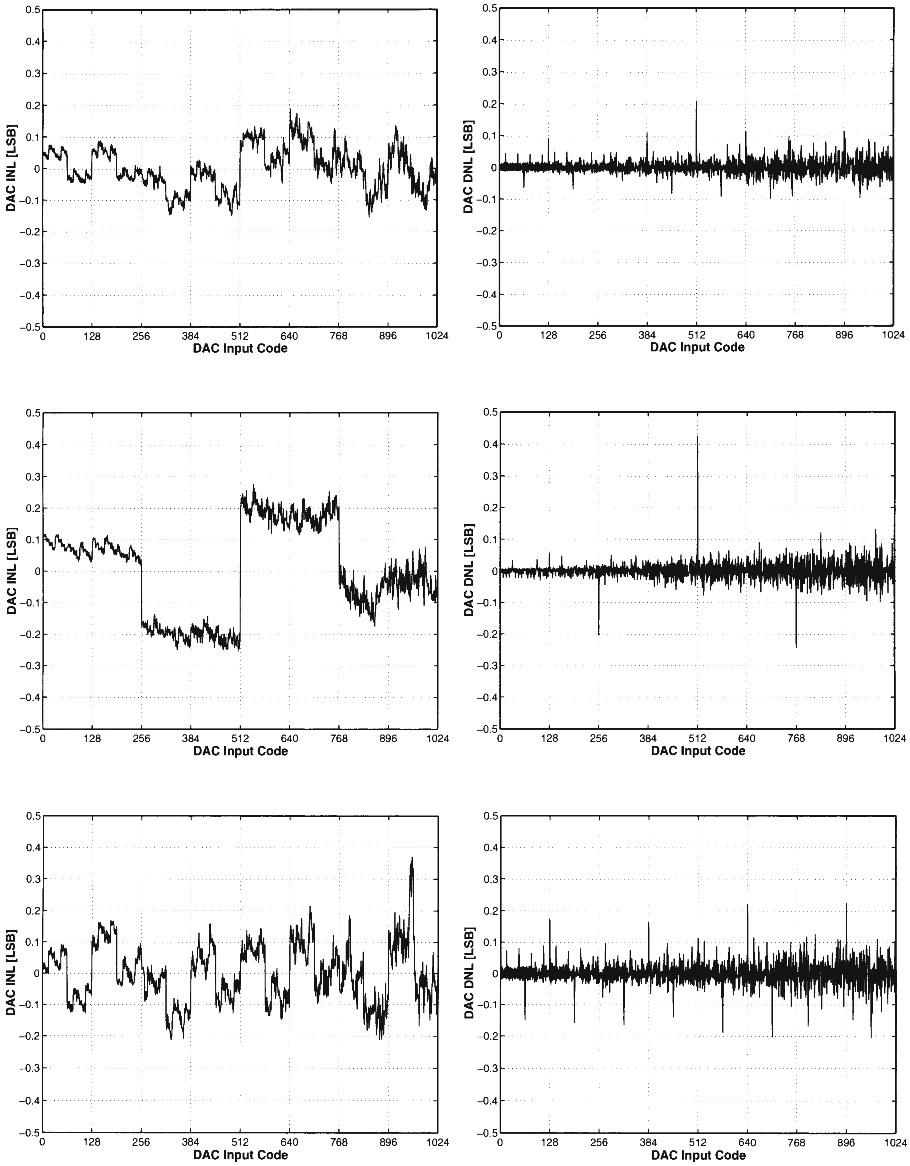


Figure 7.53: The first 3 samples of the binary 10 bit D/A converter

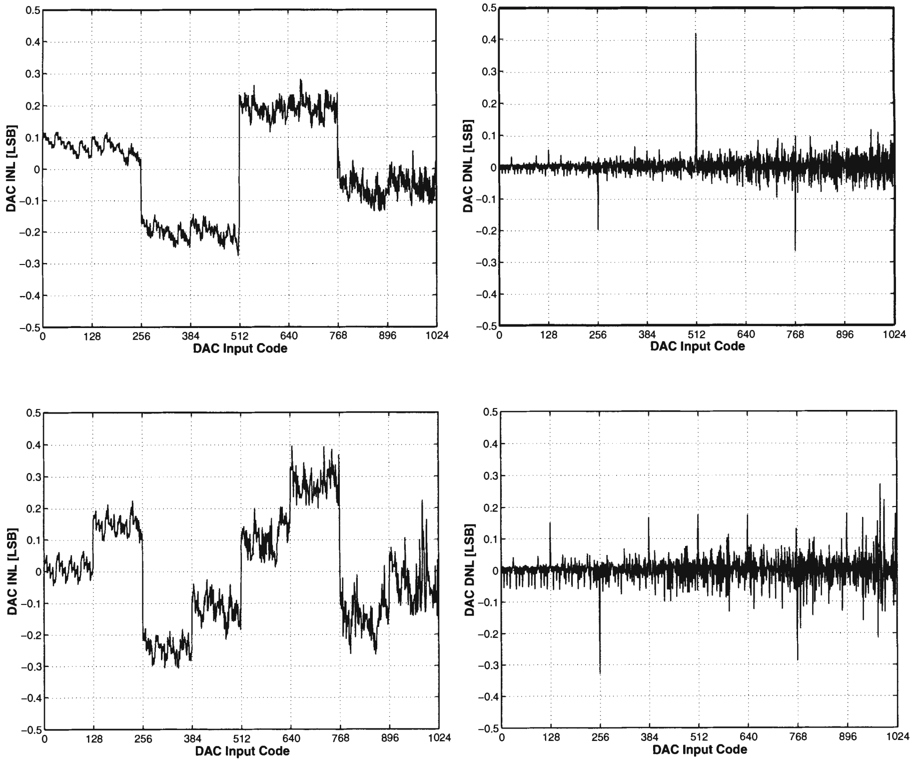
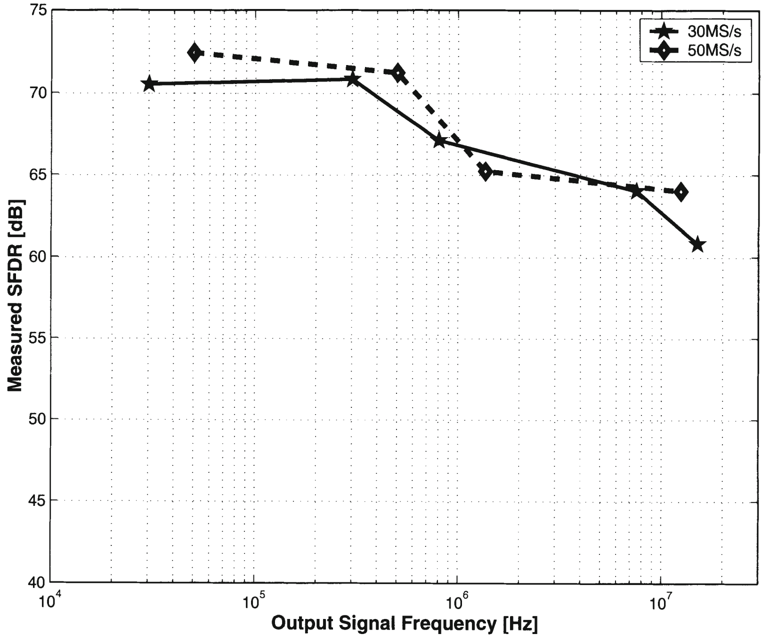


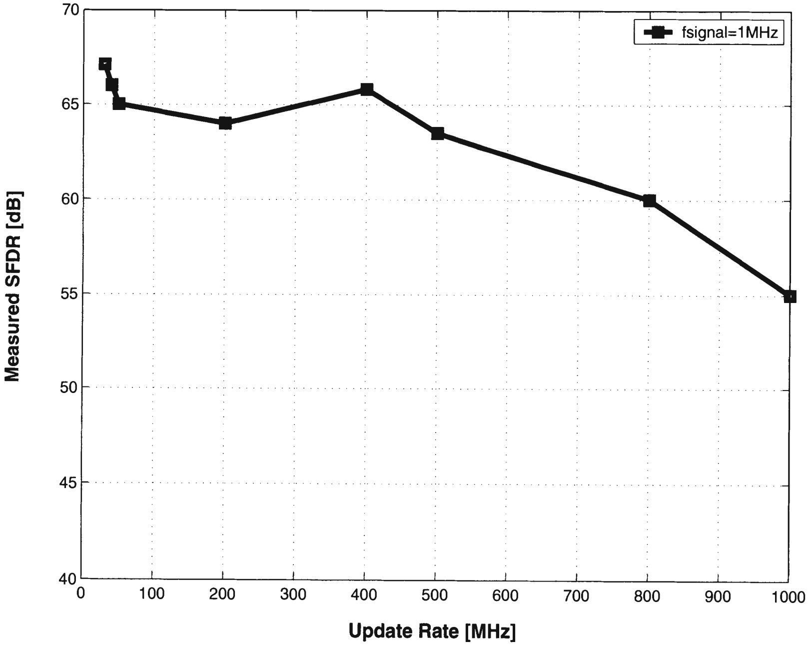
Figure 7.54: The last 2 samples of the binary 10 bit D/A converter

7.5.2.6 Conclusion

To prove that it is fundamentally possible to achieve state-of-the-art specifications at the extreme low segmentation side of the design space, the realised chip has a fully binary topology. Major important advantages of this binary implementation are the design simplicity (very fast design cycle), the reduced power consumption and the small silicon area. Measurements prove that this 10 bit D/A converter can be used in many telecommunication systems since it combines a good dynamic performance in the frequency range of several MHz (up to 15 MHz) and a low power consumption (less than 8 mW). It is the author's belief that better dynamic specifications could have been achieved if the layout of the clock interconnections had been optimised. In this design unfortunately, no clock tree has been used and as a consequence significant timing mismatches occur between the different driver circuits that normally solve the synchronisation issues for the switch control signals. An overview of the realised performance is given in table 7.15.



(a) SFDR versus signal frequency at a 30MS/s and a 50MS/s update rate



(b) The SFDR for a fixed output signal of 1MHz versus the update rate

Figure 7.55: The dynamic performance of the binary 10 bit DAC

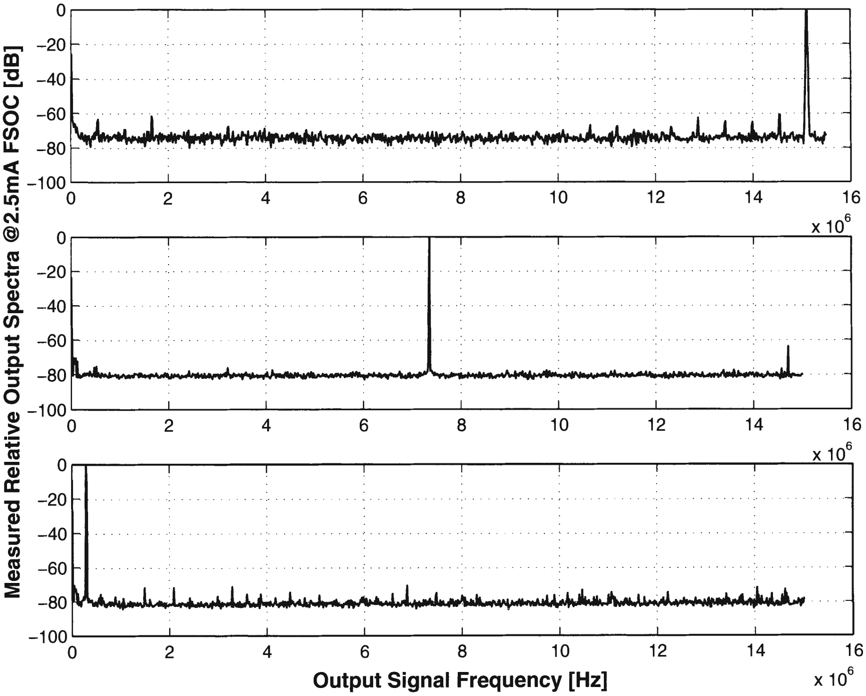


Figure 7.56: *Typical measured output spectra*

resolution	10 bit
update rate	30 MS/s
INL error	0.2 LSB
DNL error	0.2 LSB
SFDR (15MHz@30MS/s)	61 dB
SFDR (1MHz@800MS/s)	60 dB
power consumption	7.8 mW
active area	0.23 mm^2
process	0.35 μm

Table 7.15: *Summary of the binary 10 bit DAC performance*

7.6 Overview of Realised DACs

Six different implementations have been discussed in detail. Before comparing these D/A converters with recently published circuits, an overview is given of the most important specifications of the presented realisations. This is done in table 7.16.

7.7 Comparison with literature

7.7.1 The Figure of Merit

To be able to compare the performance of the presented D/A converter with recently presented current steering D/A converters, a figure of merit has been introduced [VdBos AACD01].

$$FOM = \frac{2^N * f_s@(SFDR = 6(N-1))}{P} \tag{7.6}$$

where N is the resolution and P is the power consumption of the D/A converter and f_s is the input signal frequency where the SFDR has dropped with 6 dB (=1 bit) in comparison with the expected result (6*N). For a 12 bit D/A converter, f_s is the output signal where the SFDR equals 66 dB. In fig.7.57 this figure of merit is plotted versus the inverse of the normalised area. On the same figure, the lines of equal FOM/normalized area ratio are shown. It can be concluded from this figure that the presented 12 bit D/A achieves a state-of- the-art performance in comparison to recently published 10, 12 and 14 bit D/A converters. An overview of these D/A converters is given in table 7.17. A few remarks have to be made regarding the content of this table. First, the update rate is the update rate given in the title of the paper and is as a consequence not the update rate at which Nyquist specifications are realised. The f_s is the maximal input signal frequency where the SFDR has dropped with 6 dB and is therefore not always measured at the update rate mentioned in the table. The indicated power consumption is the one needed to generate the f_s signal. The silicon area mentioned in the table represents the active area, which is the area inside the bonding pads (bonding pads excluded). For the references [Bugej JSSC99, Bugej JSSCC00], no numbers were found for the active area of the chip and as a consequence the area including the bonding pads is used to calculate the figure of merit. This implies that in reality, the points indicating the Bugeja D/A converters in fig.7.57 are shifted horizontally to the right yielding a better result.

resolution	**update rate**	**INL**	**DNL**	**SFDR**			**SFDR**			**power**	**area**
[bits]	[MS/s]	[LSB]	[LSB]	input	clock	SFDR	input	clock	SFDR	[mW]	[mm^2]
12	200	0.5	0.7	1MHz	200MS/s	65 dB	500 kHz	300MS/s	50 dB	140	14
14	150	0.3	0.2	2MHz	150MS/s	70 dB	500 kHz	200MS/s	70 dB	300	13
6	800	0.15	0.13	20MHz	800MS/s	> 30 dB	1 MHz	800MS/s	> 36 dB	86	1.4
10	1000	0.2	0.14	50MHz	100MS/s	64 dB	500 MHz	1GS/s	61 dB	110	0.35
12	500	0.3	0.25	125MHz	500MS/s	62 dB	1 MHz	500MS/s	70 dB	110	1
10	30	0.2	0.2	15MHz	30MS/s	61 dB	1 MHz	800MS/s	60 dB	7.8	0.23

Table 7.16: *Overview of the presented D/A converters*

reference	resolution [bits]	update rate [MS/s]	INL [LSB]	DNL [LSB]	f_s [MHz]	power [mW]	area [mm^2]	technology [μm]	$2^N/area$ [$1/mm^2$]	FOM [MHz/mW]
[VdPla JSSC99]	14	150	0.3	0.2	0.9	300	11	0.5	1489	49
[Bugej JSSC99]	14	100	0.5	0.5	20	750	14.4	0.35	1138	437
[Bugej JSSC00]	14	100	0.35	0.25	28	180	11.8	0.35	1388	2549
[Basto PhD]	12	250	0.5	0.5	0.2	100	1	0.7	4096	8
[VdBos CICC98]	12	200	0.5	0.7	0.9	140	14	0.5	293	26
[Marqu ISSC98]	12	300	0.6	0.3	4	320	1.9	0.5	2133	51
[VdBos ISSCC01]	12	500	0.3	0.25	75	86	1	0.35	4096	3572
[Vital AACD01]	12	200	0.65	0.3	20	180	2	0.35	2048	455
[Lin JSSC98]	10	500	0.2	0.1	220	125	0.6	0.35	1707	1802
[VdBos CICC00]	10	1000	0.2	0.14	500	110	0.35	0.35	2926	4655
[Khano VLSI99]	10	400	0.35	0.25	150	90	1.2	0.6	853	1707
[Borre CICC01]	10	30	0.2	0.2	15	8	0.23	0.35	4452	1920
[Roove AACD01]	10	200	x	x	x	130	1.2	0.25	853	x

Table 7.17: *Overview of the recently published high resolution, high speed D/A converters*

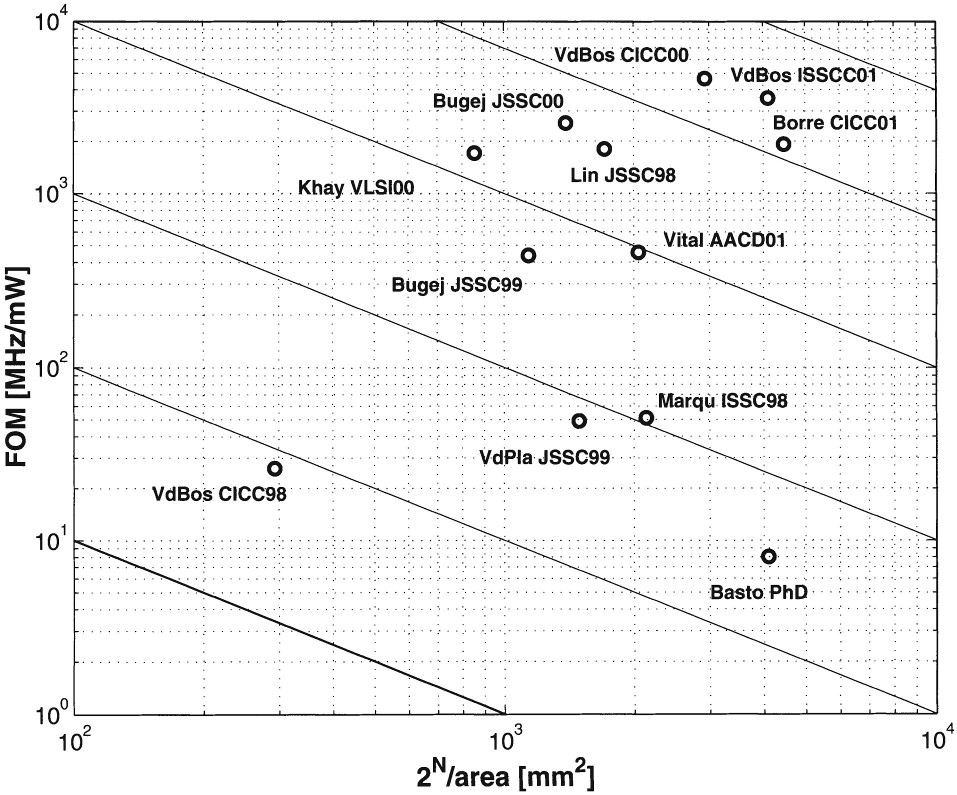


Figure 7.57: *The FOM as a function of the inverse normalised area*

7.8 Conclusion

In this chapter four classes of current steering D/A converters have been discussed. In the first section of this chapter, the design procedure to achieve a high resolution circuit has been validated by the realisation of both a 12 bit and a 14 bit D/A converter. All the measured samples (and this is valid for all the designs in this chapter) have an INL error that is smaller than 0.5 LSB. In a second stage, the limitations for the frequency domain performance have been explored by the design of a 6 bit high speed architecture. Combining both a high resolution with a high update rate following the described design considerations of chapter 6, has led to the realisation of a 10 bit and a 12 bit high speed D/A converter with a static and dynamic performance that has (to the author's knowledge) not been published elsewhere.

Since most telecommunication systems of today do not need a bandwidth of hundreds of MHz but on the contrary need a small circuit with a low power consumption, a current steering D/A converter has been processed that has an intrinsic accuracy of

10 bit and a SFDR higher than 60 dB over a frequency bandwidth of 15 MHz. This chip has a power consumption of only 7.8 mW.

According to a figure of merit based on power, speed and accuracy, three designs compare favourably to all other published D/A converters.

Chapter 8

Transistor Mismatch : Evolution and Relevance

8.1 Introduction

Measuring the electrical characteristics of a set of transistors with the same gate-length and gate-width that are processed in the same technology, will not yield the same results. During the fabrication process small variations will occur that result in a statistical variation of the transistor properties. This variation depends on the dimensions of the matched component and its biasing conditions. This phenomenon is referred to as transistor mismatch.

It is easy to understand that the design of high performance analog circuits such as D/A and A/D converters, reference sources, ... requires the availability of reliable transistor mismatch models. Designers have to be able to rely on accurate simulation tools if they want to be successful in the realisation of a circuit with a performance that lies closely to the limits of the given technology. However, the accuracy of such simulation tools is determined by the underlying models. At this moment, the analog designer has to introduce large safety margins to guarantee the required performance of the circuit leading to an unnecessary power consumption and an operation speed reduction [Kinge CICC96]. In literature, several models have been presented that describe the mismatch characteristics of the transistor by providing the designer with the standard deviation of the mismatch in a set of electrical parameters (like the threshold voltage V_T, the current factor β, the mobility degradation parameter θ, the bulk threshold parameter γ).

In the first part of this chapter, an overview of these models will be given starting with the basic models presented by Lakshmikumar and Pelgrom [Laksh JSSC86, Pelgr JSSC89]. However, going to submicron technologies, these models have to be adapted as to explain the transistor mismatch of short and narrow devices. The models presented by Abel [Abel ISCAS93] and Bastos [Basto ICMTS95] introduce the mobility degradation factor. The next two models that are discussed are used in circuit simulators that are commercially available on the market today. In the model of Gotarredonna [Gotar KLUWER], the mismatch parameters are accurately fitted by using a very general mathematical function of the channel length and width instead of the well known first order law of area. The model presented by Grünebaum [Grune KLUWER] is an extension of the spectral model of Pelgrom. Both methods have the disadvantage that the physical causes of mismatch remain hidden behind the mathematics but yield good results in terms of circuit simulations. The next model described here is the model presented by Drennan [Drenn IEDM99]. In this model, the transistor mismatch is characterised based on its underlying physical causes. The main advantage of this model is that it can be used for process diagnosis and monitoring. The last mismatch model is described by Croon [Croon JSSC02] and describes a model that is continuous from weak to strong inversion and from the linear to the saturation region.

The second part of this chapter discusses the extraction of the transistor mismatch parameters for a 0.5 and a 0.4 μm CMOS technology and the dependency of the transistor mismatch parameters on the used topology and the immediate surroundings.

The last part of this chapter discusses the possibility of using a current steering D/A converter as a test structure for the extraction of the transistor mismatch parameters in a given technology.

8.2 Model of Lakshmikumar

The first in-depth study on the characterisation and modelling of mismatch in MOS transistors has been published in 1986 by Lakshmikumar [Laksh JSSC86]. Previous work mainly focused on either experiments [Akyia IEE] or first crude attempts to unravel the physical causes of transistor mismatch [Shyu JSSC84].

8.2.1 Characterisation Methodology

Since in most analog circuits the MOS transistors operate in the saturation region, the mismatch model will be derived using the square law model :

$$I = \frac{\beta}{2}(V_{GS} - V_T)^2 \tag{8.1}$$

This model has been chosen since it gives an adequate description of the electrical behaviour of the device and at the same time uses only a limited set of parameters (threshold voltage V_T and the current factor β). The drain current mismatch in the saturation region is then given by :

$$\frac{\sigma^2(\Delta I)}{I^2} = \frac{\sigma^2(\Delta\beta)}{\beta^2} + \frac{4\,\sigma^2(\Delta V_T)}{(V_{GS} - V_T)^2} - 4 * r * \frac{\sigma(\Delta V_T)}{V_{GS} - V_T}\frac{\sigma(\Delta\beta)}{\beta} \tag{8.2}$$

where r is the coefficient that describes the correlation between the mismatches in V_T and β. However, for the investigated technology (3 μm CMOS), experimental values for this correlation factor are very small (close to zero) and as a consequence this factor can be omitted in eq.(8.2), giving the following drain current mismatch model:

$$\frac{\sigma^2(\Delta I)}{I^2} = \frac{\sigma^2(\Delta\beta)}{\beta^2} + \frac{4\,\sigma^2(\Delta V_T)}{(V_{GS} - V_T)^2} \tag{8.3}$$

The values for the threshold voltage and the current factor are extracted from measurements of the drain current as a function of the gate voltage for a small value of the drain-source voltage. The current factor is determined by the maximum slope of the I_{DS} versus the V_{GS} curve while the threshold voltage is the intercept of this slope with the V_{GS} axis. The terms on the right hand side of eq.(8.3) can be written as a function of the dimensions (W,L) of the transistor based on the underlying physical phenomena that cause the mismatch.

8.2.2 Physical causes for mismatch

8.2.2.1 Threshold Voltage Mismatch

The threshold voltage of a transistor is given by :

$$V_T = \phi_{MS} + 2\,\phi_B + \frac{Q_B - Q_f + q\,D_I}{C_{ox}} \tag{8.4}$$

where ϕ_{MS} is the gate-semiconductor work function difference, ϕ_B is the Fermi potential in the bulk, Q_B is the depletion charge density, Q_f is the fixed oxide charge density, D_I is the threshold adjust implant dose and C_{ox} is the gate oxide capacitance. The standard deviation of the threshold voltage can be calculated using the following statistical formula for a variable X that is a function of two independent parameters P_1 and P_2:

$$\sigma^2(\Delta X) = (\frac{\partial X}{\partial P_1})^2\,\sigma^2(\Delta P_1) + (\frac{\partial X}{\partial P_2})^2\,\sigma^2(\Delta P_2) \tag{8.5}$$

Since all parameters (ϕ_{MS}, ϕ_B, C_{ox}, Q_B, Q_f, D_I) are assumed to be independent, eq.(8.4) can be rewritten as :

$$\sigma^2(\Delta V_T) = (\frac{\partial V_T}{\partial \phi_{MS}})^2 \sigma^2(\Delta\phi_{MS}) + (\frac{\partial V_T}{\partial \phi_B})^2 \sigma^2(\Delta\phi_B) + (\frac{\partial V_T}{\partial C_{ox}})^2 \sigma^2(\Delta C_{ox}) + (\frac{\partial V_T}{\partial Q_B})^2 \sigma^2(\Delta Q_B) + (\frac{\partial V_T}{\partial Q_f})^2 \sigma^2(\Delta Q_f) + (\frac{\partial V_T}{\partial D_I})^2 \sigma^2(\Delta D_I) \quad (8.6)$$

Based on eq.(8.4), the partial derivatives can be calculated resulting in :

$$\sigma^2(\Delta V_T) = \sigma^2(\Delta\phi_{MS}) + \sigma^2(\Delta\phi_B) + (\frac{Q_B^2}{C_{ox}^2} + \frac{Q_f^2}{C_{ox}^2} + \frac{q^2 D_I^2}{C_{ox}^2})\frac{\sigma^2(\Delta C_{ox})}{C_{ox}^2} + \frac{1}{C_{ox}^2} * (\sigma^2(\Delta Q_B) + \sigma^2(\Delta Q_f) + \sigma^2(\Delta D_I)) \quad (8.7)$$

The Fermi potential ϕ_B and the work function ϕ_{MS} are regarded as constants not contributing to the transistor mismatch. Furthermore, it has been assumed that all the charges mentioned in eq.(8.4) have a Poisson distribution [Laksh JSSC86]. The final equation for the standard deviation of the threshold voltage is thus given by :

$$\sigma^2(\Delta V_T) = \frac{1}{\mathbf{W * L * C_{ox}^2}} [q(Q_B + Q_f + qD_I) + A_{ox}(Q_B^2 + Q_f^2 + q^2 D_I^2)] \quad (8.8)$$

where W is the effective channel width, L is the effective channel length and A_{ox} is a parameter that has to be determined from measurement data.

The most important conclusion to be drawn from this derivation is that the threshold voltage mismatch is inversely proportional to the square root of the effective channel area. Measurements have been performed on both nMOS and pMOS devices (W/L= 48/12, 24/12, 24/6, 12/6, 12/3, 6/3 $\mu m/\mu m$) to verify this equation.

8.2.2.2 Current factor mismatch

The current factor is given by :

$$\beta = \mu C_{ox} \frac{W}{L} \quad (8.9)$$

where μ is the channel mobility. The standard deviation of the current factor can be derived similar to the threshold voltage standard deviation and equals :

$$\frac{\sigma^2(\Delta\beta)}{\beta^2} = \frac{\sigma^2(\Delta L)}{L^2} + \frac{\sigma^2(\Delta W)}{W^2} + \frac{\sigma^2(\Delta\mu)}{\mu^2} + \frac{\sigma^2(\Delta C_{ox})}{C_{ox}^2} \tag{8.10}$$

The mobility mismatch has been mainly contributed to the non-uniformity of Q_f and since these charges have a Poisson distribution, the final equation for the current factor mismatch is given by:

$$\frac{\sigma^2(\Delta\beta)}{\beta^2} = \frac{\sigma^2(\Delta L)}{L^2} + \frac{\sigma^2(\Delta W)}{W^2} + \frac{1}{\mathbf{W} * \mathbf{L}}(A_\mu + A_{ox}) \tag{8.11}$$

From eq.(8.11), it can be seen that the current factor mismatch is not solely inversely proportional to the square root of the effective channel area but is also determined by a factor $\frac{1}{W^2}$ and $\frac{1}{L^2}$. Also here, measurements were performed that validated the derived equation.

8.3 Model of Pelgrom

The starting point to derive the Pelgrom model is not the wide range of possible mismatch causes, but a mathematical treatment of classes of mismatch behaviour which covers all (at that time) known area related physical causes.

8.3.1 Characterisation methodology

The value of a parameter P generally consists out of the sum of a fixed part and a random varying part so that its value differs from coordinate to coordinate. If the variations are small, the average value for the parameter can be calculated as the integral of P(x,y) over the specified area. This integral can be interpreted as the convolution of a geometry function G(x,y) and the mismatch function P(x,y) [Pelgr JSSC89] :

$$\Delta\mathcal{P}(\omega_x, \omega_y) = \mathcal{G}(\omega_x, \omega_y)\,\mathcal{P}(\omega_x, \omega_y) \tag{8.12}$$

Representing parameter fluctuations in the Fourier domain has the advantage of an easy determination of the power contents which can be interpreted as the square of the standard deviation of the parameter :

$$\sigma^2(\Delta P) = \frac{1}{4\pi^2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} |\mathcal{G}(\omega_x, \omega_y)|^2\,|\mathcal{P}(\omega_x, \omega_y)|^2\,d\omega_x d\omega_y \tag{8.13}$$

Using the Fourier analysis, the geometry function $\mathcal{G}(\omega_x, \omega_y)$ can be calculated for different transistor topologies. For further details, the reader is referred to [Pelgr JSSC89]. Concerning the contribution of the mismatch function $\mathcal{P}(\omega_x, \omega_y)$ two classes of physical mismatch causes are dealt with:

- The first class is the spatial "white noise" or short distance variations. In this class the mismatch is determined by many single events of which the mismatch contributions can be summed and with a correlation distance that is smaller than the transistor dimensions. Examples are the local mobility fluctuations, the oxide charges,...

- The second class is modelled in the Fourier domain as a fixed low frequency contribution with a spatial frequency that is inversely proportional to the wafer diameter. This class takes the effect of a circular parameter distribution into account which is caused by for example the oxidation process.

Calculating the variance for two rectangular devices leads to the following result:

$$\sigma^2(\Delta P) = \frac{A_P^2}{WL} + S_P^2\, D_x^2 \tag{8.14}$$

where A_P is the area proportionality constant and S_P is the distance coefficient that have to be determined through measurements.

8.3.2 Mismatch model

The presented drain current mismatch model is derived from the same drain current model as used by Lakshmikumar (eq.8.1) and is given by :

$$\frac{\sigma^2(\Delta I)}{I^2} = \frac{\sigma^2(\Delta\beta)}{\beta^2} + \frac{4\,[\sigma^2(\Delta V_{T0}) + A\,\sigma^2(\Delta\gamma)]}{(V_{GS} - V_T)^2} \tag{8.15}$$

with $A = \sqrt{2\phi - V_{SB}} - \sqrt{2\phi}$. This model uses three parameters to describe the mismatch, namely the threshold voltage V_{T0}, the bulk threshold parameter γ and the current factor β. This model is identical to the one proposed by Lakshmikumar in the previous section for transistors that operate in the saturation region and have a bulk-source voltage equal to zero.

The mismatch contribution of the parameters V_{T0} and γ follow the mathematical derivation in section 8.3.1 and can be written as :

$$
\begin{aligned}
\sigma^2(\Delta V_{T0}) &= \frac{A_{VT0}^2}{WL} + S_{VT0}^2\, D^2 \\
\sigma^2(\Delta\gamma) &= \frac{A_\gamma^2}{WL} + S_\gamma^2\, D^2
\end{aligned}
\tag{8.16}
$$

Also the mobility contribution can be calculated taking into account that variations in channel length and width are caused by edge roughness and can be modelled as $\sigma^2(L) \propto 1/W$ and $\sigma^2(W) \propto 1/L$:

$$
\frac{\sigma^2(\Delta\beta)}{\beta^2} = \frac{A_W^2}{W^2L} + \frac{A_L^2}{WL^2} + \frac{A_\beta^2}{WL} + S_\beta^2\, D^2 \tag{8.17}
$$

which for large size devices can be simplified to

$$
\frac{\sigma^2(\Delta\beta)}{\beta^2} = \frac{A_\beta^2}{WL} + S_\beta^2\, D^2 \tag{8.18}
$$

From eq.(8.16) and eq.(8.17), it becomes clear that - making abstraction of the distance component -, the expressions for the standard deviations of the mismatch parameters are similar to the ones derived by Lakshmikumar.

In order to validate the presented model, several modules have been designed and fabricated which contain 6 transistors of the same size : a reference transistor, a transistor at 30, 250 and 500 μm spacing, a parallel and a 90° rotated transistor (30 μm spacing). These modules were made for both nMOS and pMOS devices with different W/L ratios (2.4/1.6, 2.4/20, 3/3, 5/5, 20/5, 20/20 $\mu m/\mu m$). The mismatch parameters have been directly extracted. Measurements show a good agreement between the model and the measurements. Furthermore, measurement results on devices processed in a 2.5 μm technology have proven that the influence of the distance component is only significant for widely spaced devices with a large channel area.

Although the big advantage of this mathematical derivation is that it can also be used for other devices like capacitors, its major disadvantage is that it does not provide the reader with any physical insight. The model is only valid if the mismatch processes belong to the two mathematical classes that were discussed.

8.4 Other models

8.4.1 Model of Abel

The models derived by Lakshmikumar and Pelgrom are valid for devices with a gate-length higher than 2 μm. However, for short channel transistors, the threshold voltage and the current factor are dependent on the channel length, introducing the need for extended mismatch models to characterise the transistor mismatch of these small size devices. This model [Abel ISCAS93] determines the drain current mismatch as a function of the following four transistor parameters : V_{T0}, γ, β and θ starting from the equations for the drain current in the linear region and in the saturation region. This leads to the following mismatch models

$$\frac{\Delta I_D}{I_D} = \frac{-\Delta V_T}{V_{GS} - V_T} + \frac{\Delta V_{GS}}{V_{GS} - V_T} + \frac{\Delta\beta}{\beta} + \frac{\Delta V_{DS}}{V_{DS}} - \frac{\Delta\theta(V_{GS} - V_T)}{1 + \theta(V_{GS} - V_T)} \tag{8.19}$$

based on the drain current model for the linear region

$$I_{DS} = \frac{\beta}{1 + \theta\,(V_{GS} - V_T)}[(V_{GS} - V_T)V_{DS} - \frac{V_{DS}^2}{2}] \tag{8.20}$$

and

$$\frac{\Delta I_D}{I_D} = \frac{-2\Delta V_T}{V_{GS} - V_T} + \frac{2\Delta V_{GS}}{V_{GS} - V_T} + \frac{\Delta\beta}{\beta} + \frac{\Delta V_{DS}}{V_{DS}} - \frac{\Delta\theta(V_{GS} - V_T)}{1 + \theta(V_{GS} - V_T)} \tag{8.21}$$

based on the drain current model for the saturation region

$$I_{DS} = \frac{\beta}{2}\frac{(V_{GS} - V_T)^2}{1 + \theta\,(V_{GS} - V_T)} \tag{8.22}$$

The following two algorithms were used to extract parameter mismatch parameters from the measured I_D - V_{GS} curves from a pair of transistors in saturation. In the first algorithm, the $\sqrt{I_D}$ -V_{GS} curve is considered. The slope of the this curve equals $\sqrt{\beta/2}$ and the extrapolation to zero drain current gives a value for the V_T. The second algorithm uses the Newton-Raphson iteration method to determine the parameters by fitting the I_D - V_{GS} curves.

A test chip has been implemented in a 0.8 μm technology consisting out of five transistor pairs with different W/L ratios. From the measurements, it could be concluded that the three parameters V_{T0}, γ, θ have a standard deviation of the form :

$$(\sigma)^2(\Delta P) = \frac{a_P}{WL} \tag{8.23}$$

while the standard deviation of the current factor β is given by :

$$\frac{(\sigma)^2(\Delta\beta)}{\beta^2} = \frac{1}{WL}(a_\beta + \frac{a_L}{L}) \tag{8.24}$$

8.4.2 Model of Bastos

In [Basto ICMTS95], the influence of the mobility degradation has been taken into account. The drain current mismatch models have been derived from eq.(8.22). This model describes the transistor mismatch by using three parameters (the threshold voltage V_T, the current factor β and the mobility degradation factor θ).

A test structure containing a 10x12 transistor array has been processed in a 1.2 μm technology. Every row contains 10 identical transistors with a different spacing. The remaining positions are used to implement different layout structures and/or rotated devices. Four different algorithms have been used to extract the mismatch parameters and the results have been compared with each other through experimental data. A short overview of these algorithms is given.

In algorithm 1, the mismatch parameters ΔV_T, $\frac{\Delta\beta}{\beta}$, $\Delta\theta$ are directly extracted from the drain current mismatch model presented in eq.(8.25).

$$\frac{\Delta I_D}{I_D} = \frac{\Delta\beta}{\beta} - \frac{2\Delta V_T}{V_{GS} - V_T} - \frac{\Delta\theta(V_{GS} - V_T)}{1 + \theta(V_{GS} - V_T)} \tag{8.25}$$

This model can be reduced by combining the effective mobility degradation mismatch term with the current factor mismatch term since both expressions are significant in the same bias range (high gate-source voltage). This model will be referred to as algorithm 2 :

$$\frac{\Delta I_D}{I_D} = \frac{\Delta\beta}{\beta} - \frac{2\Delta V_T}{V_{GS} - V_T} \tag{8.26}$$

Algorithm 3 and 4 were already widely known, but have been included to compare the results with algorithm 1 and 2. In algorithm 3, a Levenberg-Marquardt iteration method has been used to fit the measured I_D versus V_{GS} curve to determine the mismatch parameters (eq.8.27).

$$I_D = \frac{\beta}{2(1 + \theta(V_{GS} - V_T))}(V_{GS} - V_T)^2 \tag{8.27}$$

In algorithm 4, the line with the maximum slope tangent to the $\sqrt{I_D}$ versus V_{GS} curve is extrapolated to zero drain current, determining hereby the value of the threshold voltage. The slope of the line is equal to $\sqrt{\frac{\beta}{2}}$.

Measurements indicate that for large transistor sizes, the four algorithms give equivalent results. For small size transistors, the threshold voltage and the current factor mismatch deviate significantly from the first order law of area as proposed by Pelgrom. An extended model is presented that incorporates the mismatch behaviour of these devices :

$$\sigma^2(\Delta V_T) = \frac{A_{1VT}}{WL} + \frac{A_{2VT}}{WL^2} - \frac{A_{3VT}}{W^2L} \tag{8.28}$$

For the current factor mismatch, the Pelgrom law stays valid for small size devices if its value is extracted using algorithm 2. This algorithm yields the best results for the drain current mismatch which can be explained by the correlation that exists between the current factor and the mobility degradation mismatch. Note that in this case, it is better to speak of an effective current factor mismatch since the current factor and the mobility degradation mismatch can not be analysed separately. If both the $\frac{\Delta\beta}{\beta}$ and the $\Delta\,\theta$ parameters have been extracted (algorithm 1), an extended model of the same form as eq.(8.28) is needed to describe the current factor mismatch :

$$\sigma^2(\frac{\Delta\beta}{\beta}) = \frac{A_{1\beta}}{WL} + \frac{A_{2\beta}}{WL^2} + \frac{A_{3\beta}}{W^2L} \tag{8.29}$$

8.4.3 Model of Gotarredona-Linares

In this section a mismatch model [Gotar ISCAS98, Gotar ICECS99, Gotar KLUWER, Gotar EDL00, Gotar ISCAS00b, Gotar ISCAS00a] is presented based on the threshold voltage mismatch $\Delta\,V_T$, the current factor mismatch $\frac{\Delta\beta}{\beta}$, the body factor mismatch $\Delta\,\gamma$ and the mismatch in the mobility degradation factor $\Delta\,\theta_{eff}$ which is split up into two components $\Delta\theta_0$ and $\Delta\theta_e$. In the linear region the value of this parameter is given by:

$$\theta_{eff}|_{lin} = \theta_0 = \theta + \beta(R_S + R_D) = \theta + \frac{2\,\mu C_{ox}\,l_d\,R_\square}{L} \tag{8.30}$$

for $V_{DS} \ll V_{GS} - V_T$ and taking both the source (R_S) and drain resistance (R_D) into account. In eq.(8.30), $R_\square$ is the diffusion sheet resistance per square and l_d is the distance between the gate diffusion edge to the source (and drain) contact region. The extracted value for θ_{eff} is also influenced by velocity saturation v_s, resulting in

$$\theta_{eff} = \theta + \beta(R_S + R_D) + (\frac{\mu}{2v_sL} - \beta R_D)\frac{V_{DS}}{V_{GS} - V_T} = \theta_0 + \theta_e * \frac{V_{DS}}{V_{GS} - V_T} \tag{8.31}$$

where V_{DS} is substituted by $V_{DSsat} = V_{GS} - V_T$ when the transistor is biased in the saturation region. Rewriting this equation as a function of the channel length L leads to the following expression :

$$\theta_{eff} = \theta + \frac{2\,\mu C_{ox}\,l_d\,R_\square}{L} + \frac{\mu}{L}(\frac{1}{2v_s} - C_{ox}l_dR_\square)\frac{V_{DS}}{V_{GS} - V_T} \tag{8.32}$$

From this equation, the significance of the extra terms besides θ for short channel devices becomes clear. The mismatch of this parameter can be expressed as:

$$\Delta\theta_{eff} = \Delta\theta_0 + \frac{V_{DS}}{V_{GS} - V_T}\Delta\theta_e \tag{8.33}$$

For a transistor operating in the linear region, the effective mobility degradation mismatch equals $\Delta\theta_0$, while for a transistor operating in the saturation region $\Delta\theta_{eff}$ equals $\Delta\theta_0 + \Delta\theta_e$. It is clear that an extraction in both the linear ($V_{DS} = 0$) and the saturation region ($V_{DS} = V_{GS} - V_T$) is necessary.

The drain current mismatch model is given by:

$$\frac{\Delta I_{DS}}{I_{DS}} = \frac{\Delta\beta}{\beta} - \frac{(1 + \frac{\theta_{eff} V_{DS}}{2})\Delta V_T}{[V_{GS} - V_T - \frac{V_{DS}}{2}][1 + \theta_{eff}(V_{GS} - V_T)]} - \frac{(V_{GS} - V_T)\Delta\theta_0}{1 + \theta_{eff}(V_{GS} - V_T)} \tag{8.34}$$

for the linear region based on

$$I_{DS} = \frac{\beta\,(V_{GS} - V_T - \frac{V_{DS}}{2})}{1 + \theta\,(V_{GS} - V_T)} * V_{DS} \tag{8.35}$$

and for the saturation region

$$\frac{\Delta I_{DS}}{I_{DS}} = \frac{\Delta\beta}{\beta} - \frac{(2 + \theta_{eff}(V_{GS} - V_T)\Delta V_T}{[V_{GS} - V_T][1 + \theta_{eff}(V_{GS} - V_T)]} - \frac{(V_{GS} - V_T)(\Delta\theta_0 + \Delta\theta_e)}{1 + \theta_{eff}(V_{GS} - V_T)} \tag{8.36}$$

based on eq.(8.22).

Furthermore,

$$\Delta V_T = \Delta V_{T0} + \Delta\gamma[\sqrt{2\phi + V_{SB}} - \sqrt{2\phi}] \tag{8.37}$$

A test chip has been designed in a 1 μm process that consists out of an array of identical cells that contain 30 pMOS and 30 nMOS transistors. Additional decoding/selection circuitry has been included as to automate the measurements as much as possible. The measurement and extraction procedure is based on 4 measurement curves both in the linear and the saturation region ($I_{DS}(V_{GS})$ and $I_{DS}(V_{SB})$). For further details, the reader is referred to [Gotar KLUWER]. The following parameters have been extracted : $\sigma(\Delta V_{T0})$, $\sigma(\frac{\Delta\beta}{\beta})$, $\sigma(\Delta\gamma)$, $\sigma(\Delta\theta_0)$, $\sigma(\Delta\theta_e)$ and the 10 correlation coefficients between the 5 mismatch parameters. From the measurement results, it could be concluded that for transistors with a channel length higher than 4 μm, only the mismatch parameters $\sigma(\Delta V_{T0})$, $\sigma(\frac{\Delta\beta}{\beta})$, $\sigma(\Delta\gamma)$ have to be extracted for one region of operation. The results are also valid in the other bias region. Also the correlation terms can be ignored which is in accordance to the Pelgrom model. For small size devices, $\Delta\theta_{eff}$ can no longer be ignored. Furthermore, to obtain a good drain current mismatch prediction three correlation coefficients are necessary: $r_{(\Delta\beta,\Delta\theta_0)}$, $r_{(\Delta\beta,\Delta\theta_e)}$ and $r_{(\Delta\beta,\Delta\gamma)}$.

The dependency of the mismatch parameters on the transistor dimensions is based on the idea that to realise a precise transistor mismatch model, more terms (beside the ones suggested by Pelgrom) have to be added. A very general mathematical function (eq.8.38) that accurately fits the measured data is used since the main goal is to use this model in a circuit simulator and not to acquire knowledge of the physical causes of the transistor mismatch.

$$\sigma^2(\Delta P) = \sum_{m,n} \frac{C_{mn}}{(W - \epsilon_w)^m \, (L - \epsilon_l)^n} \tag{8.38}$$

In [Gotar KLUWER], a verification of the characterisation results and the implementation of the transistor mismatch model in conventional electrical circuit simulators is discussed in more detail.

8.4.4 Model of Grünebaum-Oehm

8.4.4.1 The Spectral model

The starting point for this model is the spectral model presented by Pelgrom. For the derivation of the model itself, the reader is referred to section 8.3. During the years, this model has been refined [Grune ESSCIRC97, Oehm ESSCIRC98, Oehm ICECS99, Grune KLUWER] which resulted in a simulation tool called GAME (General Analysis of Mismatch Effects). The shape of the spectral density for the standard deviation $\sigma(\Delta P)$ is depicted in fig.8.1. This spectrum can be modelled by

$$\mathcal{P}(u, v) = \sqrt{A_P^2 + \frac{k_P^2}{(\sqrt{u^2 + v^2})^{2a_P}}} \tag{8.39}$$

where A_P, k_P and a_P are three parameters that have to be determined using measurement data. The white noise component A_P is identical to the well known first order area mismatch parameter. The coloured noise component is characterised by the coefficients k_P and a_P. Measurement data on different processes indicate that the value for a_P equals 1.5 for all mismatch parameters. Assuming $k_P = 0$, the standard deviation of mismatch parameter P can be reduced to the first order law of area (cfr. Lakshmikumar, Pelgrom).

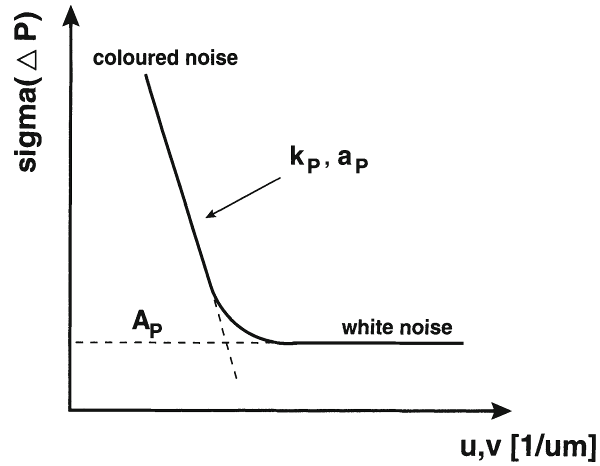


Figure 8.1: *The spectral density function*

8.4.4.2 Different mismatch effects explained

Using this model, different parameters that influence the mismatch behaviour of the transistors can be explained. A short overview is given :

- **Long distance mismatch**
  The Pelgrom model states that the value of the mismatch parameters increases linearly with the distance between the considered devices. However, the spectral model states that for larger distances a saturation effect occurs. This has been verified by measurements.

- **Layout topologies**
  Due to parameter gradients over the wafer, the layout topology of the transistor pair has a significant effect on the value of the mismatch parameters. This effect is well known from measurements on common centroid transistor structures. The advantage of the spectral model is that the influence of the topology can be correctly modelled and taken into account. Fig.8.2 shows the results from simulations with the spectral model for three different topologies. The mismatch between two transistors is the worst when the geometrical device centers are located far from each other (topology 1).

- **Geometry corrections**
  To obtain a good correspondence between the model and the measurement results, geometrical corrections of the drawn transistor dimensions are needed. This has already been indicated by [Lovet JSSC98] where the effective channel length and width are used instead of the drawn dimensions for small geometry

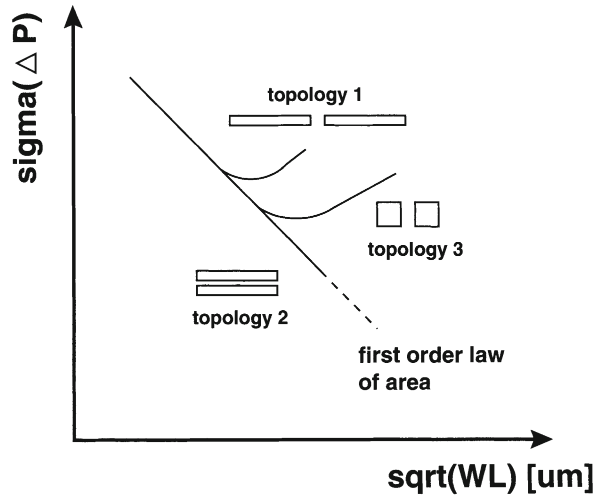


Figure 8.2: *A qualitative view of the mismatch dependence on the transistor geometry*

devices. Measurements show that introducing the variables D_L and D_W allows for precise mismatch modelling of small devices (for $\sigma(\Delta V_{T0})$).

- **Parameter correlations**
  The correlation that exists between the MOS electrical parameters like V_{T0} and β are caused by their dependence on the same underlying physical technology parameters like the gate oxide thickness t_{ox}. A strong statistical contribution of a common physical parameter increases the correlation coefficient between the model parameters while the contributions from independent physical parameters lowers the correlation.

8.4.5 Drennan-McAndrew

Most MOS transistor mismatch models that exist today are based on the results presented by Pelgrom. This model [Drenn IEDM99] approaches the mismatch problem from another angle.

- This model starts from the physical process parameters V_{fb}, t_{ox}, Δ_L, Δ_W, n_{sub}, μ_0, ρ_{sh} (drain region sheet resistance) and V_{tl} (length variation of the threshold voltage). A big advantage of this model is the analysis of the contribution of these process parameters to the overall mismatch. In this way, the most important physical causes can be identified which can be used to guide process diagnosis and improvement.

- The $\frac{1}{WL}$ dependency of the mismatch parameters is in some cases inadequate. Several parameters (f.e. Δ_L, ρ_{sh}) exhibit $\frac{1}{L^2}$, $\frac{1}{WL^2}$, $\frac{1}{W^2}$, $\frac{1}{W^2L}$, $\frac{1}{W}$ dependencies.

The variance of the mismatch parameters is not characterised directly but extracted using measurements on the drain current mismatch over different geometry and bias regions (both in linear and saturation regions) :

$$\sigma^2(\Delta I_d) = \sum_j (\frac{\partial I_d}{\partial p_j})^2 \sigma^2(\Delta p_j) \tag{8.40}$$

where the sensitivities $\frac{\partial I_d}{\partial p_j}$ are computed using SPICE models for each bias and geometry. In this work, the BSIM3 models have been used. Measurements have been performed on devices processed in a 0.28 μm technology. The results indicate that the model is able to represent the measured mismatch behaviour quantitatively and qualitatively over bias and geometry.

8.4.6 Model of Croon

In this model [Croon JSSC02] the drain current mismatch has been calculated by using a first order Taylor approximation of the drain current equation :

$$\frac{\Delta I_D}{I_D} = \frac{1}{I_D}\frac{\partial I_D}{\partial P_2}\Delta P_1 + \frac{1}{I_D}\frac{\partial I_D}{\partial P_2}\Delta P_2 + ... \tag{8.41}$$

with P_i the mismatch parameters. In this model 4 parameters are introduced. One describes the threshold voltage mismatch and the others the current factor mismatch.

For the derivation of the threshold voltage mismatch, it has been assumed that the drain current is only a function of $V_{GS} - V_T$

$$I_D = I_D(V_{GS} - V_T) \tag{8.42}$$

leading to the following mismatch model:

$$(\frac{\Delta I_D}{I_D})_{VT} = \frac{1}{I_D}\frac{\partial I_D}{\partial V_T}\Delta V_T = -\frac{g_m}{I_D}\Delta V_T \tag{8.43}$$

where $\frac{g_m}{I_D}$ is calculated from measurement data. It has been assumed that the threshold voltage mismatch is independent of the drain bias and that it is the same in weak and in strong inversion. For the bulk bias dependence a physical model has been used based on the doping concentration and the oxide capacitance instead of the frequently used body effect factor.

For the derivation of the current factor mismatch, the following drain current model has been used :

$$I_D = \beta(V_{GS} - V_T - \frac{V_{DS}}{2})V_{DS} \tag{8.44}$$

The mobility is determined by the bulk mobility, phonon scattering, surface roughness scattering and velocity saturation. These elements are integrated in the following three current factor mismatch parameters :

$$\begin{aligned} \beta_0 &= \frac{W\mu_B C_{ox}}{L} \\ \zeta_{sat} &= W C_{ox} v_{sat} \\ \frac{1}{\zeta_{sr}} &= \frac{L}{W C_{ox}}(\frac{1}{a_{ph}} + \frac{1}{a_{sr}} + R_s) \end{aligned}$$

giving the following result for the current factor mismatch :

$$(\frac{\Delta I_D}{I_D})_\beta = -\beta\Delta\frac{1}{\beta_0} - \beta(V_{GS} - V_T - \frac{V_{DS}}{2})\Delta\frac{1}{\zeta_{sr}} - \beta V_{DS}\Delta\frac{1}{\zeta_{sat}} \tag{8.45}$$

The complete mismatch model is given by :

$$(\frac{\Delta I_D}{I_D}) = (\frac{\Delta I_D}{I_D})_{VT} + (\frac{\Delta I_D}{I_D})_\beta \tag{8.46}$$

The threshold voltage mismatch can be described by the Pelgrom model and is given by :

$$\sigma^2(\Delta P) = \frac{A_0}{WL} + \frac{A_L}{WL^2} + \frac{A_W}{W^2L} + D^2 \tag{8.47}$$

where the first term on the right hand side describes the variance for a large device, the second and the third term describe the variation in short and narrow channel effects and the fourth term is contributed to gradient mismatch.

For the three other parameters, the mismatch can be described as :

$$\sigma^2(\Delta P) = \sigma^2(\Delta P)(eq.(8.47)) * (\frac{W}{L})^2 \tag{8.48}$$

For more details, the reader is referred to [Croon JSSC02].

A test chip has been made in a $0.18\mu m$ CMOS technology consisting out of 14 device pairs. From the measurements, it could be concluded that the model is continuous from weak to strong inversion and from the linear to the saturation region and achieves an accuracy within 20% in the strong inversion region.

8.5 Mismatch parameters for the 0.5 and the 0.4 μm CMOS technology

8.5.1 The 0.5 μm technology

In this paragraph, the extraction of the mismatch parameters for a 0.5 μm CMOS technology will be discussed in more detail. A test structure has been designed that is built up out of 10 rows and columns. Each row contains 10 nMOS transistors with the same dimensions. The transistor in the first column is taken as the reference device. The transistor dimensions have been chosen as to span a wide $1/\sqrt{WL}$ range (from 0.05 up to 1.6 μm^{-1}) and are given by W/L = 20/20, 10/3.2, 3.2/10, 0.8/20, 20/0.5, 1/4, 1/1.4, 2.8/0.5, 1/0.7, 0.8/0.5 $\mu m/\mu m$. The measurements have been performed on packaged devices in the saturation region ($V_{BS} = 0$) using a HP3457 multimeter with a 5 to 7 1/2 digits of resolution. The drain current mismatch model that has been used is given by

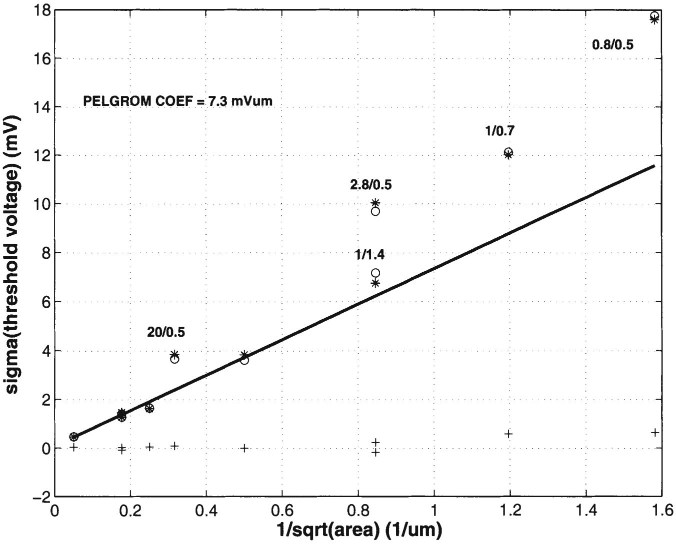
$$\frac{\sigma^2(\Delta I)}{I^2} = \frac{\sigma^2(\Delta\beta)}{\beta^2} + \frac{4\,\sigma^2(\Delta V_T)}{(V_{GS} - V_T)^2} \tag{8.49}$$

From the measurement results, it could be clearly seen that the first order law of the area does not explain the threshold voltage and current factor mismatch behaviour for short and narrow devices. Therefore, the following extended model for these mismatch parameters has been used :

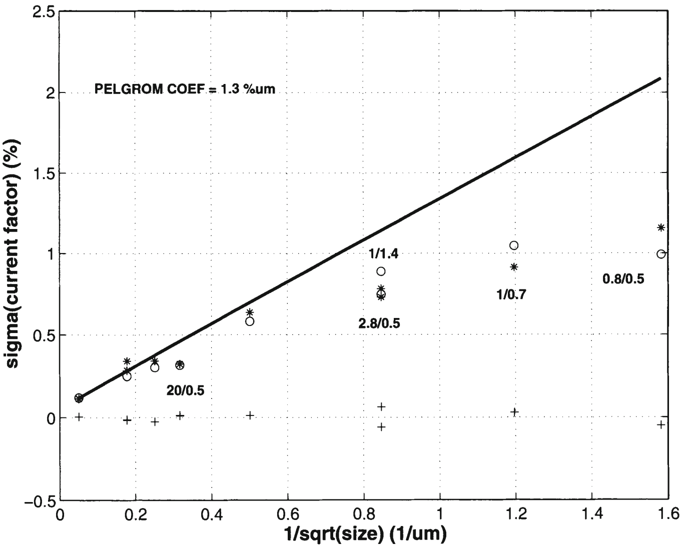
$$\sigma^2(\Delta V_T) = \frac{A_{1VT}^2}{WL} + \frac{A_{2VT}^2}{W^2L} + \frac{A_{3VT}^2}{WL^2} \tag{8.50}$$

$$\sigma^2(\frac{\Delta\beta}{\beta}) = \frac{A_{1\beta}^2}{WL} + \frac{A_{2\beta}^2}{W^2L} + \frac{A_{3\beta}^2}{WL^2} \tag{8.51}$$

The results are given in fig.8.3.a and fig.8.3.b . In these figures the standard deviation of the threshold voltage mismatch and the current factor mismatch is given as a function of the inverse of the square root of the channel area of the transistor. The measurement results are indicated with a '*'. In the figures, also the mean value for the $\sigma(\Delta\ V_T)$ and $\sigma(\frac{\Delta\beta}{\beta})$ is given (represented by '+'). The value of these standard deviations has to be close to zero to exclude any systematic effects [Basto TSM97]. The solid line indicates the Pelgrom model that for the threshold voltage underestimates the mismatch value for small size devices and for the current factor overestimates the mismatch. The results of the extended model are given by the circles on the plot. There exists a fairly good agreement between the measurement values and the model. The results are summarised in table 8.1.



(a) The standard deviation of the threshold voltage mismatch as a function of the inverse of the square root of the area



(b) The standard deviation of the current factor mismatch as a function of the inverse of the square root of the area

Figure 8.3: The extracted mismatch parameters for a 0.5 μm CMOS technology

Pelgrom coef.	**Extended Model**		
$A_{VT}[mV\,\mu m]$	$A^2_{1VT}[mV\,\mu m]^2$	$A^2_{2VT}[mV\,\mu m]^2$	$A^2_{3VT}[mV\,\mu m]^2$
7.3	45.9	43.3	-5.3
$A_\beta[\%\,\mu m]$	$A^2_{1\beta}[\%\,\mu m]^2$	$A^2_{2\beta}[\%\,\mu m]^2$	$A^2_{3\beta}[\%\,\mu m]^2$
1.3	1.8	-0.4	-0.5

Table 8.1: *Extracted mismatch parameters for a* $0.5\mu m$ *nMOS process*

Pelgrom coef.	**Extended Model**		
$A_{VT}[mV\,\mu m]$	$A^2_{1VT}[mV\,\mu m]^2$	$A^2_{2VT}[mV\,\mu m]^2$	$A^2_{3VT}[mV\,\mu m]^2$
10.4	112	18.6	-45.8
$A_\beta[\%\,\mu m]$	$A^2_{1\beta}[\%\,\mu m]^2$	$A^2_{2\beta}[\%\,\mu m]^2$	$A^2_{3\beta}[\%\,\mu m]^2$
1.6	0.8	-0.3	0.4

Table 8.2: *Extracted mismatch parameters for a* $0.4\mu m$ *process (nMOS)*

In fig.8.4, the threshold voltage and the current factor mismatch are given as a function of the distance for the transistor with dimensions W/L = 20/20 $\mu m/\mu m$. As can be seen from the plots, the mismatch parameters increase linearly with the distance (as is predicted by the Pelgrom model). The distance coefficients have been derived and equal:

$$\begin{aligned} S_{VT} &= 0.2\,\mu V\,\mu m \\ S_\beta &= 0.3\,\%\,mm \end{aligned} \tag{8.52}$$

To validate the characterisation procedure, the measured drain current mismatch is compared to the drain current mismatch calculated using the extracted values of the threshold and current factor mismatch (eq.(8.49)). The results are given in fig.8.5 for transistors with a W/L = 1/4 (top), 3.2/10 (middle) and 0.8/0.5 (bottom) $\mu m/\mu m$.

8.5.2 The 0.4 μm technology

The same extraction procedure has been followed. The test structure was buit up out of transistors with the following dimensions W/L = 8/8, 1.25/15, 4/4, 32/0.4, 2/2, 8/0.4, 1.25/1.25, 1.25/0.4 $\mu m/\mu m$. For each transistor pair 100 samples were measured. The results are given in table 8.2 for the nMOS transistors and in table 8.3 for the pMOS transistors.

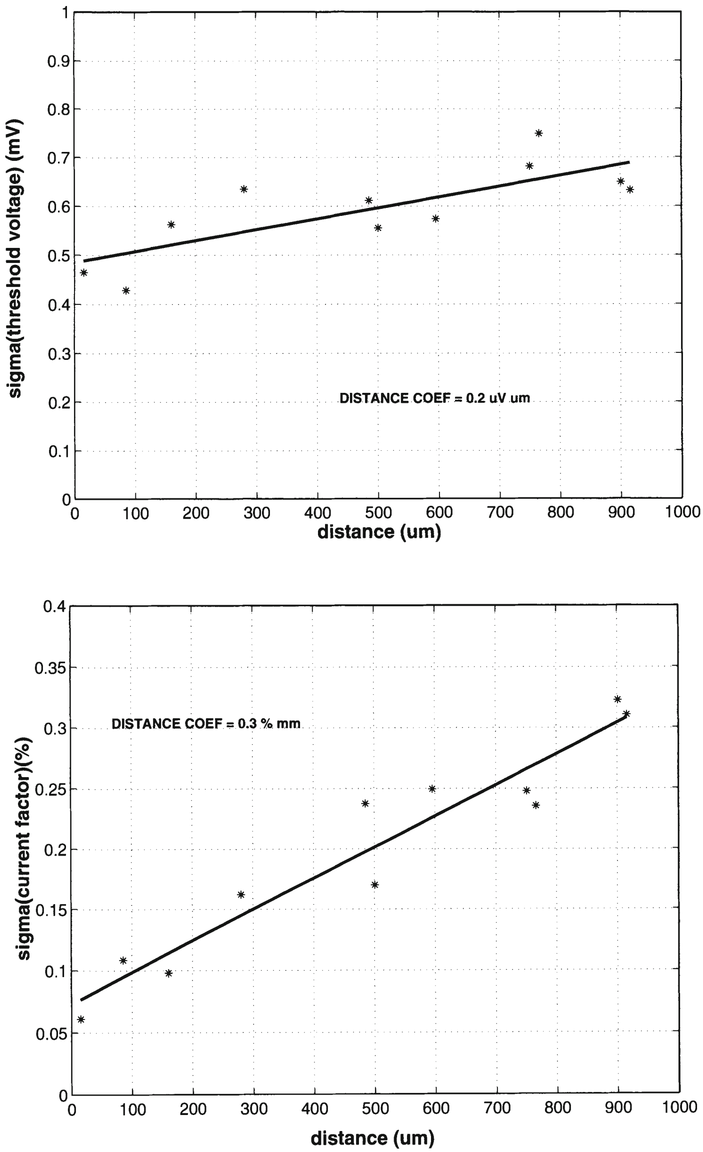


Figure 8.4: The threshold voltage and current factor mismatch as a function of the distance for a transistor with a W/L = 20/20 $\mu m/\mu m$

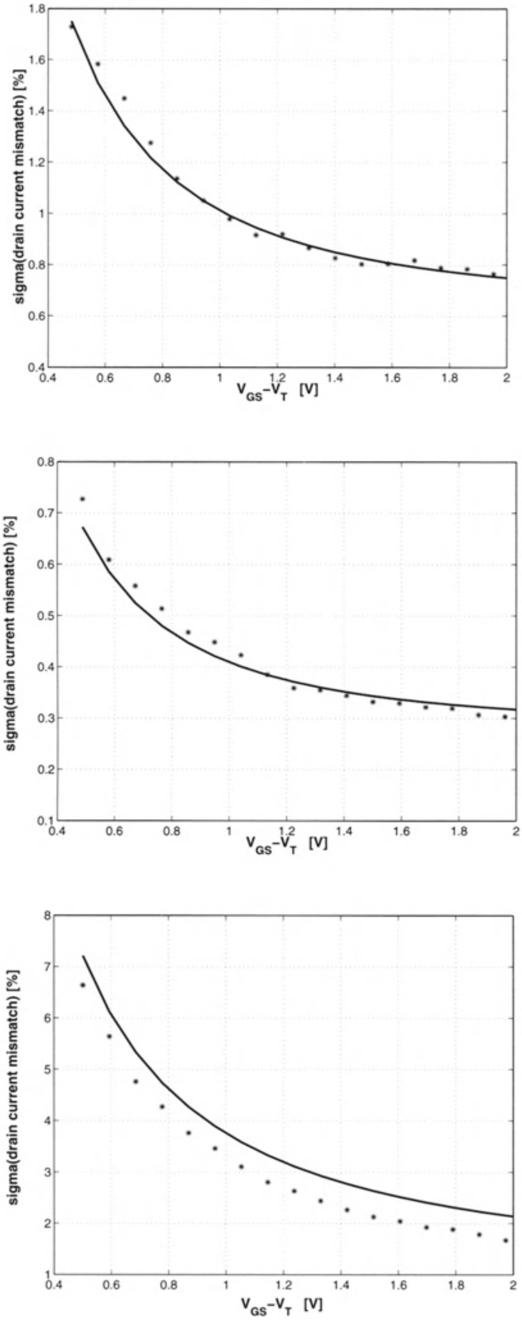


Figure 8.5: The measured (○) and calculated (solid line) drain current mismatch for transistors with a W/L = 1/4 (top), 3.2/10 (middle) and 0.8/0.5 (bottom) $\mu m/\mu m$

Pelgrom coef.	Extended Model		
$A_{VT}[mV\ \mu m]$	$A^2_{1VT}[mV\ \mu m]^2$	$A^2_{2VT}[mV\ \mu m]^2$	$A^2_{3VT}[mV\ \mu m]^2$
14.2	212.5	8.2	-2.4
$A_{\beta}[\%\ \mu m]$	$A^2_{1\beta}[\%\ \mu m]^2$	$A^2_{2\beta}[\%\ \mu m]^2$	$A^2_{3\beta}[\%\ \mu m]^2$
1.1	2.41	-0.5	0.3

Table 8.3: *Extracted mismatch parameters for a* $0.4\mu m$ *process (pMOS)*

8.5.3 Evolution of the mismatch parameters A_{VT} and A_{β}

In fig.8.6, the Pelgrom area proportionality mismatch parameters (A_{VT} and A_{β}) are given as a function of the gate-length. This data has been derived from [Pelgr IEDM98, Uyten CICC01] and from measurements at the ESAT-MICAS laboratory of the K.U. Leuven. As can be seen from this figure, the threshold voltage parameter A_{VT} scales down with technology due to its dependence on the oxide thickness. However, the current factor parameter A_{β} remains constant over the different technologies. Note that this implies that for technologies with a small gate-length, the contribution of the current factor mismatch gains in importance in comparison with the contribution of the threshold voltage mismatch.

8.6 Transistor mismatch dependency on its geometry

8.6.1 Introduction

In literature, the effect of the transistor topology on the mismatch behaviour has been investigated [Basto ICMTS96, Wong ICMTS95, Wong ICMTS96] for different layout structures. In the first part of this section, a short overview is given of the main results that have been published. In the second part, a new transistor layout structure will be discussed that has a low parasitic source and drain capacitance and a matching behaviour similar to that of the finger style topology [VdBos TSM00].

8.6.2 Different Layout Structures

In [Basto ICMTS96], the transistor mismatch dependency on topology is discussed. From measurements performed on a test structure processed in a 0.7 μm CMOS technology containing transistor pairs with an interdigitated quad, an interdigitated waffle , a finger and an interdigitated finger structure with dimensions W/L= 1050/1, 660/0.7, 336/3, 360/0.7, 100/1 60/0.7 $\mu m/\mu m$, the following conclusions have been drawn. The interdigitated waffle and quad transistor layout structures have a threshold

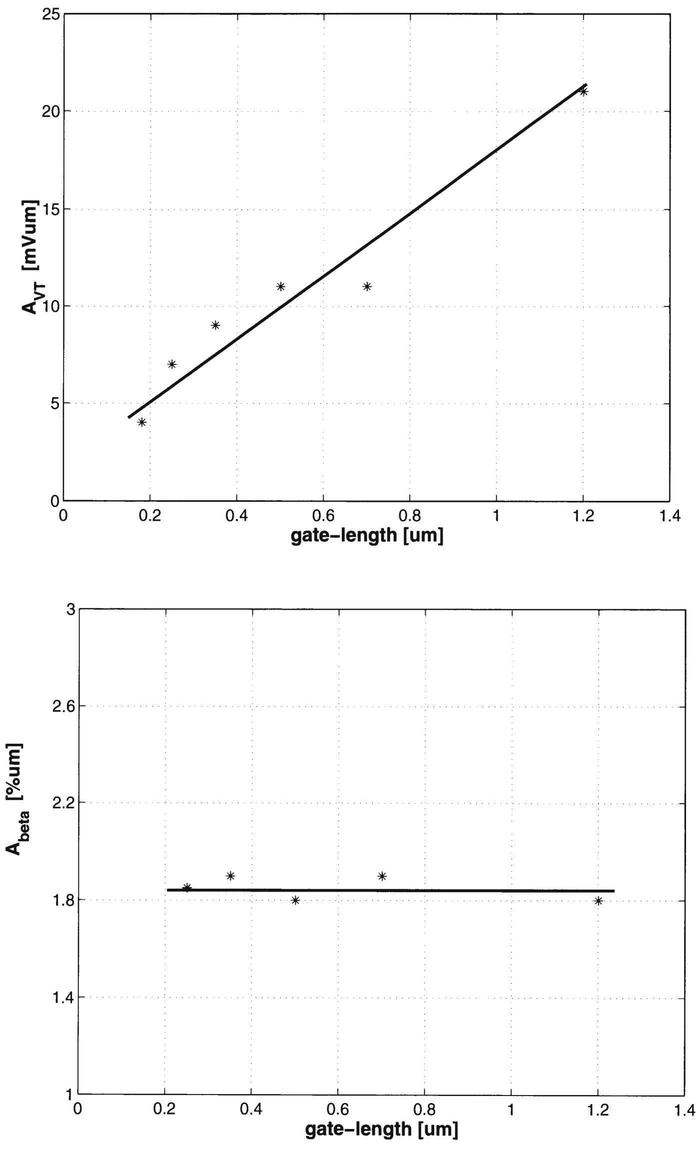


Figure 8.6: The mismatch parameters A_{VT} and A_β as a function of the gate-length for nMOS devices

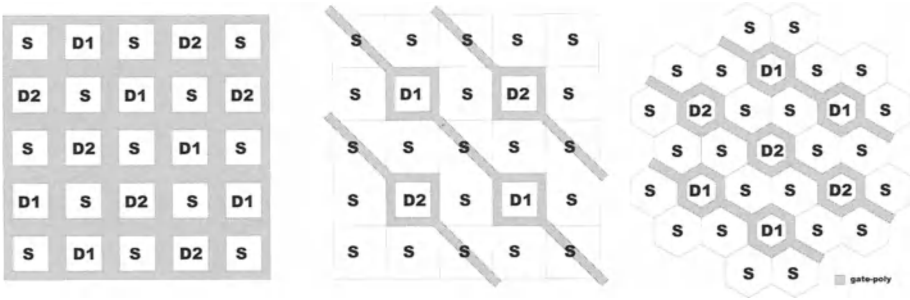


Figure 8.7: *the interdigitated waffle, the square structure and the hexagonal structure*

voltage and current factor mismatch that is inversely proportional to the square root of the area. However, for the finger style layouts, the current factor mismatch is considerably higher than that of the common centroid structures. This has been explained by stress that is induced into the silicon chip during packaging. More details can be found in [Basto TSM97].

In [Wong ICMTS95, Wong ICMTS96], the transistor mismatch has been investigated for an interdigitated quad transistor pair, a 90° rotated transistor pair, a parallel transistor pair and an interdigitated finger structure. Also here measurements prove that the best matching performance is given for the interdigitated quad topology although this behaviour can also be obtained by the parallel transistor pair if dummy transistors are placed. The 90° rotated structure shows the worst mismatch due to its inherent asymmetry.

8.6.3 The hexagonal transistor

8.6.3.1 Introduction

Since the demand for devices with a higher operation speed and a higher resolution increases, the role of the parasitic capacitance of the transistor is one of major importance. A small parasitic capacitance has not only a beneficial effect on the speed requirement but also on the power consumption of the chip which is one of the key issues in integrated design (GSM, DECT,...). Therefore alternative layout structures are implemented that reduce the parasitics as much as possible. The hexagonal transistor structure (fig.8.7.c) has a very small drain and source capacitance in comparison to the rectangle and finger style transistors that are generally used. In section 8.6.3.2, a general discussion of the hexagonal transistor is given. The structure of the transistor and the derivation of its effective W/L ratio is discussed in detail. In section 8.6.3.3, the drain and source capacitance as well as the matching parameters of the new transistor structure are calculated, measured and compared with the generally used transistors.

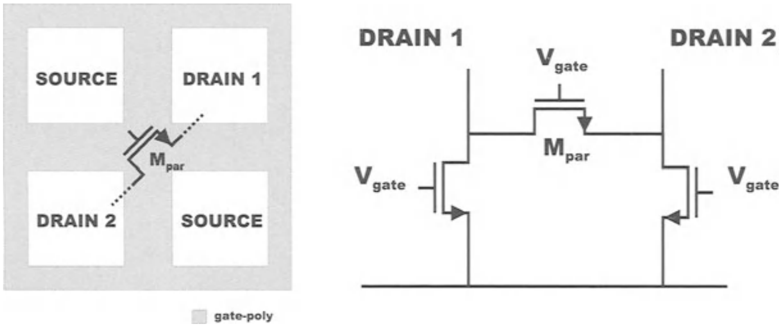


Figure 8.8: *The parasitic transistor created by the interdigitated waffle structure*

8.6.3.2 The Hexagonal Transistor

8.6.3.2.1 The transistor structure Fig.8.7.a represents a transistor pair implemented in an interdigitated waffle structure, which is recommended for optimum matching performance [Basto ICMTS96]. However, this implementation is not the equivalent of a simple MOS transistor pair circuit but is equivalent to the circuit of fig.8.8. A parasitic transistor M_{par} is formed between the drain areas of the two transistors of the transistor pair. To avoid this parasitic transistor, an alternative implementation is proposed. It is based on the principle that every drain area has to be surrounded by source areas so that no direct contact between any two drains is possible. This approach can be applied to the square drain areas of the waffle transistors (fig.8.7.b) leading however to a transistor with a large source area and thus a large source capacitance. This can be avoided by the use of a hexagonal shaped source and drain (fig.8.7.c). In section 8.6.3.3, it will be shown that this transistor structure has a source capacitance that is 25 percent lower than the source capacitance of the square transistor structure.

8.6.3.2.2 The transistor parameters Before this transistor can be effectively used in circuit design, the effective W/L ratio has to be determined. This ratio will be approximated by integrating the current-voltage relationship for an infinitesimal distance dx over the total gate length of the transistor (fig.8.9) [Gray, Laker]. The calculation is performed for a transistor operating at the transition point between the linear and the saturation region. In this case, the voltage drop along the channel is linear and $V_{gs} - V_T$ equals V_{ds}. The incremental voltage drop along the channel (with respect to the source) is given by:

$$v_{cs}(x) = \frac{v_{ds}(x - W_2 cos(30°))}{(W_1 - W_2)cos(30°)} \qquad \Rightarrow \qquad dv_{cs}(x) = \frac{v_{ds}dx}{(W_1 - W_2)cos(30°)} \tag{8.53}$$

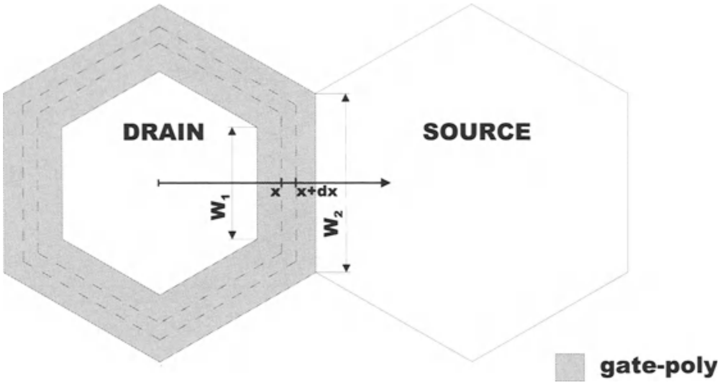


Figure 8.9: *Calculation of the effective width*

The incremental resistance depends on the mobile channel charge :

$$dR_{ds} = \frac{dx cos(30°)}{x} \frac{1}{\mu C_{ox}(v_{gs} - V_T - v_{cs}(x))} \tag{8.54}$$

The current through the channel is proportional to the voltage drop $dv_{cs}(x)$ across the resistance dR_{ds} :

$$\int_{W_1 cos(30°)}^{W_2 cos(30°)} i_{ds}\frac{dx}{x} = \int_{W_1 cos(30°)}^{W_2 cos(30°)} \frac{\mu C_{ox} v_{ds}^2}{(W_1 - W_2)^2 cos^2(30°)}(W_1 - \frac{x}{cos(30°)})dx \tag{8.55}$$

Taking into account the six sides of the hexagon, the current-voltage characteristic is given by :

$$6i_{ds} = \frac{6\mu C_{ox}}{2ln(\frac{W_2}{W_1})cos(30°)}(v_{gs} - V_T)^2 \qquad \Rightarrow \qquad (\frac{W}{L})_{eff} = \frac{6}{ln(\frac{W_2}{W_1})cos(30°)} \tag{8.56}$$

This formula links the geometrical parameters W_1 and W_2 of the gate all around the transistor to its effective W/L ratio. For a hexagonal transistor with a minimum gate-length L of $0.5\mu m$, the effective gate-width W equals $7.6\mu m$ taking into account the ground rules of the used technology. In [Grign ED82] an analytical expression for the W/L ratio can be found for transistors with a non-rectangular gate geometry. However, the analysis is quite complex. Solving the equations for the hexagonal transistor gives almost an identical result as the one obtained by eq.(8.56). In fig.8.10, the measured $I_{DS} - V_{GS}$ characteristic for different W/L 's is plotted versus Hspice simulations using the calculated W/L ratio. As can be seen from this figure, there exists a good agreement between the measured and the simulated values. For reasons of completeness, the value of the W/L ratio for the square structure is given here. The result can be easily verified by using a similar calculation as the one for the hexagonal transistor.

$$(\frac{W}{L})_{eff} = \frac{8}{ln(\frac{W_2}{W_1})} \tag{8.57}$$

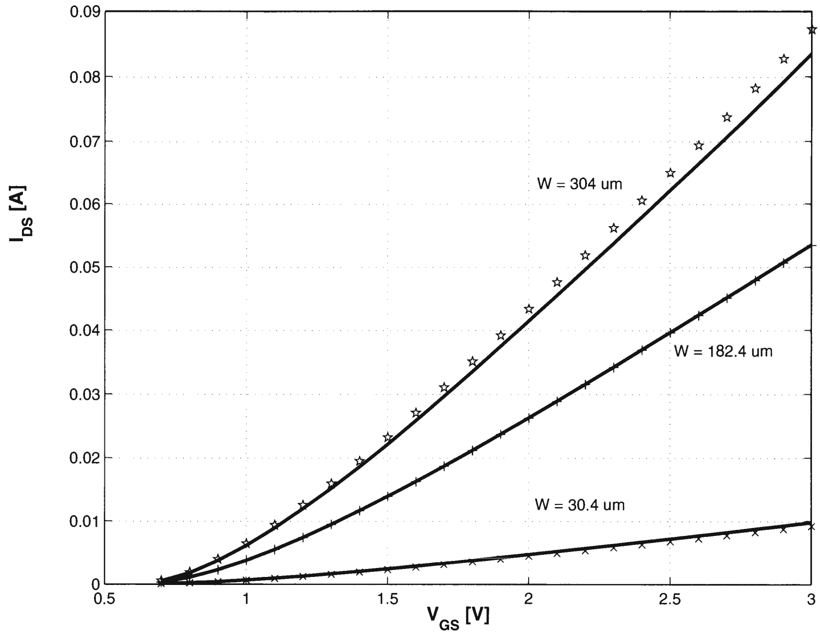


Figure 8.10: *The drain current characteristic of the hexagonal transistor*

8.6.3.3 Transistor Properties of the Hexagonal Structure

8.6.3.3.1 The drain and source capacitance The drain area of the hexagonal transistor is solely determined by the layout rules of the used technology (dimension of a contact hole, distance between a contact hole and the gate,...). As was indicated in section 8.6.3.2, the smallest possible effective W equals $7.6\mu m$ in the used $0.5\mu m$ technology. A comparison will now be made between the drain and source capacitance of the finger style and the hexagonal transistor structure. In a first step, the drain capacitance has been calculated. Using fig.8.11, the drain area of the hexagonal transistor can be easily calculated :

$$A_{D-HEX} = 3r^2 cos(30°) \tag{8.58}$$

For the finger style transistor, the drain area is minimal for an even number of fingers, since in that case no drain regions are located at the edges of the transistor. For the used technology, it can be determined that the minimal value for the transistor is given by :

$$A_{D-F} = 0.9\,W \tag{8.59}$$

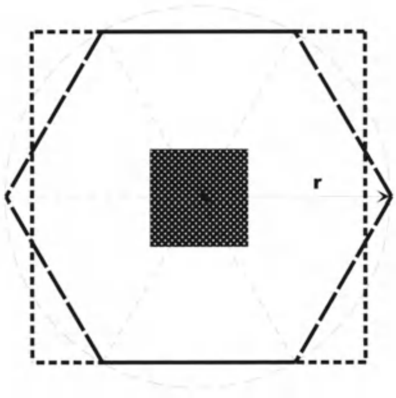


Figure 8.11: *Calculation of the drain area*

From eq.(8.58) and eq.(8.59), it follows that

$$\frac{A_{D-HEX}}{(W/L)_{HEX}} = \frac{2.9\,r^2}{W_{HEX}} \frac{A_{D-F}}{(W/L)_F} \tag{8.60}$$

where W_{HEX} represents the gate-width implemented by the hexagonal structure. For a minimum size transistor ($W_{HEX} = 7.6\,\mu m,\ L_{HEX} = 0.5\,\mu m$ and $r \simeq 1$), this ratio equals

$$\frac{A_{D-HEX}}{(W/L)_{HEX}} = 0.4\,\frac{A_{D-F}}{(W/L)_F} \tag{8.61}$$

which means a 60 % improvement for the hexagonal topology in comparison to the finger topology. Fig.8.12 shows the drain capacitance as a function of the effective width of the transistor for different layout structures.

It should be noted that in the presented structure - which will be referred to as D-HEX -, only the area of the drain has been minimised. Another interesting option is the minimisation of both the source and the drain area (SD-HEX structure) as is indicated in fig.8.13. This figure shows the hexagonal structure when respectively only the drain area (D-hex) and when both the drain and the source area (SD-hex) are optimised. Although the SD-hex structure has no influence on the drain capacitance (fig.8.12), it has a positive impact on the source area as will be explained in the next paragraph.

Since the source area of the hexagonal transistor has the same shape as the drain area, eq.(8.58) is still valid. However, when multiple unit drain hexagonals are used in a honeycomb type structure, each source is used for more than one drain. Taking into account the outer bounds of the matrix, which are source cells, every drain in the hexagonal structure has on the average three source areas. Also the value of the

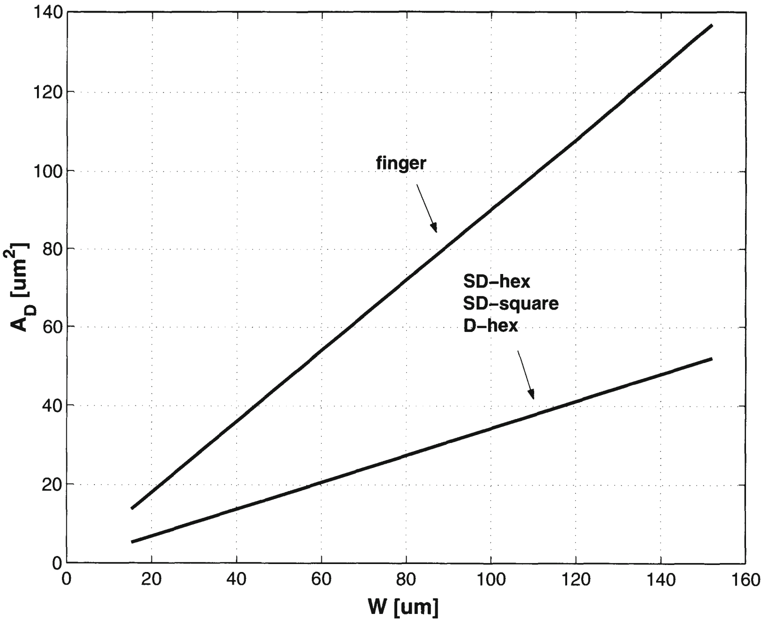


Figure 8.12: *The drain area for the D-HEX, SD-HEX and finger style structure*

parameter r^* will be different from the drain cells and equals $r^* = r + \frac{L_{HEX}}{sin(60°)}$.

$$A_{S-HEX} = 3\, r^{*2}\, cos(30°)\, \frac{\#sources}{drain} \tag{8.62}$$

For the finger style structure, the source area for an even number of fingers is given by :

$$A_{S-F} = (2\, H + (\frac{\#F}{2} - 1)\, H^*)\, \frac{W_F}{\#F} = z\, W_F \tag{8.63}$$

with $\#F$ the number of fingers (2,4,6,...), H the length of the source area at the edges of the transistor and H^* the lenght of the source area between two fingers. From eq.(8.62) and eq.(8.63), it follows that :

$$A_{S-HEX} = \frac{9\, r^{*2}\, cos(30°)}{z\, W_{HEX}}\, \frac{A_{S-F}}{(W/L)_F} \tag{8.64}$$

For a minimum size transistor this ratio reduces to ($r^* = 1.58\, \mu m$)

$$A_{S-HEX} = \frac{2.56}{z}\, \frac{A_{S-F}}{(W/L)_F} \tag{8.65}$$

which means that for a transistor with 2 fingers, the source area of the hexagonal structure is 70 % larger than that of the finger transistor. For the optimised SD-HEX

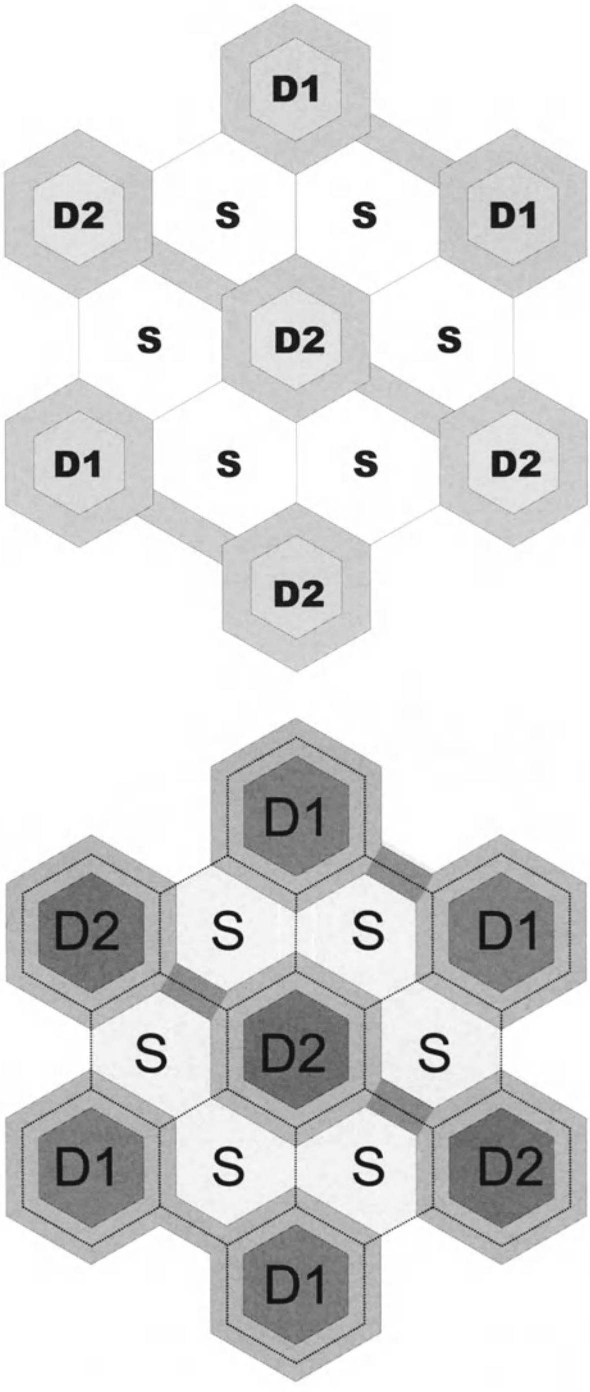


Figure 8.13: The D-HEX and the optimised SD-HEX structure

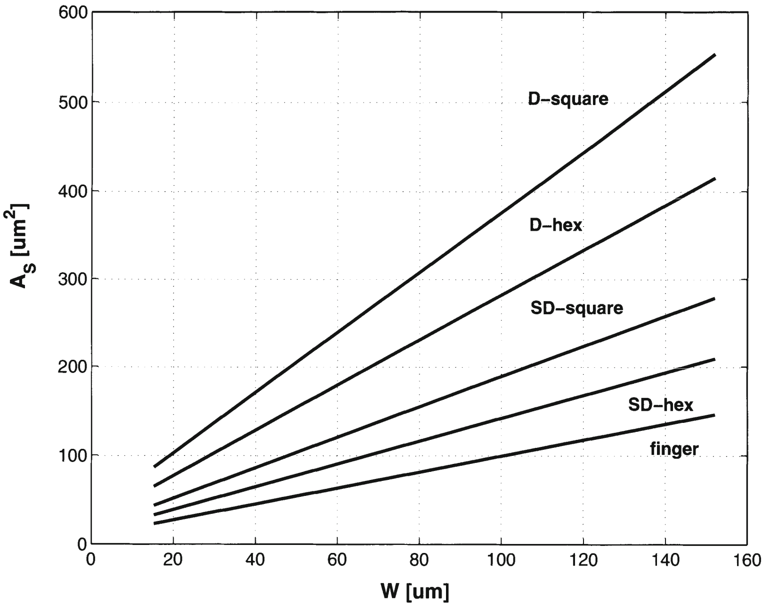


Figure 8.14: *The source area for the D-HEX, SD-HEX and finger style structure*

structure, the derivation for the source area is similar to the one of the D-HEX structure. The value of the r^* parameter changes and equals $r^* = r + \frac{L_{HEX}}{2sin(60°)}$. This implies an area decrease of 66 %. As a consequence, eq.(8.65) changes to

$$A_{S-HEX} = \frac{1.71}{z} \frac{A_{S-F}}{(W/L)_F} \tag{8.66}$$

which results in a source area for the hexagonal structure which is only 14 % larger than the finger style topology with 2 fingers. Fig.8.14 shows the source capacitance versus the effective W for different layout styles.

8.6.3.3.2 The matching performance For transistors with a large W/L ratio -as is the case for the hexagonal structure- the layout style has an influence on the matching performance of the devices [Basto PhD, Wong ICMTS95]. Common centroid geometries are recommended to reduce the effects of spatial parametric variations on the transistor mismatch [Vitto JSSC85, Elzin ICMTS96]. In this paragraph the mismatch results will be discussed for an interdigitated hexagonal transistor pair.

The structure has been implemented as a part of a test structure for a $0.5\mu m$ standard CMOS technology. Fig.8.15 shows the chip layout of this test structure. The technology has not been adapted in any way to make the realisation of this structure possible. Fifty test chips have been fabricated and measured. The transistor dimensions

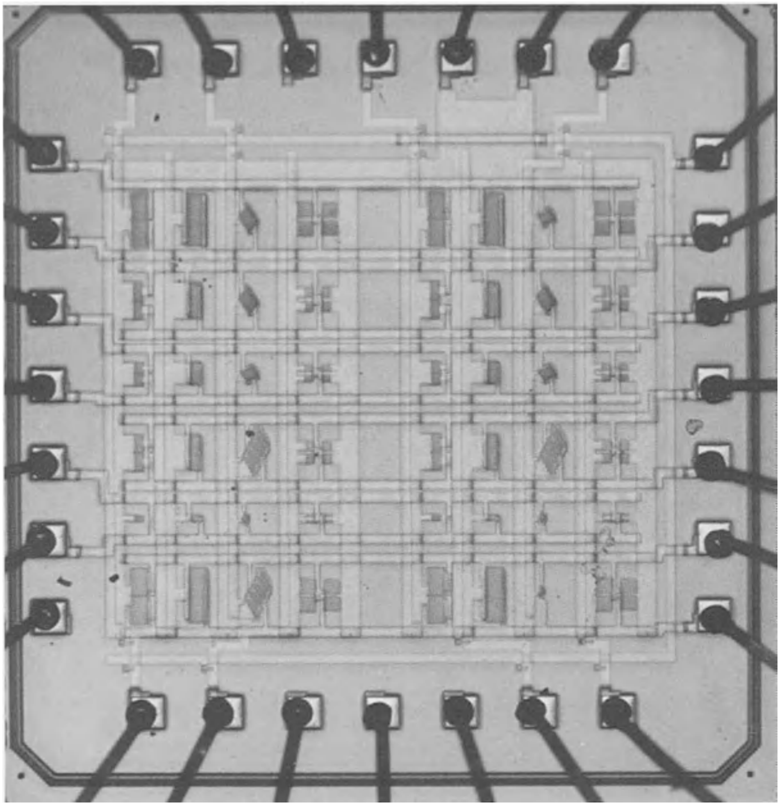


Figure 8.15: *The chip photograph of the test structure*

	$A_{VT}[mV\ \mu m]$	$A_{\beta}[\%\ \mu m]$
finger (L_{min})	11	1.2
hexagonal (L_{min})	8.5	0.93

Table 8.4: *Extracted mismatch parameters for the hexagonal structure*

of the hexagonal structure equal 304/0.5, 182.4/0.5 and 30.4/0.5$\mu m/\mu m$. The transistors have all been measured in the saturation region (V_{DS} = 3V), where mismatch is the most important for analog design. The matching parameters [Pelgr JSSC89] have been extracted using the method described in [Basto ICMTS96]. Fig.8.16.a and fig.8.16.b show the threshold voltage and the current factor standard deviation calculated with the data extracted from the test structure for the two layout styles : the interdigitated finger and the hexagonal transistor pair. It can be concluded from this figure that the mismatch behaviour of the hexagonal transistors is in the same order of magnitude as that of the finger style transistors. The values of the extracted mismatch parameters are given in table 8.4.

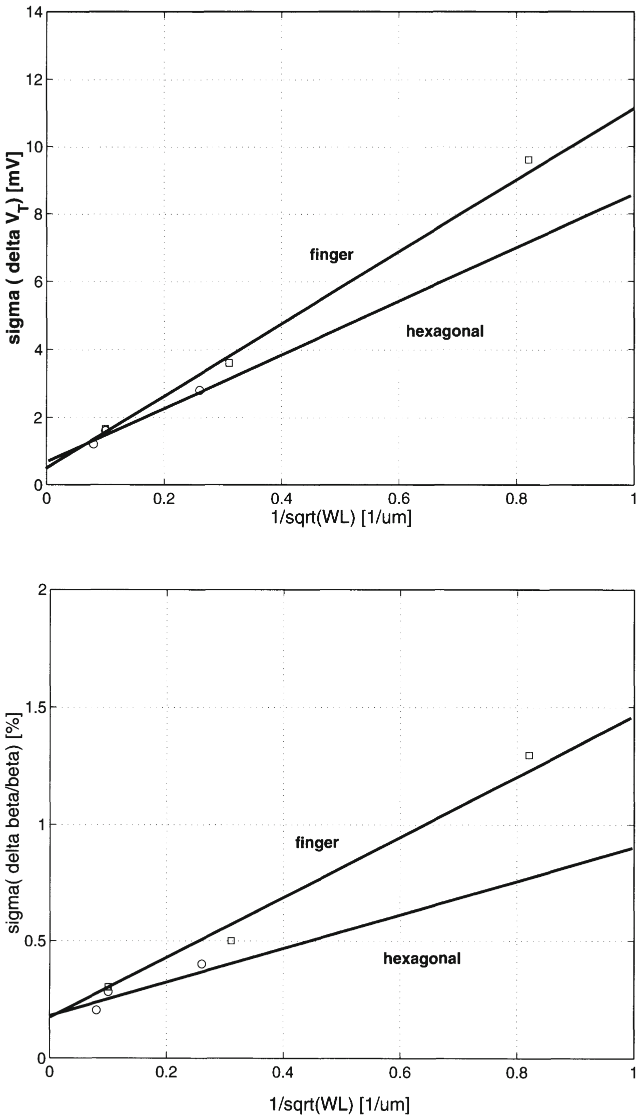


Figure 8.16: The matching behaviour of the hexagonal transistor

8.6.3.4 Conclusion

A new hexagonal transistor realised in a standard CMOS technology has been presented. The main advantages are the very low drain and source area. The I-V characteristics have been determined and compared to the simulations. The transistor pair mismatch has been investigated and compared with the results of traditional finger type transistors.

8.7 Influence of the surroundings of the transistors on the mismatch behaviour

8.7.1 Side Effects

In [Wong ICMTS95], a study has been made on the effect of dummy devices on the matching behaviour of a transistor pair. It has been shown that adding dummy structures can result in the same level of matching as the cross coupled pair while having a much better silicon area budget. In a current steering D/A converter one of the main building blocks is the current source array. To optimise the mismatch behaviour of the current sources, extra rows and columns of dummy transistors have to be added at the edges of the current source matrix.

8.7.2 Metal Coverage

To minimise the area in D/A converters, the routing needed for the complex switching schemes has been placed on top of the current source transistors. This has been done as symmetrically as possible. In [Tuinh ICMTS97], a large number of test structures is presented to evaluate the mismatch effects caused by metal lines running over matched transistor pairs. This subject gains in importance as more and more mixed signal IC's are processed in multilevel metal CMOS processes. In this paragraph, the most important results will be discussed. For more details the reader is referred to [Tuinh ICMTS97].

From measurements it seems that covering the transistor with metal lines has a significant effect. Asymmetrical metal coverage can cause drain current mismatches in the range of 3 to 50 %. Calculating the median value of the drain current mismatch distributions shows that both the reference and the symmetrically covered module have a value smaller than 0.5 %, while for the asymmetrical covered transistors this value is about 10 times larger. It has been concluded that the observed systematic mismatch is caused by mechanical stress that influences the mobility of the transistor.

8.8 The CMOS current steering D/A Converter as a test structure for transistor mismatch parameter extraction

8.8.1 Introduction

For the VLSI manufacturer, it is very important to be able to provide his customers with the necessary quantitative data of the matching quality of his technology. This can be achieved by the use of dedicated test circuits that are especially designed for this purpose (low parasitics, high measurement accuracy, ...). Furthermore, it is necessary to follow the evolution of the matching performance of the technology over different runs in time. This requires a test circuit that gives a good indication of the matching technology for a low cost. However, these circuits are of no further use to the manufacturer or the designer and therefore an alternative structure has been sought for. A high performance current steering D/A converter is highly dependent on the matching quality of the technology. Furthermore, the evaluation of the D/A converter's performance poses no problem since there exist standardised test procedures. In this chapter, the D/A converter's performance will be directly translated in the matching characteristics of the technology. In section 8.8.2 the test structures as they exist now will be shortly discussed. In the next section the D/A converter approach will be presented. In this section the transistor mismatch extraction procedure is discussed. Measurement results will be shown and evaluated for a standard 0.5 μm CMOS technology.

8.8.2 The Test Structure Approach

Until now, the test circuits are constructed using an array of about fifty to one hundred transistors with some additional digital/selection logic to be able to automate the measurements to a large extent [Basto ICMTS95, Serra ISCAS98]. Fig. 8.17 shows a simplified schematic of such a test structure. The dimensions of the transistors are chosen in such a way that the $\frac{1}{\sqrt{WL}}$ of the transistors span a wide interval. By measuring a large number of chips, an accurate determination of the mismatch parameter coefficients A_β and A_{VT} of the Pelgrom equation [Pelgr JSSC89] is possible. The test chip normally exhibits a large geometrical symmetry and consists of an array of identical cells. Furthermore, the chip area has to be kept as small as possible to reduce the cost. As is mentioned earlier, these test chips have no other function than the mismatch determination of the technology and as such are a cost overhead to the manufacturer. Therefore, both designers as manufacturers are looking for alternative structures to determine the transistor mismatch parameters.

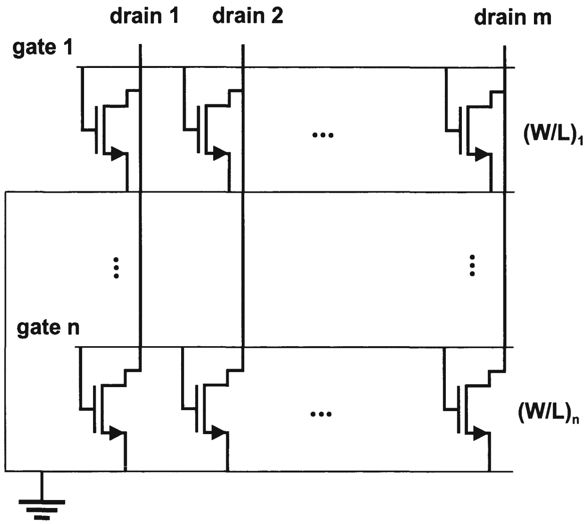


Figure 8.17: *An example of the test structure approach*

8.8.3 The D/A converter Approach

8.8.3.1 The D/A converter architecture as a test structure

A current steering D/A converter is based on an array of identical current cells. The achievable resolution is dictated by the matching properties of these cells which are determined by the used process and can only be improved by increasing the current source transistor sizes or/and adjusting the bias voltages. The close relationship between the D/A converter's performance and the transistor mismatch parameters makes this circuit the ideal process monitor. An unary architecture is chosen because in this implementation every current source can be accessed separately while in a binary implementation the current sources are grouped bit by bit. The resolution of the D/A converter is chosen to be seven or eight bits so that enough measurements are available to be able to work with gaussian distributions when processing the data. Such a D/A converter can also be used in applications in the area of e.g. video and HDTV. This leads us to the major advantage of using a current-steering D/A converter as a test structure. This chip has further use, no silicon area has gone to waste. Furthermore, the chip area can be made in the order of only a few mm^2. This enables the manufacturer to place it on every wafer having in this way a continuous monitoring of his technology.

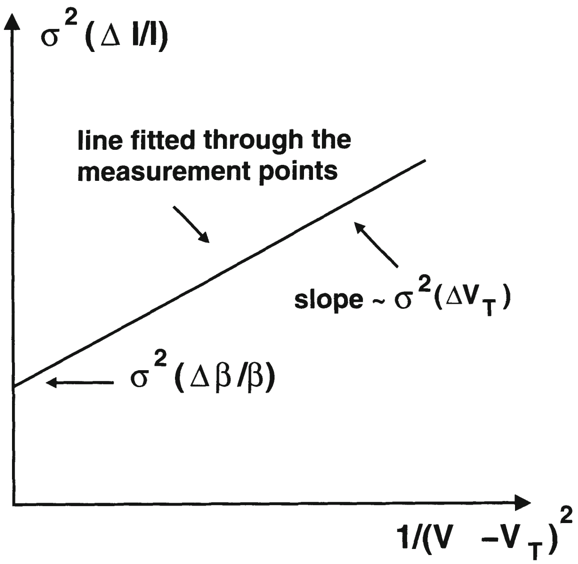


Figure 8.18: *The quadratic relative standard deviation in function of the inverse quadratic gate overdrive voltage of the current source transistors*

8.8.3.2 Measurement and extraction procedure of the transistor mismatch parameters

The measurements of the D/A converter can be fully automated. A program has been written that drives the data generator (HP80000) so that for different values for the $(V_{GS} - V_T)_{CS}$ the current at the output of the D/A converter will be measured (HP3457 multimeter) and stored. At this point, for every gate overdrive voltage of the current source transistors $2^N - 1$ measured currents are available leading to $2^N - 2$ transistor pairs (I_i, I_{i+1}) with i=1 to $2^N - 2$. Using MATLAB the values for the relative current difference $(\Delta I/I)$ and the relative current standard deviation $\sigma(\Delta I/I)$ can be easily calculated. This gives us a curve as is depicted in fig.8.18.

The formula for the relative current standard deviation is easily derived (from eq.8.3):

$$\sigma^2(\frac{\Delta I}{I}) = \sigma^2(\frac{\Delta\beta}{\beta}) + \frac{4\sigma^2(\Delta V_T)}{(V_{GS} - V_T)^2} \tag{8.67}$$

For large values of the gate overdrive voltage this formula simplifies to :

$$\sigma^2(\frac{\Delta I}{I}) = \sigma^2(\frac{\Delta\beta}{\beta}) \tag{8.68}$$

Using Pelgrom's equation a value for the current factor mismatch coefficient can then be calculated :

$$A_{\beta}^{2} = \sigma^{2}(\frac{\Delta\beta}{\beta}) * W * L = \sigma^{2}(\frac{\Delta I}{I}) * W * L \tag{8.69}$$

On the other hand, for small values of $(V_{GS} - V_T)_{CS}$ the influence of the current factor is negligible and the relative unit current standard deviation is given by :

$$\sigma^{2}(\frac{\Delta I}{I}) = \frac{4\sigma^{2}(\Delta V_T)}{(V_{GS} - V_T)^{2}} \tag{8.70}$$

Since the characteristic of fig.8.18 is a straight line, a fitting of $\sigma^2(\Delta I/I)$ calculated from the measurement data gives us the threshold voltage mismatch coefficient :

$$slope = \frac{\Delta\left[\sigma^{2}(\frac{\Delta I}{I})\right]}{\Delta\left[\frac{1}{(V_{GS}-V_T)_{CS}^{2}}\right]} = 4\sigma^{2}(\Delta V_T) \Rightarrow A_{VT}^{2} = \frac{slope * W * L}{4} \tag{8.71}$$

To avoid the influence of edge effects dummy rows and columns are used in current-steering D/A converters. This guarantees the fact that all measured transistor pairs have identical surroundings.

8.8.3.3 Measurement Results

As an example, the six bit current-steering D/A converter has been measured. The chip has been processed in a standard CMOS 0.5 μm technology. The results of the extraction procedure described in the previous paragraph are given in fig.8.19. From this figure the following values for the mismatch parameter coefficients can be obtained using eq.(8.69) and eq.(8.71) :

$$A_{VT} = 8mV\mu m$$

$$A_{\beta} = 1.2\%\mu m \tag{8.72}$$

These results are in good agreement with the results obtained from the test structures described in the previous chapter ($A_{VT} = 7.3mV\mu m$ and $A_{\beta} = 1.3\%$). This proves that the D/A converter can indeed be successfully used as a process monitor for mismatch.

8.9 Conclusion

In this chapter, different transistor mismatch models have been discussed that have been presented in literature over the last two decades. It should be noted that the

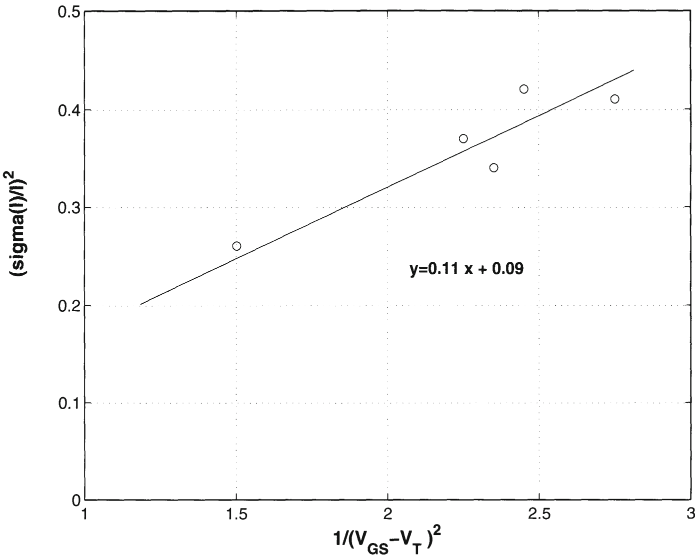


Figure 8.19: *The measured quadratic relative current standard deviation in function of the inverse of the quadratic gate overdrive voltage of the current sources*

models referred to in this text are by no means complete. A lot of research has been done on transistor mismatch models in the last few years. The main goal of presenting these models was to provide the reader with some important notions about the issue of transistor mismatch and its importance for analog designers.

Based on the Pelgrom model and the model of Bastos, the transistor mismatch parameters have been extracted for a standard 0.5 μm CMOS technology and a preliminary 0.4 μm CMOS technology. The transistor mismatch dependency on the topology has not only been addressed but a new topology has been presented. The hexagonal structure combines a very low drain capacitance with a source capacitance and a mismatch behaviour similar to the one of a finger style transistor.

In the last section of this chapter, the CMOS current steering D/A converter has been presented as a test structure for matching purposes. Using a D/A converter as a test structure has a large cost advantage. The time to implement these circuits and the silicon area they consume can now be used to design a circuit (=D/A converter) that can be further used in e.g. telecommunication applications. The extraction procedure of the transistor mismatch parameters has been described and has been verified by comparing the measurement results of a six bit current-steering D/A converter and the results obtained from the classical test structures.

Appendix 1

In chapter 4, a formula has been derived that accurately predicts the relationship between the mismatch behaviour of the current source transistors and the achievable INL_yield of the current steering D/A converter. This formula is given by :

$$\frac{\sigma(I)}{I} = \frac{1}{2\sqrt{2^N}C} \qquad (.1)$$

with $C = inv\_norm_{(-x,x)}(0.5 + \frac{INL\_yield}{2}) = inv\_norm_{(-x,x)}(z)$.

In fig.1, the value for C can be found starting from the specification for the INL_yield or starting from the calculated value for z. From the same figure, it is also possible to obtain the value for C starting from the allowed percentage of faults that can be tolerated.

z	inv_norm(z)	Yield [%]	Faults[%]		z	inv_norm(z)	Yield [%]	Faults[%]
0.50000	0.00	0.00	100.00					
0.57926	0.20	15.85	84.15		0.99931286	3.20	99.862572	1.37E-01
0.65542	0.40	31.08	68.92		0.99966307	3.40	99.932614	6.74E-02
0.72575	0.60	45.15	54.85		0.99984089	3.60	99.968178	3.18E-02
0.78814	0.80	57.63	42.37		0.99992765	3.80	99.985530	1.45E-02
0.84134	1.00	68.27	31.73		0.99996833	4.00	99.993666	6.33E-03
0.88493	1.20	76.99	23.01		0.99998665	4.20	99.997331	2.67E-03
0.91924	1.40	83.85	16.15		0.99999459	4.40	99.998917	1.08E-03
0.94520	1.60	89.04	10.96		0.99999789	4.60	99.999578	4.22E-04
0.96407	1.80	92.81	7.19		0.99999921	4.80	99.999841	1.59E-04
0.97725	2.00	95.45	4.55		0.99999971	5.00	99.999943	5.73E-05
0.98610	2.20	97.22	2.78		0.99999990	5.20	99.999980	1.99E-05
0.99180	2.40	98.36	1.64		0.99999997	5.40	99.999993	6.66E-06
0.99534	2.60	99.07	0.93		0.99999999	5.60	99.999998	2.14E-06
0.99744	2.80	99.49	0.51		1.00000000	5.80	99.999999	6.63E-07
0.99865	3.00	99.73	0.27		1.00000000	6.00	100.000000	1.97E-07

Figure .1: *The value of the parameter C as a function of the INL_yield*

Appendix 2

In this appendix, the complete derivation of the SFDR as a function of the second and third order harmonic distortion is given. The main results were given in chapter 5.

For a $sin(\omega t)$ output signal, the number of switches T that conduct current at a time t equals :

$$T(t) = S\left(\frac{1 + sin(\omega t)}{2}\right) \tag{.1}$$

with S the total number of current sources.

The output voltage of the D/A converter is determined by the product of the current and the load impedance Z_L in parallel with T(t) parallel impedances Z_{imp}:

$$V_{out}(t) = \frac{SI(1 + sin(\omega t))}{2Y_L + Y_{imp}S(1 + sin(\omega t))} \tag{.2}$$

with I the current through one switch transistor and $Y_{imp} = 1/Z_{imp}$ and $Y_L = 1/Z_L$. A Taylor series expansion of the output voltage allows to determine the influence of Y_{imp} on the dynamic performance of the D/A converter.

$$V_{out}(t) = (DC)_{cf} + Asin(\omega t) + Bsin^2(\omega t) + Csin^3(\omega t) + Dsin^4(\omega t) + \ldots \tag{.3}$$

with :

$$
\begin{aligned}
(DC)_{coef} &= \frac{SI}{2Y_L + Y_{imp}S} \\
A &= 2\,Y_L \frac{(DC)_{coef}}{2Y_L + Y_{imp}S} \\
B &= -A * \frac{Y_{imp}S}{2Y_L + Y_{imp}S} \\
C &= A * \left(\frac{Y_{imp}S}{2Y_L + Y_{imp}S}\right)^2 \\
D &= -A * \left(\frac{Y_{imp}S}{2Y_L + Y_{imp}S}\right)^3 \\
E &= A * \left(\frac{Y_{imp}S}{2Y_L + Y_{imp}S}\right)^4 \\
F &= -A * \left(\frac{Y_{imp}S}{2Y_L + Y_{imp}S}\right)^5
\end{aligned} \tag{.4}
$$

Applying goniometric relations and rearranging of the terms leads to the following result:

$$
\begin{aligned}
V_{out}(t) = (DC)_{cf} + \left[A + \frac{3C}{4}\right] sin(\omega t) + \left[\frac{-(B+D)}{2}\right] cos(2\omega t) \\
-\left[\frac{C}{4} + \frac{5E}{16}\right] sin(3\omega t) + \left[\frac{D}{8} + \frac{3F}{16}\right] cos(4\omega t)\ldots
\end{aligned} \tag{.5}
$$

Inserting the values of eq.4 in eq.5 in order to determine the influence of the second order harmonic on the SFDR performance of a current steering D/A converter, yields the following result :

$$
Q = \frac{1}{4}\frac{(2Y_L + SY_{imp})(16Y_L^2 + 16SY_LY_{imp} + 7S^2Y_{imp}^2)}{SY_{imp}(2Y_L^2 + 2SY_LY_{imp} + S^2Y_{imp}^2)} \tag{.6}
$$

The influence of the third order harmonic is described by :

$$
T = -4\frac{(2Y_L + SY_{imp})^2(16Y_L^2 + 16SY_LY_{imp} + 7S^2Y_{imp}^2)}{S^2Y_{imp}^2(16Y_L^2 + 16SY_LY_{imp} - 9S^2Y_{imp}^2)} \tag{.7}
$$

Since $Y_L \gg SY_{imp}$, both equations can be further simplified to :

$$
Q = \frac{4Y_L + 2SY_{imp}}{SY_{imp}} \approx \frac{4Y_L}{SY_{imp}} \tag{.8}
$$

and

$$T = \left(\frac{4Y_L + 2SY_{imp}}{SY_{imp}}\right)^2 \approx \left(\frac{4Y_L}{SY_{imp}}\right)^2 \tag{.9}$$

or written in function of the SFDR :

$$HD2 = 20\,log(Q) = 20\,log(\frac{4Z_{imp}}{SZ_L}) \tag{.10}$$

and

$$HD3 = 20\,log(T) = 20\,log(\frac{4Z_{imp}}{SZ_L})^2 \tag{.11}$$

From this derivation, it becomes clear that if for a 10 bit D/A converter the HD2 distortion component equals 60 dB, the HD3 distortion component equals 120dB. It can therefore be concluded that the influence of the frequency dependency of the output impedance has a negligible effect on the third order harmonic distortion component.

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